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Freier; Susan M. et al.

Compounds and Methods for Modulating UBE3A-ATS

Abstract

Provided are compounds, methods, and pharmaceutical compositions for reducing the amount or activity of UBE3A-ATS, the endogenous antisense transcript of ubiquitin protein ligase E3A (UBE3A) in a cell or subject, and in certain instances increasing the expression of paternal UBE3A and the amount of UBE3A protein in a cell or subject. Such compounds, methods, and pharmaceutical compositions are useful to ameliorate at least one symptom or hallmark of a neurogenetic disorder. Such symptoms and hallmarks include developmental delays, ataxia, speech impairment, sleep problems, seizures, and EEG abnormalities. Such neurogenetic disorders include Angelman Syndrome.

Inventors: Freier; Susan M. (San Diego, CA), Bui; Huynh-Hoa (San Diego, CA), Jafar-nejad; Paymaan (Carlsbad, CA), Rigo; Frank (Carlsbad, CA)

Applicant: Ionis Pharmaceuticals, Inc. (Carlsbad, CA)

Family ID: 72666233

Assignee: Ionis Pharmaceuticals, Inc. (Carlsbad, CA)

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Background/Summary

SEQUENCE LISTING

[0001] The present application contains a Sequence Listing which has been submitted electronically in XML format. Said XML copy, created on Sep. 29, 2023, is named "BIOL0349SEQ.xml" and is 3,790,521 bytes in size. The information in the electronic format of the sequence listing is incorporated herein by reference in its entirety.

FIELD

[0002] Provided are compounds, methods, and pharmaceutical compositions for reducing the amount or activity of UBE3A-ATS, the endogenous antisense transcript of ubiquitin protein ligase E3A (UBE3A) in a cell or subject, and in certain instances increasing the expression of paternal UBE3A and the amount of UBE3A protein in a cell or subject. Such compounds, methods, and pharmaceutical compositions are useful to ameliorate at least one symptom or hallmark of a neurogenetic disorder. Such symptoms and hallmarks include developmental delays, ataxia, speech impairment, sleep problems, seizures, and EEG abnormalities. Such neurogenetic disorders include

[0003] Angelman Syndrome.

BACKGROUND

[0004] Angelman Syndrome (AS) is a developmental disorder that affects ~1/15,000 live births and is caused by a deficiency of ubiquitin protein ligase E3A (UBE3A). Of the two copies of the UBE3A gene in neurons, the maternal gene is typically expressed, while the paternal gene is subject to genetic imprinting and silencing. Angelman Syndrome is caused when a functional copy of the UBE3A gene is not inherited from the mother, either because of a mutation, a deletion, paternal uniparental disomy of chromosome 15, or an imprinting defect. See, Buiting, et al., "Angelman Syndrome-insights into a rare neurogenetic disorder", *Nature Rev. Neuro.*, 2016, 12:584-593.

[0005] Angelman Syndrome patients experience developmental delays and speech impairment, and commonly experience sleep problems, seizures, and EEG abnormalities. The disorder is usually diagnosed in the first few years of life and the diagnosis can be confirmed by genetic testing. However, therapies for Angelman Syndrome remain limited and focus mainly on symptomatic management. See, Williams, C. A. et al., Genet. Med., 12:385-395, 2010.

[0006] Recently, topoisomerase inhibitors currently used in cancer treatment were found to “unsilence” paternal UBE3A expression in both a neuronal culture system and mice. See, Huang, H. S. et al., Nature, 481:185-189, 2012. However, the exact mechanism of unsilencing paternal UBE3A expression remains unknown and topoisomerase inhibitors are fraught with safety concerns because they are known to be non-specific and capable of inducing DNA damage, such as single and double-strand breaks.

[0007] Currently there is a lack of acceptable options for treating neurogenetic disorders such as AS. It is therefore an object herein to provide compounds, methods, and pharmaceutical compositions for the treatment of such diseases.

SUMMARY OF THE INVENTION

[0008] Provided herein are compounds, methods and pharmaceutical compositions for reducing the amount or activity of UBE3A-ATS RNA, and in certain embodiments increasing the expression of paternal UBE3A RNA or protein in a cell or subject. In certain embodiments, the subject has a neurogenetic disorder. In certain embodiments, the subject has Angelman Syndrome (AS). In certain embodiments, compounds useful for reducing expression of UBE3A-ATS RNA, or a portion thereof, are oligomeric compounds. In certain embodiments, compounds useful for reducing expression of UBE3A-ATS RNA, or a portion thereof, are modified oligonucleotides. In certain embodiments, compounds useful for increasing expression of paternal UBE3A RNA or protein are oligomeric compounds. In certain embodiments, compounds useful for increasing expression of paternal UBE3A RNA or protein are modified oligonucleotides.

[0009] Also provided are methods useful for ameliorating at least one symptom or hallmark of a neurogenetic disorder. In certain embodiments, the neurogenetic disorder is Angelman Syndrome. In certain embodiments, the symptom or hallmark includes developmental delays, ataxia, speech impairment, sleep problems, seizures, and/or EEG abnormalities.

Description

DETAILED DESCRIPTION OF THE INVENTION

[0010] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive. Herein, the use of the singular includes the plural unless specifically stated otherwise. As used herein, the use of “or” means “and/or” unless stated otherwise. Furthermore, the use of the term “including” as well as other forms, such as “includes” and “included”, is not limiting. Also, terms such as “element” or “component” encompass both elements and components comprising one unit and elements and components that comprise more than one subunit, unless specifically stated otherwise.

[0011] The section headings used herein are for organizational purposes only and are not to be construed as limiting the subject matter described. All documents, or portions of documents, cited in this application, including, but not limited to, patents, patent applications, articles, books, and treatises, are hereby expressly incorporated-by-reference for the portions of the document discussed herein, as well as in their entirety.

Definitions

[0012] Unless specific definitions are provided, the nomenclature used in connection with, and the procedures and techniques of, analytical chemistry, synthetic organic chemistry, and medicinal and pharmaceutical chemistry described herein are those well-known and commonly used in the art. Where permitted, all patents, applications, published applications and other publications and other data referred to throughout in the disclosure are incorporated by reference herein in their entirety.

[0013] Unless otherwise indicated, the following terms have the following meanings:

Definitions

[0014] As used herein, “2'-deoxynucleoside” means a nucleoside comprising a 2-H(H) deoxyribosyl sugar moiety. In certain embodiments, a 2'-deoxynucleoside is a 2-β-D-deoxynucleoside and comprises a 2'-β-D-deoxyribosyl sugar moiety, which has the β-D configuration as found in naturally occurring deoxyribonucleic acids (DNA). In certain embodiments, a 2'-deoxynucleoside may comprise a modified nucleobase or may comprise an RNA nucleobase (uracil).

[0015] As used herein, “2'-MOE” or “2''-MOE sugar moiety” means a 2'-OCH₂CH₂OCH₃ group in place of the 2'-OH group of a ribosyl sugar moiety. “MOE” means methoxyethyl. Unless otherwise indicated, a 2'-MOE has the β-D stereochemical configuration.

[0016] As used herein, “2'-MOE nucleoside” means a nucleoside comprising a 2'-MOE sugar moiety.

[0017] As used herein, “2'-OMe” or “2''-O-methyl sugar moiety” means a 2'-OCH₃ group in place of the 2'-OH group of a ribosyl sugar moiety. Unless otherwise indicated, a 2'-OMe has the β-D stereochemical configuration.

[0018] As used herein, “2'-OMe nucleoside” means a nucleoside comprising a 2'-OMe sugar moiety.

[0019] As used herein, “2'-substituted nucleoside” means a nucleoside comprising a 2'-substituted sugar moiety.

[0020] As used herein, “2'-substituted” in reference to a sugar moiety means a sugar moiety comprising at least one 2'-substituent group other than H or OH.

[0021] As used herein, “5-methyl cytosine” means a cytosine modified with a methyl group attached to the 5 position. A 5-methyl cytosine is a modified nucleobase.

[0022] As used herein, “administering” means providing a pharmaceutical agent to a subject.

[0023] As used herein, “antisense activity” means any detectable and/or measurable change attributable to the hybridization of an antisense compound to its target nucleic acid. In certain embodiments, antisense activity is a decrease in the amount or expression of a target nucleic acid or protein encoded by such target nucleic acid compared to target nucleic acid levels or target protein levels in the absence of the antisense compound.

[0024] As used herein, “antisense compound” means an oligomeric compound capable of achieving at least one antisense activity.

[0025] As used herein, “ameliorate” in reference to a treatment means improvement in at least one symptom relative to the same symptom in the absence of the treatment. In certain embodiments, amelioration is the reduction in the severity or frequency of a symptom or the delayed onset or slowing of progression in the severity or frequency of a symptom. In certain embodiments, the symptom or hallmark is developmental delays, ataxia, speech impairment, sleep problems, seizures, and EEG abnormalities.

[0026] As used herein, “bicyclic nucleoside” or “BNA” means a nucleoside comprising a bicyclic sugar moiety.

[0027] As used herein, “bicyclic sugar” or “bicyclic sugar moiety” means a modified sugar moiety comprising two rings, wherein the second ring is formed via a bridge connecting two of the atoms thereby forming a bicyclic structure. In certain embodiments, the first ring of the bicyclic sugar moiety is a furanosyl moiety. In certain embodiments, the bicyclic sugar moiety does not comprise a furanosyl moiety.

[0028] As used herein, “cleavable moiety” means a bond or group of atoms that is cleaved under physiological conditions, for example, inside a cell, a subject, or a human.

[0029] As used herein, “complementary” in reference to an oligonucleotide means that at least 70% of the nucleobases of the oligonucleotide or one or more portions thereof and the nucleobases of a target nucleic acid or one or more portions thereof are capable of hydrogen bonding with one another when the nucleobase sequence of the oligonucleotide and the other nucleic acid are aligned in opposing directions. As used herein, complementary nucleobases means nucleobases that are capable of forming hydrogen bonds with one another. Complementary nucleobase pairs include adenine (A) and thymine (T), adenine (A) and uracil (U), cytosine (C) and guanine (G), 5-methyl cytosine (mC) and guanine (G).

Complementary oligonucleotides and/or target nucleic acids need not have nucleobase complementarity at each nucleoside. Rather, some mismatches are tolerated. As used herein, “fully complementary” or “100% complementary” in reference to an oligonucleotide, or portion thereof, means that the oligonucleotide, or portion thereof, is complementary to another oligonucleotide or target nucleic acid at each nucleobase of the shorter of the two oligonucleotides, or at each nucleoside if the oligonucleotides are the same length.

[0030] As used herein, “conjugate group” means a group of atoms that is directly or indirectly attached to an oligonucleotide. Conjugate groups include a conjugate moiety and a conjugate linker that attaches the conjugate moiety to the oligonucleotide.

[0031] As used herein, “conjugate linker” means a single bond or a group of atoms comprising at least one bond that connects a conjugate moiety to an oligonucleotide.

[0032] As used herein, “conjugate moiety” means a group of atoms that is attached to an oligonucleotide via a conjugate linker.

[0033] As used herein, “contiguous” in the context of an oligonucleotide refers to nucleosides, nucleobases, sugar moieties, or internucleoside linkages that are immediately adjacent to each other. For example, “contiguous nucleobases” means nucleobases that are immediately adjacent to each other in a sequence.

[0034] As used herein, “constrained ethyl” or “cEt” or “cEt modified sugar moiety” means a β -D ribosyl bicyclic sugar moiety wherein the second ring of the bicyclic sugar is formed via a bridge connecting the 4'-carbon and the 2'-carbon of the β -D ribosyl sugar moiety, wherein the bridge has the formula 4'-CH(CH_{sub}.3)—O-2', and wherein the methyl group of the bridge is in the S configuration.

[0035] As used herein, “cEt nucleoside” means a nucleoside comprising a cEt modified sugar moiety.

[0036] As used herein, “chirally enriched population” means a plurality of molecules of identical molecular formula, wherein the number or percentage of molecules within the population that contain a particular stereochemical configuration at a particular chiral center is greater than the number or percentage of molecules expected to contain the same particular stereochemical configuration at the same particular chiral center within the population if the particular chiral center were stereorandom. Chirally enriched populations of molecules having multiple chiral centers within each molecule may contain one or more stereorandom chiral centers. In certain embodiments, the molecules are modified oligonucleotides. In certain embodiments, the molecules are compounds comprising modified oligonucleotides.

[0037] As used herein, “gapmer” means a modified oligonucleotide comprising an internal region having a plurality of nucleosides that support RNase H cleavage positioned between external regions having one or more nucleosides, wherein the nucleosides comprising the internal region are chemically distinct from the nucleoside or nucleosides comprising the external regions. The internal region may be referred to as the “gap” and the external regions may be referred to as the “wings.” Unless otherwise indicated, “gapmer” refers to a sugar motif. Unless otherwise indicated, the sugar moiety of each nucleoside of the gap is a 2'- β -D-deoxyribose sugar moiety. Thus, the term “MOE gapmer” indicates a gapmer having a gap comprising 2- β -D-deoxynucleosides and wings comprising 2-MOE nucleosides. Unless otherwise indicated, a MOE gapmer may comprise one or more modified internucleoside linkages and/or modified nucleobases and such modifications do not necessarily follow the gapmer pattern of the sugar modifications.

[0038] As used herein, “hotspot region” is a range of nucleobases on a target nucleic acid that is amenable to oligomeric compound-mediated reduction of the amount or activity of the target nucleic acid.

[0039] As used herein, “hybridization” means the pairing or annealing of complementary oligonucleotides and/or nucleic acids. While not limited to a particular mechanism, the most common mechanism of hybridization involves hydrogen bonding, which may be Watson-Crick, Hoogsteen or reversed Hoogsteen hydrogen bonding, between complementary nucleobases.

[0040] As used herein, “internucleoside linkage” means the covalent linkage between contiguous nucleosides in an oligonucleotide. As used herein, “modified internucleoside linkage” means any internucleoside linkage other than a phosphodiester internucleoside linkage. “Phosphorothioate internucleoside linkage” is a modified internucleoside linkage in which one of the non-bridging oxygen atoms of a phosphodiester internucleoside linkage is replaced with a sulfur atom.

[0041] As used herein, “linker-nucleoside” means a nucleoside that links, either directly or indirectly, an oligonucleotide to a conjugate moiety. Linker-nucleosides are located within the conjugate linker of an oligomeric compound. Linker-nucleosides are not considered part of the oligonucleotide portion of an oligomeric compound even if they are contiguous with the oligonucleotide.

[0042] As used herein, “non-bicyclic modified sugar moiety” means a modified sugar moiety that comprises a modification, such as a substituent, that does not form a bridge between two atoms of the sugar to form a second ring.

[0043] As used herein, “mismatch” or “non-complementary” means a nucleobase of a first oligonucleotide that is not complementary with the corresponding nucleobase of a second oligonucleotide or target nucleic acid when the first and second oligonucleotide are aligned.

[0044] As used herein, “motif” means the pattern of unmodified and/or modified sugar moieties, nucleobases, and/or internucleoside linkages, in an oligonucleotide.

[0045] As used herein, “neurogenetic disorder” means a condition of the nervous system caused by a heritable genetic factor, including, but not limited to, a mutation, a deletion, uniparental disomy, or an imprinting defect.

[0046] As used herein, “nucleobase” means an unmodified nucleobase or a modified nucleobase. As used herein an “unmodified nucleobase” is adenine (A), thymine (T), cytosine (C), uracil (U), or guanine (G). As used herein, a “modified nucleobase” is a group of atoms other than unmodified A, T, C, U, or G capable of pairing with at least one unmodified nucleobase. A “5-methyl cytosine” is a modified nucleobase. A universal base is a modified nucleobase that can pair with any one of the five unmodified nucleobases. As used herein, “nucleobase sequence” means the order of contiguous nucleobases in a target nucleic acid or oligonucleotide independent of any sugar or internucleoside linkage modification.

[0047] As used herein, “nucleoside” means a compound comprising a nucleobase and a sugar moiety. The nucleobase and sugar moiety are each, independently, unmodified or modified. As used herein, “modified nucleoside” means a nucleoside comprising a modified nucleobase and/or a modified sugar moiety. Modified nucleosides include abasic nucleosides, which lack a nucleobase. “Linked nucleosides” are nucleosides that are connected in a contiguous sequence (i.e., no additional nucleosides are presented between those that are linked).

[0048] As used herein, “oligomeric compound” means an oligonucleotide and optionally one or more additional features, such as a conjugate group or terminal group. An oligomeric compound may be paired with a second oligomeric compound that is complementary to the first oligomeric compound or may be unpaired. A “singled-stranded oligomeric compound” is an unpaired oligomeric compound. The term “oligomeric duplex” means a duplex formed by two oligomeric compounds having complementary nucleobase sequences. Each oligomeric compound of an oligomeric duplex may be referred to as a “duplexed oligomeric compound.”

[0049] As used herein, “oligonucleotide” means a strand of linked nucleosides connected via internucleoside linkages, wherein each nucleoside and internucleoside linkage may be modified or unmodified. Unless otherwise indicated, oligonucleotides consist of 8-50 linked nucleosides. As used herein, “modified oligonucleotide” means an oligonucleotide, wherein at least one nucleoside or internucleoside linkage is modified. As used herein, “unmodified oligonucleotide” means an oligonucleotide that does not comprise any nucleoside modifications or internucleoside modifications.

[0050] As used herein, “pharmaceutically acceptable carrier or diluent” means any substance suitable for use in administering to a subject. Certain such carriers enable pharmaceutical compositions to be formulated as, for example, tablets, pills, dragees, capsules, liquids, gels, syrups, slurries, suspension and lozenges for the oral ingestion by a subject. In certain embodiments, a pharmaceutically acceptable carrier or diluent is sterile water, sterile saline, sterile buffer solution or sterile artificial cerebrospinal fluid.

[0051] As used herein, “pharmaceutically acceptable salts” means physiologically and pharmaceutically acceptable salts of compounds.

Pharmaceutically acceptable salts retain the desired biological activity of the parent compound and do not impart undesired toxicological effects thereto.

[0052] As used herein, “pharmaceutical composition” means a mixture of substances suitable for administering to a subject. For example, a pharmaceutical composition may comprise an oligomeric compound and a sterile aqueous solution. In certain embodiments, a pharmaceutical composition shows activity in free uptake assay in certain cell lines.

[0053] As used herein, “prodrug” means a therapeutic agent in a form outside the body that is converted to a different form within a subject or cells thereof. Typically, conversion of a prodrug within the subject is facilitated by the action of an enzymes (e.g., endogenous or viral enzyme) or chemicals present in cells or tissues and/or by physiologic conditions.

[0054] As used herein, “reducing or inhibiting the amount or activity”, “reducing the amount or activity”, or “inhibiting the amount or activity” refers to a reduction or blockade of the transcriptional expression or activity relative to the transcriptional expression or activity in an untreated or control sample and does not necessarily indicate a total elimination of transcriptional expression or activity.

[0055] As used herein, “RNA” means any RNA transcript, including endogenous antisense transcripts that do not encode a protein (e.g., UBE3A-ATS), and also includes pre-mRNA and mature mRNA unless otherwise specified.

[0056] As used herein, “RNAi compound” means an antisense compound that acts, at least in part, through RISC or Ago2 to modulate a target nucleic acid and/or protein encoded by a target nucleic acid. RNAi compounds include, but are not limited to double-stranded siRNA, single-stranded RNA (ssRNA), and microRNA, including microRNA mimics. In certain embodiments, an RNAi compound modulates the amount, activity, and/or splicing of a target nucleic acid. The term RNAi compound excludes antisense compounds that act through RNase H.

[0057] As used herein, “self-complementary” in reference to an oligonucleotide means an oligonucleotide that at least partially hybridizes to itself.

[0058] As used herein, “standard cell assay” means the assay described in Example 1 and reasonable variations thereof.

[0059] As used herein, “stereorandom chiral center” in the context of a population of molecules of identical molecular formula means a chiral center having a random stereochemical configuration. For example, in a population of molecules comprising a stereorandom chiral center, the number of molecules having the(S) configuration of the stereorandom chiral center may be but is not necessarily the same as the number of molecules having the (R) configuration of the stereorandom chiral center. The stereochemical configuration of a chiral center is considered random when it is the result of a synthetic method that is not designed to control the stereochemical configuration. In certain embodiments, a stereorandom chiral center is a stereorandom phosphorothioate internucleoside linkage.

[0060] As used herein, “subject” means a human or non-human animal.

[0061] As used herein, “sugar moiety” means an unmodified sugar moiety or a modified sugar moiety. As used herein, “unmodified sugar moiety” means a 2'-OH(H) β -D-ribose moiety, as found in RNA (an “unmodified RNA sugar moiety”), or a 2-H(H) β -D-deoxyribose moiety, as found in DNA (an “unmodified DNA sugar moiety”). Unmodified sugar moieties have one hydrogen at each of the 1', 3', and 4 positions, an oxygen at the 3' position, and two hydrogens at the 5' position. As used herein, “modified sugar moiety” or “modified sugar” means a modified furanosyl sugar moiety or a sugar surrogate.

[0062] As used herein, “sugar surrogate” means a modified sugar moiety having other than a furanosyl moiety that can link a nucleobase to another group, such as an internucleoside linkage, conjugate group, or terminal group in an oligonucleotide. Modified nucleosides comprising sugar surrogates can be incorporated into one or more positions within an oligonucleotide and such oligonucleotides are capable of hybridizing to complementary oligomeric compounds or target nucleic acids.

[0063] As used herein, “symptom or hallmark” means any physical feature or test result that indicates the existence or extent of a disease or disorder. In certain embodiments, a symptom is apparent to a subject or to a medical professional examining or testing said subject. In certain embodiments, a hallmark is apparent upon invasive diagnostic testing, including, but not limited to, post-mortem tests.

[0064] As used herein, “target nucleic acid” and “target RNA” mean a nucleic acid that an antisense compound is designed to affect.

[0065] As used herein, “target region” means a portion of a target nucleic acid to which an oligomeric compound is designed to hybridize.

[0066] As used herein, “terminal group” means a chemical group or group of atoms that is covalently linked to a terminus of an oligonucleotide.

[0067] As used herein, “therapeutically effective amount” means an amount of a pharmaceutical agent that provides a therapeutic benefit to a subject. For example, a therapeutically effective amount improves a symptom or hallmark of a disease.

Certain Embodiments

[0068] The present disclosure provides the following non-limiting numbered embodiments:

[0069] Embodiment 1. An oligomeric compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides wherein the nucleobase sequence of the modified oligonucleotide is at least 90% complementary to an equal length portion of a UBE3A-ATS RNA, and wherein the modified oligonucleotide comprises at least one modification selected from a modified sugar moiety and a modified internucleoside linkage.

[0070] Embodiment 2. An oligomeric compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising at least 12, 13, 14, 15, 16, 17, 18, 19, or 20 contiguous nucleobases of any of SEQ ID NOs: 17-2762, 2786-2863, 2872-2904.

[0071] Embodiment 3. An oligomeric compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising at least 12, 13, 14, 15, 16, 17, or 18 contiguous nucleobases of any of SEQ ID NOs: 2763-2785 or 2864-2871.

[0072] Embodiment 4. An oligomeric compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, or at least 20 contiguous nucleobases complementary to: [0073] an equal length portion of nucleobases 461,413-461,487 of SEQ ID NO: 1; [0074] an equal length portion of nucleobases 468,968-469,013 of SEQ ID NO: 1; or [0075] an equal length portion of nucleobases 483,965-484,003 of SEQ ID NO: 1

[0076] Embodiment 5. An oligomeric compound comprising a modified oligonucleotide consisting of 12 to 30 linked nucleosides and having a nucleobase sequence comprising at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, or at least 18 contiguous nucleobases of a sequence selected from: [0077] SEQ ID Nos: 1053, 1329, 1501, 1576, 1873, 1949, 2025, 2096, 2245, 2512, 2591, 2680-2682, and 2844; [0078] SEQ ID Nos: 376, 377, 2751-2756, 2773-2776, 2872, 2873, 2876-2878; or [0079] SEQ ID Nos: 172, 764-770, 995, 1445, 1668, 1743, 2255, 2595, 2762-2767.

[0080] Embodiment 6. The oligomeric compound of any of embodiments 1-5, wherein the modified oligonucleotide has a nucleobase sequence that is at least 80%, 85%, 90%, 95%, or 100% complementary to the nucleobase sequence of SEQ ID NO: 1, SEQ ID NO: 2915, or SEQ ID NO: 2916 when measured across the entire nucleobase sequence of the modified oligonucleotide.

[0081] Embodiment 7. The oligomeric compound of any of embodiments 1-6, wherein the modified oligonucleotide comprises at least one modified nucleoside.

[0082] Embodiment 8. The oligomeric compound of embodiment 7, wherein the modified oligonucleotide comprises at least one modified nucleoside comprising a modified sugar moiety.

[0083] Embodiment 9. The oligomeric compound of embodiment 8, wherein the modified oligonucleotide comprises at least one modified nucleoside comprising a bicyclic sugar moiety.

[0084] Embodiment 10. The oligomeric compound of embodiment 9, wherein the modified oligonucleotide comprises at least one modified nucleoside comprising a bicyclic sugar moiety having a 2'-4 bridge, wherein the 2'-4 bridge is selected from —O—CH.sub.2—; and —O—CH(CH.sub.3)—.

[0085] Embodiment 11. The oligomeric compound of any of embodiments 7-10, wherein the modified oligonucleotide comprises at least one modified nucleoside comprising a non-bicyclic modified sugar moiety.

[0086] Embodiment 12. The oligomeric compound of embodiment 11, wherein the non-bicyclic modified sugar moiety is a 2'-MOE modified sugar moiety or 2'-OMe modified sugar moiety.

[0087] Embodiment 13. The oligomeric compound of any of embodiments 7-8, wherein the modified oligonucleotide comprises at least one modified nucleoside comprising a sugar surrogate.

[0088] Embodiment 14. The oligomeric compound of embodiment 13, wherein the modified oligonucleotide comprises at least one modified nucleoside comprising a sugar surrogate selected from morpholino and PNA.

[0089] Embodiment 15. The oligomeric compound of any of embodiments 1-8 or 11-14, wherein the modified oligonucleotide does not comprise a bicyclic sugar moiety.

[0090] Embodiment 16. The oligomeric compound of any of embodiments 1-15, wherein the modified oligonucleotide is a gapmer.

[0091] Embodiment 17. The oligomeric compound of any of embodiments 1-16, wherein the modified oligonucleotide has a sugar motif comprising: [0092] a 5'-region consisting of 1-6 linked 5'-region nucleosides; [0093] a central region consisting of 6-10 linked central region nucleosides; and [0094] a 3'-region consisting of 1-6 linked 3'-region nucleosides; wherein [0095] each of the 5'-region nucleosides and each of the 3'-region nucleosides comprises a modified sugar moiety and each of the central region nucleosides comprises a 2- β -D-deoxyribosyl sugar moiety.

[0096] Embodiment 18. The oligomeric compound of embodiment 17, wherein the modified oligonucleotide has [0097] a 5'-region consisting of 5 linked 5'-region nucleosides; [0098] a central region consisting of 10 linked central region nucleosides; and [0099] a 3'-region consisting of 5 linked 3'-region nucleosides; wherein [0100] each of the 5'-region nucleosides and each of the 3'-region nucleosides comprises a 2'-MOE modified sugar moiety, and each of the central region nucleosides comprises a 2- β -D-deoxyribosyl sugar moiety.

[0101] Embodiment 19. The oligomeric compound of embodiment 17, wherein the modified oligonucleotide has [0102] a 5'-region consisting of 4 linked 5'-region nucleosides; [0103] a central region consisting of 10 linked central region nucleosides; and [0104] a 3'-region consisting of 6 linked 3'-region nucleosides; wherein [0105] each of the 5'-region nucleosides and each of the 3'-region nucleosides a 2'-MOE modified sugar moiety, and each of the central region nucleosides comprises a 2'- β -D-deoxyribosyl sugar moiety.

[0106] Embodiment 20. The oligomeric compound of embodiment 17, wherein the modified oligonucleotide has [0107] a 5'-region consisting of 6 linked 5'-region nucleosides; [0108] a central region consisting of 10 linked central region nucleosides; and [0109] a 3'-region consisting of 4 linked 3'-region nucleosides; wherein [0110] each of the 5'-region nucleosides and each of the 3'-region nucleosides a 2'-MOE modified sugar moiety, and each of the central region nucleosides comprises a 2'- β -D-deoxyribosyl sugar moiety.

[0111] Embodiment 21. The oligomeric compound of embodiment 17, wherein the modified oligonucleotide has [0112] a 5'-region consisting of 5 linked 5'-region nucleosides; [0113] a central region consisting of 8 linked central region nucleosides; and [0114] a 3'-region consisting of 5 linked 3'-region nucleosides; wherein [0115] each of the 5'-region nucleosides and each of the 3'-region nucleosides a 2'-MOE modified sugar moiety, and each of the central region nucleosides comprises a 2'- β -D-deoxyribosyl sugar moiety.

[0116] Embodiment 22. The oligomeric compound of embodiment 17, wherein the modified oligonucleotide has [0117] a 5'-region consisting of 4 linked 5'-region nucleosides; [0118] a central region consisting of 8 linked central region nucleosides; and [0119] a 3'-region consisting of 6 linked 3'-region nucleosides; wherein [0120] each of the 5'-region nucleosides and each of the 3'-region nucleosides a 2'-MOE modified sugar moiety, and each of the central region nucleosides comprises a 2- β -D-deoxyribosyl sugar moiety.

[0121] Embodiment 23. The oligomeric compound of embodiment 17, wherein the modified oligonucleotide has [0122] a 5'-region consisting of 4 linked 5'-region nucleosides; [0123] a central region consisting of 8 linked central region nucleosides; and [0124] a 3'-region consisting of 6 linked 3'-region nucleosides; wherein [0125] each of the 5'-region nucleosides and each of the 3'-region nucleosides a 2'-MOE modified sugar moiety, and each of the central region nucleosides comprises a 2'- β -D-deoxyribosyl sugar moiety.

[0126] Embodiment 24. The oligomeric compound of any of embodiments 1-23, wherein the modified oligonucleotide comprises at least one modified internucleoside linkage.

[0127] Embodiment 25. The oligomeric compound of embodiment 24, wherein each internucleoside linkage of the modified oligonucleotide is a modified internucleoside linkage.

[0128] Embodiment 26. The oligomeric compound of embodiment 24 or 25 wherein the modified internucleoside linkage is a phosphorothioate internucleoside linkage.

[0129] Embodiment 27. The oligomeric compound of embodiment 24 or 26 wherein the modified oligonucleotide comprises at least one phosphodiester internucleoside linkage.

[0130] Embodiment 28. The oligomeric compound of any of embodiments 24, 26, or 27, wherein each internucleoside linkage is independently selected from a phosphodiester internucleoside linkage or a phosphorothioate internucleoside linkage.

[0131] Embodiment 29. The oligomeric compound of embodiments 1-24 or 27-28, wherein the modified oligonucleotide has an internucleoside linkage motif selected from among: sooooooossssssooss, sooooooossssssooss, sooooooossssssooss, sooooooossssssooss, sooooooossssssooss, or sooooooossssooss; wherein, s=a phosphorothioate internucleoside linkage and o=a phosphodiester internucleoside linkage.

[0132] Embodiment 30. The oligomeric compound of any of embodiments 1-29, wherein the modified oligonucleotide comprises a modified nucleobase.

[0133] Embodiment 31. The oligomeric compound of embodiment 30, wherein the modified nucleobase is a 5-methyl cytosine.

[0134] Embodiment 32. The oligomeric compound of any of embodiments 1-31, wherein the modified oligonucleotide consists of 12-30, 12-22, 12-20, 14-18, 14-20, 15-17, 15-25, 16-20, 18-22 or 18-20 linked nucleosides.

[0135] Embodiment 33. The oligomeric compound of any of embodiments 1-32, wherein the modified oligonucleotide consists of 18 linked nucleosides.

[0136] Embodiment 34. The oligomeric compound of any of embodiments 1-2, 4, or 6-32, wherein the modified oligonucleotide consists of 20 linked nucleosides.

[0137] Embodiment 35. The oligomeric compound of any of embodiments 1-34, consisting of the modified oligonucleotide.

[0138] Embodiment 36. The oligomeric compound of any of embodiments 1-34, comprising a conjugate group comprising a conjugate moiety and a conjugate linker.

[0139] Embodiment 37. The oligomeric compound of embodiment 36, wherein the conjugate linker consists of a single bond.

[0140] Embodiment 38. The oligomeric compound of embodiment 36, wherein the conjugate linker is cleavable.

[0141] Embodiment 39. The oligomeric compound of embodiment 36, wherein the conjugate linker comprises 1-3 linker-nucleosides.

[0142] Embodiment 40. The oligomeric compound of any of embodiments 36-39, wherein the conjugate group is attached to the modified oligonucleotide at the 5'-end of the modified oligonucleotide.

[0143] Embodiment 41. The oligomeric compound of any of embodiments 36-39, wherein the conjugate group is attached to the modified oligonucleotide at the 3'-end of the modified oligonucleotide.

[0144] Embodiment 42. The oligomeric compound of any of embodiments 1-34 or 36-41, comprising a terminal group.

[0145] Embodiment 43. The oligomeric compound of any of embodiments 1-42 wherein the oligomeric compound is a singled-stranded oligomeric compound.

[0146] Embodiment 44. The oligomeric compound of any of embodiments 1-38 or 40-43, wherein the oligomeric compound does not comprise linker-nucleosides.

[0147] Embodiment 45. An oligomeric duplex comprising an oligomeric compound of any of embodiments 1-42 or 44.

[0148] Embodiment 46. An antisense compound comprising or consisting of an oligomeric compound of any of embodiments 1-44 or an oligomeric duplex of embodiment 45.

[0149] Embodiment 47. A pharmaceutical composition comprising an oligomeric compound of any of embodiments 1-44 or an oligomeric duplex of embodiment 45 and a pharmaceutically acceptable carrier or diluent.

[0150] Embodiment 48. The pharmaceutical composition of embodiment 47, wherein the pharmaceutically acceptable diluent is artificial cerebral spinal fluid.

[0151] Embodiment 49. The pharmaceutical composition of embodiment 48, wherein the pharmaceutical composition consists essentially of the modified oligonucleotide and phosphate buffered saline.

[0152] Embodiment 50. A method comprising administering to a subject a pharmaceutical composition of any of embodiments 47-50.

[0153] Embodiment 51. A method of treating a disease associated with UBE3A-ATS comprising administering to an individual having or at risk for developing a disease associated with UBE3A-ATS a therapeutically effective amount of a pharmaceutical composition according to any of embodiments 47-49; and thereby treating the disease associated with UBE3A-ATS.

[0154] Embodiment 52. The method of embodiment 51, wherein the UBE3A-ATS-associated disease is Angelman Syndrome.

[0155] Embodiment 53. The method of any of embodiments 50-52, wherein at least one symptom or hallmark of the UBE3A-ATS-associated disease is ameliorated.

[0156] Embodiment 54. The method of embodiment 53, wherein the symptom or hallmark is developmental delays, ataxia, speech impairment, sleep problems, seizures, or EEG abnormalities.

[0157] Embodiment 55. A modified oligonucleotide according to the following chemical structure:

##STR00001##

[0158] Embodiment 56. A modified oligonucleotide according to the following chemical structure:

##STR00002##

[0159] Embodiment 57. A modified oligonucleotide according to the following chemical structure:

##STR00003##

[0160] Embodiment 58. A modified oligonucleotide according to the following chemical structure:

##STR00004##

[0161] Embodiment 59. A modified oligonucleotide according to the following chemical structure:

##STR00005##

[0162] Embodiment 60. A modified oligonucleotide corresponding to the following chemical structure:

##STR00006##

[0163] Embodiment 61. A modified oligonucleotide according to the following chemical structure:

##STR00007##

[0164] Embodiment 62. A modified oligonucleotide according to the following chemical structure:

##STR00008##

Embodiment 63. A modified oligonucleotide according to the following chemical structure:

##STR00009##

[0165] Embodiment 64. A modified oligonucleotide according to the following chemical structure:

##STR00010##

[0166] Embodiment 65. A modified oligonucleotide according to the following chemical structure:

##STR00011##

[0167] Embodiment 66. A modified oligonucleotide according to the following chemical structure:

##STR00012##

[0168] Embodiment 67. The modified oligonucleotide of any of embodiments 55, 57, 59, 61, 63, or 65, which is a sodium salt of the chemical structure.

[0169] Embodiment 68. A pharmaceutical composition comprising the modified oligonucleotide of any of embodiments 55-67 and a pharmaceutically acceptable carrier or diluent.

[0170] Embodiment 69. The pharmaceutical composition of embodiment 68, wherein the pharmaceutically acceptable diluent is artificial cerebrospinal fluid.

[0171] Embodiment 70. The pharmaceutical composition of embodiment 69, wherein the pharmaceutical composition consists essentially of the modified oligonucleotide and artificial cerebrospinal fluid.

[0172] Embodiment 71. A compound comprising a modified oligonucleotide according to the following chemical notation:

.sup.mC.sub.esA.sub.eoT.sub.eo.sup.mC.sub.eoA.sub.eoT.sub.dsG.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsG.sub.dsG.sub.dsT.sub.dsA.sub.eoA
(SEQ ID NO: 2917), wherein: [0173] A=an adenine nucleobase, [0174] mC=a 5-methyl cytosine nucleobase, [0175] G=a guanine nucleobase, [0176]
T=a thymine nucleobase, [0177] e=a 2'-MOE sugar moiety, [0178] d=a 2'-β-D-deoxyribose sugar moiety, [0179] s=a phosphorothioate
internucleoside linkage, and [0180] o=a phosphodiester internucleoside linkage.

[0181] Embodiment 72. A compound comprising a modified oligonucleotide according to the following chemical notation:

T.sub.es.sup.mC.sub.eoA.sub.eo.sup.mC.sub.eo.sup.mC.sub.eoA.sub.eoT.sub.dT.sub.dT.sub.dT.sub.dG.sub.dA.sub.ds.sup.mC.sub.ds.sup.mC.sub.dT.sub.ds (SEQ ID NO: 2918), wherein: [0182] A=an adenine nucleobase, [0183] mC=a 5-methyl cytosine nucleobase, [0184] G=a guanine nucleobase, [0185] T=a thymine nucleobase, [0186] e=a 2'-MOE sugar moiety, [0187] d=a 2'-β-D-deoxyribsyl sugar moiety, [0188] s=a phosphorothioate internucleoside linkage, and [0189] o=a phosphodiester internucleoside linkage.

[0190] Embodiment 73. A compound comprising a modified oligonucleotide according to the following chemical notation:

T.sub.esT.sub.eo.sup.mC.sub.eoA.sub.eo.sup.mC.sub.eoA.sub.dsT.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds (SEQ ID NO: 2919), wherein: [0191] A=an adenine nucleobase, [0192] mC=a 5-methyl cytosine nucleobase, [0193] G=a guanine nucleobase, [0194] T=a thymine nucleobase, [0195] e=a 2'-MOE sugar moiety, [0196] d=a 2'-β-D-deoxyribsyl sugar moiety, [0197] s=a phosphorothioate internucleoside linkage, and [0198] o=a phosphodiester internucleoside linkage.

[0199] Embodiment 74. A compound comprising a modified oligonucleotide according to the following chemical notation:

G.sub.es.sup.mC.sub.eoA.sub.eoT.sub.eoA.sub.eo.sup.mC.sub.eo.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.dsG.sub.dsG.sub.dsT.sub.dsA.sub.dsG.sub.dsC (SEQ ID NO: 2920), wherein: [0200] A=an adenine nucleobase, [0201] mC=a 5-methyl cytosine nucleobase, [0202] G=a guanine nucleobase, [0203] T=a thymine nucleobase, [0204] e=a 2'-MOE sugar moiety, [0205] d=a 2'-β-D-deoxyribsyl sugar moiety, [0206] s=a phosphorothioate internucleoside linkage, and [0207] o=a phosphodiester internucleoside linkage.

[0208] Embodiment 75. A compound comprising a modified oligonucleotide according to the following chemical notation:

A.sub.es.sup.mC.sub.eoG.sub.eo.sup.mC.sub.esA.sub.dsA.sub.dsT.sub.dsG.sub.dsT.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.eoG.sub.eoG.sub.eo.sup.mC (SEQ ID NO: 2921), wherein: [0209] A=an adenine nucleobase, [0210] mC=a 5-methyl cytosine nucleobase, [0211] G=a guanine nucleobase, [0212] T=a thymine nucleobase, [0213] e=a 2'-MOE sugar moiety, [0214] d=a 2'-β-D-deoxyribsyl sugar moiety, [0215] s=a phosphorothioate internucleoside linkage, and [0216] o=a phosphodiester internucleoside linkage.

[0217] Embodiment 76. A compound comprising a modified oligonucleotide according to the following chemical notation:

“cEt”), 4’-CH.sub.2-2’—O—CH.sub.2-2’ (4’-CH(CH.sub.2OCH.sub.3)—O-2’ (“constrained MOE” or “cMOE”) and analogs thereof (see, e.g., Seth et al., U.S. Pat. No. 7,399,845; Bhat et al., U.S. Pat. No. 7,569,686; Swayze et al., U.S. Pat. No. 7,741,457; and Swayze et al., U.S. Pat. No. 8,022,193), 4’-C(CH.sub.3)(CH.sub.3)—O-2’ and analogs thereof (see, e.g., Seth et al., U.S. Pat. No. 8,278,283), 4’-CH.sub.2—N(OCH.sub.3)-2’ and analogs thereof (see, e.g., Prakash et al., U.S. Pat. No. 8,278,425), 4’-CH.sub.2—O—N(CH.sub.3)-2’ (see, e.g., Allerson et al., U.S. Pat. No. 7,696,345 and Allerson et al., U.S. Pat. No. 8,124,745), 4’-CH.sub.2—C(H)(CH.sub.3)-2’ (see, e.g., Zhou, et al., *J. Org. Chem.*, 2009, 74, 118-134), 4’-CH.sub.2—C(=CH.sub.2)-2’ and analogs thereof (see, e.g., Seth et al., U.S. Pat. No. 8,278,426), 4’-C(R.sub.aR.sub.b)—N(R)—O-2’, 4’-C(R.sub.aR.sub.b)—O—N(R)-2’, 4’-CH.sub.2—O—N(R)-2’, and 4’-CH.sub.2—N(R)—O-2’, wherein each R, R.sub.a, and R.sub.b is, independently, H, a protecting group, or C.sub.1-C.sub.12 alkyl (see, e.g. Imanishi et al., U.S. Pat. No. 7,427,672).

[0246] In certain embodiments, such 4’ to 2’ bridges independently comprise from 1 to 4 linked groups independently selected from: —[C(R.sub.a)(R.sub.b)]_n—, —[C(R.sub.a)(R.sub.b)]_n—O—, —C(R.sub.a)=C(R.sub.b)—, —C(R.sub.a)=N—, —C(=NR.sub.a)—, —C(=O)—, —C(=S)—, —O—, —Si(R.sub.a).sub.2—, —S(=O).sub.x—, and —N(R.sub.a)—; wherein: [0247] x is 0, 1, or 2; [0248] n is 1, 2, 3, or 4; [0249] each R.sub.a and R.sub.b is, independently, H, a protecting group, hydroxyl, C.sub.1-C.sub.12 alkyl, substituted C.sub.1-C.sub.12 alkyl, C.sub.2-C.sub.12 alkenyl, substituted C.sub.2-C.sub.12 alkenyl, C.sub.2-C.sub.12 alkynyl, substituted C.sub.2-C.sub.12 alkynyl, C.sub.5-C.sub.20 aryl, substituted C.sub.5-C.sub.20 aryl, substituted heterocycle radical, substituted heterocycle radical, heteroaryl, substituted heteroaryl, C.sub.5-C.sub.7 alicyclic radical, substituted C.sub.5-C.sub.7 alicyclic radical, halogen, OJ.sub.1, NJ.sub.1J.sub.2, SJ.sub.1, N.sub.3, COOJ.sub.1, acyl (C(=O)—H), substituted acyl, CN, sulfonyl (S(=O).sub.2-J.sub.1), or sulfoxyl (S(=O)-J.sub.1); and [0250] each J.sub.1 and J.sub.2 is, independently, H, C.sub.1-C.sub.12 alkyl, substituted C.sub.1-C.sub.12 alkyl, C.sub.2-C.sub.12 alkenyl, substituted C.sub.2-C.sub.12 alkenyl, C.sub.2-C.sub.12 alkynyl, substituted C.sub.2-C.sub.12 alkynyl, C.sub.5-C.sub.20 aryl, substituted C.sub.5-C.sub.20 aryl, acyl (C(=O)—H), substituted acyl, a heterocycle radical, a substituted heterocycle radical, C.sub.1-C.sub.12 aminoalkyl, substituted C.sub.1-C.sub.12 aminoalkyl, or a protecting group.

[0251] Additional bicyclic sugar moieties are known in the art, see, for example: Freier et al., *Nucleic Acids Research*, 1997, 25(22), 4429-4443, Albaek et al., *J. Org. Chem.*, 2006, 71, 7731-7740; Singh et al., *Chem. Commun.*, 1998, 4, 455-456; Koshkin et al., *Tetrahedron*, 1998, 54, 3607-3630; Kumar et al., *Bioorg. Med. Chem. Lett.*, 1998, 8, 2219-2222; Singh et al., *J. Org. Chem.*, 1998, 63, 10035-10039; Srivastava et al., *J. Am. Chem. Soc.*, 2007, 129, 8362-8379; Wengel et al., U.S. Pat. No. 7,053,207; Imanishi et al., U.S. Pat. No. 6,268,490; Imanishi et al. U.S. Pat. No. 6,770,748; Imanishi et al., U.S. RE44,779; Wengel et al., U.S. Pat. No. 6,794,499; Wengel et al., U.S. Pat. No. 6,670,461; Wengel et al., U.S. Pat. No. 7,034,133; Wengel et al., U.S. Pat. No. 8,080,644; Wengel et al., U.S. Pat. No. 8,034,909; Wengel et al., U.S. Pat. No. 8,153,365; Wengel et al., U.S. Pat. No. 7,572,582; Ramasamy et al., U.S. Pat. No. 6,525,191; Torsten et al., WO 2004/106356; Wengel et al., WO 1999/014226; Seth et al., WO 2007/134181; Seth et al., U.S. Pat. No. 7,547,684; Seth et al., U.S. Pat. No. 7,666,854; Seth et al., U.S. Pat. No. 8,088,746; Seth et al., U.S. Pat. No. 7,750,131; Seth et al., U.S. Pat. No. 8,030,467; Seth et al., U.S. Pat. No. 8,268,980; Seth et al., U.S. Pat. No. 8,546,556; Seth et al., U.S. Pat. No. 8,530,640; Migawa et al., U.S. Pat. No. 9,012,421; Seth et al., U.S. Pat. No. 8,501,805; and U.S. Patent Publication Nos. Allerson et al., US2008/0039618 and Migawa et al., US2015/0191727.

[0252] In certain embodiments, bicyclic sugar moieties and nucleosides incorporating such bicyclic sugar moieties are further defined by isomeric configuration. For example, an LNA nucleoside (described herein) may be in the a-L configuration or in the p-D configuration.

##STR00013##

α-L-methyleneoxy (4’-CH.sub.2—O-2’) or α-L-LNA bicyclic nucleosides have been incorporated into oligonucleotides that showed antisense activity (Frieden et al., *Nucleic Acids Research*, 2003, 21, 6365-6372). Herein, general descriptions of bicyclic nucleosides include both isomeric configurations. When the positions of specific bicyclic nucleosides (e.g., LNA or cEt) are identified in exemplified embodiments herein, they are in the β-D configuration, unless otherwise specified.

[0253] In certain embodiments, modified sugar moieties comprise one or more non-bridging sugar substituent and one or more bridging sugar substituent (e.g., 5’-substituted and 4’-2’ bridged sugars).

[0254] In certain embodiments, modified sugar moieties are sugar surrogates. In certain such embodiments, the oxygen atom of the sugar moiety is replaced, e.g., with a sulfur, carbon or nitrogen atom. In certain such embodiments, such modified sugar moieties also comprise bridging and/or non-bridging substituents as described herein. For example, certain sugar surrogates comprise a 4’-sulfur atom and a substitution at the 2-position (see, e.g., Bhat et al., U.S. Pat. No. 7,875,733 and Bhat et al., U.S. Pat. No. 7,939,677) and/or the 5’ position.

[0255] In certain embodiments, sugar surrogates comprise rings having other than 5 atoms. For example, in certain embodiments, a sugar surrogate comprises a six-membered tetrahydropyran (“THP”). Such tetrahydropyrans may be further modified or substituted. Nucleosides comprising such modified tetrahydropyrans include but are not limited to hexitol nucleic acid (“HNA”), anitol nucleic acid (“ANA”), manitol nucleic acid (“MNA”) (see, e.g., Leumann, *CJ. Bioorg. & Med. Chem.* 2002, 10, 841-854), fluoro HNA:

##STR00014##

(“F-HNA”, see e.g. Swayze et al., U.S. Pat. No. 8,088,904; Swayze et al., U.S. Pat. No. 8,440,803; Swayze et al., U.S. Pat. No. 8,796,437; and Swayze et al., U.S. Pat. No. 9,005,906; F-HNA can also be referred to as a F-THP or 3-fluoro tetrahydropyran), and nucleosides comprising additional modified THP compounds having the formula:

##STR00015##

wherein, independently, for each of said modified THP nucleoside: [0256] Bx is a nucleobase moiety; [0257] T.sub.3 and T.sub.4 are each, independently, an internucleoside linking group linking the modified THP nucleoside to the remainder of an oligonucleotide or one of T.sub.3 and T.sub.4 is an internucleoside linking group linking the modified THP nucleoside to the remainder of an oligonucleotide and the other of T.sub.3 and T.sub.4 is H, a hydroxyl protecting group, a linked conjugate group, or a 5’ or 3’-terminal group;

q.sub.1, q.sub.2, q.sub.3, q.sub.4, q.sub.5, q.sub.6 and q.sub.7 are each, independently, H, C.sub.1-C.sub.6 alkyl, substituted C.sub.1-C.sub.6 alkyl, C.sub.2-C.sub.6 alkenyl, substituted C.sub.2-C.sub.6 alkenyl, C.sub.2-C.sub.6 alkynyl, or substituted C.sub.2-C.sub.6 alkynyl; and [0258] each of R.sub.1 and R.sub.2 is independently selected from among: hydrogen, halogen, substituted or unsubstituted alkoxy, NJ.sub.1J.sub.2, SJ.sub.1, N.sub.3, OC(=X)J.sub.1, OC(=X)NJ.sub.1J.sub.2, NJ.sub.3C(=X)NJ.sub.1J.sub.2, and CN, wherein X is O, S or NJ.sub.1, and each J.sub.1, J.sub.2, and J.sub.3 is, independently, H or C.sub.1-C.sub.6 alkyl.

[0259] In certain embodiments, modified THP nucleosides are provided wherein q.sub.1, q.sub.2, q.sub.3, q.sub.4, q.sub.5, q.sub.6 and q.sub.7 are each H. In certain embodiments, at least one of q.sub.1, q.sub.2, q.sub.3, q.sub.4, q.sub.5, q.sub.6 and q.sub.7 is other than H. In certain embodiments, at least one of q.sub.1, q.sub.2, q.sub.3, q.sub.4, q.sub.5, q.sub.6 and q.sub.7 is methyl. In certain embodiments, modified THP nucleosides are provided wherein one of R.sub.1 and R.sub.2 is F. In certain embodiments, R.sub.1 is F and R.sub.2 is H, in certain embodiments, R.sub.1 is methoxy and R.sub.2 is H, and in certain embodiments, R.sub.1 is methoxyethoxy and R.sub.2 is H.

[0260] In certain embodiments, sugar surrogates comprise rings having more than 5 atoms and more than one heteroatom. For example, nucleosides comprising morpholino sugar moieties and their use in oligonucleotides have been reported (see, e.g., Braasch et al., *Biochemistry*, 2002, 41, 4503-4510 and Summerton et al., U.S. Pat. No. 5,698,685; Summerton et al., U.S. Pat. No. 5,166,315; Summerton et al., U.S. Pat. No. 5,185,444; and Summerton et al., U.S. Pat. No. 5,034,506). As used here, the term “morpholino” means a sugar surrogate having the following structure:

##STR00016##

In certain embodiments, morpholinos may be modified, for example by adding or altering various substituent groups from the above morpholino structure. Such sugar surrogates are referred to herein as “modified morpholinos.”

[0261] In certain embodiments, sugar surrogates comprise acyclic moieties. Examples of nucleosides and oligonucleotides comprising such acyclic sugar surrogates include but are not limited to: peptide nucleic acid (“PNA”), acyclic butyl nucleic acid (see, e.g., Kumar et al., *Org. Biomol. Chem.*,

[0262] Many other bicyclic and tricyclic sugar and sugar surrogate ring systems are known in the art that can be used in modified nucleosides.

2. Certain Modified Nucleobases

[0263] In certain embodiments, modified oligonucleotides comprise one or more nucleosides comprising an unmodified nucleobase. In certain embodiments, modified oligonucleotides comprise one or more nucleosides comprising a modified nucleobase. In certain embodiments, modified oligonucleotides comprise one or more nucleosides that does not comprise a nucleobase, referred to as an abasic nucleoside.

[0264] In certain embodiments, modified nucleobases are selected from: 5-substituted pyrimidines, 6-azapyrimidines, alkyl or alkynyl substituted pyrimidines, alkyl substituted purines, and N-2, N-6 and O-6 substituted purines.

[0265] In certain embodiments, modified nucleobases are selected from: 2-aminopropyladenine, 5-hydroxymethyl cytosine, xanthine, hypoxanthine, 2-aminoadenine, 6-N-methylguanine, 6-N-methyladenine, 2-propyladenine, 2-thiouracil, 2-thiothymine and 2-thiocytosine, 5-propynyl ($-\text{C}\equiv\text{C}-\text{CH}_2$) uracil, 5-propynylcytosine, 6-azouracil, 6-azocytosine, 6-azothymine, 5-ribosyluracil (pseudouracil), 4-thiouracil, 8-halo, 8-amino, 8-thiol, 8-thioalkyl, 8-hydroxyl, 8-aza and other 8-substituted purines, 5-halo, particularly 5-bromo, 5-trifluoromethyl, 5-halouracil, and 5-halocytosine, 7-methylguanine, 7-methyladenine, 2-F-adenine, 2-aminoadenine, 7-deazaguanine, 7-deazaadenine, 3-deazaguanine, 3-deazaadenine, 6-N-benzoyladenine, 2-N-isobutylguanine, 4-N-benzoylcytosine, 4-N-benzoyluracil, 5-methyl 4-N-benzoylcytosine, 5-methyl 4-N-benzoyluracil, universal bases, hydrophobic bases, promiscuous bases, size-expanded bases, and fluorinated bases. Further modified nucleobases include tricyclic pyrimidines, such as 1,3-diazaphenoxazine-2-one, 1,3-diazaphenothiazine-2-one and 9-(2-aminoethoxy)-1,3-diazaphenoxazine-2-one (G-clamp). Modified nucleobases may also include those in which the purine or pyrimidine base is replaced with other heterocycles, for example 7-deaza-adenine, 7-deazaguanosine, 2-aminopyridine and 2-pyridone. Further nucleobases include those disclosed in Merigan et al., U.S. Pat. No. 3,687,808, those disclosed in *The Concise Encyclopedia Of Polymer Science And Engineering*, Kroschwitz, J. I., Ed., John Wiley & Sons, 1990, 858-859; Englisch et al., *Angewandte Chemie*, International Edition, 1991, 30, 613; Sanghvi, Y. S., Chapter 15, *Antisense Research and Applications*, Crooke, S. T. and Lebleu, B., Eds., CRC Press, 1993, 273-288; and those disclosed in Chapters 6 and 15, *Antisense Drug Technology*, Crooke S. T., Ed., CRC Press, 2008, 163-166 and 442-443.

[0266] Publications that teach the preparation of certain of the above noted modified nucleobases as well as other modified nucleobases include without limitation, Manoharan et al., US2003/0158403; Manoharan et al., US2003/0175906; Dinh et al., U.S. Pat. No. 4,845,205; Spielvogel et al., U.S. Pat. No. 5,130,302; Rogers et al., U.S. Pat. No. 5,134,066; Bischofberger et al., U.S. Pat. No. 5,175,273; Urdea et al., U.S. Pat. No. 5,367,066; Benner et al., U.S. Pat. No. 5,432,272; Matteucci et al., U.S. Pat. No. 5,434,257; Gmeiner et al., U.S. Pat. No. 5,457,187; Cook et al., U.S. Pat. No. 5,459,255; Froehler et al., U.S. Pat. No. 5,484,908; Matteucci et al., U.S. Pat. No. 5,502,177; Hawkins et al., U.S. Pat. No. 5,525,711; Haralambidis et al., U.S. Pat. No. 5,552,540; Cook et al., U.S. Pat. No. 5,587,469; Froehler et al., U.S. Pat. No. 5,594,121; Switzer et al., U.S. Pat. No. 5,596,091; Cook et al., U.S. Pat. No. 5,614,617; Froehler et al., U.S. Pat. No. 5,645,985; Cook et al., U.S. Pat. No. 5,681,941; Cook et al., U.S. Pat. No. 5,811,534; Cook et al., U.S. Pat. No. 5,750,692; Cook et al., U.S. Pat. No. 5,948,903; Cook et al., U.S. Pat. No. 5,587,470; Cook et al., U.S. Pat. No. 5,457,191; Matteucci et al., U.S. Pat. No. 5,763,588; Froehler et al., U.S. Pat. No. 5,830,653; Cook et al., U.S. Pat. No. 5,808,027; Cook et al., 6,166,199; and Matteucci et al., U.S. Pat. No. 6,005,096.

3. Certain Modified Internucleoside Linkages

[0267] In certain embodiments, nucleosides of modified oligonucleotides may be linked together using any internucleoside linkage. The two main classes of internucleoside linking groups are defined by the presence or absence of a phosphorus atom. Representative phosphorus-containing internucleoside linkages include but are not limited to phosphodiester, which contain a phosphodiester bond ($-\text{P}(\text{O})_2\text{O}-$) (also referred to as unmodified or naturally occurring linkages), phosphotriesters, methylphosphonates, phosphoramidates, and phosphorothioates ($-\text{P}(\text{O})_2\text{S}-$), and phosphorodithioates ($-\text{HS}-\text{P}(\text{S})_2-$). Representative non-phosphorus containing internucleoside linking groups include but are not limited to methylenemethylimino ($-\text{CH}_2-\text{N}(\text{CH}_2)-\text{O}-\text{CH}_2-$), thiodiester, thionocarbamate ($-\text{O}-\text{C}(=\text{O})(\text{NH})-\text{S}-$); siloxane ($-\text{O}-\text{SiH}_2-\text{O}-$); and N,N'-dimethylhydrazine ($-\text{CH}_2-\text{N}(\text{CH}_2)-\text{N}(\text{CH}_2)-$). Modified internucleoside linkages, compared to naturally occurring phosphate linkages, can be used to alter, typically increase, nuclease resistance of the oligonucleotide. In certain embodiments, internucleoside linkages having a chiral atom can be prepared as a racemic mixture, or as separate enantiomers. Methods of preparation of phosphorous-containing and non-phosphorous-containing internucleoside linkages are well known to those skilled in the art.

[0268] Representative internucleoside linkages having a chiral center include but are not limited to alkylphosphonates and phosphorothioates. Modified oligonucleotides comprising internucleoside linkages having a chiral center can be prepared as populations of modified oligonucleotides comprising stereorandom internucleoside linkages, or as populations of modified oligonucleotides comprising phosphorothioate linkages in particular stereochemical configurations. In certain embodiments, populations of modified oligonucleotides comprise phosphorothioate internucleoside linkages wherein all of the phosphorothioate internucleoside linkages are stereorandom. Such modified oligonucleotides can be generated using synthetic methods that result in random selection of the stereochemical configuration of each phosphorothioate linkage. Nonetheless, as is well understood by those of skill in the art, each individual phosphorothioate of each individual oligonucleotide molecule has a defined stereoconfiguration. In certain embodiments, populations of modified oligonucleotides are enriched for modified oligonucleotides comprising one or more particular phosphorothioate internucleoside linkages in a particular, independently selected stereochemical configuration. In certain embodiments, the particular configuration of the particular phosphorothioate linkage is present in at least 65% of the molecules in the population. In certain embodiments, the particular configuration of the particular phosphorothioate linkage is present in at least 70% of the molecules in the population. In certain embodiments, the particular configuration of the particular phosphorothioate linkage is present in at least 80% of the molecules in the population. In certain embodiments, the particular configuration of the particular phosphorothioate linkage is present in at least 90% of the molecules in the population. In certain embodiments, the particular configuration of the particular phosphorothioate linkage is present in at least 99% of the molecules in the population. Such chirally enriched populations of modified oligonucleotides can be generated using synthetic methods known in the art, e.g., methods described in Oka et al., *JACS* 125, 8307 (2003), Wan et al. *Nuc. Acid. Res.* 42, 13456 (2014), and WO 2017/015555. In certain embodiments, a population of modified oligonucleotides is enriched for modified oligonucleotides having at least one indicated phosphorothioate in the (Sp) configuration. In certain embodiments, a population of modified oligonucleotides is enriched for modified oligonucleotides having at least one phosphorothioate in the (Rp) configuration. In certain embodiments, modified oligonucleotides comprising (Rp) and/or (Sp) phosphorothioates comprise one or more of the following formulas, respectively, wherein "B" indicates a nucleobase:

##STR00017##

Unless otherwise indicated, chiral internucleoside linkages of modified oligonucleotides described herein can be stereorandom or in a particular stereochemical configuration.

[0269] Neutral internucleoside linkages include, without limitation, phosphotriesters, methylphosphonates, MMI ($3'-\text{CH}_2-\text{N}(\text{CH}_2)-\text{O}-5'$), amide-3 ($3'-\text{CH}_2-\text{C}(=\text{O})-\text{N}(\text{H})-5'$), amide-4 ($3'-\text{CH}_2-\text{N}(\text{H})-\text{C}(=\text{O})-5'$), formacetal ($3'-\text{O}-\text{CH}_2-2'-\text{O}-5'$), methoxypropyl (MOP), and thioformacetal ($3'-\text{S}-\text{CH}_2-2'-\text{O}-5'$). Further neutral internucleoside linkages include nonionic linkages comprising siloxane (dialkylsiloxane), carboxylate ester, carboxamide, sulfide, sulfonate ester and amides (See for example: *Carbohydrate Modifications in Antisense Research*; Y. S. Sanghvi and P. D. Cook, Eds., ACS Symposium Series 580; Chapters 3 and 4, 40-65). Further neutral internucleoside linkages include nonionic linkages comprising mixed N, O, S and CH_2 component parts.

B. Certain Motifs

[0270] In certain embodiments, modified oligonucleotides comprise one or more modified nucleosides comprising a modified sugar moiety. In certain embodiments, modified oligonucleotides comprise one or more modified nucleosides comprising a modified nucleobase. In certain

embodiments, modified oligonucleotides comprise one or more internucleoside linkage. In certain embodiments, the modified, unmodified, and differently modified sugar moieties, nucleobases, and/or internucleoside linkages of a modified oligonucleotide define a pattern or motif. In certain embodiments, the patterns of sugar moieties, nucleobases, and internucleoside linkages are each independent of one another. Thus, a modified oligonucleotide may be described by its sugar motif, nucleobase motif and/or internucleoside linkage motif (as used herein, nucleobase motif describes the modifications to the nucleobases independent of the sequence of nucleobases).

1. Certain Sugar Motifs

[0271] In certain embodiments, oligonucleotides comprise one or more type of modified sugar and/or unmodified sugar moiety arranged along the oligonucleotide or portion thereof in a defined pattern or sugar motif. In certain instances, such sugar motifs include but are not limited to any of the sugar modifications discussed herein.

[0272] In certain embodiments, modified oligonucleotides have a gapmer motif, which is defined by two external regions or “wings” and a central or internal region or “gap.” The three regions of a gapmer motif (the 5'-wing, the gap, and the 3'-wing) form a contiguous sequence of nucleosides wherein at least some of the sugar moieties of the nucleosides of each of the wings differ from at least some of the sugar moieties of the nucleosides of the gap. Specifically, at least the sugar moieties of the nucleosides of each wing that are closest to the gap (the 3'-most nucleoside of the 5'-wing and the 5'-most nucleoside of the 3'-wing) differ from the sugar moiety of the neighboring gap nucleosides, thus defining the boundary between the wings and the gap (i.e., the wing/gap junction). In certain embodiments, the sugar moieties within the gap are the same as one another. In certain embodiments, the gap includes one or more nucleoside having a sugar moiety that differs from the sugar moiety of one or more other nucleosides of the gap. In certain embodiments, the sugar motifs of the two wings are the same as one another (symmetric gapmer). In certain embodiments, the sugar motif of the 5'-wing differs from the sugar motif of the 3'-wing (asymmetric gapmer).

[0273] In certain embodiments, the wings of a gapmer comprise 1-6 nucleosides. In certain embodiments, each nucleoside of each wing of a gapmer comprises a modified sugar moiety. In certain embodiments, at least one nucleoside of each wing of a gapmer comprises a modified sugar moiety. In certain embodiments, at least two nucleosides of each wing of a gapmer comprises a modified sugar moiety. In certain embodiments, at least three nucleosides of each wing of a gapmer comprises a modified sugar moiety. In certain embodiments, at least four nucleosides of each wing of a gapmer comprises a modified sugar moiety. In certain embodiments, at least five nucleosides of each wing of a gapmer comprises a modified sugar moiety.

[0274] In certain embodiments, the gap of a gapmer comprises 7-12 nucleosides. In certain embodiments, each nucleoside of the gap of a gapmer comprises a 2'-β-D-deoxyribose sugar moiety. In certain embodiments, at least one nucleoside of the gap of a gapmer comprises a modified sugar moiety.

[0275] In certain embodiments, the gapmer is a deoxy gapmer. In certain embodiments, the nucleosides on the gap side of each wing/gap junction comprise 2'-deoxyribose sugar moieties and the nucleosides on the wing sides of each wing/gap junction comprise modified sugar moieties. In certain embodiments, each nucleoside of the gap comprises a 2'-β-D-deoxyribose sugar moiety. In certain embodiments, each nucleoside of each wing of a gapmer comprises a modified sugar moiety. In certain embodiments, exactly one nucleoside of the gap comprises a modified sugar moiety and each remaining nucleoside of the gap comprises a 2'-β-D-deoxyribose sugar moiety.

[0276] In certain embodiments, modified oligonucleotides comprise or consist of a portion having a fully modified sugar motif. In such embodiments, each nucleoside of the fully modified portion of the modified oligonucleotide comprises a modified sugar moiety. In certain embodiments, each nucleoside of the entire modified oligonucleotide comprises a modified sugar moiety. In certain embodiments, modified oligonucleotides comprise or consist of a portion having a fully modified sugar motif, wherein each nucleoside within the fully modified portion comprises the same modified sugar moiety, referred to herein as a uniformly modified sugar motif. In certain embodiments, a fully modified oligonucleotide is a uniformly modified oligonucleotide. In certain embodiments, each nucleoside of a uniformly modified comprises the same 2'-modification.

[0277] Herein, the lengths (number of nucleosides) of the three regions of a gapmer may be provided using the notation [# of nucleosides in the 5'-wing]-[# of nucleosides in the gap]-[# of nucleosides in the 3'-wing]. Thus, a 3-10-3 gapmer consists of 3 linked nucleosides in each wing and 10 linked nucleosides in the gap. Where such nomenclature is followed by a specific modification, that modification is the modification in each sugar moiety of each wing and the gap nucleosides comprise 2'-β-D-deoxyribose sugar moieties. Thus, a 5-10-5 MOE gapmer consists of 5 linked 2'-MOE nucleosides in the 5'-wing, 10 linked 2'-β-D-deoxynucleosides in the gap, and 5 linked 2'-MOE nucleosides in the 3'-wing. A 3-10-3 cEt gapmer consists of 3 linked cEt nucleosides in the 5'-wing, 10 linked 2'-β-D-deoxynucleosides in the gap, and 3 linked cEt nucleosides in the 3'-wing. A 5-8-5 gapmer consists of 5 linked nucleosides comprising a modified sugar moiety in the 5'-wing, 8 linked 2'-β-D-deoxynucleosides in the gap, and 5 linked nucleosides comprising a modified sugar moiety in the 3'-wing. A 5-8-5 mixed gapmer has at least two different modified sugar moieties in the 5'- and/or the 3'-wing.

[0278] In certain embodiments, modified oligonucleotides are 5-10-5 MOE gapmers. In certain embodiments, modified oligonucleotides are 4-10-6 MOE gapmers. In certain embodiments, modified oligonucleotides are 6-10-4 MOE gapmers. In certain embodiments, modified oligonucleotides are 5-8-5 MOE gapmers. In certain embodiments, modified oligonucleotides are 6-8-4 MOE gapmers. In certain embodiments, modified oligonucleotides are 4-8-6 MOE gapmers. In certain embodiments, modified oligonucleotides are X—Y—Z MOE gapmers, wherein X and Z are independently selected from 1, 2, 3, 4, 5, or 6 and Y is 7, 8, 9, 10, or 11.

2. Certain Nucleobase Motifs

[0279] In certain embodiments, oligonucleotides comprise modified and/or unmodified nucleobases arranged along the oligonucleotide or portion thereof in a defined pattern or motif. In certain embodiments, each nucleobase is modified. In certain embodiments, none of the nucleobases are modified. In certain embodiments, each purine or each pyrimidine is modified. In certain embodiments, each adenine is modified. In certain embodiments, each guanine is modified. In certain embodiments, each thymine is modified. In certain embodiments, each uracil is modified. In certain embodiments, each cytosine is modified. In certain embodiments, some or all of the cytosine nucleobases in a modified oligonucleotide are 5-methyl cytosines. In certain embodiments, all of the cytosine nucleobases are 5-methyl cytosines and all of the other nucleobases of the modified oligonucleotide are unmodified nucleobases.

[0280] In certain embodiments, modified oligonucleotides comprise a block of modified nucleobases. In certain such embodiments, the block is at the 3'-end of the oligonucleotide. In certain embodiments the block is within 3 nucleosides of the 3'-end of the oligonucleotide. In certain embodiments, the block is at the 5'-end of the oligonucleotide. In certain embodiments the block is within 3 nucleosides of the 5'-end of the oligonucleotide.

[0281] In certain embodiments, oligonucleotides having a gapmer motif comprise a nucleoside comprising a modified nucleobase. In certain such embodiments, one nucleoside comprising a modified nucleobase is in the central gap of an oligonucleotide having a gapmer motif. In certain such embodiments, the sugar moiety of said nucleoside is a 2'-β-D-deoxyribose sugar moiety. In certain embodiments, the modified nucleobase is selected from: a 2-thiopyrimidine and a 5-propynepyrimidine.

3. Certain Internucleoside Linkage Motifs

[0282] In certain embodiments, oligonucleotides comprise modified and/or unmodified internucleoside linkages arranged along the oligonucleotide or portion thereof in a defined pattern or motif. In certain embodiments, each internucleoside linking group is a phosphodiester internucleoside linkage (P(O.sub.2)=O). In certain embodiments, each internucleoside linking group of a modified oligonucleotide is a phosphorothioate internucleoside linkage (P(O.sub.2)=S). In certain embodiments, each internucleoside linkage of a modified oligonucleotide is independently selected from a phosphorothioate internucleoside linkage and phosphodiester internucleoside linkage. In certain embodiments, each phosphorothioate internucleoside linkage is independently selected from a stereorandom phosphorothioate, a (Sp) phosphorothioate, and a (Rp) phosphorothioate. In

certain embodiments, the sugar motif of a modified oligonucleotide is a gapmer and the internucleoside linkages within the gap are all modified. In certain such embodiments, some or all of the internucleoside linkages in the wings are unmodified phosphodiester internucleoside linkages. In certain embodiments, the terminal internucleoside linkages are modified. In certain embodiments, the sugar motif of a modified oligonucleotide is a gapmer, and the internucleoside linkage motif comprises at least one phosphodiester internucleoside linkage in at least one wing, wherein the at least one phosphodiester linkage is not a terminal internucleoside linkage, and the remaining internucleoside linkages are phosphorothioate internucleoside linkages. In certain such embodiments, all of the phosphorothioate linkages are stereorandom. In certain embodiments, all of the phosphorothioate linkages in the wings are (Sp) phosphorothioates, and the gap comprises at least one Sp, Sp, Rp motif. In certain embodiments, populations of modified oligonucleotides are enriched for modified oligonucleotides comprising such internucleoside linkage motifs.

[0283] In certain embodiments, modified nucleotides have an internucleoside linkage motif of sooooooossssssooss, wherein each “s” represents a phosphorothioate internucleoside linkage and each “o” represents a phosphodiester internucleoside linkage. In certain embodiments, modified nucleotides have an internucleoside linkage motif of sooooooossssssooss, wherein each “s” represents a phosphorothioate internucleoside linkage and each “o” represents a phosphodiester internucleoside linkage. In certain embodiments, modified nucleotides have an internucleoside linkage motif of sooooooossssssooss, wherein each “s” represents a phosphorothioate internucleoside linkage and each “o” represents a phosphodiester internucleoside linkage. In certain embodiments, modified nucleotides have an internucleoside linkage motif of sooooooossssssooss, wherein each “s” represents a phosphorothioate internucleoside linkage and each “o” represents a phosphodiester internucleoside linkage. In certain embodiments, modified nucleotides have an internucleoside linkage motif of sooooooossssssooss, wherein each “s” represents a phosphorothioate internucleoside linkage and each “o” represents a phosphodiester internucleoside linkage. In certain embodiments, modified nucleotides have an internucleoside linkage motif of sooooooossssssooss, wherein each “s” represents a phosphorothioate internucleoside linkage and each “o” represents a phosphodiester internucleoside linkage.

C. Certain Lengths

[0284] It is possible to increase or decrease the length of an oligonucleotide without eliminating activity. For example, in Woolf et al. (Proc. Natl. Acad. Sci. USA 89:7305-7309, 1992), a series of oligonucleotides 13-25 nucleobases in length were tested for their ability to induce cleavage of a target RNA in an oocyte injection model. Oligonucleotides 25 nucleobases in length with 8 or 11 mismatch bases near the ends of the oligonucleotides were able to direct specific cleavage of the target RNA, albeit to a lesser extent than the oligonucleotides that contained no mismatches. Similarly, target specific cleavage was achieved using 13 nucleobase oligonucleotides, including those with 1 or 3 mismatches.

[0285] In certain embodiments, oligonucleotides (including modified oligonucleotides) can have any of a variety of ranges of lengths. In certain embodiments, oligonucleotides consist of X to Y linked nucleosides, where X represents the fewest number of nucleosides in the range and Y represents the largest number nucleosides in the range. In certain such embodiments, X and Y are each independently selected from 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, and 50; provided that $X \leq Y$. For example, in certain embodiments, oligonucleotides consist of 12 to 13, 12 to 14, 12 to 15, 12 to 16, 12 to 17, 12 to 18, 12 to 19, 12 to 20, 12 to 21, 12 to 22, 12 to 23, 12 to 24, 12 to 25, 12 to 26, 12 to 27, 12 to 28, 12 to 29, 12 to 30, 13 to 14, 13 to 15, 13 to 16, 13 to 17, 13 to 18, 13 to 19, 13 to 20, 13 to 21, 13 to 22, 13 to 23, 13 to 24, 13 to 25, 13 to 26, 13 to 27, 13 to 28, 13 to 29, 13 to 30, 14 to 15, 14 to 16, 14 to 17, 14 to 18, 14 to 19, 14 to 20, 14 to 21, 14 to 22, 14 to 23, 14 to 24, 14 to 25, 14 to 26, 14 to 27, 14 to 28, 14 to 29, 14 to 30, 15 to 16, 15 to 17, 15 to 18, 15 to 19, 15 to 20, 15 to 21, 15 to 22, 15 to 23, 15 to 24, 15 to 25, 15 to 26, 15 to 27, 15 to 28, 15 to 29, 15 to 30, 16 to 17, 16 to 18, 16 to 19, 16 to 20, 16 to 21, 16 to 22, 16 to 23, 16 to 24, 16 to 25, 16 to 26, 16 to 27, 16 to 28, 16 to 29, 16 to 30, 17 to 18, 17 to 19, 17 to 20, 17 to 21, 17 to 22, 17 to 23, 17 to 24, 17 to 25, 17 to 26, 17 to 27, 17 to 28, 17 to 29, 17 to 30, 18 to 19, 18 to 20, 18 to 21, 18 to 22, 18 to 23, 18 to 24, 18 to 25, 18 to 26, 18 to 27, 18 to 28, 18 to 29, 18 to 30, 19 to 20, 19 to 21, 19 to 22, 19 to 23, 19 to 24, 19 to 25, 19 to 26, 19 to 27, 19 to 28, 19 to 29, 19 to 30, 20 to 21, 20 to 22, 20 to 23, 20 to 24, 20 to 25, 20 to 26, 20 to 27, 20 to 28, 20 to 29, 20 to 30, 21 to 22, 21 to 23, 21 to 24, 21 to 25, 21 to 26, 21 to 27, 21 to 28, 21 to 29, 21 to 30, 22 to 23, 22 to 24, 22 to 25, 22 to 26, 22 to 27, 22 to 28, 22 to 29, 22 to 30, 23 to 24, 23 to 25, 23 to 26, 23 to 27, 23 to 28, 23 to 29, 23 to 30, 24 to 25, 24 to 26, 24 to 27, 24 to 28, 24 to 29, 24 to 30, 25 to 26, 25 to 27, 25 to 28, 25 to 29, 25 to 30, 26 to 27, 26 to 28, 26 to 29, 26 to 30, 27 to 28, 27 to 29, 27 to 30, 28 to 29, 28 to 30, or 29 to 30 linked nucleosides.

D. Certain Modified Oligonucleotides

[0286] In certain embodiments, the above modifications (sugar, nucleobase, internucleoside linkage) are incorporated into a modified oligonucleotide. In certain embodiments, modified oligonucleotides are characterized by their modification motifs and overall lengths. In certain embodiments, such parameters are each independent of one another. Thus, unless otherwise indicated, each internucleoside linkage of an oligonucleotide having a gapmer sugar motif may be modified or unmodified and may or may not follow the gapmer modification pattern of the sugar modifications. For example, the internucleoside linkages within the wing regions of a sugar gapmer may be the same or different from one another and may be the same or different from the internucleoside linkages of the gap region of the sugar motif. Likewise, such sugar gapmer oligonucleotides may comprise one or more modified nucleobase independent of the gapmer pattern of the sugar modifications. Unless otherwise indicated, all modifications are independent of nucleobase sequence.

E. Certain Populations of Modified Oligonucleotides

[0287] Populations of modified oligonucleotides in which all of the modified oligonucleotides of the population have the same molecular formula can be stereorandom populations or chirally enriched populations. All of the chiral centers of all of the modified oligonucleotides are stereorandom in a stereorandom population. In a chirally enriched population, at least one particular chiral center is not stereorandom in the modified oligonucleotides of the population. In certain embodiments, the modified oligonucleotides of a chirally enriched population are enriched for β -D ribosyl sugar moieties, and all of the phosphorothioate internucleoside linkages are stereorandom. In certain embodiments, the modified oligonucleotides of a chirally enriched population are enriched for both β -D ribosyl sugar moieties and at least one, particular phosphorothioate internucleoside linkage in a particular stereochemical configuration.

F. Nucleobase Sequence

[0288] In certain embodiments, oligonucleotides (unmodified or modified oligonucleotides) are further described by their nucleobase sequence. In certain embodiments oligonucleotides have a nucleobase sequence that is complementary to a second oligonucleotide or an identified reference nucleic acid, such as a target nucleic acid. In certain such embodiments, a portion of an oligonucleotide has a nucleobase sequence that is complementary to a second oligonucleotide or an identified reference nucleic acid, such as a target nucleic acid. In certain embodiments, the nucleobase sequence of a portion or entire length of an oligonucleotide is at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, or 100% complementary to the second oligonucleotide or nucleic acid, such as a target nucleic acid.

II. Certain Oligomeric Compounds

[0289] In certain embodiments, provided herein are oligomeric compounds, which consist of an oligonucleotide (modified or unmodified) and optionally one or more conjugate groups and/or terminal groups. Conjugate groups consist of one or more conjugate moiety and a conjugate linker which links the conjugate moiety to the oligonucleotide. Conjugate groups may be attached to either or both ends of an oligonucleotide and/or at any internal position. In certain embodiments, conjugate groups are attached to the 2'-position of a nucleoside of a modified oligonucleotide. In certain embodiments, conjugate groups that are attached to either or both ends of an oligonucleotide are terminal groups. In certain such embodiments, conjugate groups or terminal groups are attached at the 3' and/or 5'-end of oligonucleotides. In certain such embodiments, conjugate groups (or terminal groups) are attached at the 3'-end of oligonucleotides. In certain embodiments, conjugate groups are attached near the 3'-end of oligonucleotides. In certain embodiments, conjugate groups (or terminal groups) are attached at the 5'-end of oligonucleotides. In certain embodiments, conjugate groups are attached near the 5'-end of oligonucleotides.

[0290] Examples of terminal groups include but are not limited to conjugate groups, capping groups, phosphate moieties, protecting groups, modified or unmodified nucleosides, and two or more nucleosides that are independently modified or unmodified.

A. Certain Conjugate Groups

[0291] In certain embodiments, oligonucleotides are covalently attached to one or more conjugate groups. In certain embodiments, conjugate groups modify one or more properties of the attached oligonucleotide, including but not limited to pharmacodynamics, pharmacokinetics, stability, binding, absorption, tissue distribution, cellular distribution, cellular uptake, charge and clearance. In certain embodiments, conjugate groups impart a new property on the attached oligonucleotide, e.g., fluorophores or reporter groups that enable detection of the oligonucleotide. Certain conjugate groups and conjugate moieties have been described previously, for example: cholesterol moiety (Letsinger et al., *Proc. Natl. Acad. Sci. USA*, 1989, 86, 6553-6556), cholic acid (Manoharan et al., *Bioorg. Med. Chem. Lett.*, 1994, 4, 1053-1060), a thioether, e.g., hexyl-S-tritylthiol (Manoharan et al., *Ann. N.Y. Acad. Sci.*, 1992, 660, 306-309; Manoharan et al., *Bioorg. Med. Chem. Lett.*, 1993, 3, 2765-2770), a thiocholesterol (Oberhauser et al., *Nucl. Acids Res.*, 1992, 20, 533-538), an aliphatic chain, e.g., do-decan-diol or undecyl residues (Saison-Behmoaras et al., *EMBO J*, 1991, 10, 1111-1118; Kabanov et al., *FEBS Lett.*, 1990, 259, 327-330; Svinarchuk et al., *Biochimie*, 1993, 75, 49-54), a phospholipid, e.g., di-hexadecyl-rac-glycerol or triethyl-ammonium 1,2-di-O-hexadecyl-rac-glycero-3-H-phosphonate (Manoharan et al., *Tetrahedron Lett.*, 1995, 36, 3651-3654; Shea et al., *Nucl. Acids Res.*, 1990, 18, 3777-3783), a polyamine or a polyethylene glycol chain (Manoharan et al., *Nucleosides & Nucleotides*, 1995, 14, 969-973), or adamantane acetic acid a palmityl moiety (Mishra et al., *Biochim. Biophys. Acta*, 1995, 1264, 229-237), an octadecylamine or hexylamino-carbonyl-oxysterol moiety (Crooke et al., *J. Pharmacol. Exp. Ther.*, 1996, 277, 923-937), a tocopherol group (Nishina et al., *Molecular Therapy Nucleic Acids*, 2015, 4, e220; and Nishina et al., *Molecular Therapy*, 2008, 16, 734-740), or a GalNAc cluster (e.g., WO2014/179620).

[0292] In certain embodiments, conjugate groups may be selected from any of a C22 alkyl, C20 alkyl, C16 alkyl, C10 alkyl, C21 alkyl, C19 alkyl, C18 alkyl, C15 alkyl, C14 alkyl, C13 alkyl, C12 alkyl, C11 alkyl, C9 alkyl, C8 alkyl, C7 alkyl, C6 alkyl, C5 alkyl, C22 alkenyl, C20 alkenyl, C16 alkenyl, C10 alkenyl, C21 alkenyl, C19 alkenyl, C18 alkenyl, C15 alkenyl, C14 alkenyl, C13 alkenyl, C12 alkenyl, C11 alkenyl, C9 alkenyl, C8 alkenyl, C7 alkenyl, C6 alkenyl, or C5 alkenyl.

[0293] In certain embodiments, conjugate groups may be selected from any of C22 alkyl, C20 alkyl, C16 alkyl, C10 alkyl, C21 alkyl, C19 alkyl, C18 alkyl, C15 alkyl, C14 alkyl, C13 alkyl, C12 alkyl, C11 alkyl, C9 alkyl, C8 alkyl, C7 alkyl, C6 alkyl, and C5 alkyl, where the alkyl chain has one or more unsaturated bonds.

1. Conjugate Moieties

[0294] Conjugate moieties include, without limitation, intercalators, reporter molecules, polyamines, polyamides, peptides, carbohydrates, vitamin moieties, polyethylene glycols, thioethers, polyethers, cholesterol, thiocholesterols, cholic acid moieties, folate, lipids, phospholipids, biotin, phenazine, phenanthridine, anthraquinone, adamantane, acridine, fluoresceins, rhodamines, coumarins, fluorophores, and dyes.

[0295] In certain embodiments, a conjugate moiety comprises an active drug substance, for example, aspirin, warfarin, phenylbutazone, ibuprofen, suprofen, fen-bufen, ketoprofen, (S)-(+)-pranoprofen, carprofen, dansylsarcosine, 2,3,5-triiodobenzoic acid, fingolimod, flufenamic acid, folic acid, a benzothiadiazide, chlorothiazide, a diazepam, indo-methicin, a barbiturate, a cephalosporin, a sulfa drug, an antidiabetic, an antibacterial or an antibiotic.

2. Conjugate Linkers

[0296] Conjugate moieties are attached to oligonucleotides through conjugate linkers. In certain oligomeric compounds, the conjugate linker is a single chemical bond (i.e., the conjugate moiety is attached directly to an oligonucleotide through a single bond). In certain embodiments, the conjugate linker comprises a chain structure, such as a hydrocarbyl chain, or an oligomer of repeating units such as ethylene glycol, nucleosides, or amino acid units.

[0297] In certain embodiments, a conjugate linker comprises one or more groups selected from alkyl, amino, oxo, amide, disulfide, polyethylene glycol, ether, thioether, and hydroxylamino. In certain such embodiments, the conjugate linker comprises groups selected from alkyl, amino, oxo, amide and ether groups. In certain embodiments, the conjugate linker comprises groups selected from alkyl and amide groups. In certain embodiments, the conjugate linker comprises groups selected from alkyl and ether groups. In certain embodiments, the conjugate linker comprises at least one phosphorus moiety. In certain embodiments, the conjugate linker comprises at least one phosphate group. In certain embodiments, the conjugate linker includes at least one neutral linking group.

[0298] In certain embodiments, conjugate linkers, including the conjugate linkers described above, are bifunctional linking moieties, e.g., those known in the art to be useful for attaching conjugate groups to parent compounds, such as the oligonucleotides provided herein. In general, a bifunctional linking moiety comprises at least two functional groups. One of the functional groups is selected to bind to a particular site on a parent compound and the other is selected to bind to a conjugate group. Examples of functional groups used in a bifunctional linking moiety include but are not limited to electrophiles for reacting with nucleophilic groups and nucleophiles for reacting with electrophilic groups. In certain embodiments, bifunctional linking moieties comprise one or more groups selected from amino, hydroxyl, carboxylic acid, thiol, alkyl, alkenyl, and alkynyl.

[0299] Examples of conjugate linkers include but are not limited to pyrrolidine, 8-amino-3,6-dioxaoctanoic acid (ADO), succinimidyl 4-(N-maleimidomethyl) cyclohexane-1-carboxylate (SMCC) and 6-aminoheptanoic acid (AHEX or AHA). Other conjugate linkers include but are not limited to substituted or unsubstituted C.sub.1-C.sub.10 alkyl, substituted or unsubstituted C.sub.2-C.sub.10 alkenyl or substituted or unsubstituted C.sub.2-C.sub.10 alkynyl, wherein a nonlimiting list of preferred substituent groups includes hydroxyl, amino, alkoxy, carboxy, benzyl, phenyl, nitro, thiol, thioalkoxy, halogen, alkyl, aryl, alkenyl and alkynyl.

[0300] In certain embodiments, conjugate linkers comprise 1-10 linker-nucleosides. In certain embodiments, conjugate linkers comprise 2-5 linker-nucleosides. In certain embodiments, conjugate linkers comprise exactly 3 linker-nucleosides. In certain embodiments, conjugate linkers comprise the TCA motif. In certain embodiments, such linker-nucleosides are modified nucleosides. In certain embodiments such linker-nucleosides comprise a modified sugar moiety. In certain embodiments, linker-nucleosides are unmodified. In certain embodiments, linker-nucleosides comprise an optionally protected heterocyclic base selected from a purine, substituted purine, pyrimidine or substituted pyrimidine. In certain embodiments, a cleavable moiety is a nucleoside selected from uracil, thymine, cytosine, 4-N-benzoylcytosine, 5-methyl cytosine, 4-N-benzoyl-5-methyl cytosine, adenine, 6-N-benzoyladenine, guanine and 2-N-isobutylguanine. It is typically desirable for linker-nucleosides to be cleaved from the oligomeric compound after it reaches a target tissue. Accordingly, linker-nucleosides are typically linked to one another and to the remainder of the oligomeric compound through cleavable bonds. In certain embodiments, such cleavable bonds are phosphodiester bonds.

[0301] Herein, linker-nucleosides are not considered to be part of the oligonucleotide. Accordingly, in embodiments in which an oligomeric compound comprises an oligonucleotide consisting of a specified number or range of linked nucleosides and/or a specified percent complementarity to a reference nucleic acid and the oligomeric compound also comprises a conjugate group comprising a conjugate linker comprising linker-nucleosides, those linker-nucleosides are not counted toward the length of the oligonucleotide and are not used in determining the percent complementarity of the oligonucleotide for the reference nucleic acid. For example, an oligomeric compound may comprise (1) a modified oligonucleotide consisting of 8-30 nucleosides and (2) a conjugate group comprising 1-10 linker-nucleosides that are contiguous with the nucleosides of the modified oligonucleotide. The total number of contiguous linked nucleosides in such an oligomeric compound is more than 30. Alternatively, an oligomeric compound may comprise a modified oligonucleotide consisting of 8-30 nucleosides and no conjugate group. The total number of contiguous linked nucleosides in such an oligomeric compound is no more than 30. Unless otherwise indicated conjugate linkers comprise no more than 10 linker-nucleosides. In certain embodiments, conjugate linkers comprise no more than 5 linker-nucleosides. In certain embodiments, conjugate linkers comprise no more than 3 linker-nucleosides. In certain embodiments, conjugate linkers comprise no more than 2 linker-nucleosides. In certain embodiments, conjugate linkers comprise no more than 1 linker-nucleoside.

[0302] In certain embodiments, it is desirable for a conjugate group to be cleaved from the oligonucleotide. For example, in certain circumstances oligomeric compounds comprising a particular conjugate moiety are better taken up by a particular cell type, but once the oligomeric compound has been taken up, it is desirable that the conjugate group be cleaved to release the unconjugated or parent oligonucleotide. Thus, certain conjugate linkers may comprise one or more cleavable moieties. In certain embodiments, a cleavable moiety is a cleavable bond. In certain embodiments, a cleavable moiety is a group of atoms comprising at least one cleavable bond. In certain embodiments, a cleavable moiety comprises a group of atoms having one, two, three, four, or more than four cleavable bonds. In certain embodiments, a cleavable moiety is selectively cleaved inside a cell or subcellular compartment, such as a lysosome. In certain embodiments, a cleavable moiety is selectively cleaved by endogenous enzymes, such as nucleases.

[0303] In certain embodiments, a cleavable bond is selected from among: an amide, an ester, an ether, one or both esters of a phosphodiester, a phosphate ester, a carbamate, or a disulfide. In certain embodiments, a cleavable bond is one or both of the esters of a phosphodiester. In certain embodiments, a cleavable moiety comprises a phosphate or phosphodiester. In certain embodiments, the cleavable moiety is a phosphate or phosphodiester linkage between an oligonucleotide and a conjugate moiety or conjugate group.

[0304] In certain embodiments, a cleavable moiety comprises or consists of one or more linker-nucleosides. In certain such embodiments, the one or more linker-nucleosides are linked to one another and/or to the remainder of the oligomeric compound through cleavable bonds. In certain embodiments, such cleavable bonds are unmodified phosphodiester bonds. In certain embodiments, a cleavable moiety is 2'-deoxynucleoside that is attached to either the 3' or 5'-terminal nucleoside of an oligonucleotide by a phosphodiester internucleoside linkage and covalently attached to the remainder of the conjugate linker or conjugate moiety by a phosphate or phosphorothioate linkage. In certain such embodiments, the cleavable moiety is 2'-deoxyadenosine.

3. Cell-Targeting Moieties

[0305] In certain embodiments, a conjugate group comprises a cell-targeting moiety. In certain embodiments, a conjugate group has the general formula:

##STR00018## [0306] wherein n is from 1 to about 3, m is 0 when n is 1, m is 1 when n is 2 or greater, j is 1 or 0, and k is 1 or 0.

[0307] In certain embodiments, n is 1, j is 1 and k is 0. In certain embodiments, n is 1, j is 0 and k is 1. In certain embodiments, n is 1, j is 1 and k is 1. In certain embodiments, n is 2, j is 1 and k is 0. In certain embodiments, n is 2, j is 0 and k is 1. In certain embodiments, n is 2, j is 1 and k is 1. In certain embodiments, n is 3, j is 1 and k is 0. In certain embodiments, n is 3, j is 0 and k is 1. In certain embodiments, n is 3, j is 1 and k is 1.

[0308] In certain embodiments, conjugate groups comprise cell-targeting moieties that have at least one tethered ligand. In certain embodiments, cell-targeting moieties comprise two tethered ligands covalently attached to a branching group. In certain embodiments, cell-targeting moieties comprise three tethered ligands covalently attached to a branching group.

B. Certain Terminal Groups

[0309] In certain embodiments, oligomeric compounds comprise one or more terminal groups. In certain such embodiments, oligomeric compounds comprise a stabilized 5'-phosphate. Stabilized 5'-phosphates include, but are not limited to 5'-phosphanates, including, but not limited to 5'-vinylphosphonates. In certain embodiments, terminal groups comprise one or more abasic nucleosides and/or inverted nucleosides. In certain embodiments, terminal groups comprise one or more 2'-linked nucleosides. In certain such embodiments, the 2'-linked nucleoside is an abasic nucleoside.

III. Oligomeric Duplexes

[0310] In certain embodiments, oligomeric compounds described herein comprise an oligonucleotide, having a nucleobase sequence complementary to that of a target nucleic acid. In certain embodiments, an oligomeric compound is paired with a second oligomeric compound to form an oligomeric duplex. Such oligomeric duplexes comprise a first oligomeric compound having a portion complementary to a target nucleic acid and a second oligomeric compound having a portion complementary to the first oligomeric compound. In certain embodiments, the first oligomeric compound of an oligomeric duplex comprises or consists of (1) a modified or unmodified oligonucleotide and optionally a conjugate group and (2) a second modified or unmodified oligonucleotide and optionally a conjugate group. Either or both oligomeric compounds of an oligomeric duplex may comprise a conjugate group. The oligonucleotides of each oligomeric compound of an oligomeric duplex may include non-complementary overhanging nucleosides.

IV. Antisense Activity

[0311] In certain embodiments, oligomeric compounds and oligomeric duplexes are capable of hybridizing to a target nucleic acid, resulting in at least one antisense activity; such oligomeric compounds and oligomeric duplexes are antisense compounds. In certain embodiments, antisense compounds have antisense activity when they reduce or inhibit the amount or activity of a target nucleic acid by 25% or more in the standard cell assay. In certain embodiments, antisense compounds selectively affect one or more target nucleic acid. Such antisense compounds comprise a nucleobase sequence that hybridizes to one or more target nucleic acid, resulting in one or more desired antisense activity and does not hybridize to one or more non-target nucleic acid or does not hybridize to one or more non-target nucleic acid in such a way that results in significant undesired antisense activity.

[0312] In certain antisense activities, hybridization of an antisense compound to a target nucleic acid results in recruitment of a protein that cleaves the target nucleic acid. For example, certain antisense compounds result in RNase H mediated cleavage of the target nucleic acid. RNase H is a cellular endonuclease that cleaves the RNA strand of an RNA:DNA duplex. The DNA in such an RNA:DNA duplex need not be unmodified DNA. In certain embodiments, described herein are antisense compounds that are sufficiently "DNA-like" to elicit RNase H activity. In certain embodiments, one or more non-DNA-like nucleoside in the gap of a gapmer is tolerated.

[0313] In certain antisense activities, an antisense compound or a portion of an antisense compound is loaded into an RNA-induced silencing complex (RISC), ultimately resulting in cleavage of the target nucleic acid. For example, certain antisense compounds result in cleavage of the target nucleic acid by Argonaute. Antisense compounds that are loaded into RISC are RNAi compounds. RNAi compounds may be double-stranded (siRNA) or single-stranded (ssRNA).

[0314] In certain embodiments, hybridization of an antisense compound to a target nucleic acid does not result in recruitment of a protein that cleaves that target nucleic acid. In certain embodiments, hybridization of the antisense compound to the target nucleic acid results in alteration of splicing of the target nucleic acid. In certain embodiments, hybridization of an antisense compound to a target nucleic acid results in inhibition of a binding interaction between the target nucleic acid and a protein or other nucleic acid. In certain embodiments, hybridization of an antisense compound to a target nucleic acid results in alteration of translation of the target nucleic acid.

[0315] Antisense activities may be observed directly or indirectly. In certain embodiments, observation or detection of an antisense activity involves observation or detection of a change in an amount of a target nucleic acid or protein encoded by such target nucleic acid, a change in the ratio of splice variants of a nucleic acid or protein and/or a phenotypic change in a cell or subject.

V. Certain Target Nucleic Acids

[0316] In certain embodiments, oligomeric compounds comprise or consist of an oligonucleotide comprising a portion that is complementary to a target nucleic acid. In certain embodiments, the target nucleic acid is an endogenous RNA molecule. In certain embodiments, the target nucleic acid encodes a protein. In certain such embodiments, the target nucleic acid is selected from: an endogenous antisense transcript that does not encode a protein (e.g., UBE3A-ATS), a mature mRNA, and a pre-mRNA, including intronic, exonic and untranslated regions. In certain embodiments, the target RNA is an antisense transcript. In certain embodiments, the target RNA is a mature mRNA. In certain embodiments, the target nucleic acid is a pre-mRNA. In certain such embodiments, the target region is entirely within an intron. In certain embodiments, the target region spans an

intron/exon junction. In certain embodiments, the target region is at least 50% within an intron. In certain embodiments, the target nucleic acid is the RNA transcriptional product of a retrogene. In certain embodiments, the target nucleic acid is a non-coding RNA. In certain such embodiments, the target non-coding RNA is selected from: a long non-coding RNA, a short non-coding RNA, an intronic RNA molecule.

A. Complementarity/Mismatches to the Target Nucleic Acid

[0317] It is possible to introduce mismatch bases without eliminating activity. For example, Gautschi et al (J. Natl. Cancer Inst. 93:463-471, March 2001) demonstrated the ability of an oligonucleotide having 100% complementarity to the bcl-2 mRNA and having 3 mismatches to the bcl-xL mRNA to reduce the expression of both bcl-2 and bcl-xL in vitro and in vivo. Furthermore, this oligonucleotide demonstrated potent anti-tumor activity in vivo. Maher and Dolnick (Nuc. Acid. Res. 16:3341-3358, 1988) tested a series of tandem 14 nucleobase oligonucleotides, and 28 and 42 nucleobase oligonucleotides comprised of the sequence of two or three of the tandem oligonucleotides, respectively, for their ability to arrest translation of human DHFR in a rabbit reticulocyte assay. Each of the three 14 nucleobase oligonucleotides alone was able to inhibit translation, albeit at a more modest level than the 28 or 42 nucleobase oligonucleotides.

[0318] In certain embodiments, oligonucleotides are complementary to the target nucleic acid over the entire length of the oligonucleotide. In certain embodiments, oligonucleotides are 99%, 95%, 90%, 85%, or 80% complementary to the target nucleic acid. In certain embodiments, oligonucleotides are at least 80% complementary to the target nucleic acid over the entire length of the oligonucleotide and comprise a portion that is 100% or fully complementary to a target nucleic acid. In certain embodiments, the portion of full complementarity is 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, or 24 nucleobases in length.

[0319] In certain embodiments, oligonucleotides comprise one or more mismatched nucleobases relative to the target nucleic acid. In certain embodiments, antisense activity against the target is reduced by such mismatch, but activity against a non-target is reduced by a greater amount. Thus, in certain embodiments selectivity of the oligonucleotide is improved. In certain embodiments, the mismatch is specifically positioned within an oligonucleotide having a gapmer motif. In certain embodiments, the mismatch is at position 1, 2, 3, 4, 5, 6, 7, or 8 from the 5'-end of the gap region. In certain embodiments, the mismatch is at position 9, 8, 7, 6, 5, 4, 3, 2, 1 from the 3'-end of the gap region. In certain embodiments, the mismatch is at position 1, 2, 3, or 4 from the 5'-end of the wing region. In certain embodiments, the mismatch is at position 4, 3, 2, or 1 from the 3'-end of the wing region.

B. UBE3A-ATS

[0320] In certain embodiments, oligomeric compounds comprise or consist of an oligonucleotide, or portion thereof, that is complementary to a target nucleic acid, wherein the target nucleic acid is UBE3A-ATS. In certain embodiments, UBE3A-ATS nucleic acid has the sequence set forth in SEQ ID NO: 1 (GENBANK Accession No: NC_000015.10_TRUNC_24821647_25441028), SEQ ID NO: 2915 (Ensemble Gene ID ENSG00000224078), or SEQ ID NO: 2916 (the cDNA of Ensemble transcript ENST00000554726.1).

[0321] In certain embodiments, an oligomeric compound complementary to SEQ ID NO: 1, SEQ ID NO: 2915, or SEQ ID NO: 2916 is capable of reducing UBE3A-ATS in a cell. In certain embodiments, an oligomeric compound complementary to SEQ ID NO: 1 is capable of increasing UBE3A RNA or protein in a cell. In certain embodiments, an oligomeric compound complementary to SEQ ID NO: 1, SEQ ID NO: 2915, or SEQ ID NO: 2916 is capable of increasing paternal UBE3A RNA or protein in a cell. In certain embodiments, the cell is in vitro. In certain embodiments, the cell is in a subject. In certain embodiments, the oligomeric compound consists of a modified oligonucleotide. In certain embodiments, an oligomeric compound complementary to SEQ ID NO: 1, SEQ ID NO: 2915, or SEQ ID NO: 2916 is capable of ameliorating one or more symptom or hallmark of a neurogenetic disorder when administered to a subject. In certain embodiments, the neurogenetic disorder is AS. In certain embodiments, the symptoms or hallmarks are selected from developmental delays, ataxia, speech impairment, sleep problems, seizures, and EEG abnormalities.

[0322] In certain embodiments, an oligomeric compound complementary to SEQ ID NO: 1, SEQ ID NO: 2915, or SEQ ID NO: 2916 is capable of reducing the detectable amount of UBE3A-ATS RNA in vitro by at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90%. In certain embodiments, an oligomeric compound complementary to SEQ ID NO: 1, SEQ ID NO: 2915, or SEQ ID NO: 2916 is capable of increasing the detectable amount of UBE3A protein in vitro by at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90%. In certain embodiments, an oligomeric compound complementary to SEQ ID NO: 1, SEQ ID NO: 2915, or SEQ ID NO: 2916 is capable of reducing the detectable amount of UBE3A-ATS RNA in the CSF of a subject by at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90%. In certain embodiments, an oligomeric compound complementary to SEQ ID NO: 1, SEQ ID NO: 2915, or SEQ ID NO: 2916 is capable of increasing the detectable amount of UBE3A protein in the CSF of a subject by at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, or at least 90%.

C. Certain Target Nucleic Acids in Certain Tissues

[0323] In certain embodiments, oligomeric compounds comprise or consist of an oligonucleotide comprising a portion that is complementary to a target nucleic acid, wherein the target nucleic acid is expressed in a pharmacologically relevant tissue. In certain embodiments, the pharmacologically relevant tissues are the cells and tissues that comprise the central nervous system. Such tissues include the cortex, hippocampus, and spinal cord.

VI. Certain Pharmaceutical Compositions

[0324] In certain embodiments, described herein are pharmaceutical compositions comprising one or more oligomeric compounds. In certain embodiments, the one or more oligomeric compounds each consists of a modified oligonucleotide. In certain embodiments, the pharmaceutical composition comprises a pharmaceutically acceptable diluent or carrier. In certain embodiments, a pharmaceutical composition comprises or consists of a sterile saline solution and one or more oligomeric compound. In certain embodiments, the sterile saline is pharmaceutical grade saline. In certain embodiments, a pharmaceutical composition comprises or consists of one or more oligomeric compound and sterile water. In certain embodiments, the sterile water is pharmaceutical grade water. In certain embodiments, a pharmaceutical composition comprises or consists of one or more oligomeric compound and phosphate-buffered saline (PBS). In certain embodiments, the sterile PBS is pharmaceutical grade PBS. In certain embodiments, a pharmaceutical composition comprises or consists of one or more oligomeric compound and artificial cerebrospinal fluid. In certain embodiments, the artificial cerebrospinal fluid is pharmaceutical grade.

[0325] In certain embodiments, a pharmaceutical composition comprises a modified oligonucleotide and artificial cerebrospinal fluid. In certain embodiments, a pharmaceutical composition consists of a modified oligonucleotide and artificial cerebrospinal fluid. In certain embodiments, a pharmaceutical composition consists essentially of a modified oligonucleotide and artificial cerebrospinal fluid. In certain embodiments, the artificial cerebrospinal fluid is pharmaceutical grade.

[0326] In certain embodiments, pharmaceutical compositions comprise one or more oligomeric compound and one or more excipients. In certain embodiments, excipients are selected from water, salt solutions, alcohol, polyethylene glycols, gelatin, lactose, amylase, magnesium stearate, talc, silicic acid, viscous paraffin, hydroxymethylcellulose and polyvinylpyrrolidone.

[0327] In certain embodiments, oligomeric compounds may be admixed with pharmaceutically acceptable active and/or inert substances for the preparation of pharmaceutical compositions or formulations. Compositions and methods for the formulation of pharmaceutical compositions depend on a number of criteria, including, but not limited to, route of administration, extent of disease, or dose to be administered.

[0328] In certain embodiments, pharmaceutical compositions comprising an oligomeric compound encompass any pharmaceutically acceptable salts of the oligomeric compound, esters of the oligomeric compound, or salts of such esters. In certain embodiments, pharmaceutical compositions comprising oligomeric compounds comprising one or more oligonucleotide, upon administration to a subject, including a human, are capable of providing (directly or indirectly) the biologically active metabolite or residue thereof. Accordingly, for example, the disclosure is also drawn to

pharmaceutically acceptable salts of oligomeric compounds, prodrugs, pharmaceutically acceptable salts of such prodrugs, and other bioequivalents. Suitable pharmaceutically acceptable salts include, but are not limited to, sodium and potassium salts. In certain embodiments, prodrugs comprise one or more conjugate group attached to an oligonucleotide, wherein the conjugate group is cleaved by endogenous nucleases within the body. [0329] Lipid moieties have been used in nucleic acid therapies in a variety of methods. In certain such methods, the nucleic acid, such as an oligomeric compound, is introduced into preformed liposomes or lipoplexes made of mixtures of cationic lipids and neutral lipids. In certain methods, DNA complexes with mono- or poly-cationic lipids are formed without the presence of a neutral lipid. In certain embodiments, a lipid moiety is selected to increase distribution of a pharmaceutical agent to a particular cell or tissue. In certain embodiments, a lipid moiety is selected to increase distribution of a pharmaceutical agent to fat tissue. In certain embodiments, a lipid moiety is selected to increase distribution of a pharmaceutical agent to muscle tissue.

[0330] In certain embodiments, pharmaceutical compositions comprise a delivery system. Examples of delivery systems include, but are not limited to, liposomes and emulsions. Certain delivery systems are useful for preparing certain pharmaceutical compositions including those comprising hydrophobic compounds. In certain embodiments, certain organic solvents such as dimethylsulfoxide are used.

[0331] In certain embodiments, pharmaceutical compositions comprise one or more tissue-specific delivery molecules designed to deliver the one or more pharmaceutical agents of the present invention to specific tissues or cell types. For example, in certain embodiments, pharmaceutical compositions include liposomes coated with a tissue-specific antibody.

[0332] In certain embodiments, pharmaceutical compositions comprise a co-solvent system. Certain of such co-solvent systems comprise, for example, benzyl alcohol, a nonpolar surfactant, a water-miscible organic polymer, and an aqueous phase. In certain embodiments, such co-solvent systems are used for hydrophobic compounds. A non-limiting example of such a co-solvent system is the VPD co-solvent system, which is a solution of absolute ethanol comprising 3% w/v benzyl alcohol, 8% w/v of the nonpolar surfactant Polysorbate 80™ and 65% w/v polyethylene glycol 300. The proportions of such co-solvent systems may be varied considerably without significantly altering their solubility and toxicity characteristics. Furthermore, the identity of co-solvent components may be varied: for example, other surfactants may be used instead of Polysorbate 80™; the fraction size of polyethylene glycol may be varied; other biocompatible polymers may replace polyethylene glycol, e.g., polyvinyl pyrrolidone; and other sugars or polysaccharides may substitute for dextrose.

[0333] In certain embodiments, pharmaceutical compositions are prepared for oral administration. In certain embodiments, pharmaceutical compositions are prepared for buccal administration. In certain embodiments, a pharmaceutical composition is prepared for administration by injection (e.g., intravenous, subcutaneous, intramuscular, intrathecal (IT), intracerebroventricular (ICV), etc.). In certain of such embodiments, a pharmaceutical composition comprises a carrier and is formulated in aqueous solution, such as water or physiologically compatible buffers such as Hanks's solution, Ringer's solution, or physiological saline buffer. In certain embodiments, other ingredients are included (e.g., ingredients that aid in solubility or serve as preservatives). In certain embodiments, injectable suspensions are prepared using appropriate liquid carriers, suspending agents and the like. Certain pharmaceutical compositions for injection are presented in unit dosage form, e.g., in ampoules or in multi-dose containers. Certain pharmaceutical compositions for injection are suspensions, solutions or emulsions in oily or aqueous vehicles, and may contain formulatory agents such as suspending, stabilizing and/or dispersing agents. Certain solvents suitable for use in pharmaceutical compositions for injection include, but are not limited to, lipophilic solvents and fatty oils, such as sesame oil, synthetic fatty acid esters, such as ethyl oleate or triglycerides, and liposomes.

[0334] Under certain conditions, certain compounds disclosed herein act as acids. Although such compounds may be drawn or described in protonated (free acid) form, or ionized and in association with a cation (salt) form, aqueous solutions of such compounds exist in equilibrium among such forms. For example, a phosphate linkage of an oligonucleotide in aqueous solution exists in equilibrium among free acid, anion and salt forms. Unless otherwise indicated, compounds described herein are intended to include all such forms. Moreover, certain oligonucleotides have several such linkages, each of which is in equilibrium. Thus, oligonucleotides in solution exist in an ensemble of forms at multiple positions all at equilibrium. The term "oligonucleotide" is intended to include all such forms. Drawn structures necessarily depict a single form. Nevertheless, unless otherwise indicated, such drawings are likewise intended to include corresponding forms. Herein, a structure depicting the free acid of a compound followed by the term "or a salt thereof" expressly includes all such forms that may be fully or partially protonated/de-protonated/in association with a cation. In certain instances, one or more specific cation is identified.

[0335] In certain embodiments, modified oligonucleotides or oligomeric compounds are in aqueous solution with sodium. In certain embodiments, modified oligonucleotides or oligomeric compounds are in aqueous solution with potassium. In certain embodiments, modified oligonucleotides or oligomeric compounds are in PBS. In certain embodiments, modified oligonucleotides or oligomeric compounds are in water. In certain such embodiments, the pH of the solution is adjusted with NaOH and/or HCl to achieve a desired pH.

[0336] Herein, certain specific doses are described. A dose may be in the form of a dosage unit. For clarity, a dose (or dosage unit) of a modified oligonucleotide or an oligomeric compound in milligrams indicates the mass of the free acid form of the modified oligonucleotide or oligomeric compound. As described above, in aqueous solution, the free acid is in equilibrium with anionic and salt forms. However, for the purpose of calculating dose, it is assumed that the modified oligonucleotide or oligomeric compound exists as a solvent-free, sodium-acetate free, anhydrous, free acid. For example, where a modified oligonucleotide or an oligomeric compound is in solution comprising sodium (e.g., saline), the modified oligonucleotide or oligomeric compound may be partially or fully de-protonated and in association with Na⁺ ions. However, the mass of the protons is nevertheless counted toward the weight of the dose, and the mass of the Na⁺ ions are not counted toward the weight of the dose. Thus, for example, a dose, or dosage unit, of 100 mg of Compound No. 1263518 equals the number of fully protonated molecules that weighs 100 mg. This would be equivalent to 106 mg of solvent-free, sodium-acetate free, anhydrous sodiated Compound No. 1263518. When an oligomeric compound comprises a conjugate group, the mass of the conjugate group is included in calculating the dose of such oligomeric compound. If the conjugate group also has an acid, the conjugate group is likewise assumed to be fully protonated for the purpose of calculating dose.

VI. Certain Compositions

1. Compound No. 1065645

[0337] In certain embodiments, Compound No. 1065645 is characterized as a 5-10-5 MOE gapmer having a sequence (from 5' to 3') of CATCATGATCTTGGTAAGGC (SEQ ID NO: 1949), wherein each of nucleosides 1-5 and 16-20 (from 5' to 3') are 2'-MOE nucleosides and each of nucleosides 6-15 are 2'-β-D-deoxynucleosides, wherein the internucleoside linkages between nucleosides 2 to 3, 3 to 4, 4 to 5, 5 to 6, 16 to 17, and 17 to 18 are phosphodiester internucleoside linkages and the internucleoside linkages between nucleosides 1 to 2, 6 to 7, 7 to 8, 8 to 9, 9 to 10, 10 to 11, 11 to 12, 12 to 13, 13 to 14, 14 to 15, 15 to 6, 18 to 19, and 19 to 20 are phosphorothioate internucleoside linkages, and wherein each cytosine is a 5-methyl cytosine.

[0338] In certain embodiments, Compound No. 1065645 is represented by the following chemical notation:

.sup.mC.sub.esA.sub.eoT.sub.eo.sup.mC.sub.eoA.sub.eoT.sub.dsG.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.dsG.sub.dsG.sub.dsT.sub.dsA.sub.eoA (SEQ ID NO: 2917), wherein: [0339] A=an adenine nucleobase, [0340] mC=a 5-methyl cytosine nucleobase, [0341] G=a guanine nucleobase, [0342] T=a thymine nucleobase, [0343] e=a 2'-MOE sugar moiety, [0344] d=a 2'-β-D-deoxyribose sugar moiety, [0345] s=a phosphorothioate internucleoside linkage, and [0346] o=a phosphodiester internucleoside linkage.

[0347] In certain embodiments, Compound No. 1065645 is represented by the following chemical structure:

##STR00019##

[0348] In certain embodiments, the sodium salt of Compound No. 1065645 is represented by the following chemical

##STR00020##

2. Compound No. 1263517

[0349] In certain embodiments, Compound No. 1263517 is characterized as a 6-10-4 MOE gapmer having a sequence (from 5' to 3') of TCACCATTTTGACCTTCTTA (SEQ ID NO: 2751), wherein each of nucleosides 1-6 and 17-20 (from 5' to 3') are 2'-MOE nucleosides and each of nucleosides 7-16 are 2'- β -D-deoxynucleosides, 10 wherein the internucleoside linkages between nucleosides 2 to 3, 3 to 4, 4 to 5, 5 to 6, 6 to 7, and 17 to 18 are phosphodiester internucleoside linkages and the internucleoside linkages between nucleosides 1 to 2, 7 to 8, 8 to 9, 9 to 10, 10 to 11, 11 to 12, 12 to 13, 13 to 14, 14 to 15, 15 to 16, 16 to 17, 18 to 19, and 19 to 20 are phosphorothioate internucleoside linkages, and wherein each cytosine is a 5-methyl cytosine.

[0350] In certain embodiments, Compound No. 1263517 is represented by the following chemical notation:

T.sub.es.sup.mC.sub.eoA.sub.eo.sup.mC.sub.eo.sup.mC.sub.eoA.sub.eoT.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.ds (SEQ ID NO: 2918), wherein: [0351] A=an adenine nucleobase, [0352] mC=a 5-methyl cytosine nucleobase, [0353] G=a guanine nucleobase, [0354] T=a thymine nucleobase, [0355] e=a 2'-MOE sugar moiety, [0356] d=a 2'- β -D-deoxyribose sugar moiety, [0357] s=a phosphorothioate internucleoside linkage, and [0358] o=a phosphodiester internucleoside linkage.

[0359] In certain embodiments, Compound No. 1263517 is represented by the following chemical structure:

##STR00021##

[0360] In certain embodiments, the sodium salt of Compound No. 1263517 is represented by the following chemical structure:

##STR00022##

3. Compound No. 1263518

[0361] In certain embodiments, Compound No. 1263518 is characterized as a 6-10-4 MOE gapmer having a sequence (from 5' to 3') of TTCACCATTTTGACCTTCTT (SEQ ID NO: 2752), wherein each of nucleosides 1-6 and 17-20 (from 5' to 3') are 2'-MOE nucleosides and each of nucleosides 7-16 are 2'- β -D-deoxynucleosides, wherein the internucleoside linkages between nucleosides 2 to 3, 3 to 4, 4 to 5, 5 to 6, 6 to 7, and 17 to 18 are phosphodiester internucleoside linkages and the internucleoside linkages between nucleosides 1 to 2, 7 to 8, 8 to 9, to 10, 10 to 11, 11 to 12, 12 to 13, 13 to 14, 14 to 15, 15 to 16, 16 to 17, 18 to 19, and 19 to 20 are phosphorothioate internucleoside linkages, and wherein each cytosine is a 5-methyl cytosine.

[0362] In certain embodiments, Compound No. 1263518 is represented by the following chemical notation:

T.sub.esT.sub.eo.sup.mC.sub.eoA.sub.eo.sup.mC.sub.eo.sup.mC.sub.eoA.sub.dsT.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.ds (SEQ ID NO: 2919), wherein: [0363] A=an adenine nucleobase, [0364] mC=a 5-methyl cytosine nucleobase, [0365] G=a guanine nucleobase, [0366] T=a thymine nucleobase, [0367] e=a 2'-MOE sugar moiety, [0368] d=a 2'- β -D-deoxyribose sugar moiety, [0369] s=a phosphorothioate internucleoside linkage, and [0370] o=a phosphodiester internucleoside linkage.

[0371] In certain embodiments, Compound No. 1263518 is represented by the following chemical structure:

##STR00023##

[0372] In certain embodiments, the sodium salt of Compound No. 1263518 is represented by the following chemical structure:

##STR00024##

4. Compound No. 1263533

[0373] In certain embodiments, Compound No. 1263533 is characterized as a 6-10-4 MOE gapmer having a sequence (from 5' to 3') of GCATACCCAGGGTAGGATTC (SEQ ID NO: 765), wherein each of nucleosides 1-6 and 17-20 (from 5' to 3') are 2'-MOE nucleosides and each of nucleosides 7-16 are 2'- β -D-deoxynucleosides, wherein the internucleoside linkages between nucleosides 2 to 3, 3 to 4, 4 to 5, 5 to 6, 6 to 7, and 17 to 18 are phosphodiester internucleoside linkages and the internucleoside linkages between nucleosides 1 to 2, 7 to 8, 8 to 9, to 10, 10 to 11, 11 to 12, 12 to 13, 13 to 14, 14 to 15, 15 to 16, 16 to 17, 18 to 19, and 19 to 20 are phosphorothioate internucleoside linkages, and wherein each cytosine is a 5-methyl cytosine.

[0374] In certain embodiments, Compound No. 1263533 is represented by the following chemical notation:

G.sub.es.sup.mC.sub.eoA.sub.eoT.sub.eoA.sub.eo.sup.mC.sub.eo.sup.mC.sub.ds.sup.mC.sub.dsA.sub.dsG.sub.dsG.sub.dsG.sub.dsT.sub.dsA.sub.dsG.sub.dsC (SEQ ID NO: 2920), wherein: [0375] A=an adenine nucleobase, [0376] mC=a 5-methyl cytosine nucleobase, [0377] G=a guanine nucleobase, [0378] T=a thymine nucleobase, [0379] e=a 2'-MOE sugar moiety, [0380] d=a 2'- β -D-deoxyribose sugar moiety, [0381] s=a phosphorothioate internucleoside linkage, and [0382] o=a phosphodiester internucleoside linkage.

[0383] In certain embodiments, Compound No. 1263533 is represented by the following chemical structure:

##STR00025##

[0384] In certain embodiments, the sodium salt of Compound No. 1263533 is represented by the chemical structure:

##STR00026##

5. Compound No. 1273039

[0385] In certain embodiments, Compound No. 1273039 is characterized as a 4-8-6 MOE gapmer having a sequence (from 5' to 3') of ACGCAATGTATCAGGCAA (SEQ ID NO: 2866), wherein each of nucleosides 1-4 and 13-18 (from 5' to 3') are 2'-MOE nucleosides and each of nucleosides 5-12 are 2'- β -D-deoxynucleosides, wherein the internucleoside linkages between nucleosides 2 to 3, 3 to 4, 13 to 14, 14 to 15, and 15 to 16 are phosphodiester internucleoside linkages and the internucleoside linkages between nucleosides 1 to 2, 4 to 5, 5 to 6, to 7, 7 to 8, 8 to 9, 9 to 10, 10 to 11, 11 to 12, 12 to 13, 16 to 17, and 17 to 18 are phosphorothioate internucleoside linkages, and wherein each cytosine is a 5-methyl cytosine.

[0386] In certain embodiments, Compound No. 1273039 is represented by the following chemical notation:

A.sub.es.sup.mC.sub.eoG.sub.eo.sup.mC.sub.esA.sub.dsA.sub.dsT.sub.dsG.sub.dsT.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsA.sub.eoG.sub.eoG.sub.eo.sup.mC (SEQ ID NO: 2921), wherein: [0387] A=an adenine nucleobase, [0388] mC=a 5-methyl cytosine nucleobase, [0389] G=a guanine nucleobase, [0390] T=a thymine nucleobase, [0391] e=a 2'-MOE sugar moiety, [0392] d=a 2'- β -D-deoxyribose sugar moiety, [0393] s=a phosphorothioate internucleoside linkage, and [0394] o=a phosphodiester internucleoside linkage.

[0395] In certain embodiments, Compound No. 1273039 is represented by the following chemical structure:

##STR00027##

[0396] In certain embodiments, the sodium salt of Compound No. 1273039 is represented by the following chemical structure:

##STR00028##

6. Compound No. 1273062

[0397] In certain embodiments, Compound No. 1273062 is characterized as a 5-10-5 MOE gapmer having a sequence (from 5' to 3') of ACCATTTTGACCTTCTTAGC (SEQ ID NO: 2873), wherein each of nucleosides 1-5 and 16-20 (from 5' to 3') are 2'-MOE nucleosides and each of nucleosides 6-15 are 2'- β -D-deoxynucleosides, wherein the internucleoside linkages between nucleosides 2 to 3, 3 to 4, 4 to 5, 5 to 6, 16 to 17, and 17 to 18 are phosphodiester internucleoside linkages and the internucleoside linkages between nucleosides 1 to 2, 6 to 7, 7 to 8, to 9, 9 to 10, 10 to 11, 11 to 12, 12 to 13, 13 to 14, 14 to 15, o to 16, 18 to 19, and 19 to 20 are phosphorothioate internucleoside linkages, and wherein each cytosine is a 5-methyl cytosine.

[0398] In certain embodiments, Compound No. 1273062 is represented by the following chemical notation:

A.sub.es.sup.mC.sub.eo.sup.mC.sub.eoA.sub.eoT.sub.eoT.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ds.sup.mC (SEQ ID NO: 2922), wherein: [0399] A=an adenine nucleobase, [0400] mC=a 5-methyl cytosine nucleobase, [0401] G=a guanine nucleobase, [0402] T=a thymine nucleobase, [0403] e=a 2'-MOE sugar moiety, [0404] d=a 2'- β -D-deoxyribose sugar moiety, [0405] s=a phosphorothioate internucleoside linkage, and [0406] o=a phosphodiester internucleoside linkage.

[0407] In certain embodiments, Compound No. 1273062 is represented by the following chemical structure:

##STR00029##

[0408] In certain embodiments, the sodium salt of Compound No. 1273062 is represented by the following chemical structure:

##STR00030##

VIII. Certain Comparator Compounds

[0409] In certain embodiments, Compound No. 1219022 (SEQ ID NO: 2923), a surrogate of which (Compound #586_9) is provided in WO2017/081223 (incorporated herein by reference), was used as a comparator compound. Compound No. 1219022 is a mixed LNA/DNA oligonucleotide having a sugar motif of (from 5' to 3') lldldldldldldldldldlll, wherein each "l" represents an LNA sugar moiety and each "d" represents a 2'-β-D-deoxyribose sugar moiety; having a sequence of (from 5' to 3') AAATTATTATACACCATCA (SEQ ID NO: 2905), wherein each "C" is a 5-methyl cytosine; and wherein each internucleoside linkage is a phosphorothioate internucleoside linkage. Compound #586_9 has cytosines at positions 13, 15, and 16 whereas Compound No. 1219022 has 5-methyl cytosines at those positions. According to Wan and Seth, "[i]ntroduction of the 5-methyl group on cytosine reduces the immunostimulatory profile of certain DNA oligonucleotides and also enhances nuclease stability" (J. Med. Chem. 2016, 59, 9645-9667).

[0410] In certain embodiments, Compound No. 1219023 (SEQ ID NO: 2924), a surrogate of which (Compound #572_7) is provided in WO2017/081223 (incorporated herein by reference), was used as a comparator compound. Compound No. 1219023 is a mixed LNA/DNA oligonucleotide having a sugar motif of (from 5' to 3') lldldldldldldldldlll, wherein each "l" represents an LNA sugar moiety and each "d" represents a 2'-B-D-deoxyribose sugar moiety; having a sequence of (from 5' to 3') TTTATCAATATCTTCTCA (SEQ ID NO: 2906), wherein each "C" is a 5-methyl cytosine; and wherein each internucleoside linkage is a phosphorothioate internucleoside linkage. Compound #572_7 has cytosines at positions 12 and 15 whereas Compound No. 1219023 has 5-methyl cytosines at those positions.

[0411] In certain embodiments, Compound No. 1219024 (SEQ ID NO: 2925), a surrogate of which (Compound #591_1) is provided in WO2017/081223 (incorporated herein by reference), was used as a comparator compound. Compound No. 1219024 is a mixed LNA/DNA oligonucleotide having a sugar motif of (from 5' to 3') lldldldldldldldldlll, wherein each "l" represents an LNA sugar moiety and each "d" represents a 2'-B-D-deoxyribose sugar moiety; having a sequence of (from 5' to 3') GCACATTCTTTCTATACCT (SEQ ID NO: 2907), wherein each "C" is a 5-methyl cytosine; and wherein each internucleoside linkage is a phosphorothioate internucleoside linkage. Compound #591_1 has cytosines at positions 2, 4, 8, 12 and 17 whereas Compound No. 1219024 has 5-methyl cytosines at those positions.

[0412] In certain embodiments, Compound No. 1219025 (SEQ ID NO: 2926), a surrogate of which (Compound #169_52) is provided in WO2017/081223 (incorporated herein by reference), was used as a comparator compound. Compound No. 1219025 is a mixed LNA/DNA oligonucleotide having a sugar motif of (from 5' to 3') lldldldldldldldlll, wherein each "l" represents an LNA sugar moiety and each "d" represents a 2'-β-D-deoxyribose sugar moiety; having a sequence of (from 5' to 3') TTATAGCCATTCTATCT (SEQ ID NO: 2908), wherein each "C" is a 5-methyl cytosine; and wherein each internucleoside linkage is a phosphorothioate internucleoside linkage. Compound #169_52 has cytosines at positions 7, 8, and 12 whereas Compound No. 1219025 has 5-methyl cytosines at those positions.

[0413] In certain embodiments, Compound No. 1219026 (SEQ ID NO: 2927), a surrogate of which (Compound #624_5) is provided in WO2017/081223 (incorporated herein by reference), was used as a comparator compound. Compound No. 1219026 is a mixed LNA/DNA oligonucleotide having a sugar motif of (from 5' to 3') lldldldldldldldlll, wherein each "l" represents an LNA sugar moiety and each "d" represents a 2'-β-D-deoxyribose sugar moiety; having a sequence of (from 5' to 3') CTCAAAGATCATTCTCA (SEQ ID NO: 2909), wherein each "C" is a 5-methyl cytosine; and wherein each internucleoside linkage is a phosphorothioate internucleoside linkage. Compound #624_5 has a cytosine at position 10 whereas Compound No. 1219026 has a 5-methyl cytosine at that position.

[0414] In certain embodiments, Compound No. 1219027 (SEQ ID NO: 2928), a surrogate of which (Compound #626_8) is provided in WO2017/081223 (incorporated herein by reference), was used as a comparator compound. Compound No. 1219027 is a mixed LNA/DNA oligonucleotide having a sugar motif of (from 5' to 3') lldldldldldldldldlll, wherein each "l" represents an LNA sugar moiety and each "d" represents a 2'-B-D-deoxyribose sugar moiety; having a sequence of (from 5' to 3') TTACACTTAATTATACTTCC (SEQ ID NO: 2910), wherein each "C" is a 5-methyl cytosine; and wherein each internucleoside linkage is a phosphorothioate internucleoside linkage. Compound #626_8 has cytosines at positions 4, 6, and 16 whereas Compound No. 1219027 has 5-methyl cytosines at those positions.

[0415] In certain embodiments, Compound No. 1219028 (SEQ ID NO: 2929), a surrogate of which (Compound #639_5) is provided in WO2017/081223 (incorporated herein by reference), was used as a comparator compound. Compound No. 1219028 is a mixed LNA/DNA oligonucleotide having a sugar motif of (from 5' to 3') lddldldldldldldlll, wherein each "l" represents an LNA sugar moiety and each "d" represents a 2'-B-D-deoxyribose sugar moiety; having a sequence of (from 5' to 3') GTTTCATCTACTATTAA (SEQ ID NO: 2911), wherein each "C" is a 5-methyl cytosine; and wherein each internucleoside linkage is a phosphorothioate internucleoside linkage. Compound #639_5 has cytosines at positions 5, 6, 9 and 12 whereas Compound No. 1219028 has 5-methyl cytosines at those positions.

[0416] In certain embodiments, Compound No. 1219029 (SEQ ID NO: 2930), a surrogate of which (Compound #642_12) is provided in WO2017/081223 (incorporated herein by reference), was used as a comparator compound. Compound No. 1219029 is a mixed LNA/DNA oligonucleotide having a sugar motif of (from 5' to 3') lldldldldldldldlll, wherein each "l" represents an LNA sugar moiety and each "d" represents a 2'-B-D-deoxyribose sugar moiety; having a sequence of (from 5' to 3') CTGTATACACCATCCCA (SEQ ID NO: 2912), wherein each "C" is a 5-methyl cytosine; and wherein each internucleoside linkage is a phosphorothioate internucleoside linkage. Compound #642_12 has cytosines at positions 8, 10, 11, 14 and 15 whereas Compound No. 1219029 has 5-methyl cytosines at those positions.

[0417] In certain embodiments, Compound No. 1219030 (SEQ ID NO: 2931), a surrogate of which (Compound #304_6) is provided in WO2017/081223 (incorporated herein by reference), was used as a comparator compound. Compound No. 1219030 is a mixed LNA/DNA oligonucleotide having a sugar motif of (from 5' to 3') lldldldldldldldlll, wherein each "l" represents an LNA sugar moiety and each "d" represents a 2'-β-D-deoxyribose sugar moiety; having a sequence of (from 5' to 3') AGTTCTACTATACTTTC (SEQ ID NO: 2913), wherein each "C" is a 5-methyl cytosine; and wherein each internucleoside linkage is a phosphorothioate internucleoside linkage. Compound #304_6 has cytosines at positions 8 and 13 whereas Compound No. 1219030 has 5-methyl cytosines at those positions.

[0418] In certain embodiments, Compound No. 1219031 (SEQ ID NO: 2932), a surrogate of which (Compound #573_8) is provided in WO2017/081223 (incorporated herein by reference), was used as a comparator compound. Compound No. 1219031 is a mixed LNA/DNA oligonucleotide having a sugar motif of (from 5' to 3') lldldldldldldldldlll, wherein each "l" represents an LNA sugar moiety and each "d" represents a 2'-β-D-deoxyribose sugar moiety; having a sequence of (from 5' to 3') TATACCTTTCTTTAACCCTT (SEQ ID NO: 2914), wherein each "C" is a 5-methyl cytosine; and wherein each internucleoside linkage is a phosphorothioate internucleoside linkage. Compound #573_8 has cytosines at positions 6, 10, 16, 17, and 18 whereas Compound No. 1219031 has cytosines at those positions.

[0419] Compound Nos. 1219022-1219031 (relating to Compound Nos. #5869, #5727, #591_1, #169_52, #6245, #6268, #639_5 #642_12, #3046, #5738, respectively) were selected as comparator compounds from Example 7 of WO2017/081223, which provides a subset of active compounds "selected for potency and efficacy testing."

[0420] In certain embodiments, compounds described herein are superior relative to compounds described in WO2017/081223 because they demonstrate one or more improved properties, such as in vivo tolerability.

[0421] For example, as described herein, certain compounds Compound No. 1263517, Compound No. 1263518, Compound No. 1263533, Compound No. 1273039, and Compound No. 1273062 achieved 3-hour FOB scores in mice of 1.0 (Table 88), 0.0 (Table 88), 1.0 (Table 88), 1.0 (Table 96), and 0.0 (Table 96), respectively, whereas each of comparator compounds Compound No. 1219022, Compound No. 1219024, Compound

No. 1219025, Compound No. 1219028, Compound No. 1219029, Compound No. 1219030, and Compound No. 1219031 achieved 3-hour FOB scores in mice of 7.0, 5.3, 4.0, 6.0, 6.3, 5.0, and 5.3 (Table 119), respectively. Therefore, certain compounds described herein are more tolerable than comparator compounds Compound No. 1219022, Compound No. 1219024, Compound No. 1219025, Compound No. 1219028, Compound No. 1219029, Compound No. 1219030, and Compound No. 1219031 in this assay.

[0422] For example, as described herein, certain compounds Compound No. 1065645, Compound No. 1263517, Compound No. 1263518, Compound No. 1263533, Compound No. 1273039, and Compound No. 1273062 each achieved 2-week FOB scores in mice of 0.0 (Example 12, Table 112), whereas Compound No. 1219023, Compound No. 1219024, Compound No. 1219025, Compound No. 1219026, Compound No. 1219028, Compound No. 1219029, Compound No. 1219030, and Compound No. 1219031 achieved 2-week delayed FOB scores in mice of 3.5, 6.5, 6.0, 6.0, 6.0, 6.0, 6.0, 5.0, and 6.0, respectively (Table 119). Therefore, certain compounds described herein are more tolerable than comparator compounds Compound No. 1219023 and Compound No. 1219026 in this assay.

[0423] For example, as described herein, certain compounds Compound No. 1065645, Compound No. 1263517, Compound No. 1263518, Compound No. 1263533, Compound No. 1273039, and Compound No. 1273062 do not cause a significant difference in body weight as compared to PBS-treated rats in an 8-week study, whereas treatment with Compound No. 1219027 leads to a greater than 10% weight loss compared to PBS-treated rats (p-value <0.05, Table 120). Therefore, certain compounds described herein are more tolerable than comparator Compound No. 1219027 in this assay.

IX. Certain Hotspot Regions

[0424] In certain embodiments, nucleobases in the ranges specified below comprise a hotspot region of UBE3A-ATS. In certain embodiments, modified oligonucleotides that are complementary to a hotspot region of UBE3A-ATS achieve an average of more than 50% reduction of UBE3A-ATS RNA in vitro in the standard cell assay.

1. Nucleobases 461,413-461,487 of SEQ ID NO: 1

[0425] In certain embodiments, nucleobases 461,413-461,487 of SEQ ID NO: 1 comprise a hotspot region. In certain embodiments, modified oligonucleotides are complementary to a portion of nucleobases 461,413-461,487 of SEQ ID NO: 1. In certain embodiments, modified oligonucleotides are 20 nucleobases in length. In certain embodiments, modified oligonucleotides are 18 nucleobases in length. In certain embodiments, modified oligonucleotides are gapmers. In certain embodiments, the gapmers are MOE gapmers. In certain embodiments, the internucleoside linkages of the modified oligonucleotides are phosphorothioate internucleoside linkages and phosphodiester internucleoside linkages. In certain embodiments, the phosphodiester (“o”) and phosphorothioate (“s”) internucleoside linkages are arranged in order from 5’ to 3’:

soooooossssssooss, soooooossssssooss, soooooossssssooss, soooooossssssooss, soooooossssssooss, or soooooossssssooss.

[0426] The nucleobase sequences of SEQ ID Nos: 1053, 1329, 1501, 1576, 1873, 1949, 2025, 2096, 2245, 2512, 2591, 2680-2682, and 2844 are complementary to a portion of nucleobases 461,413-461,487 of SEQ ID NO: 1.

[0427] The nucleobase sequence of Compound Nos: 749901-749904, 1065641-1065646, 1165562-1165563, 1165857-1165858, and 1273001 are complementary to a portion of nucleobases 461,413-461,487 of SEQ ID NO: 1.

[0428] In certain embodiments, modified oligonucleotides complementary to a portion of nucleobases 461,413-461,487 of SEQ ID NO: 1 achieve at least 36% reduction of UBE3A-ATS RNA in vitro in the standard cell assay.

[0429] In certain embodiments, modified oligonucleotides complementary to a portion of nucleobases 461,413-461,487 of SEQ ID NO: 1 achieve an average of 60% reduction of UBE3A-ATS RNA in vitro in the standard cell assay.

2. Nucleobases 468,968-469,013 of SEQ ID NO: 1

[0430] In certain embodiments, nucleobases 468,968-469,013 of SEQ ID NO: 1 comprise a hotspot region. In certain embodiments, modified oligonucleotides are complementary to a portion of nucleobases 468,968-469,013 of SEQ ID NO: 1. In certain embodiments, modified oligonucleotides are 20 nucleobases in length. In certain embodiments, modified oligonucleotides are 18 nucleobases in length. In certain embodiments, modified oligonucleotides are gapmers. In certain embodiments, the gapmers are MOE gapmers. In certain embodiments, the internucleoside linkages of the modified oligonucleotides are phosphorothioate internucleoside linkages and phosphodiester internucleoside linkages. In certain embodiments, the phosphodiester (“o”) and phosphorothioate (“s”) internucleoside linkages are arranged in order from 5’ to 3’:

soooooossssssooss, soooooossssssooss, soooooossssssooss, soooooossssssooss, soooooossssssooss, or soooooossssssooss.

[0431] The nucleobase sequences of SEQ ID Nos: 376, 377, 2751-2756, 2773-2776, 2872, 2873, 2876-2878 are complementary to a portion of nucleobases 468,968-469,013 of SEQ ID NO: 1.

[0432] The nucleobase sequence of Compound Nos: 750031-750032, 1263408-1263411, 1263426, 1263441, 1263460-1263465, 1263486-1263492, 1263517-1263523, 1273061, 1273062, and 1273065-1273067 are complementary to a portion of nucleobases 468,968-469,013 of SEQ ID NO: 1.

[0433] In certain embodiments, modified oligonucleotides complementary to a portion of nucleobases 468,968-469,013 of SEQ ID NO: 1 achieve at least 75% reduction of UBE3A-ATS RNA in vitro in the standard cell assay.

[0434] In certain embodiments, modified oligonucleotides complementary to a portion of nucleobases 468,968-469,013 of SEQ ID NO: 1 achieve an average of 78% reduction of UBE3A-ATS RNA in vitro in the standard cell assay. In certain embodiments, modified oligonucleotides complementary to a portion of nucleobases 468,968-469,013 of SEQ ID NO: 1 achieve an average of 410% upregulation of UBE3A-ATS RNA in vitro at 6.7 M in cell culture.

3. Nucleobases 483,965-484,003 of SEQ ID NO: 1

[0435] In certain embodiments, nucleobases 483,965-484,003 of SEQ ID NO: 1 comprise a hotspot region. In certain embodiments, modified oligonucleotides are complementary to a portion of 483,965-484,003 of SEQ ID NO: 1. In certain embodiments, modified oligonucleotides are 20 nucleobases in length. In certain embodiments, modified oligonucleotides are 18 nucleobases in length. In certain embodiments, modified oligonucleotides are gapmers. In certain embodiments, the gapmers are MOE gapmers. In certain embodiments, the internucleoside linkages of the modified oligonucleotides are phosphorothioate internucleoside linkages and phosphodiester internucleoside linkages. In certain embodiments, the phosphodiester (“o”) and phosphorothioate (“s”) internucleoside linkages are arranged in order from 5’ to 3’: soooooossssssooss, soooooossssssooss, soooooossssssooss, soooooossssssooss, or soooooossssssooss.

[0436] The nucleobase sequences of SEQ ID Nos: 172, 764-770, 995, 1445, 1668, 1743, 2255, 2595, 2762-2767 are complementary to a portion of nucleobases 483,965-484,003 of SEQ ID NO: 1.

[0437] The nucleobase sequence of Compound Nos: 617557, 699781, 750138-750144, 1065918-1065921, 1165621, 1165878, and 1263532-1263557 are complementary to a portion of nucleobases 483,965-484,003 of SEQ ID NO: 1.

[0438] In certain embodiments, modified oligonucleotides complementary to a portion of nucleobases 483,965-484,003 of SEQ ID NO: 1 achieve at least 24% reduction of UBE3A-ATS RNA in vitro in the standard cell assay.

[0439] In certain embodiments, modified oligonucleotides complementary to nucleobases 483,965-484,003 of SEQ ID NO: 1 achieve an average of 65% reduction of UBE3A-ATS RNA in vitro in the standard cell assay. In certain embodiments, modified oligonucleotides complementary to a portion of nucleobases 468,968-469,013 of SEQ ID NO: 1 achieve an average of 330% upregulation of UBE3A-ATS RNA in vitro at 6.7 M in cell culture.

4. Additional Hotspot Regions

[0440] In certain embodiments, the ranges described in the Table below comprise hotspot regions. Each hotspot region begins with the nucleobase of SEQ ID NO: 1 identified in the “Start Site SEQ ID NO: 1” column and ends with the nucleobase of SEQ ID NO: 1 identified in the “Stop Site SEQ ID NO: 1” column. In certain embodiments, modified oligonucleotides are complementary within any of the hotspot regions 1-61, as defined in the

table below. In certain embodiments, modified oligonucleotides are 18 nucleobases in length. In certain embodiments, modified oligonucleotides are 20 nucleobases in length. In certain embodiments, modified oligonucleotides are gapmers. In certain embodiments, modified oligonucleotides are 4-8-6, 6-8-4, 5-8-5, 4-10-6, 6-10-4, or 5-10-5 MOE gapmers.

[0441] The nucleobase sequence of compounds listed in the “Compound ID in range” column in the table below are complementary to SEQ ID NO: 1 within the specified hotspot region. The nucleobase sequence of the oligonucleotides listed in the “SEQ ID NO: in range” column in the table below are complementary to the target sequence, SEQ ID NO: 1, within the specified hotspot region.

[0442] In certain embodiments, modified oligonucleotides complementary to nucleobases within the hotspot region achieve at least “Min. % Red.” (minimum % reduction, relative to untreated control cells) of UBE3A-ATS RNA in vitro in the standard cell assay, as indicated in the table below. In certain embodiments, modified oligonucleotides complementary to nucleobases within the hotspot region achieve an average of “Avg. % Red.” (average % reduction, relative to untreated control cells) of UBE3A-ATS RNA in vitro in the standard cell assay, as indicated in the table below. In certain embodiments, modified oligonucleotides complementary to nucleobases within the hotspot region achieve a maximum of “Max. % Red.” (maximum % reduction, relative to untreated control cells) of UBE3A-ATS RNA in vitro in the standard cell assay, as indicated in the table below.

TABLE-US-00001	TABLE 1	UBE3A-ATS Hotspots	Hot- Start	Site	Stop	Site	Min.	Max.	Avg.	spot	SEQ ID	SEQ ID	%	%	%	ID NO:	1	Red.																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
Red.	Red.	Compound ID in range	SEQ ID NO: in range	1	461413	461487	36	89	60	749901-749904	1065641-1053	1329	1501	1065646	1165562-1576	1873	1949	1165563	1165857-2025	2096	2245	1165858	1273001	2512	2591	2680-2682	2844	2	468968	469013	75	81	78	750031-750032	1263408-376	377	2751-2756	1263411	1263426	2773-2776	2872	1263441	1263460-2873	2876-2878	1263465	1263486-1263492	1263517-1263523	1273061	1273062	1273065-1273067	3	483965	484003	24	91	65	617557	699781	750138-172	764-770	995	750144	1065918-1065921	1445	1668	1743	1165621	1165878	2255	2595	2762-1263532-1263557	2767	4	349103#	349150#	35	82	56	750519-750533	181-195	5	457354	457405	37	85	66	749848	1065574-1065576	552	973	1273	1795	1165519-1165521	2162	2237	2370	1179839-1179840	2537	2671	6	457899	457936	49	77	59	749852	1065581-1065582	556	1497	1945	7	457969	458013	36	78	61	749853	1065583-1065586	557	1124	1199	1165523-1165525	1647	2021	2089	2464	2538	8	458523	458558	64	77	71	749861-749862	1065595	565	566	1050	9	458560	458627	45	81	64	749863-749867	1065596-567-571	1125	1498	1065600	1272999	1573	1946	2022	2842	10	458935	458976	30	85	60	1065602-1065605	901	1425	1648	1165530-1165540	1724	2091	2092	1165853	2164	2165	2239	2240	2392-2393	2465	2511	2539	2540	11	460105	460137	43	82	60	749882	1065615-1065619	586	1126	1201	1165547-1165549	1649	1725	2023	2093	2467	2542	12	460348	460381	36	77	58	1065621-1065622	902	976	2167	2242	1165550-1165551	13	460641	460686	50	67	57	617531	1065629-1065631	107	1500	1948	2024	14	460974	461021	75	91	82	749893-749894	1165855	597	598	2219	2745-1263451-1263455	2749	1263473-1263478	1263504-1263509	15	461597	461627	50	85	68	1065651-1065655	904	978	1278	1428	1800	16	463502	463531	60	71	66	749931	1065674-1065675	1503	1578	2708	17	463826	463872	50	71	62	749938	749939	1065680	1728	2714	2715	18	464377	464428	32	77	58	749951	1065684-1065687	1280	1356	1802	1165572-1165577	1876	2097	2172	1273006-1273008	2247	2398	2471	2546	2727	2849-2851	19	464522	464575	47	77	62	749952-749954	1065690	1504	2098	2728-1165578	1263412-2730	2757-2760	1263416	1263427-2777-2781	2879-1263431	1263442-2881	1263446	1263466-1263469	1263494-1263499	1263524-1263530	1273068-1273070	20	464994	465032	27	76	52	617457	749957-749964	33	907	981	1281	1065698-1065702	1357	1877	2136	1165866	2733-2739	21	465231	465266	23	83	63	749969	1065706-1065713	318	908	1132	1207	1165583-1165587	1432	1655	1730	1263403-1263407	1953	2028	2174	1263421-1263425	2248	2400	2473	1263436-1263440	2548	2750	2768-1263456	1263479-2772	2875	1263485	1263510-1263516	1273063	1273064	22	465372	465413	33	73	54	749972	1065717-1065719	321	1058	1358	1165588-1165592	1878	2099	2137	1165867-1165869	2175	2249	2401	2475	2514	2593	23	465600	465631	65	68	67	749975	1065727-1065728	324	1433	1731	1272944-1272947	2787-2790	24	466244	466282	38	68	54	1065742-1065744	1434	1657	1732	1165595-1165598	2176	2251	2403	2549	25	466529	466570	31	80	57	617460	749989-749998	36	338-346	911	985	1065760-1065762	1435	26	466979	467014	51	75	65	617461	750000-750005	37	347-351	1806	1065764	27	467043	467095	44	93	69	617539	750006	1065765-115	352	1061	1361	1065767	1272943	1881	2786	2803	1272960	1273005	2848	28	468335	468372	42	68	56	1065781-1065783	1062	1362	1882	1165609	1165875	2139	2405	29	475827	475877	39	71	57	617470	1065830-1065834	46	1065	1365	1513	1588	1961	30	482148	482181	49	70	68	750122	1065889-1065890	748	919	993	2406	1165612	31	487589	487629	73	79	76	750172	1065953-1065955	798	997	1297	1818	1272973	1272974	2816	2817	32	487772	487826	44	73	62	750175-750177	1065957	801-803	1373	33	489873	489927	52	73	59	750195-750196	1065989	398	399	1375	34	493831	493860	44	63	56	750202-750204	405-407	35	499081	499119	47	70	55	1066036-1066038	1078	1378	1898	36	500605	500658	36	66	58	750265-750266	1066064-683	684	929	1004	1066065	37	500846	500905	30	86	55	750269-750270	1066073-687-688	1155	1230	1066077	1165670	1678	1976	2051	2562	38	501335	501375	42	67	56	1066091-1066094	1156	1231	1679	1272975-1272977	1753	2818-2820	39	502125	502157	65	79	72	750291	1066119-1066121	708	1531	1606	1979	40	502194	502228	51	88	65	750292	1066124-1066125	709	1233	1681	41	502416	502452	49	59	54	1066130-1066132	1308	1829	1904	42	502580	502618	57	59	58	750295	1066140-1066142	712	1234	1682	1756	43	503427	503461	36	73	61	750301	1066185	1165724	718	1983	2197	44	503636	503675	55	72	62	750307-750308	1066190	724	725	1759	45	503973	504034	20	81	56	617503-617505	617581-79-81	155-157	302	617582	750313-750324	731-736	2059	750383	1066202	46	504088	504122	45	85	65	750325-750327	1066206	245	246	737	1760	47	504431	504460	49	70	57	1066218-1066219	1164	2060	2119	1165736-1165740	2276	2427	2495	2574	48	506031	506071	33	66	52	750356	750357	1066274-275	276	1317	1838	1066276	1913	49	508739	508780	20	83	56	617503-617504	617580-79	80	155-157	302	617582	750313-750323	730-735	2059	750383	1066202	50	509208	509271	46	68	63	750394-750395	1066377-313-314	1995	2070	1066378	51	510203	510247	36	77	60	750400-750401	1066394-810	811	1175	2071	1165828	2440	52	510832	510872	44	70	55	1066418-1066421	1326	1402	1847	1922	53	513442	513486	66	81	72	750436-750439	846-849	#The oligonucleotides described in this section are complementary to this hotspot region at multiple sites. These sites are described in detail in Table 4b herein below (Example 1).

NONLIMITING DISCLOSURE AND INCORPORATION BY REFERENCE

[0443] Each of the literature and patent publications listed herein is incorporated by reference in its entirety. While certain compounds, compositions and methods described herein have been described with specificity in accordance with certain embodiments, the following examples serve only to illustrate the compounds described herein and are not intended to limit the same. Each of the references, GenBank accession numbers, and the like recited in the present application is incorporated herein by reference in its entirety.

[0444] Although the sequence listing accompanying this filing identifies each sequence as either “RNA” or “DNA” as required, in reality, those sequences may be modified with any combination of chemical modifications. One of skill in the art will readily appreciate that such designation as “RNA” or “DNA” to describe modified oligonucleotides is, in certain instances, arbitrary. For example, an oligonucleotide comprising a nucleoside comprising a 2'-OH sugar moiety and a thymine base could be described as a DNA having a modified sugar (2'-OH in place of one 2'-H of DNA) or as an RNA having a modified base (thymine (methylated uracil) in place of an uracil of RNA). Accordingly, nucleic acid sequences provided herein, including, but not limited to those in the sequence listing, are intended to encompass nucleic acids containing any combination of natural or modified RNA and/or DNA, including, but not limited to such nucleic acids having modified nucleobases. By way of further example and without limitation, an oligomeric compound having the nucleobase sequence “ATCGATCG” encompasses any oligomeric compounds having such nucleobase sequence, whether modified or unmodified, including, but not limited to, such compounds comprising RNA bases, such as those having sequence “AUCGAUCG” and those having some DNA bases and some RNA bases such as “AUCGATCG” and oligomeric compounds having other modified nucleobases, such as “AT.sup.mCGAUCG,” wherein .sup.mC indicates a cytosine base comprising a methyl group at the 5-position.

[0445] Certain compounds described herein (e.g., modified oligonucleotides) have one or more asymmetric center and thus give rise to enantiomers, diastereomers, and other stereoisomeric configurations that may be defined, in terms of absolute stereochemistry, as (R) or (S), as α or β such as for sugar anomers, or as (D) or (L), such as for amino acids, etc. Compounds provided herein that are drawn or described as having certain

stereoisomeric configurations include only the indicated compounds. Compounds provided herein are drawn or described with undefined stereochemistry include all such possible isomers, including their stereorandom and optically pure forms, unless specified otherwise. Likewise, tautomeric forms of the compounds herein are also included unless otherwise indicated. Unless otherwise indicated, compounds described herein are intended to include corresponding salt forms.

[0446] The compounds described herein include variations in which one or more atoms are replaced with a non-radioactive isotope or radioactive isotope of the indicated element. For example, compounds herein that comprise hydrogen atoms encompass all possible deuterium substitutions for each of the .sup.1H hydrogen atoms. Isotopic substitutions encompassed by the compounds herein include but are not limited to: .sup.2H or .sup.3H in place of .sup.1H, .sup.13C or .sup.14C in place of .sup.12C, .sup.15N in place of .sup.14N, .sup.17O or .sup.18O in place of .sup.16O, and .sup.33S, .sup.34S, .sup.35S, or .sup.36S in place of .sup.32S. In certain embodiments, non-radioactive isotopic substitutions may impart new properties on the oligomeric compound that are beneficial for use as a therapeutic or research tool. In certain embodiments, radioactive isotopic substitutions may make the compound suitable for research or diagnostic purposes such as imaging.

EXAMPLES

[0447] The following examples illustrate certain embodiments of the present disclosure and are not limiting. Moreover, where specific embodiments are provided, the inventors have contemplated generic application of those specific embodiments. For example, disclosure of an oligonucleotide having a particular motif provides reasonable support for additional oligonucleotides having the same or similar motif. And, for example, where a particular high-affinity modification appears at a particular position, other high-affinity modifications at the same position are considered suitable, unless otherwise indicated.

Example 1: Effect of 5-10-5 MOE Gapmer Modified Oligonucleotides on Human UBE3A-ATS RNA In Vitro, Single Dose

[0448] Modified oligonucleotides complementary to human UBE3A-ATS nucleic acid were tested for their effect on UBE3A-ATS RNA levels in vitro.

[0449] The modified oligonucleotides in the tables below are 5-10-5 MOE gapmers. The gapmers are 20 nucleosides in length, wherein the central gap segment consists of ten 2'- β -D-deoxynucleosides and the 5' and 3' wing segments each consists of five 2'-MOE nucleosides. The sugar motif for the gapmers is (from 5' to 3'): eeeeeeeeeeeeeee; wherein 'd' represents a 2'- β -D-deoxyribose sugar moiety, and 'e' represents a 2'-MOE sugar moiety.

[0450] Each internucleoside linkage of Compound IDs 617441-617596 (in Tables 2 and 3) is a phosphorothioate internucleoside linkage. All other compounds (Tables 4-33) have an internucleoside linkage motif of (from 5' to 3'): sooooooooooooo, wherein each "s" represents a phosphorothioate internucleoside linkage and each "o" represents a phosphodiester internucleoside linkage.

[0451] All cytosine residues are 5-methylcytosines.

[0452] "Start site" indicates the 5'-most nucleoside to which the modified oligonucleotide is complementary in the human gene sequence. "Stop site" indicates the 3'-most nucleoside to which the modified oligonucleotide is complementary in the target nucleic acid sequence. Each modified oligonucleotide listed in the Tables below is 100% complementary to SEQ ID NO: 1 (GENBANK Accession No NC_000015.10 truncated from nucleotides to 25441028). Selected compounds in the table below are complementary to the target nucleic acid sequence at more than three specific sites. For these compounds, the "start site" and "stop site" values in the Tables below are indicated with a hashtag (#) and indicate only the first site to which the compound is complementary. Additional sites to which these compounds are complementary are indicated in Table 4b below.

[0453] Human IPSC cell derived iCell GABANeurons (Cellular Dynamics) were cultured per manufacturer instructions at 20,000-60,000 cells per well (as indicated in the table heading) and were treated with 5,000-10,000 nM of modified oligonucleotide (as indicated in the table heading) by free uptake. After a treatment period of approximately 6 days, total RNA was isolated from the cells and UBE3A-ATS RNA levels were measured by quantitative real-time RT-PCR. Human UBE3A-ATS primer probe set RTS4796 (forward sequence CTCCCCAGTTCTGGAATGA, designated herein as SEQ ID NO: 2; reverse sequence TACACAGGGATTGAGCCTGCTA, designated herein as SEQ ID NO: 3; probe sequence CCCACAGATCAAGCATTCCTCCCAAGA, designated herein as SEQ ID NO: 4) was used to measure RNA levels. UBE3A-ATS RNA levels were normalized to total RNA content, as measured by RIBOGREEN®. Reduction of UBE3A-ATS RNA is presented in the tables below as percent UBE3A-ATS RNA amount relative to untreated control (UTC) cells. Each table represents results from an individual assay plate. The values marked with an asterisk (*) indicate that the modified oligonucleotide is complementary to the amplicon region of the primer probe set. Additional assays may be used to measure the potency and efficacy of the modified oligonucleotides complementary to the amplicon region. "N.D." indicates that a value was not determined in this experiment due to experimental error. However, activities of selected modified oligonucleotides, including those that are not defined in Example 1, are successfully demonstrated in dose-response studies herein below.

TABLE-US-00002 TABLE 2 Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gapmers with PS internucleoside linkages in vitro (60,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number Start Site Stop Site Sequence (5' to 3') (% UTC) NO 617441 449466 449485 ATGTTGCTTGCACTCCATCA 85 17 617442 449945 449964

ACAACATAAGGTCTTATTGT 87 18 617443 450581 450600 TTTTCAACTCCAGAATTTTC 87 19 617444 451644 451663
TTCAGTCTTATACAGAAATG 94 20 617445 452473 452492 GGAAGTACACAGAAACTGG 115 21 617446 454843 454862
AGGGTAGAAGACTAGCATAC 111 22 617447 455272 455291 GGCCATTCATTCAGCCACAC 95 23 617448 455405 455424
TAAAGATTTTATGAAATAC 104 24 617449 456040 456059 TTTAGAATCATGTGAATTTCA 109 25 617450 457025 457044
AGACATATCAGATTGCTTG 89 26 617451 457601 457620 AATATAATGTAGTATGACCA 59 27 617452 458989 459008
GAGGACACTGGCACATCTAT 59 28 617453 459376 459395 TTAAATAATAAAATATATTT 108 29 459401 459420 617454 461190 461209
ATTGGAGCAAAAAGGGATCA 45 30 617455 462327 462346 AATTATTCCTATATCCTGT 45 31 617456 463905 463924
CAACTGTATCAAGACTTCA 24 32 617457 465003 465022 ATCTGCGAATAGAGGCCCTC 44 33 617458 465460 465479
AAGGTTTTTATTTTCATCACT 68 34 617459 465872 465891 GCCCATGGATGGTTGTCAA 34 35 617460 466539 466558
TAGGTTGCATAAAGCCAGGC 23 36 617461 466981 467000 CACACATCTTGTTCCTCAA 23 37 617462 467508 467527
TTTTTTTAATCCTCTGATGA 123 38 617463 467914 467933 AGTCTCTTTTCTTCGGTGC 34 39 617464 468085 468104
GGAAAGTCTTCTCTCTCCT 69 40 617465 469204 469223 ATTTGCTTAAAGTGAAGTT 91 41 617466 470003 470022
TTTTGTGATTTTTGAATA 101 42 617467 471815 471834 TGTTAAATGCTTTTCTAAAT 80 43 617468 472985 473004
CTGATGATTGATTGTTTCCT 71 44 617469 474868 474887 TATGATAATGTGTAGTTTTT 110 45 617470 475850 475869
AAGGGTAATACGGACCTCAT 29 46 617471 476475 476494 CTATTTTTTGCTTCCCTTAT 56 47 617472 479448 479467
GAATCTTAGAAAGTTAATT 118 48 617473 479585 479604 TGCCATCTTCAAGACTAAGG 33 49 617474 480527 480546
AATGGAAAAGATGTATCACG 92 50 617475 481063 481082 AACTTTGGCAGCTATTCAAT 88 51 617476 481522 481541
GACACTAGGTTTTGCAAAAG 63 52 617477 481663 481682 TATGAATTTTCAATTCAATG 155 53 617478 483415 483434
CACTTAAGGGAACCTTCAGA 84 54 617479 483933 483952 GAGATGATGTAAAGCTTTCA 75 55 617480 484344 484363
TAAGACAAGTGGCCATTGAAG 92 56 617481 484760 484779 CTCTGTCTGTCTCAAACA 47 57 617482 485496 485515
TATAAGAAGAAGATTACAG 132 58 617483 486358 486377 AAATGTTAATAGACTGCGAT 93 59 617484 486941 486960
TTAAACTCCCCAGTCAAAAG 108 60 617485 488180 488199 ATTTACTTCAAACAGGAGCT 42 61 617486 489476 489495
TCAGGAAATTAAGCATTCA 81 62 617487 489630 489649 CCTAACCTGGATCTCAGATA 57 63 617488 493552 493571
TACAGCAACCCTAGAAACT 121 64 617489 493893 493912 ATCTTTGGCAGATGTAACCT 63 65 617490 495750 495769
GAGTATTATCCAAAAGAACA 83 66 617491 497275 497294 AAAGCCCAGGATTAGGCAGC 63 67 617492 497366 497385
CTACAAAGAGTGAATCATGA 130 68 617493 498003 498022 TAATCTTAAGTTTAAAGTGA 68 69 617494 498883 498902

AAAAAGACATAGATAGACAG 100 70 617495 499237 499256 AGATACAAATTTAAAGT 119 71 617496 499672 499691
GGTAGCCCCAATACAGATT 62 72 617497 500044 500063 CTTCACAGCAGACTTGATC 65 73 617498 501513 501532
GGTCCACACAGAGTTCAAAT 73 74 617499 501703 501722 CTTTGGGACATCCCAAAGTT 113 75 617500 502388 502407
TACACATCTTGATACAAGG 60 76 617501 503258 503277 TAATACCTTTCTTGAGTAATA 124 77 617502 503904 503923
GGGTGGTTGGATTGCTTTAT 47 78 617503 503977 503996 CTGCACACTGTACAGGAGGG 80 79 508743 508762 617504 503985 504004
TTGATTCACTGCACACTGTA 67 80 508751 508770 617505 504015 504034 GTGCAGGAAGGAGGTTTTTGT 55 81 617506 504943 504962
ATACAGTATATACATCATCC 42 82 617507 505266 505285 AAGTTTAAACAAAAAAGG 119 83 617508 507068 507087
AATGTCTGTTCTTTGTGGTA 91 84 617509 507897 507916 ATACCTGCTTTTGTGACAAT 75 85 617510 508556 508575
GAGTGAGCTCATTTAGAA 74 86 617511 509785 509804 TTTCTCAATGTCAACTCTCA 68 87 617512 510721 510740
CAAGCCGAATCTTGACATAC 73 88 617513 511227 511246 GACCAGAAAAGCTACATAGC 87 89 617514 512575 512594
ATTTTGTATATTGTAGCTTT 105 90 617515 513295 513314 CCATCTTTCTGTTATCTTGT 51 91 617516 513982 514001
CTTTGTTTCTTTTAAAGAA 108 92 617517 514259 514278 ACCTTGACAACACCCTAGCT 89 93 617518 515123 515142
AAAAAGGAACCAACTAA 117 94

TABLE-US-00003 TABLE 3 Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gapmers with PS internucleoside linkages in vitro (60,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number Start Site Stop Site Sequence (5' to 3') (% UTC) NO 617461 466981 467000 CACACATCTTGTTCCCTCAA 29 37 617519 449812 449831
GGTGTGTCAGCTGTGCTGGT 62 95 617520 450498 450517 ATCATACCTTCACTTTTTTT 101 96 617521 450896 450915
TTCATTGAGCTTCTGGATA 100 97 617522 451804 451823 TTACTTCTTTTCATTAAGT 86 98 617523 454628 454647
AGTAAACACTCCAACAAAAA 126 99 617524 455049 455068 GGTCTAAGCAAAATGTGAAG 137 100 617525 455323 455342
CAGTTAGGCTGTCCGAAT 102 101 617526 455499 455518 ACTCTGAAAATAGCCAATTT 134 102 617527 456944 456963
AGTATCTATCTACAATTA 112 103 617528 457237 457256 AATGGCTTAGCTACTCACCC 72 104 617529 458254 458273
CCTCTCTGCAAAACAAGAGA 74 105 617530 459153 459172 ATTACTTCTGGGAATGC 35 106 617531 460667 460686
GTTTTTCAGCATGATTCTAAC 46 107 617532 461259 461278 CATATATAAAAATTAGAATG 122 108 617533 463621 463640
AATGCCACAAAGCTGGCTGT 97 109 617534 464514 464533 GTGCACAGATAATGACTAGA 69 110 617535 465402 465421
CAGTGAGAAGTCAATTGTCA 67 111 617536 465580 465599 TCCTTTAGCATTTCTATTAG 28 112 617537 466423 466442
AACATGTGCTTGCAAGCCAT 46 113 617538 466701 466720 AAATAACTGGATCTCATAAC 55 114 617539 467064 467083
CTTAGGAGAGAAACACTTTC 56 115 617540 467546 467565 TACATCCTGTAGGCTTTCTT 55 116 617541 467961 467980
ACTGAGAAATCTCTGAATG 92 117 617542 468624 468643 TTGCTCTTTTACTTTGTTC 37 118 617543 469731 469750
ACTCTAAGTTTTATTAATAG 105 119 617544 470091 470110 TTTGGAAGAGCTTAATAAAG 110 120 617545 472023 472042
TAGCTAGAATTTCAACTACT 63 121 617546 474443 474462 ATGTAGATTATTTCATTT 120 122 617547 474957 474976
GCCTAATATGTGTCATCCTG 28 123 617548 475998 476017 CCTCAATTCTATGGTTAGTT 52 124 617549 479376 479395
TCTTGAGATGACTTCTCTG 75 125 617550 479580 479599 TCTTCAAGACTAAGGTAGGG 54 126 617551 479654 479673
AGTAAGTCTGTATTCTCC 35 127 617552 480998 481017 GCTATACAACAAAAAGAATT 123 128 617553 481073 481092
ACTGTGGAAGAACTTTGGCA 54 129 617554 481582 481601 ATAATCTACATGTATAGACC 85 130 617555 481984 482003
AAAAGCCAATTCTGAAATTC 44 131 617556 483689 483708 ATTCTCTACCTCTCAGCCT 107 132 617558 484654 484673
AACTGTAGCAAATATACTAC 110 133 617559 485236 485255 AAATTACTGCTCTGTAAAAG 93 134 617560 485886 485905
GATTCAATGAAATAAAAAT 144 135 617561 486894 486913 AATGACATAGCTTATGCTGT 104 136 617562 488162 488181
CTAGAATAAAGACATCCCT 63 137 617563 488563 488582 GCCACACCATCAAAAGAAC 77 138 617564 489518 489537
TGGACCACCTAAGACCTCAA 62 139 617565 489731 489750 ACCGGTTCCCAATTTTCTCC 82 140 617566 493655 493674
GAGAGAATACGGCCCTGATG 77 141 617567 494790 494809 AACTTCAATCAGAGTAATAT 117 142 617568 496045 496064
GGCAAAGGAATGAAGAGACC 69 143 617569 497328 497347 GCCAATAACAAGAAAAGAGA 96 144 617570 497416 497435
ATCAAATTGGATGACCTAAA 122 145 617571 498140 498159 CAATGGATTTCATTAACACT 54 146 617572 498893 498912
ATTCTCCAACAAAAAGACAT 93 147 617573 499276 499295 GTAGCTACAAGAAGTAATTG 117 148 617574 499958 499977
AAAGAATGCCTATAAGAATT 118 149 617575 500578 500597 CTACTGGCATAGTCAAAAC 60 150 617576 501578 501597
AAAAGGCAAGGCTAAGGAGT 69 151 617577 502305 502324 CCCACACAGGTCAATTCATTC 52 152 617578 503150 503169
TGTATACACCATCCCAAGAA 93 153 617579 503795 503814 AAAATGCCAGTTTGTGTAC 60 154 617580 503973 503992
ACACTGTACAGGAGGGTGTC 69 155 508739 508758 617581 503981 504000 TCACTGCACACTGTACAGG 70 156 508747 508766 617582
503993 504012 AAATGATGTTGATTCACTGC 36 157 508759 508778 617583 504551 504570 AGAGTTTTTCATGAATTCAGG 46 158 617584
505011 505030 GGTTCCTTTTCATTAAATAGC 63 159 617585 506396 506415 GAGGCAACTCCTGAGAGCTG 64 160 617586 507650 507669
TTAAATGTCAGGAGGTCCCC 57 161 617587 508201 508220 TGTCATCTGATCTCACACAT 86 162 617588 508772 508791
CAAAGTCCAAGGGAAATGAT 71 163 617589 510010 510029 CACAGTAGAGGCAAATGAGA 79 164 617590 511058 511077
GACATTTAGGATGATGATAT 72 165 617591 512158 512177 GTTGTATTAATGGACTTTTG 65 166 617592 513203 513222
ATTTTTTGCATAAGTGCAAT 86 167 617593 513508 513527 ACATTTCTTAGTTGAACAGT 29 168 617594 514104 514123
AGATTTCTCCCAATCCCAT 81 169 617595 514524 514543 CTCACTTGTCTCCTTTTACC 73 170 617596 515282 515301
ACAATAAGGAAGAGCAAAAC 104 171

TABLE-US-00004 TABLE 4 Reduction of UBE3A-ATS RNA by 5,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (35,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number Start Site Stop Site Sequence (5' to 3') (% UTC) NO 617456 463905 463924 CAACTGTTACCAAGACTTCA 22 32 617457 465003
465022 ATCTGCGAATAGAGGCCCTC 22 33 617460 466539 466558 TAGGTTGCATAAAGCCAGGC 35 36 617461 466981 467000
CACACATCTTGTTCCTCAA 36 37 617557 483977 483996 ATCATGTGCATACCCAGGGT 9 172 705110 229870.sup.# 229889.sup.#
TCACTCATTTTGTTCAGCTT 33 173 705112 232607.sup.# 232626.sup.# GTTTTCACTATTTTGTTC 61 174 705113 220331 220350
ACAATTATTCTCATCATCGA 113 175 456700 456719 750514 220346 220365 CCTCAGCATCCTCAGACAAAT 80 176 456715 456734 750515
220359 220378 TCTGGAATGAGTCCCTCAGC 73 177 456728 456747 750516 220371 220390 CTCAGATTGACATCTGGAAT 123 178 456740
456759 750517 229869.sup.# 229888.sup.# CACTCATTTTGTTCAGCTTT 15 179 750518 232608.sup.# 232627.sup.#
AGTTTTCACTCATTTTGTTC 67 180 750519 349103.sup.# 349122.sup.# GTCATCACCTCTCTTCAGGA 22 181 750520 349105.sup.#
349124.sup.# AAGTCATCACCTCTCTCAG 35 182 750521 349107.sup.# 349126.sup.# TTAAGTCATCACCTCTCTTC 55 183 750522
349109.sup.# 349128.sup.# TTTTAAGTCATCACCTCTCT 37 184 750523 349111.sup.# 349130.sup.# ATTTTTAAGTCATCACCTCT 45 185
750524 349113.sup.# 349132.sup.# TGATTTTTAAGTCATCACCT 25 186 750525 349115.sup.# 349134.sup.# CATGATTTTTAAGTCATCAC 55
187 750526 349117.sup.# 349136.sup.# AGCATGATTTTTAAGTCATC 65 188 750527 349119.sup.# 349138.sup.#
TGAGCATGATTTTTAAGTCA 46 189 750528 349121.sup.# 349140.sup.# ATTGAGCATGATTTTTAAGT 56 190 750529 349123.sup.#
349142.sup.# CTATTGAGCATGATTTTTAA 60 191 750530 349125.sup.# 349144.sup.# TCCTATTGAGCATGATTTTT 56 192 750531
349127.sup.# 349146.sup.# AATCCTATTGAGCATGATTT 41 193 750532 349129.sup.# 349148.sup.# GTAATCCTATTGAGCATGAT 26 194
750533 349131.sup.# 349150.sup.# GCGTAATCCTATTGAGCATG 34 195 750534 349133.sup.# 349152.sup.# CAGCGTAATCCTATTGAGCA 63
196 750535 349135.sup.# 349154.sup.# CTCAGCGTAATCCTATTGAG 50 197 750536 349137.sup.# 349156.sup.#
GCCTCAGCGTAATCCTATTG 71 198 750537 349139.sup.# 349158.sup.# GGGCCTCAGCGTAATCCTAT 48 199 750538 349141.sup.#

349160.sup.# CTGGGCTCAGCGTAATCCT 28 200 750539 349143.sup.# 349162.sup.# GGCTGGGCCTCAGCGTAATC 75 201 750540
355254.sup.# 355273.sup.# TAGGCTGGGCCTCAGCGTAA 32 202 750541 349147.sup.# 349166.sup.# CCTAGGCTGGGCCTCAGCGT 98 203
750542 349149.sup.# 349168.sup.# CACCTAGGCTGGGCCTCAGC 21 204 750543 349151.sup.# 349170.sup.# CTCACCTAGGCTGGGCCTCA
55 205 750544 349153.sup.# 349172.sup.# TTCTCACCTAGGCTGGGCCT 24 206 750545 349155.sup.# 349174.sup.#
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349178.sup.# CCAAATCTCACCTAGGCT 46 209 750548 349162.sup.# 349181.sup.# CTTCACCAAAATCTCACCTAG 95 210 750549
349165.sup.# 349184.sup.# CCTCTTCCAAAATTCTCACC 17 211 750550 349168.sup.# 349187.sup.# CATCCTCTTCCAAAATTCTC 75 212
750551 349171.sup.# 349190.sup.# CAGCATCCTCTTCCAAAATT 50 213 750552 349174.sup.# 349193.sup.# TCCCAGCATCCTCTTCCAAA
70 214 750553 349177.sup.# 349196.sup.# GGATCCCAGCATCCTCTTCC 53 215 750554 349346 349365 GCCACCCACATGCCCTGCC 25
216 409462 409481 750555 349390.sup.# 349409.sup.# AGAGCTCACTGAAAGACACA 63 217 750556 349392.sup.# 349411.sup.#
GAAGAGCTCACTGAAAGACA 85 218 750557 349442.sup.# 349461.sup.# GCAGGATCCACTCACCTATG 27 219 750558 349957.sup.#
349976.sup.# CAGAGCTCAGCCTTGACCCA 45 220 750559 350910.sup.# 350929.sup.# AGCTCAGTGCAGGAGACCAG 56 221 750560
350914.sup.# 350933.sup.# CCACAGCTCAGTGCAGGAGA 72 222 750561 350924.sup.# 350943.sup.# GATGTGCTCACCACAGCTCA 115 223
750562 351544.sup.# 351563.sup.# GGAACCTTTCTGCTGGA 56 224 750563 353287.sup.# 353306.sup.# CAATATAAGGTTCTCATCAT 42
225 750564 353293.sup.# 353312.sup.# TCAGGACAATATAAGGTTCT 115 226 750565 353305.sup.# 353324.sup.#
CATCACCTCTTTCAGGACA 37 227 750566 353356 353375 TCTCACCTAGGCTAGGCCTC 44 228 407396 407415 750567 353579.sup.#
353598.sup.# AGCTCACTGAAAGACACAAG 49 229 750568 358746.sup.# 358765.sup.# AGGAAGGGCCTGAGCTTCAG 34 230 750569
363914 363933 CTCACCACACCTCAGTGCAG 93 231 384545 384564 750570 364549.sup.# 364568.sup.# CTTTCCTGCCTGGACCACCA 68
232 750571 366064 366083 TTGGGCACCCTCCAGATGCC 63 233 373449 373468 750572 366247 366266 CACTCACCTATGCTGGTCAA 72
234 370104 370123 386794 386813 750573 366487 366506 TCAGCTGGGTGCTGCCCTGC 117 235 407918 407937 750574 366668 366687
CCATTAGGCCATGGGTTT 67 236 374053 374072 750575 391964 391983 TCCCTGCACATGCATCCAG 97 237 393859 393878 403520
403539 750576 395822 395841 ACTCCAGGACCAAGAGCTC 49 238 420812 420831 750577 397780 397799
GAGACCCACTGAGATGGCCC 26 239 415262 415281 750578 402885 402904 ACCCCATGACCCTGGAGGTG 63 240 410060 410079 750579
403932 403951 ATGACCCAGCAGGATGCAC 57 241 424209 424228 750580 422678 422697 TCACCCACCACTGTCCAGAG 95 242 426410
426429 750581 422778 422797 GTACCACTGAGATGGCCCAT 68 243 426510 426529 750582 423273 423292 AATGCAGACCTGGCAGTCGC
50 244 430180 430199

TABLE-US-00005 TABLE 4b SEQ ID NO: 1 start sites for modified oligonucleotides complementary to repeat regions # of comp. SEQ Compound
sites within ID Number SEQ ID NO: 1 SEQ ID NO: 1 start sites NO: 705110 20 229870, 232603, 235253, 237932, 240726, 243420, 246181,
248825, 251500, 173 258532, 259677, 261158, 261978, 263775, 264917, 266052, 267194, 268313, 270176, 272427 705112 16 232607, 237936,
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267193, 268312, 270175, 272426 750518 16 232608, 237937, 243425, 252519, 254324, 258537, 259682, 261163, 261983, 180 263780, 264922,
266057, 267199, 268318, 270181, 272432 750519 20 349103, 353307, 357118, 364011, 367794, 369796, 371701, 377828, 379703, 181 381607,
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TABLE-US-00006 TABLE 5 Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (35,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number
Start Site Stop Site Sequence (5' to 3') (% UTC) NO 617456 463905 463924 CAACTGTTACCAAGACTTCA 40 32 617457 465003
465022 ATCTGCGAATAGAGGCCCTC 30 33 617460 466539 466558 TAGGTTGCATAAAGCCAGGC 40 36 617461 466981 467000
CACACATCTTGTTCCCTCAA 41 37 617557 483977 483996 ATCATGTGCATACCCAGGGT 26 172 750326 504100 504119
CTTCAACTAACAGTATCTTA 20 245 750327 504103 504122 AGTCTTCAACTAACAGTATC 51 246 750328 504126 504145
CTATTTTCATTAAGTCACCCC 53 247 750329 504179 504198 GGACAGCTGTGTGGAGGGAT 26 248 750330 504218 504237
ACCACCAGGGAGACCAGCCT 68 249 750331 504228 504247 CAGCCTTTCTACCACCAGGG 55 250 750332 504229 504248
TCAGCCTTTCTACCACCAGG 32 251 750333 504334 504353 ACAGAGTGTCTTACTGTGAGC 29 252 750334 504503 504522
TCCATGGAATGGCTGTCATG 39 253 750335 504731 504750 TTCCTTCAGAGTTATTTCTT 73 254 750336 504740 504759
TTATTTCTCCTTCCTTCAGAG 70 255 750337 504741 504760 GTTATTTCTCCTTCCTTCAGA 41 256 750338 504817 504836
GAAGTAGTCTGCTCCCTTTCC 60 257 750339 504923 504942 ATATCCTTTCTCCTTCTACT 78 258 750340 504958 504977
GACTGATGTTTACCCATACA 40 259 750341 504987 505006 CATTTAAACAATTATTATCTT 27 260 750342 505003 505022
TCATTAAATAGCTAACCAAT 91 261 750343 505043 505062 CCACCTTTCGAACTCAAGATT 31 262 750344 505046 505065
CAGCCACTTTGCAACTCAAG 20 263 750345 505051 505070 GACTCCAGCCACTTTGCAAC 62 264 750346 505073 505092
CCAGAAATTTAGTCTGTTGT 59 265 750347 505096 505115 CATAAGCTGCTGGATTGTGTT 51 266 750348 505241 505260
GAAGAGAACGAGGATATAAA 35 267 750349 505306 505325 GTGATATATTAGAAGTGTAT 28 268 750350 505443 505462
AGAGGATTTAGAGTTAAAAT 22 269 750351 505569 505588 ACTGAAAGGTCTGTATGTTT 73 270 750352 505656 505675
CAGGCCTAGCTTCCACCTAA 108 271 750353 505837 505856 TCTGTCAAAGACCTGTGAGG 56 272 750354 505922 505941
TCTGCTTGTTGTTCTCCCT 71 273 750355 505985 506004 CAAATATGGAATAGCACTTG 54 274 750356 506040 506059
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ATTTCTCCATGGAATCCAGA 47 277 750359 506109 506128 CATTGGCTTCATATTTCTCC 11 278 750360 506306 506325
TATTGCCTTCAGTCTGCCT 9 279 750361 506518 506537 ACTGACTTGCTCTGCCTACT 71 280 750362 506617 506636
GAGAAGTCTCTCATGGCACC 40 281 750363 506731 506750 GGAGTGCTCCACACTTCTGT 27 282 750364 506776 506795
CTCCAGGTTGTGGAGGTTGT 75 283 750365 506783 506802 GTATATACTCCAGGTTGTGG 23 284 750366 506881 506900
ACTTCTCATTTCTCCACCA 17 285 750367 507189 507208 AAAGCATTTTCTACCAGAGC 45 286 750368 507198 507217
TAGACAAGGAAAGCATTTTC 54 287 750369 507312 507331 CACCTGCCCTGCTTTTGCTT 41 288 750370 507324 507343
TGGTCTAGCTTACCTGCC 89 289 750371 507357 507376 AGATCCAACCTGTGTGAAA 61 290 750372 507534 507553
AGAAATTAGAGCCAGGTTCC 36 291 750373 507635 507654 TCCCCCAAGGCTTGACT 71 292 750374 507755 507774
TCGCTGGCCATTTTCATATAT 44 293 750375 507978 507997 AAAACATGTTAACTGTTATC 40 294 750376 508159 508178
TTAGCCATGTGCTTTGTGAC 68 295 750377 508208 508227 TCCCTTCTGTATCTGATCT 59 296 750378 508218 508237
AAGGTAGTTCTCCCTTCTGT 93 297 750379 508426 508445 TTCCATTGCACTCCTTTCTA 113 298 750380 508530 508549
TAAAGGAAAACCCACTTGT 72 299 750381 508700 508719 TACAGAAATTACCAGTAAAG 38 300 750382 508710 508729
TCCAGATTTCTACAGAAATT 102 301 750383 503980 503999 TCACTGCACACTGTACAGGA 78 302 508746 508765 750384 508890 508909
TGCCTAGTTCCCTCCTGAAA 98 303 750385 508908 508927 TTATGTTTACTTTCATGACTG 91 304 750386 508945 508964
CTTACTATTCTAGTGCC 14 305 750387 509013 509032 CTATTCCTTCAACATGTG 44 306 750388 509060 509079
CATACCCAACATGCTTGAT 44 307 750389 509079 509098 AATGCATTACCCTACAATGC 105 308 750390 509127 509146
CTCATCACAAGTGGGTGGTA 76 309 750391 509134 509153 ATCCCAGCTCATCACAAGT 88 310 750392 509167 509186
TCCAAGTGCTTCAAGCTGAG 100 311 750393 509189 509208 CACCCACAGCAGGCAGATAA 58 312 750394 509208 509227
TACCTTGCTCCAAAATAATC 37 313 750395 509230 509249 CTGTTTCTTGAGGCAAATGA 54 314 750396 509787 509806
CTTTTCTCAATGTCAACTCT 48 315 750397 510028 510047 AGAGTTATACACGGAACCCA 71 316

TABLE-US-00007 TABLE 6 Reduction of UBE3A-ATS RNA by 8,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (40,000 cells/well) SEQ ID SEQ ID NO: 1 NO: 1 UBE3A- Compound Start Stop ATS SEQ ID
Number Site Site Sequence (5' to 3') (% UTC) NO 617456 463905 463924 CAACTGTTACCAAGACTTCA 26 32 617457 465003
465022 ATCTGCGAATAGAGGCCCTC 46 33 617460 466539 466558 TAGGTTGCATAAAGCCAGGC 8 36 617461 466981 467000
CACACATCTTGTTCCCTCAA 21 37 617557 483977 483996 ATCATGTGCATACCCAGGGT 23 172 749968 465163 465182
TTCTAGGGCTCCAGTTTATG 86 317 749969 465238 465257 TGGTCATCTCGGGTATATAA 20 318 749970 465341 465360
CTTACTATCTTCAAGAATTC 68 319 749971 465346 465365 TTTTCTTACTATCTTCAAG 56 320 749972 465394 465413
AGTCAATTGTACACTTATT 37 321 749973 465587 465606 AAGGAGTTCTTTAGCATTT 27 322 749974 465594 465613
CAGACCGAAGGAGTTCTTTT 50 323 749975 465612 465631 TCCAGTGCCTTTCTATTTC 33 324 749976 465686 465705

ACCTTGTTTAAACTTACAC 46 325 749977 465765 465784 CATCAACACAAAGTTTATAAT 47 326 749978 465892 465911
AACTCCCACAAGGTACTCTT 32 327 749979 465906 465925 ACTGTCAACTCCTGAACTCC 54 328 749980 465956 465975
CAATGAATCTATTCTTAGAT 122 329 749981 465978 465997 GTGAAACTGTTTATACCCTT 25 330 749982 465996 466015
GTTGGTGATGCAGGTAAGT 24 331 749983 466024 466043 TTGTTACTGAGCTGTGCCAC 48 332 749984 466119 466138
AAGTGGATCTTTGGGCAGG 22 333 749985 466182 466201 TTTACCTCTCCCAACCACT 51 334 749986 466217 466236
GGCTGTGTTTGAGTCCCCA 40 335 749987 466220 466239 ATGGGCTGCTGTTTGAAGTCC 58 336 749988 466221 466240
TATGGGCTGCTGTTTGAAGT 55 337 749989 466529 466548 AAAGCCAGGCCAGGTGCTGA 64 338 749990 466532 466551
CATAAAGCCAGGCCAGGTGC 30 339 749991 466534 466553 TGCATAAAGCCAGGCCAGGT 20 340 749992 466537 466556
GGTTGCATAAAGCCAGGCCA 36 341 749993 466539 466558 TAGGTTGCATAAAGCCAGGC 46 36 749994 466541 466560
TCTAGGTTGCATAAAGCCAG 52 342 749995 466544 466563 TTCTCTAGGTTGCATAAAGC 33 343 749996 466546 466565
CCTTCTCTAGGTTGCATAAA 28 344 749997 466549 466568 TCACCTTCTCTAGGTTGCAT 34 345 749998 466551 466570
TATCACCTTCTCTAGGTTGC 44 346 749999 466701 466720 AAATAACTGGATCTCATAAC 55 114 750000 466979 466998
CACATCTTGTTCCTCAAG 26 347 750001 466981 467000 CACACATCTTGTTCCTCAA 26 37 750002 466983 467002
TTCACACATCTTGTTCCTC 26 348 750003 466986 467005 TGCCTTCACACATCTTGTTC 43 349 750004 466988 467007
CCTGCTTCACACATCTTGT 49 350 750005 466991 467010 GAACCTGCTTCACACATCTT 36 351 750006 467049 467068
CTTTCATCAGTTAGTCAGGT 23 352 750007 467200 467219 GGAGTTGGTTATTGGAAAAT 48 353 750008 467289 467308
TCTATTGGTGTTCCTTTTAG 48 354 750009 467296 467315 AGAGTAGTCTATTGGTGTTC 22 355 750010 467363 467382
CTTTTAAGATAATTTTCTC 90 356 750011 467417 467436 TACGCTCCTTCATTTTCATGC 55 357 750012 467506 467525
TTTTTAATCCTCTGATGAAT 102 358 750013 467508 467527 TTTTTTTAATCCTCTGATGA 62 38 750014 467604 467623
CTCTCTTCTCCTTTATGACT 41 359 750015 467628 467647 CTTAATAAGTTTCTACCC 42 360 750016 467639 467658
AGCCATTATTTCTTAATAA 56 361 750017 467669 467688 CATATCTTTTCTAGATTG 110 362 750018 467813 467832
AGAATCCTGTCTCCCTCTTA 34 363 750019 467916 467935 TGAGTCTCTTTTCTTTTCGGT 41 364 750020 467919 467938
TGATGAGTCTCTTTTCTTTC 35 365 750021 467921 467940 TGTGATGAGTCTCTTTTCTT 30 366 750022 467962 467981
TACTGAGAAATCTCTTGAAT 77 367 750023 468277 468296 TAAAATTATTTATACACCAT 73 368 750024 468359 468378
TTTATACTGTAGTATGCATT 53 369 750025 468410 468429 TTAATCTCCCTACCTTGCAT 75 370 750026 468413 468432
CTTTTACTTCCCTACCTTG 67 371 750027 468464 468483 ATTTATTTTAAGCTGATAAC 52 372 750028 468733 468752
AAATCCATTTGTCCAGTCTG 14 373 750029 468888 468907 TATCTGCATGCAATATCTTT 70 374 750030 468924 468943
CTGTAAGTATAGATGCCTCT 15 375 750031 468968 468987 TTAGCCTTTTGATATAGTTT 25 376 750032 468988 469007
GCTTCACCAATTTGACCTTC 19 377 750033 469140 469159 TTTGAAAGGGAGGCACTGA 27 378 750034 469604 469623
TGAGGTGTAATGTTGTTTAT 32 379 750035 469717 469736 TAATAGTCTCTATTGTTTTT 50 380 750036 469884 469903
CAAGTTAACTAATATGTTGG 71 381 750037 469985 470004 TATTGATTCAATTCCCTTAT 23 382 750038 469988 470007
GAATATTGATTCAATTCCT 37 383 750039 470062 470081 AATTTTTTTAAATGGTTGGC 51 384

TABLE-US-00008 TABLE 7 Reduction of UBE3A-ATS RNA by 8,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (40,000 cells/well) SEQ ID SEQ ID NO: 1 NO: 1 UBE3A- Compound Start Stop ATS SEQ ID

Number Site Site Sequence (5' to 3') (% UTC) NO 617456 463905 463924 CAACTGTTACCAAGACTTCA 18 32 617457 465003 465022

ATCTGCGAATAGAGGCCCTC 36 33 617460 466539 466558 TAGGTTGCATAAAGCCAGGC 12 36 617461 466981 467000

CACACATCTTGTTCCTCAA 16 37 617557 483977 483996 ATCATGTGCATACCCAGGGT 27 172 750182 488205 488224

AACAGAGAATTGTATATCAA 43 385 750183 488322 488341 ACATGAGATCTTTAATAAGA 73 386 750184 488343 488362

CAAGAGAGAAGCCACTCATG 64 387 750185 488685 488704 AGCAGATTTGAGCTGGAAGA 42 388 750186 488928 488947

AACTGGCAGAAAATATCTCT 36 389 750187 488941 488960 AAAGAAGAACATAAACTGGC 70 390 750188 489021 489040

GTTGGCTCTATTTTCAAATG 33 391 750189 489449 489468 CTGTTTAGGCTGGACACTGT 90 392 750190 489478 489497

ATTCAGGAAATAAAGCATT 60 393 750191 489698 489717 AGCAGAACAGACCTTATTTG 41 394 750192 489712 489731

CCTTCAACCAGAGGAGCAGA 44 395 750193 489756 489775 GATCTTGGTTTGTAGAAGGT 35 396 750194 489824 489843

TTCAAAATGGTAGAAAATTG 64 397 750195 489873 489892 TGAAAGGCCTACAGCAACCA 46 398 750196 489893 489912

ATGGGAGCTCTGAAAAACAG 27 399 750197 490070 490089 TTCTGAGTATGTGAAACA 39 400 750198 492874 492893

ATAAGGAGATTAATTTAAGA 86 401 750199 493278 493297 GAATCAAAGAAAGGAAT 47 402 750200 493440 493459

GCTGTAATAATAATCATATT 41 403 750201 493678 493697 AGAATTCCTTCCCTACAGGTT 51 404 750202 493831 493850

CAGTAATTAATGTTCTTATA 37 405 750203 493839 493858 CCCCAATTCAGTAATTAATG 39 406 750204 493841 493860

GGCCCCAATTCAGTAATTAA 56 407 750205 493882 493901 ATGTAACCTATTCAAGATGA 47 408 750206 493908 493927

TTATTTGGAAGAAGAATCTT 98 409 750207 494002 494021 TATAAAATACTTTTATGCG 101 410 750208 494398 494417

GACTGTGATAAAGATGTATA 84 411 750209 494443 494462 TAAGGGTCATGTACTATACA 33 412 750210 494494 494513

GTGTGTGCAATAGCCTAAAT 27 413 750211 494554 494573 CAGAAGCAAAAGATAGCAGC 53 414 750212 494873 494892

TATTAATGTACTTGAAGATG 52 415 750213 495043 495062 ATCAGTGTGTGCAGATTCTG 45 416 750214 495050 495069

TCCCATCATCAGTGTGTGCA 27 417 750215 495201 495220 TATGAAAATAATGTTCAAG 80 418 750216 495233 495252

AAAAAATTTTAAACCCCTAG 76 419 750217 495441 495460 TCATGAAAGTAGAGAGTAAG 68 420 750218 495659 495678

TGAAACAATTTAAGTGCCCA 64 421 750219 495718 495737 TTGAAGGGATATCTTCATTC 106 422 750220 495795 495814

TCTGAAAAAATAGAACAAC 64 423 750221 496202 496221 AATAAAAGACCTGCACAGCA 66 424 750222 496205 496224

CAAAATAAAAGACCTGCACA 48 425 750223 496379 496398 AGACCCTTATTTACACCATA 65 426 750224 496557 496576

CTATTAATGAGCTATCTGAA 37 427 750225 496561 496580 TCTACTATTAATGAGCTATC 45 428 750226 496567 496586

TTTCCATCTACTATTAATGA 39 429 750227 496661 496680 GTCTCCACCAGAACAAAACCT 69 430 750228 496691 496710

ATGACATAACCATATACAAA 35 431 750229 496697 496716 TTACTGATGACATAACCATA 71 432 750230 496869 496888

CAAAATTCACCTGTGTTTCA 39 433 750231 496927 496946 ATGGAGGGTGTAATCAAAT 36 434 750232 497148 497167

TGTTACACTGTTATTAAGC 28 435 750233 497197 497216 AAAACAATGAAGCAGAGGGA 37 436 750234 497299 497318

TAATAAAAAGTATCCACCA 72 437 750235 497429 497448 GAAGTTATGCACCATCAAAT 46 438 750236 497477 497496

TTTACTACTGATATCACCAA 41 439 750237 497680 497699 AGAACAGACTAAGCCCGAAA 40 440 750238 497917 497936

AATTCTGGTGGAAAATACAG 71 441 750239 497984 498003 ACAGAAATCATATCAAATCC 69 442 750240 498096 498115

ATATATGAAACATTCCATCC 39 443 750241 498127 498146 TTACACTTTAGACCAAACGG 37 444 750242 498179 498198

GGGCAGATTGATCACCTAGA 25 445 750243 498208 498227 CGGGACCTCAATACTCTACT 37 446 750244 498432 498451

TAAATGGAAATGAGAGCAG 61 447 750245 498842 498861 AAGTCACAACCACATCTGC 57 448 750246 498894 498913

GATTCTCCAACAAAGAGACA 50 449 750247 498942 498961 GACACTAGCAGTTTCTTACC 30 450 750248 499132 499151

GCTTAAAGTAGCCTAAGGAT 32 451 750249 499286 499305 ACAATAAGTTGTAGCTACAA 61 452 750250 499304 499323

TTAAAAACAACGTAGCTAC 62 453 750251 499335 499354 TTCCTATTAGGAGGTTTAAA 43 454 750252 499395 499414

ATAAGTAATTCATAGTCAGA 44 455 750253 499452 499471 GGCTTAAATAAAAGACTGCT 63 456

TABLE-US-00009 TABLE 8 Reduction of UBE3A-ATS RNA by 8,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (35,000 cells/well) SEQ ID SEQ ID NO: 1 NO: 1 UBE3A- Compound Start Stop ATS SEQ ID

Number Site Site Sequence (5' to 3') (% UTC) NO 617456 463905 463924 CAACTGTTACCAAGACTTCA 34 32 617457 465003 465022
ATCTGCGAATAGAGGCCCTC 44 33 617460 466539 466558 TAGGTTGCATAAAGCCAGGC 45 36 617461 466981 467000
CACACATCTTGTTCCTCAA 19 37 617557 483977 483996 ATCATGTGCATACCCAGGGT 37 172 749753 448285 448304
TAAGATTCCATTGCCAGAAT 115 457 749754 448296 448315 AAGTTTCGCAATAAGATTCC 103 458 749755 448500 448519
TTTCAGTCAGCAAAGGCAGC 80 459 749756 448681 448700 TCTCAACCCTGAAAACAATC 87 460 749757 449889 449008
CATTTGGCATTATTACAGCA 103 461 749758 449031 449050 CTGACTGTCATTTCATATAAG 69 462 749759 449067 449086
TGCATAACGATAATCATGTG 161 463 749760 449112 449131 TTGTAGATCCTGGCAAGTAT 132 464 749761 449130 449149
AAAAATAAAAGTTTCCCTCTT 64 465 749762 449167 449186 CAAAACCTTGTCACATTATA 129 466 749763 449218 449237
CTATATTTAGACCAAATGAG 91 467 749764 449254 449273 CTGACATATATAAATTAGAA 104 468 749765 449263 449282
CATAGCAACCTGACATATAT 109 469 749766 449279 449298 ACAAACATTGTAAAAACATA 92 470 749767 449337 449356
TTAGAACATGTAGCCATAAT 69 471 749768 449382 449401 AATTGTATTTAATTTAATGA 102 472 749769 449439 449458
ATCATACTGGAGCCAGGTGA 85 473 749770 449545 449564 AATGCAGAAATAAGACCTTC 88 474 749771 449765 449784
CAGTCATCGCTTACTAAGGG 63 475 749772 449786 449805 AAGCCAAAGAGTACTCTTCC 83 476 749773 449820 449839
GATATCTGGGTGTGTACGCT 67 477 749774 449944 449963 CAACATAAGGTCTTATTGTT 74 478 749775 450083 450102
GGTCTTCTAGAAGCTAATAG 77 479 749776 450092 450111 TTGATAAGTGGTCTTCTAGA 64 480 749777 450204 450223
CAATTCTTTTGAGTGTGTAC 66 481 749778 450205 450224 TCAATTCTTTTGAGTGTGTA 70 482 749779 450511 450530
GTGTGACATTGTATCATAC 87 483 749780 450710 450729 CTTTATGCTTTCTGTTCTTT 76 484 749781 450733 450752
GAGACTCTCTTTGTTTCTTT 96 485 749782 450906 450925 TTCATTGGAATTCATTGAGC 94 486 749783 450978 450997
CAAGACTTTCTTCTTGTTTT 115 487 749784 451042 451061 AGTCAGCTGTTAATCTTCCT 85 488 749785 451104 451123
TTTTTCTTTGAGCAITTTTA 104 489 749786 451250 451269 TTTAAGTATTTAGAGAACTC 122 490 749787 451286 451305
TACCTAGGCTCACTTGCTTT 91 491 749788 451306 451325 TAGTTGTTGATTTGAATAAC 87 492 749789 451308 451327
TGTAGTTGTTGATTTGAATA 93 493 749790 451389 451408 TTAATTTTAAAATCACTTCAG 96 494 749791 451466 451485
TCTTCTTTGGCAAATATATT 126 495 749792 451604 451623 TCCCTTTTGGCATCTTTTG 58 496 749793 451673 451692
AATAAGTTTCTAATTTACAC 113 497 749794 451834 451853 TCTCATGTGTTTTTGTTCCT 49 498 749795 451954 451973
TAAACCTGGAGATTTTATCC 128 499 749796 452167 452186 GGAACACACCATTTCAGCAG 38 500 749797 452222 452241
AATCTTGTTGACTAAAAGTA 114 501 749798 452255 452274 GTCTCACGCTGTGTGAATCA 104 502 749799 452257 452276
AGGTCTCACGCTGTGTGAAT 89 503 749800 452285 452304 TGCTCCTATTTCAATATGAG 100 504 749801 452324 452343
ATAAGGTTCAAAAGCCGGTG 182 505 749802 452459 452478 AACTGGTAACATAAATGCAG 99 506 749803 452485 452504
TCAGAGCAAAATGGACTGAC 142 507 749804 452517 452536 TACAATAACCCAGTCTGCAG 65 508 749805 452590 452609
AAGGCCCTTTCATATGCCA 96 509 749806 452661 452680 TCCTATAAAGAAGATATTT 65 510 749807 452715 452734
CTGTGAGAGCTCTGCCCTCA 93 511 749808 453217 453236 GTTAATAGTGTTCCTTACATC 106 512 749809 453233 453252
CAGTGGAGTCAAGTTGGTTA 135 513 749810 453350 453369 TCTACTAGCTAACTTTATCA 109 514 749811 453453 453472
AGGATCAAAGCAGAAATTAA 102 515 749812 453814 453833 AAAACATAGACTTTACTAA 139 516 749813 453899 453918
CACAAAAAACTGATTATATA 92 517 749814 453933 453952 GATCTGTGTTGTTCTTAAGT 73 518 749815 453978 453997
AAGGTAGATTTTATGGCTAC 68 519 749816 454002 454021 GGGCAAAAATAGCAACATAA 47 520 749817 454067 454086
AAGACATAGATGATCCCAT 89 521 749818 454085 454104 TATGAGATGTAAGGAGACAA 86 522 749819 454303 454322
GATAGGAGTAATCTTTTTT 86 523 749820 454611 454630 AAACAAAACCTCCTAAGT 77 524 749821 454657 454676
AGAATAACAGAATTAAGAG 89 525 749822 454658 454677 CAGAATAACAGAATTAAGA 111 526 749823 454661 454680
ACACAGAATAACAGAATTA 71 527 749824 454759 454778 AAAAAGAGGAAAACCGAAAG 88 528
TABLE-US-00010 TABLE 9 Reduction of UBE3A-ATS RNA by 8,000 nM 5-10-5 MOE gapmers with a mixed PO/PS
internucleoside linkages in vitro (35,000 cells/well) SEQ ID SEQ ID NO: 1 NO: 1 UBE3A- Compound Start Stop ATS SEQ ID
Number Site Site Sequence (5' to 3') (% UTC) NO 617456 463905 463924 CAACTGTTACCAAGACTTCA 40 32 617457 465003 465022
ATCTGCGAATAGAGGCCCTC 49 33 617460 466539 466558 TAGGTTGCATAAAGCCAGGC 28 36 617461 466981 467000
CACACATCTTGTTCCTCAA 33 37 617557 483977 483996 ATCATGTGCATACCCAGGGT 42 172 749825 454828 454847
CATACAAATGTCACTAATAT 114 529 749826 454836 454855 AAGACTAGCATACAAATGTC 85 530 749827 454864 454883
ACAGGTGATATTCCCTATC 84 531 749828 454998 455017 ATATATCTACTGAATATGAT 108 532 749829 455062 455081
CTCAGAACCCTTTGGTCTAA 66 533 749830 455110 455129 CTGACTTGACTTTAATGAAA 68 534 749831 455148 455167
GCATTGAGCTTATAGCAGAA 58 535 749832 455150 455169 CTGCATTGAGCTTATAGCAG 84 536 749833 455193 455212
AAACATTCCTCATACCACAC 73 537 749834 455230 455249 CTTATCCTTGATCCAGACAA 78 538 749835 455436 455455
AATGGGAAATGCTAAAGTTG 72 539 749836 455459 455478 AGTGACACAGTAGTTGTATC 70 540 749837 455485 455504
CAATTTCTCTCTACCAAGC 71 541 749838 455542 455561 CTTTGTCAITTCATTTATAA 70 542 749839 455603 455622
ATACAGGCATCTCAGCCCTC 61 543 749840 455767 455786 AAAGTACCAAAGTGGCTGCT 71 544 749841 455768 455787
AAAAGTACCAAAGTGGCTGC 68 545 749842 455829 455848 GAAATATGAACCTCACAGTC 70 546 749843 455960 455979
CACAGTTGTTTAGGTCACTC 119 547 749844 457070 457089 CATGCCATATTTCTTCTT 91 548 749845 457076 457095
ATAGAGGCATGCCATATTTT 86 549 749846 457151 457170 AGCAAAAATAATCTTAGAAA 91 550 749847 457194 457213
GATTCCAGGCCTTCTATCTC 80 551 749848 457354 457373 GGTCTCAATGAGGAAAAGGA 34 552 749849 457424 457443
ATAAAAGCAAAAGAGAGTTA 84 553 749850 457556 457575 TCTATTCTGAAACCCCAATA 45 554 749851 457705 457724
GACATGCCATCAAGAGAAGA 59 555 749852 457917 457936 ATGTATTAACCAATATTTTG 51 556 749853 457993 458012
AGAACCCACTTGATCTATTA 53 557 749854 458199 458218 TAAAGAATTGAGTACCAAAA 64 558 749855 458268 458287
TATCACATTAATCCCTCTC 37 559 749856 458289 458308 TAGAATAGAAAAGCATGAAG 71 560 749857 458328 458347
AACAACATATATTTGCTTGA 39 561 749858 458346 458365 ATAATCACACTAAATCTTAA 70 562 749859 458391 458410
CCTCTATTAAGATCTATGAG 75 563 749860 458438 458457 ACTTCATCAATATTTCCCA 31 564 749861 458537 458556
GCACTTAAATTTATCAGTTG 28 565 749862 458539 458558 AAGCACTTAAATTTATCAGT 36 566 749863 458567 458586
TTAGTATGTCGAGAACTCAA 24 567 749864 458588 458607 AAGTTGAAACACATTTTAGC 32 568 749865 458596 458615
AGGATTAAGTTGAAACAC 30 569 749866 458597 458616 CAGGATTAAGTTGAAACA 49 570 749867 458608 458627
AATCAGGGAAGCAGGATTAA 55 571 749868 458668 458687 TGATTACTCTTGGCAGTAAT 58 572 749869 458823 458842
TGATAGATACTTGATTAGC 14 573 749870 458896 458915 CAAATCTAAAGCTCATTTAC 72 574 749871 458908 458927
CTACCAAAATGTCAAATCTA 56 575 749872 459220 459239 GTAACCTGAATATTTTCATGA 37 576 749873 459228 459247
AAATATGGTAACCTGAATA 56 577 749874 459240 459259 AATACATTACTGAAATAATG 86 578 749875 459256 459275
TTTTACCTTGACATTCTAATA 40 579 749876 459450 459469 TTAACCTATTTTAATAATAT 89 580 749877 459567 459586
TTAAGTGATTGGAAATAAAA 82 581 749878 459580 459599 CAATTAGGAATATTTAAGTG 97 582 749879 459633 459652
CCAGGCAATGGCTCTTTCAA 66 583 749880 459645 459664 TGTATAGTTTACCCAGGCAA 37 584 749881 460066 460085
CTTTATCAATCTAATCAATT 65 585 749882 460115 460134 ACTTTATAGTGTGGATGGTA 22 586 749883 460146 460165
TTTTAGGGAATTGTCCTGAT 39 587 749884 460211 460230 AAGGAAACACACATAATACC 46 588 749885 460213 460232
GGAAGGAAACACACATAATA 26 589 749886 460272 460291 CACTAAGGACAAAGATATGG 75 590 749887 460303 460322

TATTGTTTCAATCTCAAGAA 56 591 749888 460545 460545 CAACCTGGGCTCTCATCTAA 41 592 749889 460709 460728
AATGCTTGATCTGTGGGCTC 32* 593 749890 460749 460768 ATTGTGTACACAGGGATTG 22* 594 749891 460792 460811
AACTGTATACTTTGAAAGTA 68 595 749892 460870 460889 GAAGCAAGTAAGTAAATAAT 81 596 749893 460981 461000
CTTAGATGTGTTTATCCAAA 25 597 749894 460999 461018 GTGTTTTTCCATTTTTCTCT 9 598 749895 461027 461046
TGGCTCTATAAGGCTTCCC 39 599 749896 461039 461058 AATCCTATATTTTGGCTCTA 34 600
TABLE-US-00011 TABLE 10 Reduction of UBE3A-ATS RNA by 10,000 nM 5-10-5 MOE gapmers with mixed PO/PS
internucleoside linkages in vitro (40,000 cells/well) SEQ ID SEQ ID NO: 1 NO: 1 UBE3A- Compound Start Stop ATS SEQ ID
Number Site Site Sequence (5' to 3') (% UTC) NO 617456 463905 463924 CAACTGTTACCAAGACTTCA 47 32 617457 465003 465022
ATCTGCGAATAGAGGCCCTC 70 33 617460 466539 466558 TAGGTTGCATAAAGCCAGGC 51 36 617461 466981 467000
CACACATCTTGTTCCCTCAA 45 37 617557 483977 483996 ATCATGTGCATACCCAGGGT 29 172 699780 479654 479673
AGTAAGGTCTGTTATTCTCC 122 127 750040 470321 470340 GAACTATCCTGCATCCGAGG 41 601 750041 470398 470417
CAACCATCGAGATGATCATA 70 602 750042 471812 471831 TAAATGCTTTTCTAAATCTA 100 603 750043 471892 471911
GCTTTTATAGTGTGAGGCA 81 604 750044 472153 472172 TTAAGCTCTAATTAACAG 113 605 750045 473241 473260
TATTTCACTGGAGCTTTGAT 107 606 750046 473564 473583 AATCATCTACTGATGAACAC 99 607 750047 473573 473592
TCTTTATCTAATCATCTACT 90 608 750048 473576 473595 TTCTCTTTATCTAATCATCT 111 609 750049 473697 473716
AGTTCCTGGAAACCGCCATT 63 610 750050 473789 473808 ATGCTGTTGTACACTAGATC 82 611 750051 473955 473974
GTACACTATTGTTTTGATAT 44 612 750052 474053 474072 CACCCCTAATTTATATTACT 56 613 750053 474069 474088
CATAGGTCAATTCCTTCACC 47 614 750054 474103 474122 TTAATTTAAATAGTTTACAA 100 615 750055 474123 474142
CTAGCTTGAATGGATACCAA 71 616 750056 474169 474188 TAGTGGTTGCCTTAGTATTA 52 617 750057 474198 474217
CAAGTGCTATTTTTTTTAA 95 618 750058 474230 474249 TATATATAATTGAGGGCCAC 116 619 750059 474414 474433
ATAATTATAAGATAGGTTTT 127 620 750060 474618 474637 ATTAGTATTGCTGCTCTAGC 91 621 750061 474786 474805
CCAACTGTAATCATTGATTT 79 622 750062 474865 474884 GATAATGTGTAGTTTTTTTAT 82 623 750063 474878 474897
TAAGGTGTTGTATGATAATG 78 624 750064 474953 474972 AATATGTGTATCCTGAAGA 93 625 750065 475066 475085
TCATTGTGAATTTCCACAT 81 626 750066 475233 475252 TTAGGGATATACTGTTATAC 72 627 750067 475268 475287
AAAATTAGGTTATTGGATTG 103 628 750068 475293 475312 CTTTCACTCTCATTCTTAA 91 629 750069 475397 475416
ACTTTCAAGTTTATTAATTT 115 630 750070 475495 475514 TACTTTCATTTATGTCTAGT 78 631 750071 475503 475522
GATATCTATACTTTCATTTA 125 632 750072 475628 475647 CATGTTTTTACACCAGCTGT 87 633 750073 475693 475712
TTACTTACTCAATTTCTTCT 97 634 750074 476072 476091 GTTTCAGCAGTTTCTGCTCC 85 635 750075 476164 476183
GGGCACTAGGAGTTCCTAA 80 636 750076 476337 476356 CTCTGAGAGTGTTTAGAAAT 55 637 750077 476350 476369
GTGAAAGTAAGATGCTCTGAG 62 638 750078 476405 476424 TACATCAGACAAGGCTCAGG 91 639 750079 476845 476864
TTCTATAATCTTATGGTTAA 77 640 750080 476851 476870 GAAGTGTCTATAATCTTAT 74 641 750081 477296 477315
GTGAAAATTTGTAGGTTGC 110 642 750082 478347 478366 TACAATCAGTGTCTTTCACC 78 643 750083 478472 478491
ATGGTTAGGGCTATGTTATG 96 644 750084 478542 478561 GAAATAAACGCAATGTATC 118 645 750085 478710 478729
CTTGCATGTTGCCTTTTTCT 129 646 750086 479037 479056 AACTCTCTGCCATTATTACT 81 647 750087 479330 479349
TAATCTCACTGGATACAGAA 107 648 750088 479468 479487 AAACATTTTACCATTTTATA 133 649 750089 479644 479663
GTTATTCTCCCTCTTGAACC 82 650 750090 479647 479666 TCTGTTATTCTCCCTCTTGA 86 651 750091 479649 479668
GGTCTGTATTCTCCCTCTT 51 652 750092 479652 479671 TAAGCTCTGTTATTCTCCCT 52 653 750093 479656 479675
AAGTAGAGGTCTGTTATTCT 92 654 750094 479659 479678 CTAAAGCTAAGGTCTGTTAT 74 655 750095 479661 479680
GACTAAAAGTAAGGTCTGTT 70 656 750096 479664 479683 AATGACTAAAAGTAAGGTCT 133 657 750097 480219 480238
AAAGATATAGAAGTGAAG 128 658 750098 480286 480305 GTTGGAGACTTCAATTTCTC 89 659 750099 480361 480380
AAACAACAGAGCCTCAAATA 114 660 750100 480441 480460 AAAATACCACAAAGATAGGC 38 661 750101 480530 480549
AAGAATGGAAGATGTATC 57 662 750102 480533 480552 TAAAAGAATGGAAGATGT 99 663 750103 480648 480667
AATTTTACCTCAGTATAAAA 106 664 750104 480698 480717 ATTACCAAATGCCACTGTAT 59 665 750105 480714 480733
GTTGCATAACATGTGTATTA 65 666 750106 480787 480806 AATTAAGTCTTAATGAGTA 145 667 750107 480898 480917
ATAAAGGTCACTATTGTAT 89 668 750108 480968 480987 AATAAACTGATACATACTA 101 669 750109 481010 481029
ATCAACTTTTACTGATACATA 119 670 750110 481027 481046 TATCATTATGACCTAGGAAT 69 671
TABLE-US-00012 TABLE 11 Reduction of UBE3A-ATS RNA by 8,000 nM 5-10-5 MOE gapmers with mixed PO/PS
internucleoside linkages in vitro (40,000 cells/well) SEQ ID SEQ ID NO: 1 NO: 1 UBE3A- Compound Start Stop ATS SEQ ID
Number Site Site Sequence (5' to 3') (% UTC) NO 617456 463905 463924 CAACTGTTACCAAGACTTCA 16 32 617457 465003 465022
ATCTGCGAATAGAGGCCCTC 21 33 617460 466539 466558 TAGGTTGCATAAAGCCAGGC 24 36 617461 466981 467000
CACACATCTTGTTCCCTCAA 15 37 617557 483977 483996 ATCATGTGCATACCCAGGGT 12 172 750254 499480 499499
ACTGGAATTACTAAAAGGGA 62 672 750255 499621 499640 AGTAGAGAGTGGGATGGTAT 53 673 750256 499691 499710
TAAGAAACATCATTCAAG 67 674 750257 500139 500158 TGAACAGAATGAGAAGTTTA 82 675 750258 500141 500160
TCTGAACAGAAATGAAAGTT 69 676 750259 500285 500304 TGGTGTACATTGGATATGAA 24 677 750260 500319 500338
TTTCTAACATTTACTGTGGA 83 678 750261 500367 500386 TTCTAACATACTAATTAGCA 41 679 750262 500384 500403
CACCAATGCAAGCCAGCTTC 28 680 750263 500526 500545 TGTTTAGAGGTCAAGCCCTG 75 681 750264 500570 500589
ATCAGTCAAAACATGTTCTG 56 682 750265 500629 500648 TATTACACAAGGTATTGGTA 35 683 750266 500639 500658
ATGAAAGGTTTATTACACAA 34 684 750267 500760 500779 AATCAATTTGTGCCACAGGC 34 685 750268 500786 500805
GCAGCTTATAAAGAGAGCCA 78 686 750269 500860 500879 TTTGGTAGGTAACACTACGGGT 35 687 750270 500882 500901
TCGGATATAGCTTTTACATA 14 688 750271 501004 501023 AAATAAGCACATCCGATAG 60 689 750272 501060 501079
ATCATCTTTGATTGACTTT 37 690 750273 501091 501110 TTCTTTATGAATCTTTGAAA 51 691 750274 501151 501170
GTAAAGAGCCACTAAGGGA 40 692 750275 501165 501184 ATGAGATGGGCACAGTAAAG 33 693 750276 501391 501410
TCCCCAGATAATGCATAGAT 49 694 750277 501437 501456 GGTGGCATAGAAGGCAGCAC 74 695 750278 501549 501568
GGATGCAGCAGGAGAAGAAA 42 696 750279 501576 501595 AAGGCAAGGCTAAGGAGTGC 27 697 750280 501679 501698
AGAGAGCTAGAGCTAGGACT 17 698 750281 501694 501713 ATCCCAAAGTTACACAGAGA 58 699 750282 501703 501722
CTTTGGGACATCCCAAAGTT 73 75 750283 501752 501771 CACCTGTATCCAAAATTCAA 47 700 750284 501788 501807
CATATCTGAAGCACAGAGAG 39 701 750285 501860 501879 AGTCTGCTCTGCTGCTCTG 66 702 750286 501862 501881
TCCAGTCTGCTCTGTGCTC 67 703 750287 501882 501901 AGCTCAGAGGCAGCAGGAGC 47 704 750288 502030 502049
CTAAGTCCATATTTAAATC 69 705 750289 502049 502068 TTCAGGCTTCCTTCACAGCC 69 706 750290 502052 502071
TTCTTCAGGCTTCCTTACAA 44 707 750291 502138 502157 GGAATCAGTGTCTACCCATTA 23 708 750292 502194 502213
TTAGCCATCATTTTATTCTC 12 709 750293 502341 502360 GGCCCATTTTTTCAATCTCA 43 710 750294 502404 502423
CATGGTCTTCCTTGACTACA 40 711 750295 502580 502599 AAGCCAATGCGCAAGAAAAG 42 712 750296 503121 503140
CATCCAGTTAATCTCTGACA 31 713 750297 503125 503144 CTCGCATCCAGTTAATCTCT 38 714 750298 503145 503164
ACACCATCCAGAAATGGTC 75 715 750299 503180 503199 TTTCTGACTCCCTATCCAGT 44 716 750300 503376 503395
TTCCCCAGGGTCATAGGAGT 71 717 750301 503442 503461 GCACTGTCCAGTTGGATTA 27 718 750302 503490 503509

ACCGACCAAAATGTGGGTCT 119 719 750303 503504 503523 TTGGTTGTGGTGAAACCCAG 79 720 750304 503533 503552
TGTTAAAAGAGAAAAGAATC 77 721 750305 503550 503569 TTGCAGTGATGTACTGATGT 26 722 750306 503599 503618
TTGTTTTTATAAGCAATTAG 59 723 750307 503645 503664 TATAAATATGCCCATATGCT 28 724 750308 503656 503675
CTGCCTGCAACTATAAATAT 45 725 750309 503825 503844 GTATCTCTTACGCCAGTGCC 32 726 750310 503828 503847
ACTGTATCTCTAGCCCAAT 64 727 750311 503895 503914 GATTGCTTTATTGCAACTAA 63 728 750312 503936 503955
CTTAAATAGTAGGAAAGCCA 14 729 750313 503973 503992 ACACGTGTACAGGAGGGTGTCT 25 155 508739 508758 750314 503975 503994
GCACACTGTACAGGAGGGTG 30 730 508741 508760 750315 503977 503996 CTGCACACTGTACAGGAGGG 17 79 508743 508762 750316
503979 503998 CACTGCACACTGTACAGGAG 41 731 508745 508764 750317 503981 504000 TTCACTGCACACTGTACAGG 33 156 508747
508766 750318 503983 504002 GATTCACTGCACACTGTACA 34 732 508749 508768 750319 503985 504004 TTGATTCACTGCACACTGTA
19 80 508751 508770 750320 503987 504006 TGTTGATTCACTGCACACTG 28 733 508753 508772 750321 503991 504010
ATGATGTTGATTCACTGCAC 19 734 508757 508776 750322 503993 504012 AAATGATGTTGATTCACTGC 43 157 508759 508778 750323
503995 504014 GGAAATGATGTTGATTCACT 29 735 508761 508780 750324 503998 504017 TGTGGAAATGATGTTGATTTC 50 736 750325
504090 504109 CAGTATCTTAATCGGTGAAC 15 737

TABLE-US-00013 TABLE 12 Reduction of UBE3A-ATS RNA by 8,000 nM 5-10-5 MOE gapmers with mixed PO/PS
internucleoside linkages in vitro (40,000 cells/well) SEQ ID SEQ ID NO: 1 NO: 1 UBE3A- Compound Start Stop ATS SEQ ID
Number Site Site Sequence (5' to 3') (% UTC) NO 617456 463905 463924 CAACTGTTACCAAGACTTCA 25 32 617457 465003 465022
ATCTGCGAATAGAGGCCCTC 27 33 617460 466539 466558 TAGGTTGCATAAAGCCAGGC 25 36 617461 466981 467000
CACACATCTTGTTCCTCAA 17 37 617557 483977 483996 ATCATGTGCATACCCAGGGT 13 172 699781 483977 483996
ATCATGTGCATACCCAGGGT 20 172 750111 481036 481055 ACATAGAATTATCATTTATGA 49 738 750112 481073 481092
ACTGTGGAAAACTTTGGCA 38 129 750113 481398 481417 ATGGGTAAATACTTATAA 79 739 750114 481451 481470
TAAACCTTTGTGCATAATG 32 740 750115 481487 481506 CAACAACAAAAGCGGATAAA 50 741 750116 481524 481543
GTACACATAGGTTTGGCAAA 34 742 750117 481529 481548 ACCTCATGACACTAGGTTTT 26 743 750118 481571 481590
GTATAGACCCAAACTATAAA 35 744 750119 481573 481592 ATGTATAGACCCAAACTATA 55 745 750120 481725 481744
TGATCTCTTCAACAAATTGT 82 746 750121 482139 482158 AATGTTGCCGAAAGAAAAGA 73 747 750122 482156 482175
ACAAGTGTCATATACTAAAT 51 748 750123 482174 482193 AAATTTAACCAAGGAGTTAC 48 749 750124 482246 482265
AGCCATCTAAAAGAGAAATT 38 750 750125 482396 482415 GATCAAGAAGTAAAATTATC 74 751 750126 482685 482704
CACATGATAATCTCAATTAT 52 752 750127 482916 482935 GAAAAATTCACCTCTATAGGT 58 753 750128 482988 483007
CTATAGAGAGATAGATATTA 52 754 750129 483017 483036 AGAAAAATGTCTTTCCAAAT 43 755 750130 483089 483108
CAAATACTTAAATCAACTGT 50 756 750131 483255 483274 GGAGGGAGGAAAATTATTTC 14 757 750132 483297 483316
AGCAAACAACAGCAACATGC 56 758 750133 483323 483342 AAAATACTTTAGAAAAAGTCA 73 759 750134 483365 483384
CAGAATTCAATGGACCCACA 33 760 750135 483809 483828 TCGGCAAAGGCATTATTATT 32 761 750136 483920 483939
GCTTTACCTATAGGTGGCC 45 762 750137 483928 483947 GATGTTAAGCTTTACCTAT 25 763 750138 483967 483986
TACCCAGGGTAGGATTCATG 17 764 750139 483970 483989 GCATACCCAGGGTAGGATTC 13 765 750140 483972 483991
GTGCATACCCAGGGTAGGAT 9 766 750141 483975 483994 CATGTGCATACCCAGGGTAG 16 767 750142 483979 483998
AGATCATGTGCATACCCAGG 34 768 750143 483982 484001 AGAAGATCATGTGCATACCC 36 769 750144 483984 484003
TAAGAAGATCATGTGCATAC 44 770 750145 483987 484006 AATTAAGAAGATCATGTGCA 119 771 750146 484085 484104
GTTCAGATGAACATGAAGCA 45 772 750147 484311 484330 ACTCATGCTGGTTACTAGGG 30 773 750148 484598 484617
TCATATGGGTGGTCGCTAAT 24 774 750149 484719 484738 ACATAACGATGAACCTCTAA 55 775 750150 484721 484740
TAACACTAACGATGAACCTCT 42 776 750151 485136 485155 TTAAACTCAACCCAGTGCA 29 777 750152 485344 485363
ATTCTAGAGAAAACCTAGG 82 778 750153 485478 485497 AGAACAATGTTTCTTATGAA 72 779 750154 485865 485884
GAACTGGGATATTACCAAT 45 780 750155 485871 485890 AAAATGGAACTGGGATATT 37 781 750156 486052 486071
AAAAATTAAAGGCCAACAGA 46 782 750157 486157 486176 TTATGGCAGGTAGTGAAAGG 47 783 750158 486265 486284
ACAATATGCAAAAATTAAAT 74 784 750159 486400 486419 CATTCTCCAACATAGATCCT 41 785 750160 486568 486587
CAATAATAGAGACTTTACCA 79 786 750161 486812 486831 ATATATTATGTAAATGTAAC 78 787 750162 486824 486843
AGGATGGAAAGATATATTA 76 788 750163 486842 486861 AAACAGGTGAAAATGAAAG 79 789 750164 487049 487068
AAAGGCAATATTGAAGGAAA 72 790 750165 487267 487286 GAATTGAGTTAATAATTCCT 57 791 750166 487304 487323
TCTCACAGAGAAAGAGGTGG 21 792 750167 487314 487333 TTTTCTATAATCTCACAGAG 43 793 750168 487341 487360
GGCTCAGAAAACATCCTTTT 35 794 750169 487437 487456 ATTTAATTTACACTTAATTA 93 795 750170 487464 487483
ACTTTCTCTGCTTATAACT 22 796 750171 487469 487488 ACTATACTTTCTCTGCTTA 28 797 750172 487589 487608
AGCAATAGAAATCACATGA 21 798 750173 487724 487743 AAAGCAGTAAACAATAAGTG 53 799 750174 487739 487758
TCATATGTAAATCCAAAAGC 57 800 750175 487772 487791 CATTGTAAGGATAAGAGATA 27 801 750176 487784 487803
CAGTTTGAATTACATTGTAA 29 802 750177 487804 487823 AATTGAACTTAAATTGGCAT 32 803 750178 487831 487850
CTGCATAGGAGTAGAGTTTT 31 804 750179 487877 487896 AAGATGGAATTTGGACATC 50 805 750180 487921 487940
ACATAAAAATGTAATATCC 70 806 750181 488018 488037 ACTGATAAAGGTAAGCACAT 31 807

TABLE-US-00014 TABLE 13 Reduction of UBE3A-ATS RNA by 8,000 nM 5-10-5 MOE gapmers with mixed PO/PS
internucleoside linkages in vitro (40,000 cells/well) SEQ ID SEQ ID NO: 1 NO: 1 UBE3A- Compound Start Stop ATS SEQ ID
Number Site Site Sequence (5' to 3') (% UTC) NO 617456 463905 463924 CAACTGTTACCAAGACTTCA 13 32 617457 465003 465022
ATCTGCGAATAGAGGCCCTC 38 33 617460 466539 466558 TAGGTTGCATAAAGCCAGGC 16 36 617461 466981 467000
CACACATCTTGTTCCTCAA 11 37 617557 483977 483996 ATCATGTGCATACCCAGGGT 13 172 750398 510078 510097
CAGAGAGGAGAGAAGGACAA 45 808 750399 510104 510123 GGCTAGGAAAATTCTCTCTGG 39 809 750400 510207 510226
AGGAGAGCTGTCTTAAGGC 27 810 750401 510214 510233 GATATCCAGGGAGAGCTGTC 23 811 750402 510269 510288
TTGAAGGGAATAATTATTAAT 43 812 750403 510319 510338 AGCTGGACGGACAGTGTTGC 47 813 750404 510407 510426
CAAGACATGCTAGGGACTCT 38 814 750405 510408 510427 CCAAGACATGCTAGGGACTC 30 815 750406 510416 510435
GCCACAGTCCAAGACATGCT 49 816 750407 510442 510461 GCAGATAATCTGAAGATAGA 43 817 750408 510515 510534
TTTATTACAACTGAAAGAG 42 818 750409 510609 510628 GGTCAGATAATGAGGGCCTT 20 819 750410 510701 510720
AGGATAGGTGTATGTGAGTG 31 820 750411 510729 510748 ATGAAGGTCAAGCCGAATCT 50 821 750412 510798 510817
AAAAGAAGTATGTGAAAGTA 45 822 750413 510997 511016 TGTCTCTAATCCTATTCCAA 12 823 750414 511100 511119
AGTCTTCATGATTTGGTAA 37 824 750415 511341 511360 AGTAAGAGGCAGAATTATGT 54 825 750416 511441 511460
GGGCAGCAACAGCTCTTGAA 16 826 750417 511503 511522 TAATAATTAACAGTGACTGG 24 827 750418 511529 511548
AAGTCATCCCCACCTTGGC ND 828 750419 511555 511574 TCCAGCAGGGTGTATTCCAG 47 829 750420 511567 511586
TCAGCTGTGGTTTCCAGCAG 34 830 750421 511579 511598 GCTTAGAGAATCTCAGCTGT 49 831 750422 511594 511613
GGTTGGAAGCTGCCAGCTTA 22 832 750423 511668 511687 AATGGACTGCCGGCCTGGAG 47 833 750424 511893 511912
TGAATACTTGGAGACCTTC 55 834 750425 511895 511914 AGTGAATACTTGGAGACCT 45 835 750426 512282 512301
TCTTTTATTAAACCTAGTTT 60 836 750427 512318 512337 TTAGCTAGCTGTGGGTTTGC 25 837 750428 512392 512411
ACAAGAGCAGACATCTTTTT 60 838 750429 512564 512583 TGTAAGCTTTTCTAAAATTCT 34 839 750430 512570 512589

GTAATTGATTGAGTTTCTTAA 16 840 750431 513029 CTTGATGGCTGTAGCTTGGT 21 841 750432 513074 513093
TTTCTGGATTCTCAGTCTTA 28 842 750433 513189 513208 TGCATTGGTATCTGTATATC 23 843 750434 513219 513238
GTTTCCTATTTTCATCCATTT 25 844 750435 513378 513397 CTGGAGTGTAAGTAAATTCA 33 845 750436 513442 513461
ATACTCTTTCCACATTTTCAG 19 846 750437 513450 513469 TCCTCTTGATACTCTTTCCA 34 847 750438 513459 513478
CTTAGGGTCTCCTCTTGATA 32 848 750439 513467 513486 CAGAGTGCTTTAGGGTCTCC 27 849 750440 513620 513639
ATATTTCTCAAATCTCTTC 48 850 750441 513760 513779 TGTGCTAGAACACTATCTTG 62 851 750442 513793 513812
TGACTCTCATCTCCCTCTTC 70 852 750443 513796 513815 CCCTGACTCTCATCTCCCTC 50 853 750444 513875 513894
CCTTCCTTCTTTATACTGCC 46 854 750445 513944 513963 CATCAAAATCTCTCATTCCT 22 855 750446 513947 513966
AATCATCAAAATCTCTCATT 35 856 750447 513951 513970 CCCCATCATCAAAATCTCT 70 857 750448 514039 514058
TAGTCAGTAGTCCTAAAAAC 58 858 750449 514073 514092 AGACCTCTGTCTCTGGGATT 54 859 750450 514159 514178
AAAGTCCTTACCTGTTCTTG 36 860 750451 514326 514345 TTCACCGAAAGTACAGTCTT 54 861 750452 514330 514349
TAGTTTACCGAAAGTACAG ND 862 750453 514439 514458 AGTCTTTTCTCCCTCCCC 28 863 750454 514504 514523
TCAACAGTGTTAGTGTGTA 45 864 750455 514513 514532 CCTTTTACCTCAACAGTGT 35 865 750456 514673 514692
AGACTAGTCCTGTGTGATC 27 866 750457 514676 514695 CCAAGACTAGTCCCTGTGTA 40 867 750458 514701 514720
AACTGTCAGAGGAAGAAATG 62 868 750459 514861 514880 CGAACAGAACTTGCCGATGT 37 869 750460 514920 514939
TTGTATAGGGAGATGATAGA 39 870 750461 515069 515088 TAACCTTTTATTATTATAGA 55 871 750462 515111 515130
TAAACTAAAGTAACTGGTTT 55 872 750463 515143 515162 CCAGGAAGCTCTGGAGGGAA 23 873 750464 515187 515206
ACTCTATTATATATTTTGGT 28 874 750465 515215 515234 GTTATAAAAGTACTTTTTTT 78 875 750466 515242 515261
GGATACTTCTCTACCCCAA 30 876 750467 515247 515266 ACAGTGGATACTTCTCTAC 40 877 750468 515318 515337
CTGCCTTTTCATTACTATTT 30 878 750469 515500 515519 GTTTTAAAGTGGTAATTGAA 44 879

TABLE-US-00015 TABLE 14 Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (37,000 cells/well) SEQ ID SEQ ID NO: 1 NO: 1 UBE3A- Compound Start Stop ATS SEQ ID Number Site Site Sequence (5' to 3') (% UTC) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 44 172 699781 483977 483996 ATCATGTGCATACCCAGGGT 51 172 749882 460115 460134 ACTTTATAGTGTGGATGGTA 39 586 750519 349103* 349122* GTCATCACCTCTCTTCAGGA 17 181 1065269 432253 432272 TCATCATCAACCTAACCGAG 92 880 1065285 433960 433979 CCATCTTGGAACCGGATGAC 65 881 1065301 442413 442432 GTAATTGAGGTGGCCATATC 101 882 1065317 443132 443151 GAGTCCTAAATGTCTCATAC 81 883 1065333 444649 444668 GGTTTTAGTCCACTGGTTCA 44 884 1065349 445540 445559 CTAGTCATGTATCTACAGCC 88 885 1065365 446609 446628 GTATATGCGCTTACACAGGCT 82 886 1065381 447599 447618 GAGATTTAGATCACTGTTCA 64 887 1065397 448125 448144 ATTGACCCAGCCCATGGCAC 97 888 1065413 448713 448732 GCGCATTGAGCAAATTTCCA 73 889 1065429 449066 449085 GCATAACGATAATCATGTGG 116 890 1065445 449627 449646 GGCTATAGGCCATTTATTCA 111 891 1065461 451295 451314 TTGAATAACTACCTAGGCTC 86 892 1065477 452310 452329 CCGGTGCAGGAATCCAGGAT 98 893 1065493 453797 453816 TAACAATAAGATAAAGGCGC 123 894 1065509 454580 454599 GGCAAGCAAAGACTACACCG 55 895 1065525 455169 455188 GCTCCTAACATTGTATCCCC 94 896 1065541 455476 455495 CTCTACCAAGCTATCCAAGT 78 897 1065557 456768 456787 GTCCTATTGGAGGGCCGCAC 39 898 1065573 457320 457339 GTCCTCCAATAAGCTTCTGA 48 899 1065589 458165 458184 GGGACAATACTGCAATCCTT 63 900 1065605 458957 458976 AGAGACCTCAAGACCTATAG 15 901 1065621 460348 460367 GACTACTTCAACCTGATACC 30 902 1065637 460957 460976 GGGCATTAATAAGATAAGCT 69 903 1065652 461602 461621 ACAACGGGCTATGTGAGAT 50 904 1065667 463052 463071 CACTGATTTTTGTAGTTTCG 32 905 1065682 463885 463904 GGCATGTGTGATAGTCAACA 75 906 1065698 464994 465013 TAGAGGCCCTCTTGTTTCAA 44 907 1065713 465247 465266 GGATTTTATTGGTCATCTCG 30 908 1065729 465664 465683 GTAGTTACTTATACTGGTTC 62 909 1065745 466276 466295 GTGGGTTCTTGATGGAGTCC 79 910 1065761 466540 466559 CTAGGTTGCATAAAGCCAGG 64 911 1065777 467942 467961 GCCTTAAAAGGGTTCCCTGT 64 912 1065793 470287 470306 GGTCATGATGTATGCCATTA 39 913 1065809 474125 474144 ATCTAGCTTGAATGGATACC 55 914 1065825 475454 475473 GTTCTTTCTCGGTCAAACCTA 41 915 1065841 476084 476103 TACGAGTTGCTGGTTTCAGC 46 916 1065857 478470 478489 GGTTAGGGCTATGTTATGTT 31 917 1065873 480382 480401 GATTCCACTTGTGTATGCAC 33 918 1065889 482148 482167 CATATACTAAATGTTGCCGA 30 919 1065905 483368 483387 ATACAGAATTCATGGACCC 55 920 1065936 485657 485676 GCCTAGGACCGATTGGTTTCA 48 921 1065952 487409 487428 GGCTGTTGTACATTCCTAGT 65 922 1065968 489096 489115 CCCTAAGCTTAGATATACCC 88 923 1065984 489740 489759 AGGTGGGAAACCGGTTCCCA 100 924 1066000 493977 493996 GGTCGGTCCCCACCAAAGGA 58 925 1066016 496817 496836 GCCATGTACGACCCTCATCA 76 926 1066032 498218 498237 TACTATAATACGGGACCTCA 70 927 1066048 499683 499702 ATCACATTCAAGGTAGCCCC 48 928 1066064 500605 500624 TGACCAGCAATAGGCTCTGC 37 929 1066080 500895 500914 ACTGGCCTTTGGGTTCGATA 81 930 1066096 501395 501414 GGAGTCCCCAGATAATGCAT 91 931 1066112 501767 501786 GGATGGGCTAAGAGTCACCT 74 932 1066128 502370 502389 GGTATTAAGGCCCTTGGCCA 104 933 1066144 502741 502760 GGCCCGATGACACCAGCCAC 87 934 1066160 502949 502968 TAACAGTCCTGTAACAGAC 90 935 1066176 503331 503350 CCCCCTGAGATCCATGAGGT 125 936 1066192 503867 503886 CAGTTGTGCTAGTTGTTAACT 86 937 1066208 504141 504160 AACCAGTGGCTGAGCTATT 89 938 1066224 504518 504537 TTAGGCCATACCTCTTCCAT 77 939 1066240 505065 505084 TTAGTCTGTTGTGTGACTCC 73 940 1066256 505644 505663 CCACCTAAGAGCTATCCGCT 75 941 1066272 505945 505964 GGCCGACCAGCTCTGAAAGT 120 942 1066288 506275 506294 ACCAGGGTGGTATTATAAAG 102 943 1066304 506773 506792 CAGGTTGTGGAGGTTGTTCC 96 944 1066320 507371 507390 GTTGTCTGGCATGGAGATCC 60 945 1066336 507929 507948 GTTGATTGGAGGCACTGCAG 62 946 1066352 508488 508507 TGGTGAGTAAATCAATCACG 106 947 1066368 509007 509026 CCCTTCAACATGTGTTAACG 62 948 1066384 509917 509936 CGTGGCTGTACCTGCAGTGC 49 949 1066400 510393 510412 GACTCTGATTGATTCACTGC 93 950 1066416 510817 510836 AGTACAAACCAATAAGGTA 114 951 1066432 511629 511648 GTGCCATATAATGTTGAAGC 71 952 1066448 511804 511823 CGTGTCTGAACCTGGCCTGTG 75 953

TABLE-US-00016 TABLE 15 Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (37,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number Start Site Stop Site Sequence (5' to 3') (% UTC) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 28 172 749882 460115 460134 ACTTTATAGTGTGGATGGTA 25 586 750519 349103# 349122# GTCATCACCTCTCTTCAGGA 17 181 1065270 432284 432303 CATAACCTCTCCTCGAGACA 96 954 1065286 434182 434201 TCTGTACGCTTCCCTTGGC 61 955 1065302 442470 442489 TCACTGGTGTGGTATCATGC 49 956 1065318 443199 443218 GATTGCATCTCTTGACGAC 63 957 1065334 444722 444741 CTCACCTGTGTATGATGAC 93 958 1065350 445583 445602 ACATGTTAGAGCCCCACA 71 959 1065366 446657 446676 GCATAGGAATGGGACCATAG 88 960 1065382 447672 447691 GACTGATACATAGTGTATGC 80 961 1065398 448219 448238 CTTGTCTCACAATTGTTACC 83 962 1065414 448721 448740 TAACTGTAGCGCATTGAGCA 91 963 1065430 449092 449111 GCTAGTCACTCATTGAGAAG 110 964 1065446 449633 449652 TAGATAGGCTATAGGCCATT 85 965 1065462 451694 451713 GCATATATCTTAACTACCC 56 966 1065478 452315 452334 AAAAGCCGGTGCAGGAATCC 82 967 1065494 453918 453937 TAAGTGGCCATGGCATAGTC 65 968 1065510 454719 454738 GAGTGCTGAAGAACATCCCC 75 969 1065526 455213 455232 CAAGTCTACTACCAATAAGT 89 970 1065542 455567 455586 CGTCAGCATTGCTTGATCTC 89 971 1065558 456773 456792

ATTGTTGCTTATTGGAGGGG ND 972 1065574 457363 457382 CATGTGAGGTTCTCAATGA 40 973 1065590 458258 458257
GAGACATGTGCCAGTTC AAG 19 974 1065606 458987 459006 GGACACTGGCACATCTATAG 64 975 1065622 460362 460381
CTGTATAACTGATTGACTAC 51 976 1065638 461068 461087 GCTCATCTATGGATTGCATT 49 977 1065653 461604 461623
GTACAAACGGGCTATGTGAG 30 978 1065668 463144 463163 AGCTATAGGTACCTGAAGTT 49 979 1065683 464365 464384
CTTAGCTTGCCCAAGCATA 57 980 1065699 464999 465018 CCGAATAGAGGCCCTCTTGT 49 981 1065714 465287 465306
CATGTGTATAGGTGTTGGTC 55 982 1065730 465808 465827 CCAATTTGGTGCTATAACCC 43 983 1065746 466283 466302
CTTATAGGTGGGTTCTTGAT 40 984 1065762 466543 466562 TCTCTAGGTTGCATAAAGCC 69 985 1065778 468007 468026
GCATTGTATAGTTCATCCC 37 986 1065794 470316 470335 ATCCTGCATCCGAGGCATGA 76 987 1065810 474167 474186
GTGGTTGCCCTTAGTATTACA 24 988 1065826 475512 475531 CTGTAAGGTGATATCTATAC 28 989 1065842 476090 476109
TGTTTCATACGAGTTGCTGGT 52 990 1065858 478533 478552 CGCAATGTATCAGGCAACAG 29 991 1065874 480393 480412
GAGGGCACAAATGATTCCACT 65 992 1065890 482162 482181 GGAGTTACAAGTGTCATATA 37 993 1065906 483424 483443
GGCTCTATGCACTTAAGGGA 42 994 1065921 483978 483997 GATCATGTGCATACCCAGGG 35 995 1065937 485769 485788
GGTCGATTCCCTAAATACGC 26 996 1065953 487600 487619 GTGTCATGTAGCAATTAG 25 997 1065969 489319 489338
CACCTGTATAGGAGAATTGT 100 998 1065985 489776 489795 CCACTCCCGTGGCAACATGA 85 999 1066001 493985 494004
GGGTAAGAGGTCGGTCCCCA 48 1000 1066017 496821 496840 GGAGGCCATGTACGACCCTC 80 1001 1066033 498225 498244
GCTGCAATACTATAATACGG 56 1002 1066049 499792 499811 AGCTGAGGTCACCGATCAGA 83 1003 1066065 500610 500629
AACAGTGACCAGCAATAGGC 64 1004 1066081 500898 500917 CTAAGTGGCCTTTGGGTCCG 107 1005 1066097 501399 501418
GGGAGGAGTCCCCAGATAAT 24 1006 1066113 501802 501821 GGCCCAAGTGATGACATATC 50 1007 1066129 502410 502429
GTAAGGCATGGTCTTCCTTG 79 1008 1066145 502746 502765 AATATGGCCCGATGACACCA 85 1009 1066161 502956 502975
GCAAGATTAACAGTCCCTGA 53 1010 1066177 503342 503361 TCAGTCAACACCCCCTGAG 96 1011 1066193 503872 503891
AGGGTCAGTTGTGCAAGTTGT 39 1012 1066209 504260 504279 GCAAGAAATCATGTGGCTCC 70 1013 1066225 504526 504545
ACCAAGATTTAGGCCATACC 55 1014 1066241 505106 505125 GTAACCTTACATAAGCTGC 48 1015 1066257 505654 505673
GGCCTAGCTTCCACCTAAGA 104 1016 1066273 505949 505968 CGCTGGCCGACCAGCTCTGA ND 1017 1066289 506280 506299
CGTCTACCAGGGTGGTATTA 41 1018 1066305 506779 506798 ATACTCCAGGTTGTGGAGGT 84 1019 1066321 507481 507500
CAAATTGGTGAATGTTCCCC 44 1020 1066337 508000 508019 TCTCATGACCACCTAATTGA 72 1021 1066353 508571 508590
CCGTGCTGCTTTCTTGAGTG 92 1022 1066369 509052 509071 ACATGCTTGCATCCAGGCC 53 1023 1066385 510017 510036
CGGAACCCACAGTAGAGGCA 46 1024 1066401 510398 510417 CTAGGGACTCTGATTGATTC 45 1025 1066417 510827 510846
ACCAGGCATAAGTACAAACC 62 1026 1066433 511633 511652 CTGAGTGCCATATAATGTTG 71 1027 1066449 511822 511841
AAGGCTCTCAGGTAAGACG 42 1028

TABLE-US-00017 TABLE 16 Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro(37,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number
Start Site Stop Site Sequence (5' to 3')(% UTC) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 52 172 749882 460115
460134 ACTTTATAGTGTGGATGGTA 22 586 750519 349103.sup.# 349122.sup.# GTCATCACCTCTCTTCAGGA 12 181 1065259 430439
430458 CGTGAACACTGACCCATCA 73 1029 1065275 432915 432934 CAGCATGTAACCTCATGTTGT 75 1030 1065291 441514 441533
GTTCTAAAGTCGGTTGTGTC 57 1031 1065307 442648 442667 AGAGTAGGCCAGGTGTCAA 60 1032 1065323 443577 443596
TACCAGTGAGTAAACTGGCC 89 1033 1065339 444914 444933 GTACTCCCCATGCACACTTG 48 1034 1065355 445958 445977
ACGGATTATCATACCTCCCT 115 1035 1065371 446805 446824 GTCTTGTAACCCCTTACCAG 83 1036 1065387 447818 447837
CCGTTACTTAAGCTGTGGTA 88 1037 1065403 448458 448477 TAGTGCCAGTAGGACTTACT 75 1038 1065419 448827 448846
GGACCCTTAAGTCATAAAGA 78 1039 1065435 449209 449228 GACCAATGAGCATCACATC 69 1040 1065451 449704 449723
CCTGTATGTATGGCTTATGC 73 1041 1065467 452181 452200 ATCCATAGTGATCTGGGAAC 77 1042 1065483 453133 453152
ACTATAATTGGGAATGGTCA 77 1043 1065499 454160 454179 GATTTGACCCTTATGGAGAC 104 1044 1065515 454855 454874
TATTCCCTATCCAGGGTAGA 96 1045 1065531 455296 455315 TTATCCTGTTGCAATAGAAC 81 1046 1065547 456499 456518
ATTAGATTAAAATCTGGCCG 93 1047 1065563 456886 456905 GACTCATAAGACAGGATGCC 87 1048 1065579 457738 457757
CCAGCCAGGTGCTTATATC 18 1049 1065595 458523 458542 CAGTTGTGTATTGACTAAGT 23 1050 1065611 459549 459568
AAGACTATCCTGGTATGACT 56 1051 1065627 460624 460643 TCGATTTCTTGCTTCCCTG 32 1052 1065642 461446 461465
GGCAGCTCCCTGACAAATTAG 22 1053 1065658 461694 461713 AGACCCCTTAAATGTAA 53 1054 1065673 463249 463268
ACCTGGGTATTGCTGTCCAA 41 1055 1065688 464458 464477 CTCACCCTTTACTAAGGTGC 75 1056 1065703 465140 465159
CCATTGGTATGAGAGATGCT 72 1057 1065719 465386 465405 GTCAGACTTATTGAGGATGG 27 1058 1065735 465932 465951
TTACTCCTCTGGTGTGCTGA 29 1059 1065751 466372 466391 GCGCACTTGGGAGCCAGCCA 68 1060 1065767 467076 467095
TTACTGACTGGCCTTAGGAG 38 1061 1065783 468351 468370 GTAGTATGCATTGACAAGCT 32 1062 1065799 471923 471942
GAGTATATTACCTCCAGGTT 27 1063 1065815 474614 474633 GTATTGCTGCTCTAGCTCTA 49 1064 1065831 475841 475860
ACGGACCTCATACGTAGAGT 31 1065 1065847 476213 476232 GGGCTCTTGGAAGTCTAGTT 59 1066 1065863 479124 479143
ACATACCTGGTTGCAGACGC 26 1067 1065879 480825 480844 TAGGTTTGGGATGCCAGTG 50 1068 1065895 482771 482790
TACACATGGGATTGACCC 59 1069 1065911 483853 483872 ATGGCTGACTAGGCAATTGA 76 1070 1065926 484302 484321
GGTTACTAGGGCCAGAGAAT 42 1071 1065942 486355 486374 TGTTAATAGACTGCGATTAT 56 1072 1065958 487839 487858
GGGCGGAGCTGCATAGGAGT 51 1073 1065974 489522 489541 ACAATGGACCACCTAAGACC 96 1074 1065990 489953 489972
CAACCCTTACTCTGCCAGGG 80 1075 1066006 494491 494510 TGTGCAATAGCCTAAATGCC 82 1076 1066022 497436 497455
CTACTATGAAGTTATGCACC 69 1077 1066038 499100 499119 GTTAGCCTTACAGCAAATAC 52 1078 1066054 500359 500378
TACTAATTAGCAAGCCACTG 52 1079 1066070 500766 500785 GTGTCAAATCAATTTGTGCC 47 1080 1066086 501001 501020
CTAAGCACATCCGATAGTCA 49 1081 1066102 501563 501582 GGAGTGCTCTTTGTGGATGC 39 1082 1066118 502117 502136
ACCCATGGCTCATCAGTGGG 91 1083 1066134 502447 502466 GGCAGCTCTTTGTAGGCCCA 60 1084 1066150 502782 502801
GGAGTGGGTTCTCATAAGGA 53 1085 1066166 503100 503119 GGACTAATAGGCTTTCTAC 61 1086 1066182 503399 503418
CTACAGTACCAGGTCATTTG 45 1087 1066198 503946 503965 CACCACCAACCTTAAATAGT 80 1088 1066214 504395 504414
CCATGCCACAGATTGGCTTG 64 1089 1066230 504654 504673 CGGAGCCTTACGCTTGGCTG 66 1090 1066246 505185 505204
CAGTCTGTCTCTGTGTACCG 46 1091 1066262 505741 505760 GGTTGACAGGACATGCTGTC 58 1092 1066278 506119 506138
GTTAGCCGAGCATTGGCTTC 84 1093 1066294 506589 506608 CCCATGGTGGTGGAAATGCTG 102 1094 1066310 506992 507011
TCCCGCACCATGCATTCTGA 43 1095 1066326 507684 507703 TCAGCTTCCTTCAGTGGGCG 65 1096 1066342 508193 508212
GATCTCACACATGGCAGTCG 71 1097 1066358 508691 508710 TACCAGTAAAGGGTAGTATA 49 1098 1066374 509110 509129
GTAAATACACTTGGGCC 73 1099 1066390 510117 510136 TAGCTGAACCTGTGGCTAGG 60 1100 1066406 510654 510673
AGTGTCAAGCTGTAAAGGAT 59 1101 1066422 510901 510920 GCCATAGGTTTCAGGCTGAT 46 1102 1066438 511707 511726
GAGATGGGCTATGAGCCATC 84 1103

TABLE-US-00018 TABLE 17 Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (37,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number
Start Site Stop Site Sequence (5' to 3')(% UTC) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 32 172 749882 460115
460134 ACTTTATAGTGTGGATGGTA 21 586 750519 349103# 349122# GTCATCACCTCTCTTCAGGA 8 181 1065264 430601 430620

GGACGACGCGGGCCGCAACGCA 101 1104 1065280 433410 433429 ATGCAAAAGGTGTCTGATTGGT 47 1105 1065296 441815 441834
GGTCCATAGATGTCAAGTTAC 27 1106 1065312 442767 442786 ATTAATGTTCTATGGTGGAT 46 1107 1065328 443839 443858
GGAGTAGAATGCCCCTGGG 53 1108 1065344 445212 445231 GAGACCTGAATATACCTTAC 55 1109 1065360 446498 446517
GGTATGCCAGGTTTTTGCAC 49 1110 1065376 447152 447171 CCGTGTAATGACACACCTCT 67 1111 1065392 447922 447941
AGGCTATAGTCAAAGGGTGG 46 1112 1065408 448614 448633 GTGTCAGTCAATCATTTAGA 75 1113 1065424 449001 449020
GGTCTTATTTACCATTGGCA 63 1114 1065440 449470 449489 TAACATGTTGCTTGCACCTC 85 1115 1065456 450017 450036
GCTCAAACCTAGAATACCCCA 81 1116 1065472 452261 452280 GTCTAGGTCTCACGCTGTGT 96 1117 1065488 453235 453254
GACAGTGGAGTCAAGTTGGT 106 1118 1065504 454462 454481 GGACTCCACACACCTACTAG 107 1119 1065520 455019 455038
GTGGGATGGCTGTCAATGCT 56 1120 1065536 455341 455360 GATAATTGAGGAGTTACCA 81 1121 1065552 220313 220332
GATCCAAACAAGCACCTCC 85 1122 456682 456701 1065568 457020 457039 TATCACAGTTGCTTGACCCT 70 1123 1065584 457982
458001 GATCTATTATGAGGGCATCA 39 1124 1065600 458571 458590 AGCTTTAGTATGTCGAGAAC 20 1125 1065616 460107 460126
GTGTGGATGGTATCCTGGTC 26 1126 1065632 460711 460730 GGAATGCTTGATCTGTGGGC 25* 1127 1065647 461518 461537
GCTAGACTAGTTGAAATCGG 44 1128 1065663 462056 462075 CCCTCCCATCAGTGAAGATT 78 1129 1065677 463767 463786
GCTTTAGAGGTCAAATAGTG 50 1130 1065693 464784 464803 CTGGTCTTCTCGAGGATAG 69 1131 1065708 465236 465255
GTCATCTCGGGTATATAAAT 30 1132 1065724 465561 465580 GGACAAGTCTTATAGAGAAC 52 1133 1065740 466229 466248
GTTCCAAGTATGGGCTGCTG 44 1134 1065756 466426 466445 GTCAACATGTGCTTGCAAGC 35 1135 1065772 467425 467444
GCTGTTGATACGCTCCTTCA 37 1136 1065788 469130 469149 GAGGCACTGAAGTTCCTACTAC 37 1137 1065804 472937 472956
AGGGTCCCACTTGTTTAGGT 44 1138 1065820 474883 474902 CGTATTAAGGTGTTGTATGA 40 1139 1065836 475943 475962
CCTTG TGACAATAGTAGTGA 54 1140 1065852 476443 476462 GGCTAGATGGAGCTTGAGCC 94 1141 1065868 479997 480016
CGTGAGCTATCTGACAAAA 24 1142 1065884 481441 481460 GTGCATAATGGTCTACCACA 32 1143 1065900 482857 482876
TGAAGATAGGTCAGTAAACG 58 1144 1065916 483925 483944 GTTAAGCTTTACCTTATAGG 33 1145 1065931 484601 484620
TGATCATATGGGTGGTCGCT 46 1146 1065947 487134 487153 ATTGTCTCTCGTAACCAC 28 1147 1065963 488572 488591
GGGACTGGTGCCACACCATC 55 1148 1065979 489629 489648 CTAACCTGGATCTCAGATAG 75 1149 1065995 493844 493863
CTGGGCCCCCAATTCAGTAAT 98 1150 1066011 495399 495418 TTGGTAAAGGAGGGAATCGG ND 1151 1066027 498030 498049
TCCAGGATATATGTTAGTCC 46 1152 1066043 499291 499310 TAGCTACAATAAGTTGTAGC 94 1153 1066059 500494 500513
GAGGGCCATGTTAAAGGCCT 119 1154 1066075 500861 500880 TTTTGGTAGGTAACCTACGGG 41 1155 1066091 501335 501354
TGTAGCTCAGCTCAATGTGT 58 1156 1066107 501628 501647 GAACCTGCACTGGGTTGTCTC 58 1157 1066123 502162 502181
GTATTGCACTACATCTGT 61 1158 1066139 502579 502598 AGCCAATGCGCAAGAAAGCT 69 1159 1066155 502824 502843
CATAGGCATAAAGCCTCTTA 79 1160 1066171 503139 503158 TCCCAAGAAATGGTCTCGCA 86 1161 1066187 503480 503499
ATGTGGGTCTGCACCAAGTT 86 1162 1066203 504077 504096 GGTGAACCTCCTGTGACTGAT 52 1163 1066219 504441 504460
GTGGAGTAGGTATATTAGTC 32 1164 1066235 504825 504844 GTATTCCTGAAGTAGTCCTG 36 1165 1066251 505330 505349
AGAAATTGGGCGCCTCTGT 58 1166 1066267 505817 505836 GTGCTTAGTGAAGTGTGGGC 55 1167 1066283 506185 506204
CCGGCATGCATCAGCTCTGA 34 1168 1066299 506693 506712 TAAGAAGCTTGCCTTTTCGAT 96 1169 1066315 507250 507269
GCAGTGCTACTGTGCCCTTA 58 1170 1066331 507818 507837 TCCAGCCCTCAGTATATAGA 62 1171 1066347 508285 508304
CCGAAGTGGAAGTAGTCATG 63 1172 1066363 508790 508809 TGAATGCCACCGTGATTGCA 50 1173 1066379 509416 509435
CGTAGTATCATCACCATAAA 45 1174 1066395 510227 510246 GTTGTGTCTGAGGGATATCC 45 1175 1066411 510718 510737
GCCAGATCTTGACATACGAG 55 1176 1066427 511241 511260 GATTACCAAAAAGGGACCAG 48 1177 1066443 511766 511785
CTCCTACCAATAATGGAGTC 76 1178

TABLE-US-00019	TABLE	18 Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gappers with mixed PO/PS internucleoside linkages in vitro (37,000 cells/well)	SEQ ID	SEQ ID	UBE3A-Compound NO:	1 NO:	1 ATS	SEQ ID	Number					
Start	Site	Stop	Site	Sequence (5' to 3') (%)	UTC)	NO	617557	483977	483996	ATCATGTGTCATACCCAGGGT	38	172	749882	460115
460134	ACTTTATAGTGTGGATGGTA	32	586	750519	349103#	349122#	GTCATCACCTCTCTTCAGGA	17	181	1065265	430612	430631		
	CGGACACATCGGGACACGCG	90	1179	1065281	433463	433482	GCTCCTATCAAGCTTTTCCC	46	1180	1065297	441909	441928		
	TGAGGTGCATTTTACCAGCCC	78	1181	1065313	442830	442849	ATCTATGGAGTCACTCCCC	76	1182	1065329	443979	443998		
	GGTAGAGGTTAATCTATGTG	59	1183	1065345	445310	445329	CGTCGGTAAGGAAGCTACT	58	1184	1065361	446510	446529		
	TCATAAGGACAAGGTATGCC	73	1185	1065377	447310	447329	GGACCTCATTAGATCAGTCA	90	1186	1065393	448007	448026		
	GGCAGTTGAGTGGTGTGAGT	99	1187	1065409	448628	448647	GGATTGCATCACATGTGTCA	55	1188	1065425	449017	449036		
	TATAAGTGTCCATAAAGGTC	90	1189	1065441	449475	449494	GGCATTAAACATGTTGCTTGC	66	1190	1065457	450095	450114		
	GAATTGATAAGTGGTCTTCT	86	1191	1065473	452266	452285	GGGAAGTCTAGGTCTCACGC	ND	1192	1065489	453243	453262		
	CTACATATGACAGTGGAGTC	89	1193	1065505	454467	454486	TGAGTGGACTCCACACACCT	84	1194	1065521	455028	455047		
	GATATTGACATGGGATGGCT	60	1195	1065537	455360	455379	CGTGATAGTGCTATTGTGAG	60	1196	1065553	220317	220336		
	CATCGATCCAAACAAGCACC	74	1197	1065686	456705	1065569	457233	457252	GCTTAGCTACTCACCCCTGT	70	1198	1065585	457989	
	458008	CCCACTTGATCTATTATGAG	64	1199	1065601	458790	458809	CCCTAATATAGGGCAGATGA	59	1200	1065617	460108	460127	
	AGTGTGGATGGTATCCTGGT	44	1201	1065633	460736	460755	GGATTTGAGCCTGCTATGTC	15*	1202	1065648	461524	461543		
	CATTACAGCTAGACTAGTTGA	70	1203	1065664	462061	462080	TATGGCCCTCCCATCAGTGA	49	1204	1065678	463775	463794		
	TAGATTGGGCTTTAGAGGTC	33	1205	1065694	464788	464807	ACATCTGGTCTTCCTCGAGG	61	1206	1065709	465237	465256		
	GGTCATCTCGGGTATATAAA	17	1207	1065725	465591	465610	ACCGAAGGAGTTCCTTTAGC	40	1208	1065741	466233	466252		
	GGCAGTTCCAAGTATGGGCT	76	1209	1065757	466438	466457	GTGGAGCCAGTGTCAACAT	66	1210	1065773	467718	467737		
	TGGAATTCTCCTGTACGGG	52	1211	1065789	469377	469396	GTTGTATACATTGTGTGTC	42	1212	1065805	472944	472963		
	GTTTGTAGGGTCCCACCTTG	49	1213	1065821	474901	474920	TCTGCTATTGTTGGATATCG	33	1214	1065837	475988	476007		
	ATGGTTAGTTTAAGAAATCG	65	1215	1065853	478295	478314	GAGTCCTGGTTGATGTGGTG	65	1216	1065869	480141	480160		
	ACGGGCATGCTTTATAATTA	73	1217	1065885	481448	481467	AACCTTTGTGCATAATGGTC	82	1218	1065901	482908	482927		
	CACTCTATAGGTTCAAGCAG	37	1219	1065917	483964	483983	CCAGGGTAGGATTCATGGTC	63	1220	1065932	484673	484692		
	TGTTGTATGCAGGTTTCATCA	31	1221	1065948	487139	487158	GATACATTGTCCTCCTGGTA	35	1222	1065964	488637			

GCTTGACGCTGCTGCTGCTGCTG 95 1245 1066332 507900 507919 TCGATACCTGCTTTTGTGAC 69 1246 1066338 508290 508309
CATAACCGAAGTGGAAGTAG 71 1247 1066364 508901 508920 TACTTCATGACTGCCTAGTT 78 1248 1066380 509434 509453
AAGAAGGGCATATATCTACG 72 1249 1066396 510312 510331 CGGACAGTGTGCTGTTAGG 39 1250 1066412 510724 510743
GGTCAAGCCGAATCTTGACA 68 1251 1066428 511490 511509 TGACTGGGTAAGGCAGGATC 50 1252 1066444 511770 511789
AGTACTCCTACCAATAATGG 77 1253

TABLE-US-00020 TABLE 19 Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (37,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number
Start Site Stop Site Sequence (5' to 3') (% UTC) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 17 172 749882 460115
460134 ACTTTATAGTGTGGATGGTA 26 586 750519 349103# 349122# GTCATCACCTCTCTTCAGGA 12 181 1065271 432577 432596
GGCAGTGCATTCAACTGTAG 75 1254 1065287 440156 440175 CGTGGGTAGTCCAAGTAAC 49 1255 1065303 442546 442565
GCCCATTATGTAGTCTATAG 37 1256 1065319 443234 443253 GACGAAGGCAGTCACTCTGC 88 1257 1065335 444735 444754
GGGAGAACTACATCTCACCT 63 1258 1065351 445658 445677 CCCCATACCACGGTCTCTT 63 1259 1065367 446675 446694
AGCAGTCTTTATCTACAGGC 85 1260 1065383 447745 447764 TTGATGACCTGGTAAGGGAT 76 1261 1065399 448290 448309
GCGAATAAGATTCCATTGCC 83 1262 1065415 448725 448744 ACGATAACTGTAGCGCATTG 70 1263 1065431 449100 449119
GCAAGTATGCTAGTCACTCA 82 1264 1065447 449637 449656 CCATTAGATAGGCTATAGGC 77 1265 1065463 452066 452085
TCGCCGCTGATTTAAAGAT 97 1266 1065479 452320 452339 GGTTCAAAGCCGGTGCAGG 83 1267 1065495 453926 453945
GTTGTTCTTAAGTGCCCATG 61 1268 1065511 454740 454759 GCACTTTGCATGAGAATAGG 60 1269 1065527 455219 455238
TCCAGACAAAGTCTACTACCA 69 1270 1065543 455585 455604 TCTACTCCTTGCTACAAACG 84 1271 1065575 457378 457397
CAATAGTGTAGGTGACATGG 36 1273 1065591 458400 458419 GCATGTGGGCCTCTATTAAG 8 1274 1065607 459079 459098
GTATGGTATCTTGCAGTAGC 15 1275 1065623 460427 460446 ATAGCCCTTTCAGCACACTT 16 1276 1065639 461343 461362
GCTGGCAATATGTCAACCTT 36 1277 1065654 461605 461624 GTTACAACCGGCTATGTGA 30 1278 1065669 463151 463170
CATATGAAGCTATAGGTACC 22 1279 1065684 464383 464402 TGTATTTCACAACCTTCTCT 59 1280 1065700 465002 465021
TCTGCGAATAGAGGCCCTCT 49 1281 1065715 465301 465320 GACACGCTGACTATGATGTG 77 1282 1065731 465875 465894
CTTGCCCATGGATGGTTGTC 71 1283 1065747 466287 466306 GTGGCTTATAGGTGGGTTCC ND 1284 1065763 466945 466964
GGTTGTATGCCTTCTGCATT 44 1285 1065779 468014 468033 CCTTGCAGCACTTGTATAGT 44 1286 1065795 470319 470338
ACTATCCTGCATCCGAGGCA 23 1287 1065811 474227 474246 ATATAATTGAGGGCCACCAT 74 1288 1065827 475597 475616
CCTTTAGAGGGATTTGTGTA 34 1289 1065843 476097 476116 CTTAACATGTTTCATACGAGT 47 1290 1065859 478733 478752
GCTGATCACAATTACCATCC 24 1291 1065875 480518 480537 GATGTATCAGCAACAACAATT 81 1292 1065891 482167 482186
ACCAAGGAGTTACAAGTGTG 62 1293 1065907 483587 483606 GCGCAGGATGGTCAACCTTA 50 1294 1065922 484119 484138
TTTTTGCCACAGCCGCTTGG 193 1295 1065938 485811 485830 CGAATAATTGCATGCCAACT 68 1296 1065954 487606 487625
GGTTTAGTGTCAATGTAGC 23 1297 1065970 489324 489343 GCACACACCTGTATAGGAGA 43 1298 1065986 489780 489799
CCTGCCACTCCCGTGGCAAC 95 1299 1066002 493989 494008 TTATGGGTAAGAGGTCCGTC ND 1300 1066018 496826 496845
GGTATGGAGGCCATGTACGA 84 1301 1066034 498510 498529 GCGCCCGCAAGAGATTAC ND 1302 1066050 499912 499931
TTGCATAATAGGAGGTCTT 51 1303 1066066 500662 500681 GGGCTTACAACCTCTCAATT 74 1304 1066082 500899 500918
CCTAACTGGCCTTTGGGTCG 80 1305 1066098 501446 501465 GTCTGCACAGGTGGCATAGA 53 1306 1066114 501817 501836
GTCCAGAGGTCATATGGCCC 54 1307 1066130 502416 502435 GCATAAGTAAGGCATGGTCT 41 1308 1066146 502755 502774
CTTCGGGCAAATAGCCCCG 76 1309 1066162 503036 503055 GGTGCTACAGTCATACTATC 62 1310 1066178 503346 503365
GTGATCAGCTCAACACCCCC 65 1311 1066194 503878 503897 TAATAAAGGTCAGTTGTGC 39 1312 1066210 504270 504289
CACTGCCTAAGCAAGGAATC 86 1313 1066226 504530 504549 TGCCCCAAAGATTTAGGCCA 69 1314 1066242 505115 505134
TCAGACATGGTAACCTTCAC 40 1315 1066258 505660 505679 ATTGCAGGCCTAGCTTCAC 51 1316 1066274 506031 506050
GGTATTGAGTAGGACTTCTC 34 1317 1066290 506402 506421 CTTAAGGAGGCAACTCCTGA 51 1318 1066306 506781 506800
ATATACTCCAGGTTGTGGAG 66 1319 1066322 507526 507545 GAGCCAGGTTCTGTTCACG 33 1320 1066338 508110 508129
CCGTCTTTAGGAACTTAAAT 43 1321 1066354 508581 508600 AGCACAACCCCCGTGCTGCT 88 1322 1066370 509058 509077
TACCCAACATGCTTGCATCC 58 1323 1066386 510022 510041 ATACACGGAACCCACAGTAG 70 1324 1066402 510402 510421
CATGCTAGGAGCTGTGATTG 69 1325 1066418 510832 510851 GTTAGACCAGGCATAAGTAC 44 1326 1066434 511638 511657
TGCAGCTGAGTGCCATATA 56 1327 1066450 511890 511909 ACTACTTGGAGACCTTCACC 57 1328

TABLE-US-00021 TABLE 20 Reduction of UBE3A-ATS RNA by 7,000 nm 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (37,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number
Start Site Stop Site Sequence (5' to 3') (% UTC) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 35 172 749882 460115
460134 ACTTTATAGTGTGGATGGTA 26 586 749902 461442 461461 GCTCCTGACAAATTAGCACT 61 1329 750519 349103.sup.#
349122.sup.# GTCATCACCTCTCTTCAGGA 20 181 1065258 430419 430438 CGGATGCAGGGCACCCCTTG 96 1330 1065274 432770 432789
CATGATATAAGGAGAAACGC 63 1331 1065290 440729 440748 ATCTTGCGGCTGTCTCCAG 64 1332 1065306 442613 442632
TCCCATAGTTATAGGCTCA 60 1333 1065322 443476 443495 CTGCACTGTATAGTTGGCAT 48 1334 1065338 444879 444898
GCTGTGATGCTAGGATGCAT 84 1335 1065354 445874 445893 AGACATAGACCCAGACGGT 83 1336 1065370 446795 446814
CCCTTACCAGAATAGCATCC 81 1337 1065386 447803 447822 TGGTAGACCACTGATTCCCC 63 1338 1065402 448415 448434
AGCAAGGTCTTAGTGCCAA 60 1339 1065418 448742 448761 GTCTGTATATTTAGACCACG 83 1340 1065434 449159 449178
GTCAACATTATATCGATGCA 64 1341 1065450 449697 449716 GTATGGCTTATGCATGCTAT 86 1342 1065466 452137 452156
TGGGCAGGTCAACTTGGTAT 66 1343 1065482 453037 453056 AAAAGGTGATTAGGCGGCCG 88 1344 1065498 454156 454175
TGACCCTTATGGAGACTTAT 74 1345 1065514 454846 454865 TCCAGGGTAGAAGACTAGCA 63 1346 1065530 455286 455305
GCAATAGAACAGATGGCCAT 97 1347 1065546 455764 455783 GTACCAAAGTGGCTGCTCAC 89 1348 1065562 456866 456885
ACTGCCCTCTTCGAAGAGAT 71 1349 1065578 457715 457734 GATTATCTAGACATGCCAT 20 1350 1065594 458488 458507
CCTTAAGCAAGGAGTTACT 39 1351 1065610 459543 459562 ATCCTGGTATGACTGTCACT 45 1352 1065626 460538 460557
GGCTCTCATCTAAATAAGCC 75 1353 1065657 461674 461693 AACCTAAGGTGAAGTCTGT 65 1354 1065672 463245 463264
GGGTATTGCTGTCCAAATGG 20 1355 1065687 464409 464428 GTGTTATGGTTCCCATTCAG 32 1356 1065702 465012 465031
CCTGCTCAAATCTGCGAATA 58 1357 1065718 465382 465401 GACTTATTGAGGATGGTGTG 53 1358 1065734 465894 465913
TGAACCTCCACAAGGTACTC 52 1359 1065750 466341 466360 AGGCATTTGGAGCATTCGGG 16 1360 1065766 467050 467069
ACTTTCATCAGTTAGTCAGG 30 1361 1065782 468343 468362 CATTGACAAGCTATTGCAGC 44 1362 1065798 470982 471001
TTTTATGGCTTATAGCAGCG 50 1363 1065814 474563 474582 CACGCCCAAATGGAATCTA 38 1364 1065830 475827 475846
GTGAGTTATAGAGTGTTCCC 39 1365 1065846 476165 476184 TGGGCACTTAGGAGTTCCCTA 73 1366 1065862 479069 479088
GCTGTATCTGAGGTTTAGCA 67 1367 1065878 480818 480837 GCGGATGCCAGTGCCGAGA 40 1368 1065894 482519 482538
GCTATACCCCTAGGATCAGA 31 1369 1065910 483843 483862 TAGGCATTGAATGAGGGCCC 71 1370 1065925 484245 484264
GTATGGACTTGTCTTATGGC 57 1371 1065941 486351 486370 AATAGACTGCCATTATACAA 48 1372 1065957 487807 487826
CGCAATTGAACCTAAATTGG 56 1373 1065973 489517 489536 GGACCACCTAAGACCTCAAG 50 1374 1065989 489908 489927
GGCTAAAGTAATCTTATGGG 48 1375 1066005 494490 494509 GTGCAATAGCCTAAATGCCA 69 1376 1066021 497267 497286
GGATTAGGCAGCTTCACTAC 44 1377 1066037 499088 499107 GCAAATACAATGGTAATCGC 30 1378 1066053 500286 500305

CTGGTGATGACATTGGATATGA 57 1379 1066069 500694 500713 GCAAGAGGACTGTAAAGCCC 70 1380 1066085 500993 501012
ATCCGATAGTCAAACCTATGA 52 1381 1066101 501532 501551 AAACACCCTTCCAATGAGGG 75 1382 1066117 501986 502005
GTTCACTAGCATCTGCTACA 46 1383 1066133 502443 502462 GCTCTTTGTAGGCCCAAGGG 38 1384 1066149 502776 502795
GGTTCTCTATAAGGAATAGGC 73 1385 1066165 503096 503115 TAATAGGCCTTTCTACAGCT 77 1386 1066181 503374 503393
CCCCAGGGTCAAGGAGTGT 80 1387 1066197 503944 503963 CCACCAACCTTAAATAGTAG 70 1388 1066213 504390 504409
CCACAGATTGGCTTGGAATG 74 1389 1066229 504650 504669 GCCTTACGCTTGGCTGACAT 73 1390 1066245 505179 505198
GTCTCTGTGTACCGAGCTCA 63 1391 1066261 505724 505743 GTCTGGTGGCCAAGTGCTTC 70 1392 1066277 506116 506135
AGCCGAGCATTGGCTTCATA 73 1393 1066293 506584 506603 GGTGGTGGAATGCTGTCCAC 89 1394 1066309 506869 506888
CTCCACCAAATACTTAGCCC 67 1395 1066325 507649 507668 TAAATGTCAGGAGGTCCCCC 55 1396 1066341 508126 508145
CCTTGGGTATTCTTACCGT 55 1397 1066357 508680 508699 GGTAGTATAAGAATGGTTCC 54 1398 1066373 509104 509123
ATACACTTGGGCCCCAAATGG 78 1399 1066389 510039 510058 GCTAGCATTTGAGAGTTATA 43 1400 1066405 510647 510666
GGCTGTAAGGGATTAAGATG 79 1401 1066421 510853 510872 CTGTTACAGGGAGACAATCT 56 1402 1066437 511678 511697
ACCTGSCACAAATGGACTGC 103 1403

TABLE-US-00022 TABLE 21 Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (37,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number
Start Site Stop Site Sequence (5' to 3') (% UTC) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 42 172 749882 460115
460134 ACTTTATAGTGTGGATGGTA 33 586 750519 349103.sup.# 349122.sup.# GTCATCACCTCTCTTCAGGA 21 181 1065268 432234
432253 GGGCACCCCCGTCACCGTGC 82 1404 1065284 433765 433784 AGATTTTCCGACCTATGTCA 62 1405 1065300 442386 442405
ATCGCCATGGTCACTTGCCA 63 1406 1065316 443032 443051 GGCAAAGCAAGTTCGCCACA 91 1407 1065332 444610 444629
ATCATACAGGTCACTTGGCT 63 1408 1065348 445505 445524 ATTGGACTAGGTAACAGTGA 103 1409 1065364 446595 446614
CAGGCTATTTCTTAGTGAC 59 1410 1065380 447562 447581 CAAGGGTAACCTATAGCTGA 116 1411 1065396 448099 448118
GCATATGCACACTTGGATCA 89 1412 1065412 448676 448695 ACCGTGAAACAAATCAATCG 102 1413 1065428 449061 449080
ACGATAATCATGTGGCCTGA 85 1414 1065444 449593 449612 GACTAACTATACAGGTCTCT 107 1415 1065460 451288 451307
ACTACCTAGGCTCACTTGCT 83 1416 1065476 452297 452316 CCAGGATGCTAATGCTCCTA 69 1417 1065492 453781 453800
GCGCTTTCACCAAGAAATTT 78 1418 1065508 454575 454594 GCAAAGACTACACCGTGACA 109 1419 1065524 455153 455172
CCCCTGCATTACGCTTATAG 85 1420 1065540 455467 455486 GCTATCCAAGTGACACAGTA 77 1421 1065556 456757 456776
GGGCCGCACATCTGGACCTC 104 1422 1065572 457293 457312 TACAAAACGACCTAAAGACC 83 1423 1065588 458156 458175
CTGCAATCCCTTAAGACTTGA 37 1424 1065604 458952 458971 CCTCAAGACCTATAGGACCT 37 1425 1065620 460284 460303
ATCATGCTCCAGCATAAGG 60 1426 1065636 460849 460868 GATTTACCACACATACGAGGC 36 1427 1065651 461597 461616
CGGGCTATGTGAGATAATTC 15 1428 1065666 462686 462705 GGCTGACATGCCCTTTTAA 142 1429 1065681 463881 463900
GTTGTGATAGTCAACAATTG 89 1430 1065697 464991 465010 AGGCCCTCTTGTTCATTTG 63 1431 1065712 465246 465265
GATTTTATTTGGTCATCTCG 30 1432 1065728 465606 465625 GCCTTTCTATTTTCAGACCGA 35 1433 1065744 466263 466282
GGAGTCCATGAAGTAACTGG 62 1434 1065760 466538 466557 AGGTTGCATAAAGCCAGGCC 55 1435 1065776 467801 467820
CCCTCTTAGTGATTGGTGGT 63 1436 1065792 470049 470068 GGTTGGCAGTCTTCACCAGT 62 1437 1065808 474120 474139
GCTTGAATGGATACCAATTA 40 1438 1065824 475447 475466 CTCGGTCAAACCTAATAATAC 53 1439 1065840 476015 476034
CCAGTTTTTCGCGCTTACCT 32 1440 1065856 478440 478459 GTAGCTATAGGTGTCACATA 21 1441 1065872 480351 480370
GCCTCAAATATGTGATGCAC 45 1442 1065888 482108 482127 GAATGGCAATACATCCGTGT 80 1443 1065904 483359 483378
TCAATGACCCCATGTGATCA 63 1444 1065920 483971 483990 TGCATACCCAGGGTAGGATT 29 1445 1065935 485084 485103
TCTAATACTACAACGATGGA 84 1446 1065951 487204 487223 AACCTTCCTAAATCCCCGA 52 1447 1065967 489085 489104
GATATACCCAGTTAATCAGT 63 1448 1065983 489731 489750 ACCGGTTCCTCAATTTTCTCC 78 1449 1065999 493889 493908
TTGGCAGATGTAACCTATTC 52 1449 1066015 496809 496828 CGACCCTCATCACTTTTGA 73 1450 1066031 498209 498228
ACGGGACCTCAATACTCTAC 56 1451 1066047 499678 499697 ATCAAGGTAGCCCCAATAC 89 1452 1066063 500578 500597
CTACTGGCATCAGTCAAAAC 84 150 1066079 500894 500913 CTGGCCTTTGGGTCCGATAT 86 1453 1066095 501390 501409
CCCCAGATAATGCATAGATC 98 1454 1066111 501763 501782 GGGCTAAGAGTCACCTGTAT 59 1455 1066127 502360 502379
CCCTTGGCCACCTGACTTCG 77 1456 1066143 502629 502648 TTGGCTCAGTGTTCACTTCG 52 1457 1066159 502939 502958
GTAACCGACCCCAAGGCACT 62 1458 1066175 503325 503344 GAGATCCATGAGGTATATAC 55 1459 1066191 503815 503834
GCCCAGTGCCCTATAGGTCA 67 1460 1066207 504127 504146 GCTATTTTCATTAAGTCACCC 39 1461 1066223 504502 504521
CCATGGAATGGCTGTCATGC 72 1462 1066239 504974 504993 TTATCTTCTTAGGGTCGACT 38 1463 1066255 505635 505654
AGCTATCCGCTTCCCAAGGG 47 1464 1066271 505906 505925 CCCTGAGATGCTAGTTGGGC 85 1465 1066287 506256 506275
GTATGTCCTTGAGGTGAGC 88 1466 1066303 506744 506763 GCCTTTCATTTTGGGAGTGC 48 1467 1066319 507364 507383
GGCATGGAGATCCAACCTGT 96 1468 1066335 507919 507938 GGCATGTCAGGATAGCCATT 95 1469 1066351 508470 508489
CGTTATCCTAAGAAGTGACT 58 1470 1066367 509003 509022 TCAACATGTGTTAACGGAAC 63 1471 1066383 509910 509929
GTACCTGCAGTGTCATAGC 52 1472 1066399 510387 510406 GATTGATTACGTGCTCATGC 49 1473 1066415 510747 510766
ACTCAGATAGGTAGGTGAT 43 1474 1066431 511593 511612 GTTGAAGCTGCCAGCTTAG 96 1475 1066447 511795 511814
ACTGGCCTGTGGCAGTTAAC 80 1476

TABLE-US-00023 TABLE 22 Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gapmers with with mixed PO/PS internucleoside linkages in vitro (37,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID
Number Start Site Stop Site Sequence (5' to 3') (%UTC) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 28 172 749882
460115 460134 ACTTTATAGTGTGGATGGTA 24 586 750519 349103.sup.# 349122.sup.# GTCATCACCTCTCTTCAGGA 17 181 1065261
430489 430508 GCGCTGGCTCACCGCATGCTC 65 1477 1065277 432989 433008 CCTAGGATTTTCTTCTCCCG 79 1478 1065293 441703
441722 GGGATGCAATGCCCTCCAAC 97 1479 1065309 442707 442726 TAAATGCCCTGGGTGCAGC 71 1480 1065325 443778 443797
GCCCTTTCAGGACCAACTGG 107 1481 1065341 445116 445135 GTTGGCAACAAAGTTACCC 92 1482 1065357 446183 446202
GGGACCCCAATTTACTACGC 112 1483 1065373 446912 446931 CTTCTTGCAAGGATTGGCAC 69 1484 1065389 447847 447866
ACTGAGATGTGTGATACTCC 70 1485 1065405 448471 448490 CCACTCAATAGAATAGTGCC 123 1486 1065421 448838 448857
CCCAATTGCAAGGACCCTTA 80 1487 1065437 449334 449353 GAACATGTAGCCATAATGCC 65 1488 1065453 449867 449886
GAGTGGCATCCTTAAATCCT 66 1489 1065469 452204 452223 TAATTGGTCAAGTAAACAGC 73 1490 1065485 453199 453218
TCCACCGTTACTGATTATCT 51 1491 1065501 454230 454249 CAAGTCCCACCATGTTAATC 87 1492 1065517 454913 454932
GTTAGGATCTATTTGACAGC 67 1493 1065533 455318 455337 AGGCCTGCTCCGAATGTGTT 107 1494 1065549 456627 456646
GTGACTGTTTACTATAATGCA 87 1495 1065565 456917 456936 TTGGTGCACTAGGCACACTA 110 1496 1065581 457899 457918
TGACTAGTTCGATAAGTTTTT 48 1497 1065597 458561 458580 TTGCTGACTACTCAAAAGGTG 31 1498 1065613 459646 459665
CTGTATAGTTTACCCAGGCA 24 1499 1065629 460641 460660 CAGAGTGCATTGGCAACTCG 50 1500 1065644 461457 461476
GATCTTGGAAGGCAGCTCC 29 1501 1065660 461891 461910 CAATTTGGTCCCATTTAGTAGT 84 1502 1065675 463503 463522
TCTATCAAGGCTGTTATTGG 40 1503 1065690 464528 464547 GCTCAGTTAGGTTAGTGAC 23 1504 1065705 465162 465181
TCTAGGGCTCCAGTTTATGT 78 1505 1065721 465442 465461 CTTTGGTTATGTACATTGCC 73 1506 1065737 466098 466117
GCCTGTATGTCTTGAGAAAC 78 1507 1065753 466388 466407 CTCGCTTTTCACAGGAGCGC 36 1508 1065769 467300 467319

TCACAGAGAGAGTATTGTTGTT 43 1509 1065785 468922 468941 GTAAGTATAGATGCCCTCTCC 33 1510 1065801 472475 472494
GCAAACTTTAGGAGTGTGTT 60 1511 1065817 474799 474818 GGTATTGACACCTCCAACTG 18 1512 1065833 475854 475873
ATATAAGGGTAATACGGACC 39 1513 1065849 476362 476381 GCCCCCTGCCGTGTGAAAGA 72 1514 1065865 479579 479598
CTTCAAGACTAAGGTAGGGA 38 1515 1065881 480923 480942 GTATGCGAAGCGAACGAAGC 75 1516 1065897 482786 482805
CCTAAAAAACCCGTGTACAC 62 1517 1065913 483907 483926 GTGGCCCTTCAGTCAGTACA 57 1518 1065928 484317 484336
GCCCCAGCTCATGCTGGTTA 96 1519 1065944 486448 486467 ATACTTCACCTAATAGCAC 86 1520 1065960 488086 488105
GGAGTTCTTTAGGTTGGAAC 62 1521 1065976 489542 489561 GCAACTATGGGTGGAGACAT 53 1522 1065992 493191 493210
TCGGGCCAGGTCCAGGCGCA 52 1523 1066008 495127 495146 GCTGTCATATGGGAACTACG 57 1524 1066024 497743 497762
CAAACCTACGCCAATAAAGA 86 1525 1066040 499138 499157 TTTATCGCTTAAAGTAGCCT 83 1526 1066056 500407 500426
CGTATATCGAATACCCTCAA 33 1527 1066072 500818 500837 GTAAGAGTTAGCTATTCCCC 30 1528 1066088 501153 501172
CAGTAAAGAGCCACCTAAGG 71 1529 1066104 501574 501593 GGCAAGGCTAAGGAGTGCTC 54 1530 1066120 502129 502148
GCTACCCATTAGACCCATGG 34 1531 1066136 502502 502521 GGAGTCCCTGTGTCAATTGGA 37 1532 1066152 502803 502822
GGATGTAGCCCCATAACCTT 77 1533 1066168 503118 503137 CCAGTTAATCTCTGACATGG 57 1534 1066184 503422 503441
AGGTGATGACACCCCTACCA 57 1535 1066200 503960 503979 GGGTGTCTCTTTGACACCAC 76 1536 1066216 504419 504438
AGGCAGTCAGGGTAATGCTA 76 1537 1066232 504667 504686 CCTTCTGTCCCAACGGAGCC 111 1538 1066248 505213 505232
CTCTCGCTGAGGACACATCA 64 1539 1066264 505791 505810 ACCACAGGGTACTATTCTGG 64 1540 1066280 506126 506145
CTGCTCTGTTAGCCGAGCAT 75 1541 1066296 506634 506653 GCTTGTATGCCCCACTGGAG 59 1542 1066312 507213 507232
GCTTGCCATGTTTTATAGAC 80 1543 1066328 507762 507781 CTCAGGATCGCTGGCCATTT 62 1544 1066344 508223 508242
GCCTAAAGGTAGTTCTCCCT 57 1545 1066360 508731 508750 CAGGAGGGTGTCAACCAGAC 75 1546 1066376 509178 509197
GGCAGATAACCTCCAAGTGC 95 1547 1066392 510128 510147 CACATAGACCATAGCTGAAC 56 1548 1066408 510668 510687
GTCTAGATGTCTGAGTGTC 87 1549 1066424 511065 511084 GACTCGGGACATTTAGGATG 72 1550 1066440 511730 511749
ATAGGCCACGCTGCTACATG 48 1551

TABLE-US-00024 TABLE 23 Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gapmers with with mixed PO/PS internucleoside linkages in vitro (37,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID
Number Start Site Stop Site Sequence (5' to 3') (% UTC) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 39 172 749882
460115 460134 ACTTTATAGTGTGGATGGTA 16 586 750519 349103# 349122# GTCATCACCTCTCTTCAGGA 14 181 1065260 430471
430490 TCATCCTCTATGGACCCACG 83 1552 1065276 432973 432992 CCCGACTTACATTACCTACC 69 1553 1065292 441621 441640
ATGATCTGCTGTTTCAGTGC 45 1554 1065308 442687 442706 AGGCCAAGTGGTGCTCTTCC 58 1555 1065324 443691 443710
ATTACAGATACTAGTTTACCC 36 1556 1065340 445011 445030 TCTAGGTAGGCTGAGCCTCA 89 1557 1065356 445972 445991
TTGTCTTGGAACATACGGAT 81 1558 1065372 446901 446920 TAGTGGCACTATTTTGAGCC 113 1559 1065388 447835 447854
GATACTCCAAATATGACCCG 80 1560 1065404 448465 448484 AATAGAATAGTGCCAGTAGG 70 1561 1065420 448833 448852
TTGCAAGGACCCCTTAAGTCA 87 1562 1065436 449328 449347 GTAGCCATAATGCCATAGTC 61 1563 1065452 449862 449881
GCATCCTTAAATCCTGGTTG 97 1564 1065468 452186 452205 GCAATATCCATAGTGATCTG 84 1565 1065484 453194 453213
CGTTACTGATTATCTTATCC 72 1566 1065500 454225 454244 CCCACCATGTTAATCAAGGC 69 1567 1065516 454863 454882
CAGGTGTATATTCCCTATCC 70 1568 1065532 455309 455328 CCGAATGTGTTATTTATCCT 58 1569 1065548 456621 456640
GGTTACTAATGCATCATCA 71 1570 1065564 456909 456928 CTAGGCACACTAAAGTGAC 84 1571 1065580 457802 457821
GATGCAATCAGACGAAGTAC 68 1572 1065596 458560 458579 CTCGAGAAGTCAAAAGGTGA 51 1573 1065612 459639 459658
GTTTACCCAGGCAATGGCTC 70 1574 1065628 460629 460648 GCAACTCGATGTTCTTGTCTT 79 1575 1065643 461453 461472
TTGGTAAGGCAGCTCCTGAC 58 1576 1065659 461886 461905 TGGTCCCATTGTAGTTGTGT 50 1577 1065674 463502 463521
CTATCAAGGCTGTTATTGGT 29 1578 1065689 464477 464496 GCTTCTAACTAACATGCCTC 61 1579 1065704 465158 465177
GGGCTCCAGTTTATGTATCC 79 1580 1065720 465418 465437 TAGCCTGCATGGTTTACAGT 72 1581 1065736 466030 466049
GGGCTCTTGTTACTGAGCTG 79 1582 1065752 466382 466401 TTTCACAGGAGCGCACTTGG 55 1583 1065768 467295 467314
GAGTAGTCTATTGGTGTTCC 27 1584 1065784 468404 468423 CCCTACCCCTGTCATGCTATG 75 1585 1065800 471928 471947
CCATTGAGTATATTACCTCC 45 1586 1065816 474789 474808 CCTCCAAGTGAATCATTTGA 65 1587 1065832 475846 475865
GTAATACGGACCTCATACAG 61 1588 1065848 476316 476335 ACTACTTGTCCTTGGGCTT 45 1589 1065864 479256 479275
CTTGCTTGATGGTCTGATG 63 1590 1065880 480830 480849 AAAAGTAGGTTGCGGATGC 61 1591 1065896 482778 482797
ACCCGTGTACACAGTGGGAT 60 1592 1065912 483896 483915 GTCAGTACACAAGCAGGTAG 68 1593 1065927 484309 484328
TCATGCTGGTTACTAGGGCC 42 1594 1065943 486378 486397 GTTAGGCCACAAGACTTAAT 51 1595 1065959 488046 488065
CTCAAACCCGTATAAAGATG 58 1596 1065975 489527 489546 GACATACAATGGACCACCTA 70 1597 1065991 489961 489980
GGAAGTTCCAACCCCTACTC 64 1598 1066007 494921 494940 GAATCCACTACATGGGATTA 110 1599 1066023 497674 497693
GACTAAGCCCGAAAGTTAGC 79 1600 1066039 499133 499152 CGCTTAAAGTAGCCTAAGGA 41 1601 1066055 500400 500419
CGAATACCCCTCAACTTACC 59 1602 1066071 500797 500816 CAACCTTGATAGCAGCTTAT 38 1603 1066087 501007 501026
GTGAAACTAAGCACATCCGA 77 1604 1066103 501569 501588 GGCTAAGGAGTGCTCTTTGT 66 1605 1066119 502125 502144
CCATTAGACCCATGGCTCA 21 1606 1066135 502465 502484 TACAACAGTCCCAGGCTAGG 51 1607 1066151 502799 502818
GTAGCCCATCAACCTGGGA 72 1608 1066167 503107 503126 CTGACATGGACTAATAGGCC 47 1609 1066183 503415 503434
GACACCCCTACCATGGCTAC 54 1610 1066199 503956 503975 GTCTCTTTGACACCACCAAC 62 1611 1066215 504408 504427
GTAATGCTATCTTCCATGCC 47 1612 1066231 504661 504680 GTCCCAACGGAGCCTTACGC 62 1613 1066247 505200 505219
CACATCAGCTGTAGCAGTC 64 1614 1066263 505749 505768 ACTCTGTAGGTTGACAGGAC 59 1615 1066279 506121 506140
CTGTTAGCCGAGCATTGGCT 88 1616 1066295 506628 506647 ATGCCCCACTGGAGAAGTCT 77 1617 1066311 506997 507016
CGTCTTCCCGCACCATGCAT 40 1618 1066327 507699 507718 CGTGTCACCTTCAGGTCAGC 59 1619 1066343 508216 508235
GGTAGTTCTACCTTCTGTCA 73 1620 1066359 508726 508745 GGGTGTCAACACAGACTTCCA ND 1621 1066375 509128 509147
GCTCATCACTCACTGGTGGT 35 1622 1066391 510121 510140 ACCATAGCTGAACCTGTGGC 62 1623 1066407 510659 510678
GTCTGAGTGTACAGGCTGTAA 68 1624 1066423 510905 510924 GTGTGCCATAGGTTTCAGGC 18 1625 1066439 511725 511744
CCACGCTGGTCACTGAAAGA 70 1626

TABLE-US-00025 TABLE 24 Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (37,000 cells/well) SEQ SEQ ID ID NO: 1 NO: 1 UBE3A- SEQ Compound Start Stop Sequence ATS
ID Number Site Site (5' to 3') (% UTC)) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 49 172 749882 460115 460134
ACTTTATAGTGTGGATGGTA 25 586 750519 349103# 349122# GTCATCACCTCTCTTCAGGA 10 181 1065266 432182 432201
CCGCTCAATGCAGGACACCA 9 1627 1065282 433495 433514 TGCTCCATAGTAAGATGGT 82 1628 1065298 442149 442168
TAGGCATGCGCTTTGTTGCG 87 1629 1065314 442877 442896 GTCGAAGACAATAAACATC 59 1630 1065330 444418 444437
ATCAGGGCACATGGAGTTGT 21 1631 1065346 445359 445378 CCCTTAGGTGGGAGTCTTCC 128 1632 1065362 446536 446555
GGCTCTAGACCATTGACTCA 72 1633 1065378 447498 447517 GTAGTGCCTCAATGATCCAC 62 1634 1065394 448042 448061
GGTAAAGTGGACTGGATGCC 66 1635 1065410 448635 448654 CTCATGAGGATTGCATCACA 71 1636 1065426 449041 449060
AGGAACGGCCCTGACTGTCA 110 1637 1065442 449492 449511 CCATATCAATAACCAATTGGC 111 1638 1065458 450273 450292
GGAACCTCAGCCAGTGAATA 96 1639 1065474 452270 452289 ATGAGGGAAGTCTAGGTCTC 80 1640 1065490 453360 453379

ACATAGGACTACTACTAGCT 101 1641 1065506 454514 454533 TATCAGTCCATAGCTACACAC 104 1642 1065522 455119 455138
GGTGGATGTCTGACTTGACT 61 1643 1065538 455426 455445 GCTAAAGTTGTGGTCTACCA 83 1644 1065554 220324 220343
TTCTCATCATCGATCCAAAC 60 1645 456693 456712 1065570 457238 457257 TAATGGCTTAGCTACTCACC 90 1646 1065586 457994
458013 GAGAACCCACTTGATCTATT 24 1647 1065602 458938 458957 GGACCTCAGGAGATTGTACA 35 1648 1065618 460111 460130
TATAGTGTGGATGGTATCCT 31 1649 1065634 460744 460763 GTACACAGGGATTGAGCCT 12* 1650 1065649 461528 461547
GGAACATCTCAGCTAGACTAG 40 1651 1065665 462567 462586 TAAATTGGTCTGGTTACAAG 56 1652 1065679 463779 463798
CCTATAGATTGGGCTTTAGA 48 1653 1065695 464905 464924 GTCTATATTTGGTAAGACAC 69 1654 1065710 465240 465259
ATTGGTCATCTCGGGTATAT 18 1655 1065726 465596 465615 TTCAGACCGAAGGAGTTCCT 60 1656 1065742 466244 466263
GTAGGATCTATGGCAGTTCC 59 1657 1065758 466448 466467 GTTCTACAGAGTGGAGCCAG 65 1658 1065774 467791 467810
GATTGGTGGTTTATTCATCG 47 1659 1065790 469893 469912 TACCCATTGCAAGTTAACTA 73 1660 1065806 473260 473279
TCCTAATAGTTGGAGGTGGT ND 1661 1065822 474959 474978 CAGCCTAATATGTGTCATCC 40 1662 1065838 476003 476022
CGTTACCTCAATTCTATGGT 49 1663 1065854 478299 478318 ACTGGAGTCTTGTTGATGT 67 1664 1065870 480146 480165
ACAAACGGGCATGCTTTAT 63 1665 1065886 481476 481495 GCGGATAAATTGGATTATC 37 1666 1065902 483108 483127
AGGCAATGGACCTTAGTACAC 21 1667 1065918 483965 483984 CCCAGGGTAGGATTCATGGT 45 1668 1065933 484683 484702
CAAGTGATGATGTTGTATGC 87 1669 1065949 487192 487211 ATCCCCGAATGGGAGAGATT 107 1670 1065965 489067 489086
GTGAACGAATACTGTGGCAT 59 1671 1065981 489671 489690 CCAACCTGTCATGGGACTGC 45 1672 1065997 493859 493878
TAGGCCTAATCTATGCTGGG 44 1673 1066013 496023 496042 TCTCAAAGACATTCCGTAC 137 1674 1066029 498183 498202
CCATGGGCAGATTGATCACC 102 1675 1066045 499515 499534 TGCCTAAGGGAGTTTGTCAC 73 1676 1066061 500527 500546
ATGTTTAGAGGTCAAGCCCT 91 1677 1066077 500886 500905 TGGGTCGGATATAGCTTTTA 58 1678 1066093 501347 501366
GACATCTGCTTATGTAGCTC 45 1679 1066109 501708 501727 AGTTACTTTGGGACATCCCA 68 1680 1066125 502209 502228
GTTTGAACATCATGTTAGC 49 1681 1066141 502595 502614 TCCAAGTATGCATCAAGCC 43 1682 1066157 502904 502923
GCCTATATCAACCAGCTAC 73 1683 1066173 503217 503236 CGTTGGAAACCTCAAAGGCA 69 1684 1066189 503506 503525
CCTTGGTTGTGGTGAACCC 72 1685 1066205 504087 504106 TATCTTAAGTGGTGAAGTCC 73 1686 1066221 504477 504496
GGATGGTTGACCTCAAAATT 30 1687 1066237 504952 504971 TGGTTACCCATACAGTATAT 42 1688 1066253 505549 505568
GCTTTTCAGTCAGGTACAGG 29 1689 1066269 505828 505847 GACCTGTGAGGGTGCTTAGT 81 1690 1066285 506235 506254
GTTTGACCAGCTCCTTGTTG 76 1691 1066301 506734 506753 TTGGGAGTGCTCCACACTTC 72 1692 1066317 507331 507350
GATGCCCTGGTCTAGCTTC 60 1693 1066333 507909 507928 GATAGCCATTCGATACCTGC 56 1694 1066349 508297 508316
GCTTCTACATAACCGAAGTG 116 1695 1066365 508933 508952 CTAGTGCCTATTACAATCTG 61 1696 1066381 509882 509901
CATGTGCTTTGTGGACCCAT 47 1697 1066397 510318 510337 GCTGGACGGACAGTGTGCT 61 1698 1066413 510733 510752
GGTGTAAGGTCGAAGCCGA 103 1699 1066429 511561 511580 GTGGTTTCCAGCAGGGTGTA 20 1700 1066445 511777 511796
ACTAAAGAGTACTCCTACCA 110 1701
TABLE-US-00026 TABLE 25 Reduction of UBE3A-ATS RNA by 7,000 5-10-5 MOE gapmers with mixed PO/PS
internucleoside linkages in vitro (37,000 cells/well) SEQ ID SEQ ID NO: 1 NO: 1 UBE3A- SEQ Compound Start Stop Sequence ATS
ID Number Site Site (5' to 3') (% UTC)) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 34 172 749882 460115 460134
ACTTTATAGTGTGGATGGTA 30 586 749921 462626 462645 GTAATGATTTGCCCTCCTAC 32 1702 750519 349103# 349122#
GTCATCACCTCTCTTCAGGA 17 181 1065267 432208 432227 CCAGCGGGACCTGGACAAGC 62 1703 1065283 433699 433718
GGTGTAGTGGCAAGAAGTTC 59 1704 1065299 442231 442250 CCCCTAGCTTTCCAATGGG 82 1705 1065315 442923 442942
TGCTCCACATAAATCCTATTG 48 1706 1065331 444553 444572 ACATCCCATTAAATCATCGCT 49 1707 1065347 445388 445407
TCAGATTACCTGAAGGTGTA 104 1708 1065363 446582 446601 AGTGACAGATACGGTCCGT 95 1709 1065379 447508 447527
TCAGTTGTGAGTAGTGCCCTC 75 1710 1065395 448074 448093 GCTTACCAGAGATGGTTTCC 111 1711 1065411 448658 448677
CGTAGGGCAAAAGTAAGGAT 71 1712 1065427 449050 449069 GTGGCCTGAAGGAACGGCCC 92 1713 1065443 449532 449551
GACCTTCAATTGATCAGTCT 112 1714 1065459 450791 450810 CCTGTCTGGTACTCCCATAT 69 1715 1065475 452275 452294
TCAATATGAGGGGAAGTCTAG 73 1716 1065491 453366 453385 TTAAGTACATAGGCACTCTA 76 1717 1065507 454568 454587
CTACACCGTGACAAAAGAGC 76 1718 1065523 455125 455144 AGCACAGGTGGATGTCTGAC 82 1719 1065539 455450 455469
GTAGTTGTATCTTGAATGGG 65 1720 1065555 220380 220399 CATCTGGACCTCAGATTGAC 64 1721 456749 456768 1065571 457288
457307 AATGACCTAAGACCTAGCA 36 1722 1065587 458044 458063 CTTGAGAACTACTCATGAC 83 1723 1065603 458946 458965
GACCTATAGGACCTCAGGAG 33 1724 1065619 460118 460137 CTAAGTATATAGTGTGGATG 18 1725 1065635 460843 460862
CCACACATATAGGCTAGCCA 9 1726 1065650 461573 461592 GGTCATCATATTGAACCAAT 44 1727 1065680 463826 463845
GATTTCTTACTGAGTGAGC 29 1728 1065696 464909 464928 GGTAGTCTATATTTGGTAAG 17 1729 1065711 465242 465261
TTATTGGTCATCTCGGGTAT 47 1730 1065727 465600 465619 CTATTTCAGACCGAAGGAGT 32 1731 1065743 466251 466270
GTAAGTGGTAGGATCTATGG 49 1732 1065759 466496 466515 TTTTATAGGGTAGTGTCTCTGA 92 1733 1065775 467797 467816
CTTAGTGATTGGTGGTTTAT 54 1734 1065791 469899 469918 GGGATTTACCCATTGCAAGT ND 1735 1065807 473264 473283
GCCCTCTAATAGTTGGAGG 44 1736 1065823 475306 475325 GGTTAAGTCTGCTCTTTTAC 25 1737 1065839 476008 476027
TCGCCCCGTACCTCAATTCT 60 1738 1065855 478307 478326 GATGACCTGAGTCCCTG 45 1739 1065871 480150 480169
CAATACAAAACGGGCATGCT 94 1740 1065887 481747 481766 GATCATTCCCTGTGGTAAAG 68 1741 1065903 483112 483131
GCTTAGGCAATGGACTTAGT 11 1742 1065919 483969 483988 CATACCCAGGGTAGGATTCA 43 1743 1065934 484722 484741
ATAACACTAACGATGAAGTC 37 1744 1065950 487198 487217 CCTAAATCCCCGAATGGGA 98 1745 1065966 489079 489098
CCCAGTTAATCAGTGAACGA 77 1746 1065982 489676 489695 ATGTTCCAACCTGTCATGGG 71 1747 1065998 493867 493886
GATGAGATTAGGCCTAATCT 43 1748 1066014 496564 496583 CCATCTACTATTAATGAGCT 58 1749 1066030 498186 498205
TCACCATGGGCAGATTGATC 109 1750 1066046 499672 499691 GTAGCCCCAATACAGATTTC 26 72 1066062 500564 500583
CAAAACATGTTCTGACCTCG 93 1751 1066078 500892 500911 GGCCTTTGGGTCGGGATATAG 93 1752 1066094 501356 501375
CGTCAAACTGACATCTGCTT 38 1753 1066110 501733 501752 AGTTAAGACCTCAAGTTG 58 1754 1066126 502355 502374
GGCCACCTGACTTCGGCCCA 115 1755 1066142 502599 502618 GACTTCCAAGTATGCATCA 42 1756 1066158 502919 502938
GACCTTCTGTGAAAGCCTA 61 1757 1066174 503279 503298 GACTGCATCTCAATCCTATG 62 1758 1066190 503636 503655
GCCCATATGCTTGAACAATT 40 1759 1066206 504088 504107 GTATCTTAAGTGGTGAAGTTC 55 1760 1066222 504484 504503
GCAGTTTGGATGGTTGACCT 43 1761 1066238 504963 504982 GGGTCGACTGATGGTTACCC 80 1762 1066254 505594 505613
ATAGGAGCTGAATAGTAGGG 37 1763 1066270 505833 505852 TCAAAGACCTGTGAGGGTGC 78 1764 1066286 506243 506262
GGTGAGCTGTTTGACAGCT 49 1765 1066302 506739 506758 TCATTTTGGGAGTGCTCCAC 53 1766 1066318 507359 507378
GGAGATCCAACCTGTGTGGA 79 1767 1066334 507915 507934 CTCGAGTATGCCATTTCAT 69 1768 1066350 508354 508373
CATTATTCAATTAAGGGTGG 15 1769 1066366 508999 509018 CATGTGTTAAACGGAAGTGCAG 85 1770 1066382 509901 509920
GTGCATAGAGCAGACTGTAC 103 1771 1066398 510328 510347 GGTTTTGAGAGCTGGACGGA 57 1772 1066414 510740 510759
ATAGGTAGGTGATGAAGGTC 75 1773 1066430 511586 511605 GCTGCCAGCTTAGAGAATCT 95 1774 1066446 511781 511800
GTTAACTAAAGAGTACTCCT 92 1775
TABLE-US-00027 TABLE 26 Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gapmers with mixed PO/PS
internucleoside linkages in vitro (37,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number

Start Site Stop Site Sequence (5' to 3') (% UTC)) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 35 172 749882 460115
460134 ACTTTATAGTGTGGATGGTA 42 586 749960 465003 465022 ATCTGCGAATAGAGGCCCTC 56 33 750519 349103* 349122*
GTCATCACCTCTCTTCAGGA 16 181 1065272 432587 432606 GTAGGAGATAGGCACTGTCAT 25 1776 1065288 440174 440193
AATTATACTGTCCTACTTTCG 65 1777 1065304 442560 442579 GTACCCAGAAATAGCCCAT 45 1778 1065320 443271 443290
CATCATAGCTAGCTAGTATCT 114 1779 1065336 444826 444845 GGGTGCTGTCAACTGTCCCA 45 1780 1065352 445690 445709
GGCAAAGTACATACTGATCC 54 1781 1065368 446725 446744 CCTGTGCGGTAGCACTTGCC 80 1782 1065384 447772 447791
GGGCCAAGCACATAGGTATC 74 1783 1065400 448306 448325 CGTGCAAAACAAGTTTCGCA 91 1784 1065416 448729 448748
GACCACGATAACTGTAGCGC 62 1785 1065432 449105 449124 TCCTGGCAAGTATGCTAGTC 73 1786 1065448 449643 449662
ACTGATCCATTAGATAGGCT 53 1787 1065464 452074 452093 AGAACACTTCGCCGCTGATT 76 1788 1065480 452327 452346
AACATAAGGTTCAAAAGCCG 86 1789 1065496 454063 454082 CATAGATGATCCCATAATAG 77 1790 1065512 454746 454765
CCGAAAGCACTTTGCATGAG 70 1791 1065528 455227 455246 ATCCTTGATCCAGACAAAGTC 76 1792 1065544 455606 455625
ATCATACAGGCATCTCAGCC 68 1793 1065560 220447 220466 TGTCACTGAGGGATCCCCAA 83 1794 456816 456835 1065576 457383
457402 GGATTCAATAGTGTAGGTGA 15 1795 1065592 458407 458426 GTACAATGCATGTGGGCCTC ND 1796 1065608 459102 459121
ATTGGCATATTCGATTATA 24 1797 1065624 460467 460486 CGTATTGGACCTTTCAATGA 28 1798 1065640 461349 461368
ATTGAAGCTGGCAATATGTC 67 1799 1065655 461608 461627 GCATGTACAAACGGGCTATG 33 1800 1065670 463188 463207
GGAATCTTGTAGAGGATTGG 51 1801 1065685 464393 464412 TCAGCACTTAGGTAGTTCAC 27 1802 1065716 465309 465328
TGTCTTTTGACACGCTGACT 60 1803 1065732 465880 465899 GTACTCTTGCCCATGGATGG 42 1804 1065748 466292 466311
CCCCTGTGGCTTATAGGTGG 93 1805 1065764 466995 467014 GGATGAACCTGCTTCACACA 45 1806 1065780 468051 468070
CAGATTTGCCAGGTAAAGCG 58 1807 1065796 470320 470339 AACTATCCTGCATCCGAGGC 33 1808 1065812 474393 474412
GGTCAACCAATTTGCTATTC 23 1809 1065828 475722 475741 GGTCTTGGGATTATAGTTT 32 1810 1065844 476155 476174
GGAGTTCCTAACTCTCAATC 78 1811 1065860 478826 478845 TCTACTGAGGAACCCATCAC 86 1812 1065876 480715 480734
CGTTGCATAACATGTGTATT 46 1813 1065892 482259 482278 TACACTGCAACAAGCCATC 85 1814 1065908 483816 483835
GAGCAAATCGGCAAAGGCAT 43 1815 1065923 484156 484175 CCAAGTGGCTTGTGGTGAAA 87 1816 1065939 485817 485836
GAACTACGAATAATTGCATG 49 1817 1065955 487610 487629 CAATGGTTTGTAGTTCATATG 27 1818 1065971 489454 489473
AGTAGCTGTTTAGGCTGGAC 58 1819 1065987 489784 489803 TTGCCCTGCCACTCCCGTGG 76 1820 1066003 494419 494438
GGGATATATCACTTGAGCT ND 1821 1066019 496833 496852 CAGATTAGGTATGGAGGCCA 49 1822 1066035 498902 498921
ACGGATTAGATTCTCCAACA 36 1823 1066051 499936 499955 AGGACACATCAAGCTAGCTA 55 1824 1066067 500680 500699
AAGCCCTTTGCTTTTTCGGG 81 1825 1066083 500908 500927 GTGTTTTGACCTAACTGGCC 65 1826 1066099 501472 501491
CGTGGGTATGTTTCTCTCT 75 1827 1066115 501905 501924 TCCAACGAGTGATAATTCAC 81 1828 1066131 502423 502442
AATTGATGCATAAGTAAGGC 51 1829 1066147 502762 502781 ATAGGCACTCGGGCAAATA 67 1830 1066163 503043 503062
GGATTTAGGTGCTACAGTCA 38 1831 1066179 503356 503375 GTTAGCCCAGGTGATCAGCT 57 1832 1066195 503886 503905
ATTGCAACTAATAAAGGGTC 79 1833 1066211 504295 504314 GCTCTCATGTGAGAGTATGA 81 1834 1066227 504544 504563
TCATGAATTCAGGTTGCCCC 42 1835 1066243 505149 505168 CTTTGTTTGAAGTTAGACC 46 1836 1066259 505666 505685
CTGCTTATTGCAGGCCTAGC 69 1837 1066275 506042 506061 CAGTTGAGAACGGTATTGAG 67 1838 1066291 506447 506466
GTCTGTCTTTAGGGTCACCC 47 1839 1066307 506828 506847 GCTTTAGGATGGTGTGGATC 60 1840 1066323 507587 507606
CTCTCACTGACTAGCTTTTCG 57 1841 1066339 508114 508133 CTTACCGTCTTTAGGAACCT 47 1842 1066355 508585 508604
GGACGACACAACCCCGCTGC 94 1843 1066371 509070 509089 CCCTACAATGCATACCCAAC 58 1844 1066387 510027 510046
GAGTTATACGGAACCCAC 56 1845 1066403 510410 510429 GTCCAAGACATGCTAGGGAC 71 1846 1066419 510839 510858
CAATCTAGTTAGACCAGGCA 31 1847 1066435 511667 511686 ATGGAAGTCCGGCCTGGAGC 49 1848 1066451 511894 511913
GTGAACTACTTGAGACCTT 50 1849

TABLE-US-00028 TABLE 27 Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (37,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number
Start Site Stop Site Sequence (5' to 3') (% UTC)) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 31 172 749882 460115
460134 ACTTTATAGTGTGGATGGTA 17 586 750519 349103# 349122# GTCATCACCTCTCTTCAGGA 16 181 1065273 432754 432773
ACGCAACTGAGACAATCCAT 33 1850 1065289 440229 440248 GTGGCTGTGCTGGATATTAC 48 1851 1065305 442574 442593
GAGCAAAATGGTCAGTACCC 43 1852 1065321 443357 443376 GATAGTCTAGTCACTCACTG 40 1853 1065337 444867 444886
GGATGCATTGGTGTCAGAG 47 1854 1065353 445763 445782 GGTTTACACATGATAGGAAC 80 1855 1065369 446772 446791
AACAACGGTTGCAGGGACAG 29 1856 1065385 447787 447806 CCCCCTGTACAGTAAGGGCC 73 1857 1065401 448408 448427
GTCTTAGTGCCAAATATCCA 52 1858 1065417 448735 448754 TATTTAGACCACGATAACTG 100 1859 1065433 449118 449137
TCCCTCTTGTAGATCCTGGC 40 1860 1065449 449666 449685 GAGATATGATTAGTACTGGA 63 1861 1065465 452079 452098
GAGATAGAACACTTCGCCGC 39 1862 1065481 453032 453051 GTGATTAGGCGGCCGGGCGC 119 1863 1065497 454151 454170
CTTATGGAGACTTATATACC 61 1864 1065513 454842 454861 GGGTAGAAGACTAGCATACA 22 1865 1065529 455233 455252
GTACTTACTCTTGATCCAGA 48 1866 1065545 455636 455655 AGCACAAGATTGGTTCCATT 47 1867 1065561 456827 456846
GGGTACTATGTTGTCACTGA N.D. 1868 1065577 457571 457590 ATGAACCATGGAGTCTCTAT 77 1869 1065593 458456 458475
GGCTTTATACTTTACCCAAC 15 1870 1065609 459265 459284 GTAGATAACTTTACCTTGAC 14 1871 1065625 460511 460530
GTTTATAGGGTGGTTGGTTC N.D. 1872 1065641 461413 461432 GGGTTACTTTCCAATAGAGT 11 1873 1065656 461610 461629
AGGCATGTACAAACGGGCTA 73 1874 1065671 463193 463212 GTTCTGGAATCTTGTAGAGG 11 1875 1065686 464398 464417
CCCATTCACTAGCTAGGTAG 23 1876 1065701 465011 465030 CTGCTCAAATCTGCGAATAG 73 1877 1065717 465374 465393
GAGGATGGTGTGTATGTTAT 31 1878 1065733 465889 465908 TCCCACAAGGTACTCTTGCC 61 1879 1065749 466296 466315
GCATCCCCTGTGGCTTATAG 65 1880 1065765 467043 467062 TCAGTTAGTCAAGTTAGGGA 7 1881 1065781 468335 468354
AGCATTTGAGCTATGGGTT 35 1882 1065797 470401 470420 GTTCAACCATCGAGATGATC 64 1883 1065813 474404 474423
GATAGGGTTTAGGTCAACCA 20 1884 1065829 475726 475745 ATCTGGTCTTGGGATTATAG 26 1885 1065845 476159 476178
CTTAGGAGTTTCTAACTCTC 56 1886 1065861 478935 478954 GTTTTGGAGATTGTGTTACG 57 1887 1065877 480733 480752
GGAGCCAGGTAGAGTTAACG 41 1888 1065893 482489 482508 GGATAGTTGTGAACAACCTT 70 1889 1065909 483839 483858
CATTGAATGAGGGCCCAAAC 77 1890 1065924 484241 484260 GGAATTGTCTTATGGCTGCC N.D. 1891 1065940 485829 485848
TGATTATTCAGAGAACTACG 47 1892 1065956 487697 487716 CGTTGGCATTAAAGTCTGAG 35 1893 1065972 489491 489510
CTGGGTTGAGACTATTCAGG 64 1894 1065988 489844 489863 CCATTGAAGAAAAACCCGCG 52 1895 1066004 494452 494471
GCACTACTGTAAAGGTCATG 48 1896 1066020 497158 497177 ACAAGGCTAGTGTTACACTG 42 1897 1066036 499081 499100
CAATGGTAATCGCATACT 53 1898 1066052 500210 500229 GACTAATCTCTTAAGTTCA 45 1899 1066068 500689 500708
AGGTACTGTAAAGCCCTTTGC 80 1900 1066084 500972 500991 AACTTTTCACTAGTGGG 68 1901 1066100 501490 501509
CCCTTTGTTCACTGAACG 69 1902 1066116 501913 501932 AGCTTCCCTCCAACGAGTGA 78 1903 1066132 502433 502452
GGCCCAAGGGAATTGATGCA 45 1904 1066148 502766 502785 AGGAATAGGCACTTCGGGCA 58 1905 1066164 503051 503070
GGAACCCAGGATTTAGGTGC 50 1906 1066180 503368 503387 GGTATAGGAGTGTAGCCC 44 1907 1066196 503900 503919
GGTTGGATTGCTTTATTGCA 85 1908 1066212 504299 504318 GCCTGCTCTCATGTGAGAGT 47 1909 1066228 504644 504663
CGCTTGGCTGACATTTAGGG 38 1910 1066244 505169 505188 ACCGAGCTCAAGAACTGTGA 59 1911 1066260 505670 505689

TGGCTGCTTATTGACAGGCC 73 1912 1066277 506047 506066 GTTTCAGTTAGAGAACGGTA 41 1913 1066292 506493 506512
GGGACCTCCTTATATTCACC 35 1914 1066308 506833 506852 CCCCTGCTTTAGGATGGTGT 134 1915 1066324 507637 507656
GGTCCCCCAGAAGGCTTGA 56 1916 1066340 508118 508137 TATTCTTACCGTCTTTAGGA 39 1917 1066356 508640 508659
GGATGGAGCTGTTGTGGCAT 57 1918 1066372 509076 509095 GCATTACCCTACAATGCATA 84 1919 1066388 510035 510054
GCATTTGAGAGTTATACACG 33 1920 1066404 510596 510615 GGGCCTTATAGACCATAACCA 65 1921 1066420 510844 510863
GGAGACAATCTAGTTAGACC 30 1922 1066436 511673 511692 GCACAAATGGACTGCCGGCC 89 1923 1066452 511964 511983
GAGAGTCATACTCCTGACAG 53 1924

TABLE-US-00029 TABLE 28 Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (37,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number
Start Site Stop Site Sequence (5' to 3') (% UTC)) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 24 172 749882 460115
460134 ACTTTATAGTGTGGATGGTA 27 586 750519 349103# 349122# GTCATCACCTCTCTTCAGGA 7 181 1065262 430574 430593
CTCCGGGCTCACACCTGGT 82 1925 1065278 433033 433052 TCTATACACTGCTGTTGACG 52 1926 1065294 441732 441751
GCTTGCTAAACCTGTCCATT 40 1927 1065310 442717 442736 CTGGATGGTATAAAATTGCC 43 1928 1065326 443811 443830
TTTGCCGCTTTGGTGAACCT 69 1929 1065342 445129 445148 GTGCATGTATACAGTTGGGC 31 1930 1065358 446360 446379
CCATCAAGTAGGCCAGTCAC 37 1931 1065374 446952 446971 GGAATAACCCATACCCCCCA 97 1932 1065390 447883 447902
GCATGACTCCTAATTGCTGT 59 1933 1065406 448539 448558 GAGCTCAGTCTTCACTAGTT 46 1934 1065422 448919 448938
GTCAGGCACCAGATTGCTCA 70 1935 1065438 449442 449461 ATGATCATACTGGAGCCAGG 8 1936 1065454 449875 449894
ACCTACCAGAGTGGCATCCT 47 1937 1065470 452250 452269 ACGCTGTGTGAATCAAAAGC 85 1938 1065486 453211 453230
AGTGTTCCTACATCCACCGT 62 1939 1065502 454279 454298 CATGCCAATCGCATGCAAGC 71 1940 1065518 454967 454986
GTCAGGAGTTAACCTATGAAC 95 1941 1065534 455326 455345 CACCAAGTTAGGCCTGCTCCG 94 1942 1065550 456651 456670
CGTTTCCACAGCCAGGCTTA 83 1943 1065566 456921 456940 AGTCTTGGTCTACTAGGCAC 81 1944 1065582 457908 457927
CCAATATTTGACTAGTTTCG 23 1945 1065598 458568 458587 TTTAGTATGTCGAGAACTCA 31 1946 1065614 459652 459671
TGAAACCTGTATAGTTTACC 70 1947 1065630 460652 460671 CTAACCATAGACAGAGTGCA 33 1948 1065645 461463 461482
CATCATGATCTTGGAAGGC 18 1949 1065661 461898 461917 GACCCACAATTTGGTCCCA 62 1950 1065676 463668 463687
GACTATCTCAGCACTAGGGA 26 1951 1065691 464558 464577 GCCCTGTAGTATGTGGATAC 46 1952 1065706 465231 465250
CTCGGGTATATAAATTAATG 43 1953 1065722 465480 465499 GGATCTAGAATGATTGGTTG 75 1954 1065738 466218 466237
GGGCTGCTGTTGAGTCCCC 65 1955 1065754 466397 466416 AGATACTCACTCGCTTTTCA 20 1956 1065770 467416 467435
ACGCTCCTTCATTTTCATGCA 46 1957 1065786 468925 468944 GCTGTAAGTATAGATGCCCTC 35 1958 1065802 472661 472680
ATTAGTTGACTCTATAAACG 70 1959 1065818 474805 474824 AAGTTTGGTATTGACACCTC 36 1960 1065834 475858 475877
CTTGATATAAGGGTAATACG 58 1961 1065850 476393 476412 GCCTCAGGCAAGTAGCCTTC 93 1962 1065866 479584 479603
GCCATCTTCAAGACTAAGGT 50 1963 1065882 480932 480951 CGAAAAATAGTATGCGAAGC 32 1964 1065898 482792 482811
GGAGACCCTAAAAAACCCGT 67 1965 1065914 483913 483932 CCTATAGGTGGCCTTCAGTC 30 1966 1065929 484322 484341
GGCAAGCCCAGACTCATGCT 86 1967 1065945 486491 486510 CGTCAACACTATAGATGAAT 40 1968 1065961 488332 488351
CCACTCATGTACATGAGATC 44 1969 1065977 489547 489566 GGCAAGCAACTATGGGTGGA 26 1970 1065993 493198 493217
GTTTACTTCGGGCCAGGTCC 97 1971 1066009 495221 495240 ACCCCTAGGAATAATGTTGC 36 1972 1066025 497750 497769
AGCGATACAAACCTACGCCA 71 1973 1066041 499146 499165 GGCAAGTTTTTATCGCTTAA 48 1974 1066057 500425 500444
AGTATTTTAGCCGGGATACG 53 1975 1066073 500846 500865 ACGGGTTCCTATTTGTCAGG 54 1976 1066089 501208 501227
GTGAGTTTFACTCTGCAATA 31 1977 1066105 501601 501620 GCCTCATTGGAGAGAGAACT 91 1978 1066121 502134 502153
TCAGTGCTACCCATTAGACC 35 1979 1066137 502512 502531 CATACATGGAGGAGTCCCTG 59 1980 1066153 502809 502828
TCCTAAGGATGTAGCCCATC 55 1981 1066169 503123 503142 CGCATCCAGTTAATCTCTGA 84 1982 1066185 503437 503456
GTCCCAGTTGGATTAAGGTG 64 1983 1066201 503970 503989 CTGTACAGGAGGGTGTCTCT N.D. 1984 1066217 504426 504445
TAGTCTAAGGCAGTCAGGGT 28 1985 1066233 504757 504776 ATCTAGACTCTGCCTGGTTA 51 1986 1066249 505218 505237
GGACACTCTCGCTGAGGACA 21 1987 1066265 505800 505819 GGCATATGACCACAGGGTA 35 1988 1066281 506127 506146
TCTGCTCTGTTAGCCGAGCA 77 1989 1066297 506641 506660 GCACAATGCTTGTATGCCCC 57 1990 1066313 507239 507258
GTGCCCTTATGCTTAAATCC 48 1991 1066329 507766 507785 TTTGCTCAGGATCGCTGGCC 93 1992 1066345 508230 508249
AGCCTCTGCCTAAGGTAGT 64 1993 1066361 508780 508799 CGTGATTGCAAAAGTCCAAG 47 1994 1066377 509247 509266
CAGTAGTGGTATCAAATCTG 36 1995 1066393 510133 510152 TCCAGCACATAGACCATAGC 60 1996 1066409 510676 510695
GGTGAAAGTCTAGTATGTC 73 1997 1066425 511077 511096 GACCCATGGAAAGACTCGGG 119 1998 1066441 511737 511756
GTTTCCCATAGGCCACGCTG 81 1999

TABLE-US-00030 TABLE 29 Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (37,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number
Start Site Stop Site Sequence (5' to 3') (% UTC)) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 35 172 749882 460115
460134 ACTTTATAGTGTGGATGGTA 26 586 749935 463695 463714 AGTGGTTGCTATCCTGCTAA 65 2000 750519 349103# 349122*
GTCATCACCTCTCTTCAGGA 16 181 1065263 430586 430605 ACGCAAGTCCTCCCGGGC 73 2001 1065279 433138 433157
TCAAGGGAACCTAATCCTCC 91 2002 1065295 441764 441783 GATTGATCATGTTATGCCCT 34 2003 1065311 442754 442773
GGTGGAATTGGAGTTGCTACA 74 2004 1065327 443823 443842 TGGGACACATATGTTGCCGC 81 2005 1065343 445173 445192
CTGAAGCAAAGCTTATAGGC 91 2006 1065359 446483 446502 TGACACTGAGTAATCTGCGA 71 2007 1065375 446994 447013
TGTAAGATTCCCCAAGGGT 103 2008 1065391 447907 447926 GGTGGGATTACATCTCTTAT 97 2009 1065407 448582 448601
CGTGATATTCATGTTGTCTG 89 2010 1065423 448947 448966 CGAAGCCAAAGGTAAGT 100 2011 1065439 449461 449480
GCTTGCACTCCATCATATAA 126 2012 1065455 449988 450007 AGGACCAAGCAGTCTAGTTT 47 2013 1065471 452256 452275
GGTGTCAAGCTGTGTGAATC 90 2014 1065487 453229 453248 GGAGTCAAGTTGGTTAATAG 136 2015 1065503 454339 454358
GGGCGAAAGTATATCAGGCA N.D. 2016 1065519 455012 455031 GGCTGTCAATGCTGATATAT 82 2017 1065535 455332 455351
GGAGTTCACCAAGTTAGGCCT 88 2018 1065551 456669 456688 ACCCTCCCAACAAATAAGCG 160 2019 1065567 456925 456944
AGCAAGTCTTGGGTCACTAG 72 2020 1065583 457969 457988 GGCATCAATCAACAAGCACC 49 2021 1065599 458570 458589
GCTTTAGTATGTCGAGAACT 19 2022 1065615 460105 460124 GTGGATGGTATCCTGGTCAA 46 2023 1065631 460660 460679
GCATGATTCTAACCATAGAC 31 2024 1065646 461468 461487 GTACTCATCATGATCTTGGT 26 2025 1065662 462030 462049
GGAATAGGGCTCTGCTTATT 60 2026 1065692 464776 464795 CCTCGAGGATAGTTTCACTG 76 2027 1065707 465234 465253
CATCTCGGGTATATAAATTA 77 2028 1065723 465484 465503 TGTGTGATAGATAATGATTG 71 2029 1065739 466222 466241
TATGGGCTGCTGTTTGAAGT 55 2030 1065755 466403 466422 CTCAGAGATACTACTCGC 55 2031 1065771 467421 467440
TTGATACGCTCCTTCAATTC 75 2032 1065787 468927 468946 ATGCTGTAAGTATAGATGCC 86 2033 1065803 472669 472688
CGTTGAATATTAGTTGACTC 74 2034 1065819 474875 474894 GGTGTTGTATGATAATGTGT 72 2035 1065835 475935 475954
CAATAGTAGTGATGACTTCC 41 2036 1065851 476439 476458 AGATGGAGCTTGAGCCATCC 76 2037 1065867 479607 479626
GCCCCACAGTAGAAATGTGG 87 2038 1065883 481338 481357 AGTCAACAAGACAACCTCGAT 62 2039 1065899 482839 482858
CGTTATTCAGTCTCAGGGAG 28 2040 1065915 483919 483938 CTTTACCTATAGGTGGCCT 67 2041 1065930 484442 484461
GAGTAATGGACTTCTGGTCT 133 2042 1065946 486772 486791 GTGTGAGGGAATCTAAGATC 54 2043 1065962 488566 488585

GGTGCACACACACACACACACACAAA 63 2044 1065978 489581 489600 GCTGCAGTGGTACCACACAGAC 91 2045 1065994 493203 493222
AAGAAAGTTTACTTCGGGGCCA 59 2046 1066010 495383 495402 TCGGGAGTTGTTAGTCCAAG 62 2047 1066026 497873 497892
GCTCCCAAAACAACTAATAGT 76 2048 1066042 499204 499223 TCAAAATCACGAAAGGCGGG 71 2049 1066058 500430 500449
TTAGAAGTATTTTACCGGG 44 2050 1066074 500857 500876 GGTAGGTAACCTACGGGTTCC 59 2051 1066090 501222 501241
CACCCCTGTAAACAGTCAGC 96 2052 1066106 501609 501628 CTGCTCAAGCCTTATTGGAG 120 2053 1066122 502140 502159
GTGGAATCAGTGCCTACCCAT 77 2054 1066138 502527 502546 AGCTGGGCACCTACATACATAC 55 2055 1066154 502813 502832
AGCCTCCTAAGGATGTAGCC 108 2056 1066170 503131 503150 ATGGTCCTCGCATCCAGTTA 54 2057 1066186 503444 503463
CAGCACTGTCCCAGTTGGAT 61 2058 1066202 503986 504005 GTTGATTCACTGCACACTGT 80 2059 508752 508771 1066218 504433
504452 GGTATATTAGTCTAAGGCAG 43 2060 1066234 504789 504808 GCGAAGCCTTCTCAAAGACT 92 2061 1066250 505323 505342
GGGCCGCTCTGTTATTGTG 88 2062 1066266 505808 505827 GAACTGTGGGCACATGACC 86 2063 1066282 506165 506184
CCTTTGGAGCTTTGACTGGC 76 2064 1066298 506677 506696 CGATTTCTGCCAGTGGCTG 79 2065 1066314 507244 507263
CTACTGTGCCCTTACTGTTA 83 2066 1066330 507795 507814 GAAGACTTAGGTCTCAAGCT 62 2067 1066346 508274 508293
GTAGTCATGTAGAACAGCAC 78 2068 1066362 508784 508803 CCACCGTGATTGCAAAGTCC 69 2069 1066378 509252 509271
GGGCACAGTAGTGGTATCAA 22 2070 1066394 510203 510222 AGAGCTGTCTTAAGGCAATT 54 2071 1066410 510711 510730
CTTGACATACAGGATAGGTG 52 2072 1066426 511088 511107 TTGGTAAATTAGACCCATGG 110 2073 1066442 511744 511763
GATGTATGTTTCCCATAGGC 56 2074

TABLE-US-00031 TABLE 30 Reduction of UBE3A-ATS RNA by 8,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (20,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number
Start Site Stop Site Sequence (5' to 3') (% UTC)) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 51 172 749882 460115
460134 ACTTTATAGTGTGGATGGTA 46 586 750519 349103* 349122* GTCATCACCTCTCTTCAGGA 28 181 1165442 449002 449021
AGGTCCTTATTTACCATTTGGC 72 2377 1165447 449444 449463 TAATGATCATACTGGAGCCA 87 2378 1165449 452253 452272
CTCACGCTGTGTGAATCAAA 101 2379 1165455 452265 452284 GGAAGTCTAGGTCTCACGCT 88 2380 1165467 453362 453381
CTACATAGGCACCTCTACTAG 96 2381 1165471 454154 454173 ACCCTTATGGAGACTTATAT 96 2382 1165477 454461 454480
GACTCCACACACCTACTAGA 97 2383 1165483 454578 454597 CAAGCAAAGACTACACCGTG 106 2384 1165487 454848 454867
TATCCAGGGTAGAAGACTAG 83 2385 1165493 454861 454880 GGTGTATATTCCTATCCAG 89 2386 1165499 455020 455039
AGTGGGATGGCTGTCAATGC 82 2387 1165510 220439 220458 AGGGATCCCCAAATAGAGCC 96 2388 456808 456827 1165516 457018
457037 TCACAGTTGCTTGACCCTTA 79 2389 1165522 457572 457591 TATGAACCATGGAGTCTCTA 69 2390 1165528 458791 458810
TCCCTAATATAGGCGAGAT 88 2391 1165534 458942 458961 TATAGGACCTCAGGAGATTG 52 2392 1165539 458950 458969
TCAAGACCTATAGGACCTCA 37 2393 1165546 459550 459569 AAAGACTACCTGGTATGAC 65 2394 1165557 460735 460754
GATTTGACCTGCTATGTCT 16* 2395 1165565 461525 461544 ACATTCAGCTAGACTAGTTG 58 2396 1165570 461901 461920
TGAGACCCCAACAATTTGGTC 89 2397 1165572 464385 464404 TAGGTAGTTCACAACTCTTC 68 2398 1165582 464562 464581
ATATGCCCTGTAGTATGTGG 77 2399 1165587 465245 465264 ATTTTATTTGGTCATCTCGGG 47 2400 1165592 465387 465406
TGTCAGACTTATTGAGGATG 67 2401 1165594 465891 465910 ACTCCCACAAGGTACTCTTG 84 2402 1165598 466249 466268
AACTGGTAGGATCTATGGCA 40 2403 1165603 467301 467320 TTCACAGAGTAGTCTATTGG 84 2404 1165609 468352 468371
TGTAGTATGCATTGACAAGC 58 2405 1165612 482161 482180 GAGTTACAAGTGTCTATATAC 48 2406 1165618 483960 483979
GGTAGGATTCATGGTCCAAA 64 2407 1165625 485656 485675 CTAGGACCAGTTGGTTCAC 99 2408 1165633 489492 489511
ACTGGGTTGAGACTATTCAG 68 2409 1165638 489521 489540 CAATGGACCACCTAAGACCT 104 2410 1165641 493854 493873
CTAATCTATGCTGGGCCCA 89 2411 1165647 493864 493883 GAGATTAGGCCTAATCTATG 116 2412 1165653 496830 496849
ATTAGGTATGGAGGCCATGT 69 2413 1165664 499674 499693 AAGGTAGCCCCAATACAGAT 87 2414 1165668 500690 500709
GAGGTACTGTAAGCCCTTTG 115 2415 1165673 500906 500925 GTTTTGACCTAACTGGCCTT 86 2416 1165678 501567 501586
CTAAGGAGTGCTCTTTGTGG 59 2417 1165684 502371 502390 AGGTATTAAGGCCCTTGCC 95 2418 1165690 502378 502397
GTATACAAGGTATTAAGGCC 69 2419 1165695 502449 502468 TAGGCAGCTCTTTGTAGGCC 82 2420 1165701 503155 503174
AGTTCTGTATACACCATCCC 51 2421 1165707 503361 503380 GGAGTGTTAGCCCAGGTGAT 95 2422 1165713 503367 503386
GTCATAGGAGTGTTAGCCCCA 84 2423 1165719 503421 503440 GGTGATGACACCCCTACCAT 106 2424 1165725 503507 503526
GCCTTGCTGTGGTGAAACC 64 2425 1165729 504076 504095 GTGAACCTCTGTGACTGATA 52 2426 1165736 504431 504450
TATATTAGTCTAAGGCAGTC 47 2427 1165747 505173 505192 GTGTACCGAGCTCAAGAACT 51 2428 1165753 505181 505200
CTGTCTCTGTGTACCGAGCT 64 2429 1165766 505320 505339 CCGCCTCTGTTATTGTGATA 78 2430 1165772 505327 505346
AATTGGGCCGCTCTGTTAT 74 2431 1165778 505542 505561 AGTCAGGTACAGGTGTTGGA 56 2432 1165783 505745 505764
TGTAGGTTGACAGGACATGC 64 2433 1165789 505821 505840 GAGGGTGCTTAGTGAAGTGT 87 2434 1165795 505830 505849
AAGACCTGTGAGGGTGCTTA 87 2435 1165800 506448 506467 AGTCTGTCTTTAGGGTCACC 57 2436 1165806 507907 507926
TAGCCATTCGATACCTGCTT 76 2437 1165812 507914 507933 TGCAGGATAGCCATTTCGATA 75 2438 1165823 508791 508810
ATGAATGCCACCGTGATTGC 94 2439 1165828 510228 510247 AGTTGTGTCTGAGGGATATC 64 2440 1165839 448411 448430
AGGGCTTAGTGCCAAATAT 102 2441 1165852 455763 455782 TACCAAAGTGGCTGCTCACC 76 2442 1165854 460426 460445
TAGCCCTTTAGCACACATTC 44 2443 1165856 461342 461361 CTGGCAATATGTCAACCTTA 55 2444 1165864 464515 464534
AGTGCACAGATAATGACTAG 91 2445 1165880 484240 484259 GACTTGTCTTATGGCTGCCT 90 2446 1165884 486440 486459
CCTAATAGCACCAGAATAGT 104 2447 1165891 499666 499685 CCAATACAGATTAGTGGC 89 2448 1165901 504818 504837
TGAAGTAGTCTGCCCTTTC 86 2449 1165902 505216 505235 ACACCTCTCGCTGAGGACACA 74 2450 1165908 506966 506985
GGGCATTCAAGTCTCCCTCC 106 2451

TABLE-US-00032 TABLE 31 Reduction of UBE3A-ATS RNA by 8,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (20,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number
Start Site Stop Site Sequence (5' to 3') (% UTC)) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 53 172 749882 460115
460134 ACTTTATAGTGTGGATGGTA 50 586 750519 349103* 349122* GTCATCACCTCTCTTCAGGA 31 181 1165439 448413 448432
CAAGGGTCTTAGTGCCAAAT 106 2452 1165444 449095 449114 TATGCTAGTCACTCATTGAG 113 2453 1165451 452258 452277
TAGGTCTCACGCTGTGTGAA 97 2454 1165457 452268 452287 GAGGGAAGTCTAGGTCTCAC 89 2455 1165468 453365 453384
TAACATACATAGGCACTCTAC 79 2456 1165473 454161 454180 AGATTTGACCCTTATGGAGA 115 2457 1165479 454465 454484
AGTGGACTCCACACACCTAC 108 2458 1165485 454843 454862 AGGGTAGAAGACTAGCATAC 96 22 1165489 454857 454876
TATATTCCTATCCAGGGTA 92 2459 1165495 454865 454884 GACAGGTGTATATTCCTAT 116 2460 1165505 456628 456647
TGTGCATGGTTATCTAATGC 74 2461 1165512 457013 457032 GTTGCTTGACCCCTAATTCA 108 2462 1165518 457022 457041
CATATCACAGTTGCTTGACC 132 2463 1165524 457984 458003 TTGATCTATTATGAGGCGAT 22 2464 1165530 458935 458954
CCTCAGGAGATTGTACAACA 47 2465 1165542 459545 459564 CTATCCTGGTATGACTGTCA 39 2466 1165548 460110 460129
ATAGTGTGGATGGTATCCTG 57 2467 1165553 460508 460527 TATAGGGTGGTTGGTTCAA 29 2468 1165559 460844 460863
ACCACACATATAGGCTAGCC 39 2469 1165566 461527 461546 GAACATTCAGCTAGACTAGT 64 2470 1165574 464388 464407
ACTTAGGTAGTTTCAACTC 55 2471 1165579 464557 464576 CCCTGTAGTATGTGGATACT 97 2472 1165583 465239 465258
TTGGTCATCTCGGGTATATA 43 2473 1165589 465380 465399 CTTATTGAGGATGGTGTGTA 64 2474 1165605 467802 467821
TCCCTCTTAGTGATTGGTGG 80 2475 1165614 483844 483863 CTAGGCATTGAATGAGGGCC 93 2476 1165620 483962 483981

AGGGTAGGATTATCATGGTCCA 95 2477 1165627 486353 486372 TTAATAGACTGCGATTATAC 100 2478 1165630 489097 489116
 TCCCTAAGCTTAGATATACC 76 2479 1165634 489516 489535 GACCACCTAAGACCTCAAGG 73 2480 1165643 493860 493879
 TTAGGCCTAATCTATGCTGG 84 2481 1165649 493866 493885 ATGAGATTAGGCCTAATCTA 108 2482 1165655 496832 496851
 AGATTAGGTATGGAGGCCAT 85 2483 1165660 499668 499687 GCCCCAATACAGATTCAAGT 94 2484 1165675 500909 500928
 GGTGTTTTACCTAAGTGGC 81 2485 1165680 501762 501781 GGCTAAGAGTCACCTGTATC 85 2486 1165686 502373 502392
 CAAGGTATTAAGGCCCTTGG 92 2487 1165691 502438 502457 TTGTAGGCCCAAGGGAATTG 105 2488 1165696 502451 502470
 GCTAGGCAGCTCTTTGTAGG 65 2489 1165703 503357 503376 TGTTAGCCCAGGTGATCAGC 69 2490 1165709 503363 503382
 TAGGAGTGTTAGCCCAGGTG 56 2491 1165715 503370 503389 AGGGTCATAGGAGTGTTAGC 65 2492 1165721 503424 503443
 TAAGGTGATGACACCCCTAC 109 2493 1165726 503871 503890 GGGTCAGTTGTGCAGTTGTT 46 2494 1165738 504434 504453
 AAGTATATTAGTCTAAGGCA 49 2495 1165743 504977 504996 CTATTATCTTCTTAGGGTCG 57 2496 1165749 505175 505194
 CTGTGTACCGAGCTCAAGAA 65 2497 1165755 505198 505217 CATCAGCTGTTAGCAGTCTG 82 2498 1165760 505219 505238
 AGGACACTCTCGCTGAGGAC 64 2499 1165768 505322 505341 GGCCGCCTCTGTTATTGTGA 91 2500 1165774 505329 505348
 GAAATTGGGCCCGCTCTGTT 91 2501 1165779 505651 505670 CTAGCTTCCACCTAAGAGCT 111 2502 1165785 505750 505769
 AACTCTGTAGGTTGACAGGA 78 2503 1165791 505824 505843 TGTGAGGGTGCTTAGTGAAC 77 2504 1165796 505836 505855
 CTGTCAAAGACCTGTGAGGG 78 2505 1165808 507910 507929 GGATAGCCATTGATACCTG 101 2506 1165814 508119 508138
 TTATCTTACCCTCTTTAGG 70 2507 1165825 509129 509148 AGCTCATCACAACTGGGTGG 86 2508 1165842 449667 449686
 AGAGATATGATTAGTACTGG 105 2509 1165850 455030 455049 GTGATATTGCAGTGGGATGG 89 2510 1165853 458945 458964
 ACCTATAGGACCTCAGGAGA 70 2511 1165857 461416 461435 ACTGGGTTACTTTCCAATAG 61 2512 1165861 463093 463112
 CAATTTACTTGATACAGGGC 43 2513 1165869 465393 465412 GTCAATTGTCAGACTTATTG 54 2514 1165873 466100 466119
 GGGCCTGTATGTCTTGAGAA 88 2515 1165874 467294 467313 AGTAGTCTATTGGTGTTCCT 68 2516 1165876 468356 468375
 ATACTGTAGTATGCATTGAC 79 2517 1165882 484300 484319 TTAGTAGGGCCAGAGAATCC 94 2518 1165889 493623 493642
 CAGATGACTAGCCTCCAAAC 102 2519 1165892 499910 499929 GCATAATAGGAGGTCTTAA 46 2520 1165894 500764 500783
 GTCAAATCAATTTGTGCCAC 72 2521 1165900 504422 504441 CTAAGGCAGTCAGGGTAATG 55 2522 1165905 506694 506713
 TTAAGAAGCTTGTCTTTTCGA 115 2523 1165909 507901 507920 TTCGATACCTGCTTTTGTGA 111 2524 1165914 511254 511273
 CTGATGATTTGTGATTACC 84 2525

TABLE-US-00033	TABLE 32	Reduction of UBE3A-ATS RNA by 8,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (20,000 cells/well)	SEQ ID	SEQ ID	UBE3A-Compound NO:	1 NO:	1 ATS	SEQ ID	Number
Start	Site	Stop	Site	Sequence (5' to 3') (% UTC))	NO 617557	483977	483996	ATCATGTGCATACCCAGGGT	49 172 749882 460115
460134	ACTTTATAGTGTGGATGGTA	33 586 750519	349103*	349122*	GTCATCACCTCTCTTCAGGA	22 181 1165440	448459	448478	
	ATAGTGCCAGTAGGACTTAC	98 2526 1165445	449097	449116	AGTATGCTAGTCACTCATTG	68 2527 1165452	452259	452278	
	CTAGGTCTCACGCTGTGTGA	84 2528 1165458	452271	452290	TATGAGGGAAGTCTAGGTCT	64 2529 1165474	454162	454181	
	AAGATTTGACCCCTTATGGAG	96 2530 1165480	454466	454485	GAGTGGACTCCACACACCTA	80 2531 1165490	454858	454877	
	GTATATTCCTATCCAGGGT	93 2532 1165496	454866	454885	TGACAGGTGTATATTCCCTA	55 2533 1165501	455031	455050	
	AGTGATATTGCAGTGGGATG	77 2534 1165506	220320	220339	CATCATCGATCCAAACAAGC	102 2535 456689	456708	1165513 457015	
	457034 CAGTTGCTTGACCCTTAATT	76 2536 1165519	457364	457383	ACATGGTGAGGGTCTCAATG	33 2537 1165525	457986	458005	
	ACTTGATCTATTATGAGGGC	35 2538 1165531	458936	458955	ACCTCAGGAGATTGTACAAC	42 2539 1165536	458947	458966	
	AGACCTATAGGACCTCAGGA	28 2540 1165543	459546	459565	ACTATCCTGGTATGACTGTC	40 2541 1165549	460112	460131	
	TTATAGTGTGGATGGTATCC	49 2542 1165554	460509	460528	TTATAGGGTGGTTGGTTCAA	24 2543 1165560	460845	460864	
	TACCACACATATAGGCTAGC	43 2544 1165567	461671	461690	CCTAAGGTGAAGTCTGTGTA	58 2545 1165575	464389	464408	
	CATTAGGTAGTTTCAACAAT	59 2546 1165580	464559	464578	TGCCCTGTAGTATGTGGATA	69 2547 1165584	465241	465260	
	TATTGGTCATCTCGGTAATA	51 2548 1165595	466246	466265	TGGTAGGATCTATGGCATT	35 2549 1165600	467297	467316	
	CAGAGTAGTCTATTGGTGTT	83 2550 1165606	467803	467822	CTCCCTCTTAGTGATTGGTG	84 2551 1165610	468403	468422	
	CCTACCCTTGCATGCTATGT	88 2552 1165615	483850	483869	GCCTGACTAGGCATTGAATG	79 2553 1165622	484301	484320	
	GTTACTAGGGCCAGAGAATC	105 2554 1165628	486354	486373	GTTAATAGACTGCGATTATA	43 2555 1165631	489098	489117	
	CTCCCTAAGCTTAGATATAC	88 2556 1165635	489518	489537	TGGACCACCTAAGACCTCAA	73 139 1165644	493861	493880	
	ATTAGGCCTAATCTATGCTG	89 2557 1165650	493868	493887	AGATGAGATTAGGCCTAATC	136 2558 1165656	496834	496853	
	ACAGATTAGGTATGGAGGCC	57 2559 1165661	499670	499689	TAGCCCCAATACAGATTCAG	55 2560 1165666	499911	499930	
	TGCATAATAGGAGTCTCTTA	54 2561 1165670	500864	500883	TACTTTTGGTAGGTAACCTA	70 2562 1165676	500910	500929	
	AGGTGTTTTGACCTAATGG	80 2563 1165681	501764	501783	TGGGCTAAGAGTCACTGTGA	109 2564 1165687	502374	502393	
	ACAAGGTATTAAGGCCCTTG	106 2565 1165692	502439	502458	TTTGTAGGCCCAAGGGAATT	119 2566 1165697	502740	502759	
	GCCCGATGACACCAGCCACT	61 2567 1165704	503358	503377	GTGTTAGCCCAGGTGATCAG	66 2568 1165710	503364	503383	
	ATAGGAGTGTTAGCCAGGT	62 2569 1165716	503418	503437	GATGACACCCCTACCATGGC	63 2570 1165722	503425	503444	
	TTAAGGTGATGACACCCTTA	102 2571 1165727	503873	503892	AAGGGTCAGTTGTGCAGTTG	43 2572 1165733	504427	504446	
	TTAGTCTAAGGCAGTCAGGG	50 2573 1165739	504435	504454	TAGGTATATTAGTCTAAGGC	50 2574 1165744	505167	505186	
	CGAGCTCAAGAACTGTGACT	65 2575 1165750	505176	505195	TCTGTGTACCGAGCTCAAGA	41 2576 1165756	505199	505218	
	ACATCAGCTGTTAGCAGTCT								

TABLE-US-00034 TABLE 33 Reduction of UBE3A-ATS RNA by 8,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro (20,000 cells/well) SEQ ID SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number

Start	Site	Stop	Site	Sequence (5' to 3') (% UTC))	NO	1065438	449442	449461	ATGATCATACTGGAGCCAGG	95	1936	1165509
220406	220425	CTTATTTGTCCTATTGGAGG	91	2600	456775	456794	1178377	220315	220334	TCGATCCAAACAAGCACCT	98	2601
456684	456703	1178379	220318	220337	TCATCGATCCAAACAAGCAC	132	2602	456687	456706	1178384	456753	456772
CGCACATCTGGACCTCAGAT	98	2603	1178386	456755	456774	GCCGACATCTGGACCTCAG	129	2604	1178388	456758	456777	
AGGGCCGCACATCTGGACCT	115	2605	1178390	456760	456779	GGAGGGCCGCACATCTGGAC	70	2606	1178392	456762	456781	
TTGGAGGGCCGCACATCTGG	106	2607	1178394	456764	456783	TATTGGAGGGCCGCACATCT	123	2608	1178396	456766	456785	

CTATTGGAGGGAGGGCCGCACAT 136 2609 1178398 456769 456788 TGTTCCTATTGGAGGGCCGCA 100 2610 1178400 456771 456790
TTTGTCTATTGGAGGGCCG 104 2611 1178404 220455 220474 TACTATGTTGTCACTGAGGG 83 2612 456824 456843 1178406 220457
220476 GGTACTATGTTGTCACTGAG 68 2613 456826 456845 1178408 456864 456883 TGCCCTCTTCGAAGAGATAG 87 2614 1178411
456914 456933 GGTCACCTAGGCACACTAAAG 87 2615 1178413 456916 456935 TGGGTCACTAGGCACACTAA 87 2616 1178415 456923
456942 CAAGTCTTGGGTCACTAGGC 93 2617 1178417 456926 456945 AAGCAAGTCTTGGGTCACTA 72 2618 1178418 457232 457251
CTTAGTACTCACTCCCTGTT 111 2619 1178420 457237 457256 AATGGCTTAGTACTCACTCC 79 104 1178422 457240 457259
GATAATGGCTTAGCTACTCA 77 2620 1178423 457286 457305 CGACCTAAAGACCTAGCAAA 55 2621 1178425 457289 457308
AAACGACCTAAAGACCTAGC 50 2622 1178426 457294 457313 GTACAAAACGACCTAAAGAC 77 2623 1178428 457574 457593
ATTATGAACCATGGAGTCTC 47 2624 1179735 448291 448310 TCGCAATAAGATTCCATTGC 106 2625 1179742 448305 448324
GTGCAAAACAAGTTTCGCAA 119 2626 1179745 448453 448472 CCAGTAGGACTTACTAAATC 79 2627 1179747 448455 448474
TGCCAGTAGGACTTACTAAA 99 2628 1179749 448457 448476 AGTGCCAGTAGGACTTACTA 76 2629 1179751 448630 448649
GAGGATTGCATCACATGTGT 118 2630 1179753 448715 448734 TAGCGCATTGAGCAAAATTC 75 2631 1179755 448717 448736
TGTAGCGCATTGAGCAAAAT 75 2632 1179757 448719 448738 ACTGTAGCGCATTGAGCAAA 83 2633 1179759 448722 448741
ATAACTGTAGCGCATTGAGC 103 2634 1179761 448724 448743 CGATAACTGTAGCGCATTGA 68 2635 1179763 448727 448746
CCACGATAACTGTAGCGCAT 91 2636 1179765 448730 448749 AGACCACGATAACTGTAGCG 84 2637 1179767 448732 448751
TTAGACCACGATAACTGTAG 104 2638 1179769 448734 448753 ATTTAGACCACGATAACTGT 114 2639 1179770 448737 448756
TATATTTAGACCACGATAAC 106 2640 1179772 448740 448759 CTGTATATTTAGACCACGAT 78 2641 1179774 448828 448847
AGGACCCCTTAAGTCATAAAG 75 2642 1179776 448834 448853 ATTGCAAGGACCCCTTAAGTC 75 2643 1179778 448841 448860
TATCCCAATTGCAAGGACCC 131 2644 1179781 448948 448967 ACGAAGCCAAAGGTACTTGA 124 2645 1179783 449034 449053
GCCCTGACTGTCATTATAT 86 2646 1179785 449037 449056 ACGGCCCTGACTGTCATTCA 98 2647 1179787 449039 449058
GAACGGCCCTGACTGTCATT 171 2648 1179789 449042 449061 AAGGAACGGCCCTGACTGTC 78 2649 1179791 449044 449063
TGAAGGAACGGCCCTGACTG 101 2650 1179793 449046 449065 CCTGAAGGAACGGCCCTGAC 108 2651 1179795 449049 449068
TGGCCTGAAGGAACGGCCCT 119 2652 1179797 449052 449071 ATGTGGCCTGAAGGAACGGC 76 2653 1179799 449058 449077
ATAATCATGTGGCCTGAAGG 104 2654 1179801 449060 449079 CGATAATCATGTGGCCTGAA 79 2655 1179803 449063 449082
TAACGATAATCATGTGGCCT 61 2656 1179805 449065 449084 CATAACGATAATCATGTGGC 81 2657 1179807 449101 449120
GGCAAGTATGCTAGTCACTC 73 2658 1179809 449103 449122 CTGGCAAGTATGCTAGTCAC 105 2659 1179811 449106 449125
ATCCTGGCAAGTATGCTAGT 83 2660 1179813 449108 449127 AGATCCTGGCAAGTATGCTA 103 2661 1179815 449110 449129
GTAGATCCTGGCAAGTATGC 68 2662 1179817 449115 449134 CTCTTGCTAGATCCTGGCAAG 78 2663 1179818 449157 449176
CAACATTATATCGATGCAAT 89 2664 1179820 449160 449179 TGTCAACATTATATCGATGC 109 2665 1179822 449163 449182
AATTGTCAACATTATATCGA 78 2666 1179825 448297 448316 CAAGTTTCGCAATAAGATTTC 72 2667 1179833 456498 456517
TTAGATTAAATCTGGCCGG 92 2668 1179834 456912 456931 TCACTAGGCACACTAAAGTG 107 2669 1179838 457298 457317
AGATGTACAAAACGACCTAA 99 2670 1179840 457386 457405 TAAGGATTCAATAGTGTAGG 34 2671 1179842 457503 457522
GGCACACCTGGATTTCTACTA 12 2672 1179845 448416 448435 GAGCAAGGGTCTTAGTGCCA 108 2673 1179847 448625 448644
TTGCATCACATGTGTCACTC 102 2674 1179849 448843 448862 TTTATCCCAATTGCAAGGAC 86 2675

Example 2: Effect of 5-10-5 MOE Gapmer Modified Oligonucleotides on Human UBE3A-ATS RNA In Vitro, Single Dose
[0454] Modified oligonucleotides complementary to human UBE3A-ATS nucleic acid were tested for their effect on UBE3A-ATS RNA levels in vitro.

[0455] The modified oligonucleotides in the tables below are 5-10-5 gapmers. The gapmers are 20 nucleosides in length, wherein the central gap segment consists of ten 2'- β -D-deoxynucleosides and the 5' and 3' wing segments each consists of five 2'-MOE nucleosides. The sugar motif for the gapmers is (from 5' to 3'): eeeeeeeeeeeeeeeeeee; wherein 'd' represents a 2'- β -D-deoxyribose sugar moiety, and 'e' represents a 2'-MOE sugar moiety.

[0456] Each internucleoside linkage of Compound IDs 617456, 617457, 617460, 617461, and 617557 is a phosphorothioate internucleoside linkage. All other compounds have an internucleoside linkage motif of: sooooooooooooo, wherein each "s" represents a phosphorothioate internucleoside linkage and each "o" represents a phosphodiester internucleoside linkage.

[0457] All cytosine residues are 5-methylcytosines.

[0458] "Start site" indicates the 5'-most nucleoside to which the modified oligonucleotide is complementary in target nucleic acid sequence. "Stop site" indicates the 3'-most nucleoside to which the modified oligonucleotide is complementary in the target nucleic acid sequence. Each modified oligonucleotide listed in the Tables below is 100% complementary to SEQ ID NO: 1. Start and stop sites for Compound 750519 are marked with a hashtag (#) in several tables below. The complete list of start sites for Compound 750519 are listed in Table 4b.

[0459] Human IPS cell derived ReproNeuro™ Neurons (ReproCELL) were cultured per manufacturer instructions at 40,000 cells per well and were treated with 8,000 nM of modified oligonucleotide by free uptake. After a treatment period of approximately 5 days, total RNA was isolated from the cells and UBE3A-ATS RNA levels were measured by quantitative real-time RTPCR. Human UBE3A-ATS primer probe set RTS4796 (described herein in Example 1) was used to measure RNA levels. UBE3A-ATS RNA levels were normalized to total RNA content, as measured by RIBOGREEN®. Reduction of UBE3A-ATS is presented in the tables below as percent UBE3A-ATS RNA amount relative to untreated control (UTC) cells. Each table represents results from an individual assay plate. The values marked with an asterisk (*) indicate that the modified oligonucleotide is complementary to the amplicon region of the primer probe set. Additional assays may be used to measure the potency and efficacy of the modified oligonucleotides complementary to the amplicon region.

TABLE-US-00035 TABLE 34 Reduction of UBE3A-ATS RNA by 8,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro SEQ ID UBE3A- Compound NO: 1 NO: 1 ATS SEQ ID Number Start Site Stop Site Sequence (5' to 3') (% UTC)) NO 617557 483977 483996 ATCATGTGCATACCCAGGGT 38 172 749882 460115 460134
ACTTTATAGTGTGGATGGTA 30 586 750519 349103* 349122* GTCATCACCTCTCTTCAGGA 23 181 1165438 448412 448431
AAGGGTCTTAGTGCCAAATA 75 2075 1165443 449094 449113 ATGCTAGTCACTCATTTGAGA 65 2076 1165448 449641 449660
TGATCCATTAGATAGGCTAT 80 2077 1165450 452254 452273 TCTCACGCTGTGTGAATCAA 81 2078 1165456 452267 452286
AGGGAAGTCTAGGTCTCAG 54 2079 1165472 454155 454174 GACCCCTTATGGAGACTTATA 77 2080 1165478 454464 454483
GTGGACTCCACACACCTACT 62 2081 1165484 454579 454598 GCAAGCAAAGACTACACCGT 48 2082 1165488 454856 454875
ATATCCCTATCCAGGGTAG 91 2083 1165494 454862 454881 AGGTGTATATCCCTATCCA 76 2084 1165500 455029 455048
TGATATTGCAGTGGGATGGC 59 2085 1165504 456625 456644 GCATGGTTATCTAATGCATC 60 2086 1165511 220440 220459
GAGGGATCCCCAAATAGAGC 51 2087 456809 456828 1165517 457019 457038 ATCACAGTTGCTTGACCCTT 63 2088 1165523 457983
458002 TGATCTATTATGAGGGCATC 25 2089 1165529 458792 458811 TTCCCTAATATAGGGCAGAT 49 2090 1165535 458943 458962
CTATAGGACCTCAGGAGATT 42 2091 1165540 458951 458970 CTCAGACCTATAGGACCTC 50 2092 1165547 460109 460128
TAGTGTGGATGGTATCCTGG 55 2093 1165552 460506 460525 TAGGGTGGTTGGTTCAAAT 24 2094 1165558 460842 460861
CACACATATAGGCTAGCCAA 44 2095 1165562 461414 461433 TGGGTTACTTTCCAATAGAG 27 2096 1165573 464387 464406
CTTAGGTAGTTCACAACCTCT 37 2097 1165578 464556 464575 CCTGTAGTATGTGGATACTG 40 2098 1165588 465372 465391
GGATGGTGTGTATGTTATGA 32 2099 1165599 467293 467312 GTAGTCTATTGGTGTTCCTT 45 2100 1165604 467418 467437
ATACGCTCCTTCATTTTCATG 49 2101 1165613 483363 483382 GAATTCAATGGACCCACATG 67 2102 1165619 483961 483980

[illegible]

TCATCATCCATCCAAACCAAG 58 2235 456690 456709 1165514 457035 ACAGTTGGCTTGACCTTAAT 126 2236 1165520 457379
457398 TCAATAGTGTAGGTGACATG 63 2237 1165526 458398 458417 ATGTGGGCCTCTATTAAGAT 47 2238 1165532 458937 458956
GACCTCAGGAGATTGTACAA 45 2239 1165537 458948 458967 AAGACCTATAGGACCTCAGG 53 2240 1165544 459547 459566
GACTATCCTGGTATGACTGT 57 2241 1165550 460352 460371 GATTGACTACTTCAACCTGA 23 2242 1165555 460510 460529
TTTATAGGGTGGTGGTTCA 30 2243 1165561 460850 460869 GGATTTACCACACATATAGG 39 2244 1165563 461451 461470
GGTAAGGCAGCTCCTGACAA 32 2245 1165568 461899 461918 AGATCCCAACAATTTGGTCCC 51 2246 1165576 464390 464409
GCACTTAGGTAGTTTACAAAC 41 2247 1165585 465243 465262 TTTATTGGTCATCTCGGGTA 44 2248 1165590 465384 465403
CAGACTTATTGAGGATGGTG 30 2249 1165593 465564 465583 TTAGGACAAGTCTTATAGAG 20 2250 1165596 466247 466266
CTGGTAGGATCTATGGCAGT 32 2251 1165601 467298 467317 ACAGAGTAGTCTATTGGTGT 67 2252 1165607 467804 467823
TCTCCCTCTTAGTGATTGGT 45 2253 1165616 483851 483870 GGCCTGACTAGGCATTGAAT 16 2254 1165621 483976 483995
TCATGTGCATACCCAGGGTA 76 2255 1165623 485654 485673 TAGGACCAGTTGGTTCCTG 66 2256 1165629 486356 486375
ATGTTAATAGACTGCGATTA 102 2257 1165632 489099 489118 GCTCCCTAAGCTTAGATATA 85 2258 1165636 489519 489538
ATGGACCCTAAGACCTCA 79 2259 1165639 493852 493871 AATCTATGCTGGGCCCCAAT 52 2260 1165645 493862 493881
GATTAGGCCTAATCTAGTCT 71 2261 1165651 496565 496584 TCCATCTACTATTAAATGAGC 81 2262 1165657 497266 497285
GATTAGGCAGCTTCACTACT 52 2263 1165662 499671 499690 GTAGCCCCAATACAGATTCA 35 2264 1165671 500900 500919
ACCTAACTGGCCTTTGGGTC 111 2265 1165677 500913 500932 TCAAGGTGTTTTGACCTAAC 64 2266 1165682 501766 501785
GATGGGCTAAGAGTCACCTG 99 2267 1165688 502375 502394 TACAAGGTATTAAGGCCCTT 98 2268 1165693 502442 502461
CTCTTTGTAGGCCCAAGGGA 66 2269 1165698 502763 502782 AATAGGCACTTCGGGCAAAT 75 2270 1165705 503359 503378
AGTGTTAGCCCAGGTGATCA 71 2271 1165711 503365 503384 CATAGGAGTGTTAGCCCAGG 61 2272 1165717 503419 503438
TGATGACACCCCTACCATGG 96 2273 1165723 503426 503445 ATTAAGGTGATGACACCCCT 81 2274 1165734 504428 504447
ATTAGCTTAAGGCATCAGG 96 2275 1165740 504436 504455 GTAGGTATATTAGTCTAAGG 51 2276 1165745 505171 505190
GTACCGAGCTCAAGAACTGT 71 2277 1165751 505178 505197 TACTCTGTACCGAGCTCAA 58 2278 1165757 505214 505233
ACTCTCGCTGAGGACACATC 73 2279 1165762 505223 505242 AATGAGGACACTCTCGCTGA 76 2280 1165770 505325 505344
TTGGGCCGCTCTGTTATTG 51 2281 1165776 505332 505351 GTAGAAATTGGGCCGCTCT 35 2282 1165781 505743 505762
TAGGTTGACAGGACATGCTG 66 2283 1165787 505815 505834 GCTTAGTGAAGTGTGGGCAC 72 2284 1165793 505827 505846
ACCTGTGAGGGTGCTTAGTG 62 2285 1165798 506277 506296 CTACCAGGGTGGTATTATAA 33 2286 1165810 507912 507931
CAGGATAGCCATTCGATACC 44 2287 1165821 508786 508805 TGCCACCGTGATTGCAAAGT 61 2288 1165826 510127 510146
ACATAGACCATAAGCTGAACC 65 2289 1165844 449863 449882 GGCATCCTTAAATCCTGGTT 66 2290 1165849 454845 454864
CCAGGGTAGAAGACTAGCAT 74 2291 1165851 455058 455077 GAACCCCTTTGGTCTAAGCAA 80 2292 1165863 463777 463796
TATAGATTGGGCTTTAGAGG 76 2293 1165865 464560 464579 ATGCCCTGTAGTATGTGGAT 74 2294 1165877 473792 473811
GTAATGCTGTTGTACACTAG 59 2295 1165893 499914 499933 TTTTGCATAATAGGAGGTCC 62 2296 1165899 503942 503961
ACCAACCTTAAATAGTAGGA 61 2297 1165907 506780 506799 TATACTCCAGGTTGTGGAGG 55 2298 1165910 507903 507922
CATTCGATACCTGCTTTTGT 71 2299

TABLE-US-00038 TABLE 37 Reduction of UBE3A-ATS RNA by 8,000 nM 5-10-5 MOE gapmers with mixed PO/PS internucleoside linkages in vitro SEQ ID NO: 1 NO: 1 UBE3A- SEQ Compound Start Stop ATS ID Number Site Site Sequence (5' to 3') (% UTC) NO 1065438 449442 449461 ATGATCATACTGGAGCCAGG 65 1936 1178376 220314 220333 CGATCCAAACAAGCACCCCT 80 2300 456683 456702 1178378 220316 220335 ATCGATCCAAACAAGCACCC 69 2301 456685 456704 1178380 220319 220338 ATCATCGATCCAAACAAGCA 55 2302 456688 456707 1178385 456754 456773 CCGCACATCTGGACCTCAGA 97 2303 1178387 456756 456775 GGCCGCACATCTGGACCTCA 89 2304 1178389 456759 456778 GAGGGCCGCACATCTGGACC 94 2305 1178391 456761 456780 TGGAGGGCCGCACATCTGGA 83 2306 1178393 456763 456782 ATTGGAGGGCCGCACATCTG 93 2307 1178395 456765 456784 CTATTGGAGGGCCGCACATC 63 2308 1178397 456767 456786 TCCTATTGGAGGGCCGCACA 90 2309 1178399 456770 456789 TTGTCTATTGGAGGGCCGC 64 2310 1178401 220403 220422 ATTTGTCTATTGGAGGGCC 101 2311 456772 456791 1178403 220452 220471 TATGTTGTCCTGAGGGATC 78 2312 456821 456840 1178405 220456 220475 GTACTATGTTGTCCTGAGG 80 2313 456825 456844 1178407 456863 456882 GCCCTCTTCGAAGAGATAGA 90 2314 1178409 456865 456884 CTGCCCTCTTCGAAGAGATA 58 2315 1178410 456913 456932 GTCATAGGCACACTAAAGT 62 2316 1178412 456915 456934 GGGTCACTAGGCACACTAAA 76 2317 1178414 456922 456941 AAGTCTTGGGTCACTAGGCA 63 2318 1178416 456924 456943 GCAAGTCTTGGGTCACTAGG 81 2319 1178419 457236 457255 ATGGCTTAGCTACTACCCC 68 2320 1178421 457239 457258 ATAATGGCTTAGCTACTCAC 60 2321 1178424 457287 457306 ACGACCTAAAGACCTAGCAA 49 2322 1178427 457297 457316 GATGTACAAAACGACCTAAA 60 2323 1179741 448298 448317 ACAAGTTTCGCAATAAGATT 55 2324 1179743 448414 448433 GCAAGGGTCTTAGTGCCAAA 78 2325 1179744 448450 448469 GTAGGACTTACTAAATCATC 75 2326 1179746 448454 448473 GCCAGTAGGACTTACTAAAT 57 2327 1179748 448456 448475 GTGCCAGTAGGACTTACTAA 62 2328 1179750 448629 448648 AGGATTGCATCACATGTGTC 56 2329 1179752 448714 448733 AGCGCATGTAGCAAAATTCC 82 2330 1179754 448716 448735 TAGCGCATTGAGCAAAATT 76 2331 1179756 448718 448737 CTGATCGCATTGAGCAAAA 83 2332 1179758 448720 448739 AACTGTAGCGCATTGAGCAA 52 2333 1179760 448723 448742 GATAACTGTAGCGCATTAG 85 2334 1179762 448726 448745 CACGATAAGTGTAGCGCAT 38 2335 1179764 448728 448747 ACCACGATAACTGTAGCGCA 67 2336 1179766 448731 448750 TAGACCACGATAACTGTAGC 80 2337 1179768 448733 448752 TTTAGACCACGATAACTGTA 54 2338 1179771 448738 448757 GTATATTTAGACCACGATAA 58 2339 1179773 448741 448760 TCTGTATATTAGACCACGA 49 2340 1179775 448831 448850 GCAAGGACCCTTAAGTCATA 72 2341 1179777 448836 448855 CAATTGCAAGGACCCTTAAG 99 2342 1179779 448842 448861 TTATCCCAATTGCAAGGACC 49 2343 1179780 448920 448939 AGTCAGGCACCAGATTGCTC 112 2344 1179782 448949 448968 AACGAAGCCAAAGGTACTTG 80 2345 1179784 449036 449055 CGGCCCTGACTGTCATTCA 71 2346 1179786 449038 449057 AACGGCCCTGACTGTCATT 53 2347 1179788 449040 449059 GGAACGGCCCTGACTGTCAT 95 2348 1179790 449043 449062 GAAGGAACGGCCCTGACTGT 70 2349 1179792 449045 449064 CTGAAGGAACGGCCCTGACT 105 2350 1179794 449047 449066 GCCTGAAGGAACGGCCCTGA 111 2351 1179796 449051 449070 TGTGGCCTGAAGGAACGGCC 100 2352 1179798 449054 449073 TCATGTGGCCTGAAGGAACG 68 2353 1179800 449059 449078 GATAATCATGTGGCCTGAAG 49 2354 1179802 449062 449081 AACGATAATCATGTGGCCTG 86 2355 1179804 449064 449083 ATAACGATAATCATGTGGCC 52 2356 1179806 449096 449115 GTATGCTAGTCACTCATTGA 91 2357 1179808 449102 449121 TGGCAAGTATGCTAGTCACT 12 2358 1179810 449104 449123 CCTGGCAAGTATGCTAGTCA 76 2359 1179812 449107 449126 GATCCTGGCAAGTATGCTAG 103 2360 1179814 449109 449128 TAGATCCTGGCAAGTATGCT 71 2361 1179816 449111 449130 TGTAGATCCTGGCAAGTATG 79 2362 1179819 449158 449177 TCAACATTATATCGATGCAA 66 2363 1179821 449162 449181 CTTGTCAACATTATATCGAT 61 2364 1179823 449164 449183 AACTGTCAACATTATATCG 79 2365 1179824 448294 448313 GTTTCGCAATAAGATTCCAT 98 2366 1179835 457231 457250 TTAGTACTACCCCTGTTT 74 2367 1179836 457241 457260 AGATAATGGCTTAGCTACTC 63 2368 1179837 457290 457309 AAAACGACCTAAAGACCTAG 92 2369 1179839 457384 457403 AGGATTCAATAGTGTAGGTG 25 2370 1179841 457502 457521 GCACACCTGGATTTCCTACT 33 2371 1179843 457505 457524 TTGGCACACCTGGATTTCAC 28 2372 1179844 457674 457693 ATAACCTACCTTGTGTCAC 65 2373 1179846 448600 448619 TTGAGAAGAAAACCTATCG 82 2374 1179848 448736 448755 ATATTTAGACCACGATAACT 83 2375 1179850 449117 449136 CCCTCTTGTAGATCCTGGCA 51 2376

Example 3: Effect of 5-10-5 MOE Gapmer Modified Oligonucleotides on Human UBE3A-ATS RNA In Vitro, Single Dose
[0460] Modified oligonucleotides complementary to human UBE3A-ATS nucleic acid were tested for their effect on UBE3A-ATS RNA levels in vitro, essentially as described in Example 1, except that human UBE3A-ATS primer probe set LTS01075 (forward sequence GCCCGAAGTGCCTATTCTT, designated herein as SEQ ID NO: 5; reverse sequence TGGTCAGGAGAACATAGGCATAAA, designated herein as SEQ ID NO: 6; probe sequence ACTCCCAGGGTTGATGGGCTACATCC, designated herein as SEQ ID NO: 7) was used to measure RNA levels.

[0461] The modified oligonucleotides in the tables below are 5-10-5 gapmers. The gapmers are 20 nucleosides in length, wherein the central gap segment consists of ten 2'-β-D-deoxynucleosides and the 3' and 5' wing segments each consists of five 2'-MOE nucleosides. The sugar motif for the gapmer is (from 5' to 3'): eeeeeeeeeeeeeeeee; wherein 'd' represents a 2'-β-D-deoxyribose sugar moiety, and 'e' represents a 2'-MOE sugar moiety. Each internucleoside linkage of Compound IDs 617456, 617457, 617460, 617461, and 617557 is a phosphorothioate internucleoside linkage. All other compounds have an internucleoside linkage motif of: sooooooooooooo, wherein each "s" represents a phosphorothioate internucleoside linkage and each "o" represents a phosphodiester internucleoside linkage.

[0462] All cytosine residues are 5-methylcytosines.

[0463] "Start site" indicates the 5'-most nucleoside to which the modified oligonucleotide is complementary in the target nucleic acid. "Stop site" indicates the 3'-most nucleoside to which the modified oligonucleotide is complementary in the target nucleic acid. Each modified oligonucleotide listed in the Tables below is 100% complementary to SEQ ID NO: 1.

[0464] Reduction of UBE3A-ATS RNA is presented in the tables below as percent UBE3A-ATS RNA amount relative to untreated control (UTC) cells. The values marked with an asterisk (*) indicate that the modified oligonucleotide is complementary to the amplicon region of the primer probe set. Additional assays may be used to measure the potency and efficacy of the modified oligonucleotides complementary to the amplicon region.

TABLE-US-00039	TABLE	38	Reduction of UBE3A-ATS RNA by 7,000 nM 5-10-5 MOE gapmers with PS internucleoside linkages in vitro (35,000 cells/well)	SEQ ID	SEQ ID	UBE3A- SEQ Compound NO:	1 NO:	1	ATS ID	Number	Start	Site	Stop	Site
	Sequence (5' to 3')	(% UTC)	NO	617456	463905	463924	CAACTGTTACCAAGACTTCA	25	32	617457	465003	465022		
	ATCTGCGAATAGAGGCCCTC	14	33	617460	466539	466558	TAGGTTGCATAAAGCCAGGC	22	36	617461	466981	467000		
	CACACATCTTGTTCCTCAA	15	37	617557	483977	483996	ATCATGTGCATACCCAGGGT	17	172	699779	463905	463924		
	CAACTGTTACCAAGACTTCA	42	32	749897	461087	461106	AATAATATTACTGCCAAATG	85	2676	749898	461222	461241		
	TGAATCAATTTTCAATATTT	104	2677	749899	461325	461344	TTATAAACTTCAGCAGAGTC	41	2678	749900	461397	461416		
	GAGTTGTTATTTGCATTAAG	44	2679	749901	461431	461450	ATTAGCACTTCTATAACTGG	49	2680	749902	461442	461461		
	GCTCCTGACAAATTAGCACT	91	1329	749903	461455	461474	TCTTGTAAGGCAGCTCCTG	64	2681	749904	461467	461486		
	TACTCATGATCTTGGTA	54	2682	749905	461516	461535	TAGACTAGTTGAAATCGGAA	46	2683	749906	461556	461575		
	AATTCTAAAAGTCTCTCTT	69	2684	749907	461590	461609	TGTGAGATAATTCAGGAGGT	11	2685	749908	461810	461829		
	ATTCACCTTATAAACACTGA	68	2686	749909	461895	461914	CCCACAATTTGGTCCCATTG	39	2687	749910	462029	462048		
	GAATAGGGCTCTGCTTATTT	57	2688	749911	462064	462083	TTTTATGGCCCTCCCATCAG	81	2689	749912	462098	462117		
	CATGAATTTAATCTTTAAA	80	2690	749913	462126	462145	CATTGTGGAATTAATTAAC	54	2691	749914	462141	462160		
	TTTCTAATTCATATCATTTG	59	2692	749915	462149	462168	TCCTCTTATTCTAATTCAA	54	2693	749916	462159	462178		
	CAAGAGATATTCCTCTTATT	41	2694	749917	462212	462231	CAATAAATAGGTCAGAAATG	84	2695	749918	462406	462425		
	CTAAGTTTCTTAAGGTAAAA	65	2696	749919	462607	462626	CATTTTCAAATATTGGTATT	77	2697	749920	462625	462644		
	TATGATTTTCCCTCCTACA	51	2698	749921	462626	462645	GTAATGATTTGCCCTCCTAC	26	1702	749922	462674	462693		
	CTTTTAAATAATTTTCTCT	66	2699	749923	462992	463011	AAAATGTTGGCATACTTTT	70	2700	749924	462993	463012		
	AAAAATGTTGGCATACTTT	69	2701	749925	463248	463267	CCTGGGTATTGCTGTCCAAA	36	2702	749926	463307	463326		
	TATGTTCTTAAGGAATAATG	113	2703	749927	463318	463337	TTCTTTGCATTTATGTTTCT	40	2704	749928	463319	463338		
	TTTCTTTGCATTTATGTTCC	36	2705	749929	463453	463472	TTACTCTGACTTTCCAGAAG	70	2706	749930	463474	463493		
	GGAGTAGATTTTGGAGTTT	43	2707	749931	463512	463531	TCAACTATTTCTATCAAGGC	22	2708	749932	463519	463538		
	ATTAATTTCAACTATTTCTA	87	2709	749933	463601	463620	TCTTACTGATTCAGCCATT	27	2710	749934	463663	463682		
	TCTCAGCACTAGGGAGAAAA	56	2711	749935	463695	463714	AGTGGTTGCTATCCTGCTAA	71	2000	749936	463788	463807		
	AATATAAATCCTATGATTG	71	2712	749937	463805	463824	CTGAGTCAGTCCAAATGAAT	40	2713	749938	463840	463859		
	TACCTTGAAATTGAGATTT	50	2714	749939	463853	463872	TCTTTTGACCAATACCTTG	34	2715	749940	463871	463890		
	TCAACAATTGCCATGGATT	78	2716	749941	463895	463914	CAAGACTTCAGGCAGTTGTG	54	2717	749942	463898	463917		
	TACCAAGACTTCAGGCAGTT	51	2718	749943	463900	463919	GTTACCAAGACTTCAGGCAG	35	2719	749944	463903	463922		
	ACTGTTACCAAGACTTCAGG	37	2720	749945	463907	463926	CCCACTGTTACCAAGACTT	27	2721	749946	463910	463929		
	TATCCCACTGTTACCAAGA	37	2722	749947	463912	463931	TTTATCCCACTGTTACCAA	29	2723	749948	463915	463934		
	TTGTTTATCCCACTGTTAC	42	2724	749949	463926	463945	TGATCAGCTTCTTGTTTATC	29	2725	749950	464355	464374		
	CCAGAGCATAAAAGGAAAGC	87	2726	749951	464377	464396	TCACAACCTTCTCTTAGCTT	28	2727	749952	464525	464544		
	CAGTTAGGTTAGTGACAGA	44	2728	749953	464530	464549	CTGCTCAGTTAGGTTAGTGC	31	2729	749954	464542	464561		
	ATACTGAAGTCTCTGCTCAG	53	2730	749955	464738	464757	TGTGCCATATTTTCTATTT	41	2731	749956	464993	465012		
	AGAGGCCCTCTTGTTTCAAT	36	2732	749957	464996	465015	AATAGAGGCCCTCTTGTTTC	42	2733	749958	464998	465017		
	CGAATAGAGGCCCTCTTGTT	59	2734	749959	465001	465020	CTGCGAATAGAGGCCCTCTT	35	2735	749960	465003	465022		
	ATCTGCGAATAGAGGCCCTC	30	33	749961	465005	465024	AAATCTGCGAATAGAGGCC	33	2736	749962	465008	465027		
	CTCAAATCTGCGAATAGAGG	34	2737	749963	465010	465029	TGCTCAAATCTGCGAATAGA	67	2738	749964	465013	465032		
	GCCTGCTCAAATCTGCGAAT	24	2739	749965	465050	465069	AGTTGACATATCTTCAAGTT	69	2740	749966	465057	465076		
	TAATCTCAGTTGACATATCT	55	2741	749967	465059	465078	GATAATCTCAGTTGACATAT	42	2742					

Example 4: Effect of Modified Oligonucleotides on Human UBE3A-ATS RNA In Vitro, Multiple Doses

[0465] Modified oligonucleotides selected from the examples above were tested at various doses in human IPS derived iCell GABANeurons (Cellular Dynamics). Cells were plated at a density of 35,000-60,000 cells per well, maintained per manufacturer instructions, and treated with modified oligonucleotides by free uptake at various concentrations as specified in the tables below. After a treatment period of approximately 6 days, total RNA was isolated from the cells and UBE3A-ATS RNA levels were measured by quantitative real-time PCR. Human UBE3A-ATS primer probe set RTS4796, described hereinabove, was used to measure RNA levels. UBE3A-ATS RNA levels were normalized according to total RNA content, as measured by RIBOGREEN®. Reduction of UBE3A-ATS RNA is presented in the tables below as percent UBE3A-ATS RNA amount relative to untreated control (UTC) cells.

[0466] Where possible, the half maximal inhibitory concentration (IC₅₀) of each modified oligonucleotide was calculated using a linear regression on a log/linear plot of the data in excel. In some cases, an IC₅₀ could not be reliably calculated and the data point is marked as "N.C.". Values marked with "N.D." indicate that a value was not determined in this experiment.

TABLE-US-00040	TABLE	39	Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (60,000 cells/well)	Compound	UBE3A-ATS	expression (% UTC)	Number	740.7 nM	2,222 nM	6,666 nM	20,000 nM	617456	43	35	16	11	617457	38	32	27	18	617460	28	29	24	15	617461	40	19	11	12	617557	23	18	20	16
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TABLE-US-00041	TABLE	40	Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (60,000 cells/well)	Compound	UBE3A-ATS
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expression (% UTC) Number 740.7 nM 2,222 nM 6,666 nM 20,000 nM 617456 48 38 17 15 617457 35 27 18 18 617460 29 24 18 617461 38 24 13 12 617557 23 17 14 13

TABLE-US-00042 TABLE 41 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (60,000 cells/well) UBE3A-ATS expression (% UTC) Compound 2,222 IC.sub.50 Number 740.7 nM nM 6,666 nM 20,000 nM (μM) 617456 41 26 15 20 N.C. 617557 69 43 24 30 1.9 749969 46 15 12 5 N.C. 749991 43 27 19 13 N.C. 750028 54 31 24 15 N.C. 750030 57 37 23 14 1.0 750032 40 18 14 7 N.C. 750326 76 60 38 25 3.8 750329 60 57 33 16 2.1 750344 63 77 49 29 5.6 750350 67 71 59 35 8.6 750359 110 56 38 41 6.7 750360 69 47 39 38 3.3 750365 58 60 38 22 2.4 750366 87 58 44 24 4.6 750386 76 57 42 40 5.3 750517 30 25 12 8 N.C. 750519 45 23 10 8 N.C. 750542 96 71 55 41 9.9 750549 92 54 40 24 4.3

TABLE-US-00043 TABLE 42 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (60,000 cells/well) Compound UBE3A-ATS expression (% UTC) IC.sub.50 Number 740.7 nM 2,222 nM 6,666 nM 20,000 nM (μM) 617557 85 57 37 25 4.1 749861 47 31 25 14 N.C. 749863 83 52 34 23 3.5 749869 46 30 18 11 N.C. 749882 84 60 40 27 4.5 749885 61 53 31 20 2.0 749893 63 36 25 13 1.3 749894 28 21 10 11 N.C. 750040 78 59 46 32 5.1 750051 47 18 21 23 N.C. 750092 80 54 40 28 3.9 750100 82 67 59 51 N.C. 750270 43 34 25 21 N.C. 750292 58 37 28 26 1.0 750312 57 54 41 30 2.3 750325 101 94 68 46 18.4 750413 98 60 35 28 4.9 750416 94 48 38 15 3.7 750430 85 47 33 24 3.3 750431 136 125 90 62 N.C.

TABLE-US-00044 TABLE 43 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (35,000 cells/well) Compound UBE3A-ATS expression (% UTC) IC.sub.50 Number 740.7 nM 2,222 nM 6,666 nM 20,000 nM (μM) 617459 118 69 33 20 5.4 617470 84 51 28 13 2.9 617473 44 33 30 20 N.C. 617536 100 54 57 24 5.9 617547 122 52 29 18 4.6 617593 81 75 32 21 4.2 749794 89 102 66 52 N.C. 750418 135 132 105 51 N.C. 750439 98 60 44 26 5.4 750452 113 94 78 51 N.C.

TABLE-US-00045 TABLE 44 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (35,000 cells/well) Compound UBE3A-ATS expression (% UTC) IC.sub.50 Number 740.7 nM 2,222 nM 6,666 nM 20,000 nM (μM) 617557 129 53 32 28 5.5 749796 84 31 29 17 2.3 749816 94 56 46 22 4.8 749907 65 57 21 16 2.1 749921 50 47 25 18 1.0 749931 246 39 32 12 6.3 749933 51 41 26 11 0.9 749937 74 46 27 16 2.3 749944 116 N/A 74 40 15.3 749956 136 74 18 27 5.7 749964 75 80 35 17 4.2 750131 73 50 36 25 2.9 750139 35 29 21 15 N.C. 750140 27 27 13 11 N.C. 750141 52 61 25 24 1.6 750196 107 95 129 47 N.C. 750210 53 38 26 23 0.8 750214 34 70 47 41 N.C. 750228 109 105 95 70 N.C. 750242 77 70 36 15 3.8

TABLE-US-00046 TABLE 45 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (35,000 cells/well) Compound UBE3A-ATS expression (% UTC) IC.sub.50 Number 740.7 nM 2,222 nM 6,666 nM 20,000 nM (μM) 617557 80 59 31 13 3.1 750519 87 67 41 27 5.2 1065438 120 90 88 95 N.C. 1065582 65 61 41 29 3.4 1065597 142 135 111 103 N.C. 1065599 63 47 47 26 2.5 1065613 96 70 65 39 11.5 1065631 60 61 48 26 3.4 1065644 64 65 42 27 3.8 1065645 68 50 36 19 2.4 1065646 52 55 34 16 1.5 1065676 50 38 33 23 N.C. 1065690 56 53 38 24 1.9 1065754 74 56 57 27 5.1 1065817 94 76 42 38 7.6 1065899 59 56 46 30 3.0 1066072 133 108 86 63 N.C. 1066249 73 60 70 48 N.C. 1066378 116 114 87 52 N.C.

TABLE-US-00047 TABLE 46 Dose-dependent percent reduction of human UBE3A-ATS RNA by modified oligonucleotides in vitro (42,000 cells/well) Compound UBE3A-ATS expression (% UTC) IC.sub.50 Number 740.7 nM 2,222 nM 6,666 nM 20,000 nM (μM) 617557 83 54 31 13 3.1 750519 63 48 43 26 2.4 1065578 77 36 36 24 2.4 1065579 55 58 30 66 N.C. 1065595 75 60 32 19 3.1 1065642 52 58 40 27 2.0 1065672 85 67 42 18 4.5 1065674 64 69 40 22 3.5 1065719 76 62 46 38 6.1 1065750 38 28 25 28 N.C. 1065766 78 52 32 37 3.7 1065768 86 66 43 30 5.5 1065799 66 52 41 34 3.2 1065863 54 60 46 27 2.6 1065894 82 100 52 52 N.C. 1066037 66 70 56 62 N.C. 1066119 82 116 101 72 N.C. 1066375 116 98 68 77 N.C. 1066423 39 70 63 46 N.C.

TABLE-US-00048 TABLE 47 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (42,000 cells/well) Compound UBE3A-ATS expression (% UTC) IC.sub.50 Number 740.7 nM 2,222 nM 6,666 nM 20,000 nM (μM) 617557 65 52 23 8 1.9 1065324 86 90 67 43 18.4 1065369 107 91 108 83 N.C. 1065465 97 96 97 100 N.C. 1065513 156 104 66 70 N.C. 1065558 116 125 92 112 N.C. 1065592 85 84 50 23 6.5 1065624 87 71 39 20 4.7 1065667 64 59 35 21 2.7 1065747 91 65 55 N/A N.C. 1065955 84 55 43 23 4.1 1066002 94 75 68 50 N.C. 1066003 90 73 49 43 9.5 1066034 137 120 101 67 N.C. 1066201 98 110 84 79 N.C. 1066273 101 95 108 104 N.C. 1066359 111 96 101 81 N.C. 1066420 70 43 27 33 2.2

TABLE-US-00049 TABLE 48 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (42,000 cells/well) Compound UBE3A-ATS expression (% UTC) IC.sub.50 Number 740.7 nM 2,222 nM 6,666 nM 20,000 nM (μM) 617557 70 46 23 10 1.9 1065654 96 87 54 43 11.9 1065680 78 63 41 28 4.5 1065686 63 64 62 24 5.1 1065735 102 66 41 41 7.4 1065785 81 74 52 50 14.7 1065829 64 59 45 31 3.7 1065858 69 49 28 23 2.2 1065859 52 36 28 20 N.C. 1065901 89 69 73 55 N.C. 1065914 108 111 82 74 N.C. 1065977 106 87 101 76 N.C. 1066009 88 71 130 84 N.C. 1066046 84 59 43 36 5.5 1066089 46 71 58 41 N.C. 1066217 67 64 46 41 6.5 1066221 59 49 39 31 2.0 1066311 93 75 73 49 N.C. 1066377 94 73 51 35 7.9 1066396 100 75 82 54 N.C.

TABLE-US-00050 TABLE 49 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (42,000 cells/well) Compound UBE3A-ATS expression (% UTC) IC.sub.50 Number 740.7 nM 2,222 nM 6,666 nM 20,000 nM (μM) 617557 66 47 21 6 1.7 750519 75 58 36 25 3.5 1065272 90 63 38 33 5.4 1065576 21 28 19 13 N.C. 1065590 74 49 31 27 2.8 1065591 83 60 29 13 3.2 1065607 68 47 28 17 2.1 1065608 72 38 18 13 1.7 1065623 46 28 21 13 N.C. 1065669 89 75 57 28 7.6 1065685 64 34 19 20 1.2 1065795 100 86 56 38 10.8 1065810 62 46 23 22 1.6 1065812 77 67 36 24 4.1 1065826 108 93 54 43 12.1 1065937 86 61 60 37 8.7 1065953 77 46 34 29 3.0 1065954 78 62 45 38 6.1 1066097 133 121 119 125 N.C.

TABLE-US-00051 TABLE 50 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (42,000 cells/well) Compound UBE3A-ATS expression (% UTC) IC.sub.50 Number 740.7 nM 2,222 nM 6,666 nM 20,000 nM (μM) 617557 39 43 25 13 N.C. 750519 44 36 34 17 N.C. 1065605 50 48 29 19 N.C. 1065619 95 73 47 5 4.6 1065621 5 63 44 21 4.6 1065635 48 29 19 4 N.C. 1065651 13 8 29 10 N.C. 1065696 77 71 61 7 4.7 1065712 58 40 34 20 1.3 1065713 4 4 7 6 N.C. 1065728 73 54 42 25 3.5 1065823 7 9 10 N/A N.C. 1065840 73 61 41 32 4.4 1065856 78 49 37 24 3.2 1065857 10 61 49 28 5.1 1065889 58 81 61 62 N.C. 1065903 85 65 60 6 4.6 1065920 53 42 40 26 1.0 1066350 89 101 78 21 11.3

TABLE-US-00052 TABLE 51 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (42,000 cells/well) Compound UBE3A-ATS expression (% UTC) IC.sub.50 Number 740.7 nM 2,222 nM 6,666 nM 20,000 nM (μM) 617557 14 16 3 0 N.C. 750519 31 27 22 12 N.C. 1065296 78 56 45 7 3.2 1065330 13 5 4 3 N.C. 1065586 5 44 22 11 N.C. 1065600 79 55 36 4 2.9 1065616 84 75 65 5 5.4 1065708 9 8 6 N/A N.C. 1065709 6 4 4 4 N.C. 1065710 52 33 16 9 N.C. 1065821 110 58 29 16 4.3 1065868 73 42 28 6 2.0 1065902 8 60 35 24 3.6 1065932 51 42 35 17 0.9 1065947 65 48 38 10 2.1 1066076 81 76 41 22 5.0 1066092 80 49 40 27 3.6 1066253 0 71 31 21 4.5 1066429 28 17 6 6 N.C.

TABLE-US-00053 TABLE 52 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (42,000 cells/well) Compound UBE3A-ATS expression (% UTC) IC.sub.50 Number 740.7 nM 2,222 nM 6,666 nM 20,000 nM (μM) 617557 37 17 29 12 N.C. 750519 50 40 40 20 N.C. 1065295 78 73 74 49 N.C. 1065473 31 29 16 11 N.C. 1065503 105 107 107 77 N.C. 1065561 66 44 21 13 1.7 1065593 59 45 30 6 1.5 1065609 59 50 34 6 1.7 1065625 70 56 41 22 3.2 1065641 43 31 18 7 N.C. 1065671 55 49 30 8 1.4 1065678 75 78 54 33 8.0 1065765 64 46 28 12 1.8 1065791 7 58 44 30 4.1 1065806 24 72 82 74 N.C. 1065813 12 11 11 7 N.C. 1065924 107 94 79 41 18.4 1066011 94 85 83 50 N.C. 1066220 24 83 75 62 N.C.

TABLE-US-00054 TABLE 53 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (40,000 cells/well) Compound UBE3A-ATS expression (% UTC) IC.sub.50 Number 740.7 nM 2,222 nM 6,666 nM 20,000 nM (μM) 750519 83 53 21 14 2.8 1165521 55 38 32 27 0.9 1165524 42 33 26 14 N.C. 1165536 47 39 32 21 N.C. 1165545 62 49 32 33 2.0 1165552 53 35 24 16 0.8 1165553 55 34 24 18 0.8 1165554 63 55 29 19 2.1 1165555 53 43 21 22 0.9 1165562 72 69 35 23 3.8 1165577 65 59 33 28 2.8 1165588 78 57 33 38 4.3 1165590 44 41 30 21 N.C. 1165593 85 92 83 50 N.C. 1165611 47 39 25 21 N.C. 1165724 80 71 57 54 N.C. 1165788 66 71 42 25 4.2 1165798 73 81 86 52 N.C. 1179842 78 37 27 20 2.2

Example 5: Effect of Modified Oligonucleotides on Human UBE3A-ATS RNA In Vitro, Multiple Doses

[0467] Modified oligonucleotides selected from the examples above were tested at various doses in human IPS derived ReproNeuro™ Neurons (ReproCELL). Cells were plated at a density of 20,000 cells per well, maintained per manufacturer instructions, and treated with modified oligonucleotides by free uptake at various concentrations as specified in the tables below. After a treatment period of approximately 5 days, total

RNA was isolated from the cells and UBE3A-ATS RNA levels at quantified real-time PCR. Human UBE3A-ATS primer probe set RTS4796, described hereinabove, was used to measure RNA levels. UBE3A-ATS RNA levels were normalized according to total RNA content, as measured by RIBOGREEN®. Reduction of UBE3A-ATS RNA is presented in the tables below as percent UBE3A-ATS RNA amount relative to untreated control (UTC) cells.

[0468] The half maximal inhibitory concentration (IC.sub.50) of each modified oligonucleotide was calculated using a linear regression on a log/linear plot of the data in excel. In cases where IC.sub.50 could not be calculated, IC.sub.50s are marked as N.C. (Not Calculated).
TABLE-US-00055 TABLE 54 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro Compound UBE3A-ATS expression (% UTC) IC.sub.50 Number 740.7 nM 2,222 nM 6,666 nM 20,000 nM (μM) 750519 73 54 19 11 2.3 1165523 75 60 43 33 4.7 1165533 60 50 18 33 1.5 1165538 64 47 34 26 2.1 1165550 65 62 47 32 4.5 1165563 69 51 41 29 3.1 1165586 62 50 35 32 2.3 1165596 88 72 69 43 16.4 1165608 88 89 62 54 N.C. 1165616 89 88 80 74 N.C. 1165694 71 57 55 50 15.0 1165737 71 55 53 46 9.1 1165827 72 59 65 42 13.6 1165855 60 60 37 19 2.5 1165897 78 76 54 50 N.C. 1179808 101 100 85 75 N.C. 1179839 58 41 26 23 1.2 1179841 68 45 38 26 2.4 1179843 52 57 36 21 1.7

Example 6: Effect of Modified Oligonucleotides on Human UBE3A-ATS RNA and UBE3A RNA In Vitro, Multiple Doses
[0469] Modified oligonucleotides selected from the examples above were tested at various doses in 10-week differentiated human neuronal cells derived from Angelman Syndrome patient derived IPS cells (Protocols and cells described in Chamberlain S.J., et al., *Induced pluripotent stem cell models of the genomic imprinting disorders Angelman and Prader-Willi syndromes*. PNAS, 2010. 41: 17668-17673). At the end of the 10-week differentiation period, cells were treated with modified oligonucleotides by free uptake at various concentrations as specified in the tables below. After a treatment period of approximately 6 days, total RNA was isolated from the cells. Both UBE3A-ATS RNA and UBE3A RNA levels were measured by quantitative real-time PCR. Human UBE3A-ATS primer probe set RTS4796 was used to measure UBE3A-ATS RNA levels as described above. Human UBE3A primer probe set RTS35984 (forward sequence CACCCTGATGTCACCGAATG, designated herein as SEQ ID NO: 8; reverse sequence GCGTTCATTAGATGCTTTGCAG, designated herein as SEQ ID NO: 9; probe sequence ACTGAGGTTCTCTGATCTTTACAAGCTG, designated herein as SEQ ID NO: 10) was used to measure UBE3A RNA levels. RNA levels were adjusted according to total RNA content, as measured by RIBOGREEN®. Reduction of UBE3A-ATS RNA or induction of UBE3A RNA is presented in the tables below as percent UBE3A-ATS RNA amount or percent UBE3A RNA relative to untreated control (UTC) cells. Values marked with “N.D.” indicate that a value was not determined in this experiment.

[0470] Several modified oligonucleotides were found to reduce UBE3A-ATS RNA accompanied by concurrent increase in UBE3A RNA in Angelman's patient IPS cell derived neurons.
TABLE-US-00056 TABLE 55 Reduction of UBE3A-ATS RNA and increase of UBE3A RNA in Angelman Syndrome IPS cell derived neurons UBE3A-ATS RNA UBE3A RNA (% UTC) (RTS4796) (% UTC) (RTS35984) Compound 741 2222 6667 741 2222 6667 ID nM nM nM nM nM nM 617557 17 16 N.D. 142 151 N.D. 750140 7 N.D. N.D. 226 N.D. N.D. 750139 35 10 3 104 196 194 750131 29 25 N.D. 49 53 N.D. 750141 22 N.D. N.D. 89 N.D. N.D. 750242 42 N.D. N.D. 45 N.D. N.D. 750210 49 14 6 122 101 166 750214 53 56 13 87 154 85 749907 16 13 7 133 298 205 749931 35 17 11 152 271 260 749964 27 15 6 144 160 202 749921 25 12 N.D. 73 98 N.D. 749933 42 10 13 116 92 210 749956 51 33 N.D. 94 103 N.D. 749937 52 27 7 104 131 137 749796 73 67 N.D. 81 79 N.D. 749816 75 N.D. 19 63 N.D. 27 749969 33 14 10 171 249 305 749991 32 15 15 193 165 276

TABLE-US-00057 TABLE 56 Reduction of UBE3A-ATS RNA and increase of UBE3A RNA in Angelman Syndrome IPS cell derived neurons UBE3A-ATS RNA UBE3A RNA (% UTC) (RTS4796) (% UTC) (RTS35984) Compound 741 2222 6667 741 2222 6667 ID nM nM nM nM nM nM 617557 54 51 39 144 365 252 750140 22 37 N.D. 345 328 N.D. 750517 34 30 19 269 328 418 750549 60 72 72 337 225 309 750542 64 70 22 98 113 234 617456 58 35 26 142 176 253 750519 33 26 18 374 440 447 750360 93 113 59 151 221 204 750359 32 31 27 267 134 229 750386 35 40 17 182 282 320 750366 77 45 37 87 122 185 750344 51 50 30 91 139 150 750326 67 49 31 106 108 166 750350 68 66 102 111 132 138 750365 74 63 37 122 160 183 750329 88 82 N.D. 154 148 N.D. 750028 81 N.D. 14 324 N.D. 448 750030 87 30 22 221 307 373 750032 47 17 13 202 323 254

TABLE-US-00058 TABLE 57 Reduction of UBE3A-ATS RNA and increase of UBE3A RNA in Angelman Syndrome IPS cell derived neurons UBE3A-ATS RNA UBE3A RNA (% UTC) (RTS4796) (% UTC) (RTS35984) Compound 741 2222 6667 741 2222 6667 ID nM nM nM nM nM nM 617557 28 54 N.D. 318 411 N.D. 750140 19 20 10 619 542 132 750040 38 49 25 171 192 318 750051 21 21 13 106 544 416 750092 40 32 12 175 507 118 749894 12 19 6 143 2085 605 749869 10 11 13 376 802 1504 749882 34 31 19 502 907 926 749863 45 42 22 326 2584 927 749893 34 10 7 215 428 1023 749885 22 12 10 105 233 614 749861 19 11 8 209 236 466 750292 28 15 13 123 318 645 750270 18 14 10 267 578 584 750312 39 19 12 87 289 420 750413 46 49 20 123 375 143 750416 49 N.D. N.D. 119 N.D. N.D. 750430 33 21 12 329 359 241 750439 40 31 19 188 197 183

TABLE-US-00059 TABLE 58 Reduction of UBE3A-ATS RNA and increase of UBE3A RNA in Angelman Syndrome IPS cell derived neurons UBE3A-ATS RNA UBE3A RNA (% UTC) (RTS4796) (% UTC) (RTS35984) Compound 741 2222 6667 741 2222 6667 ID nM nM nM nM nM nM 617557 60 39 N.D. 266 161 N.D. 617557 65 23 8 229 204 122 750140 18 29 N.D. 312 222 N.D. 617470 36 34 23 141 191 317 617473 29 37 20 91 162 87 617459 38 18 8 80 155 191 617547 31 19 28 52 145 226 617536 83 28 109 165 108 265 617593 67 142 107 100 73 83 750544 98 107 51 221 258 307 750554 65 72 88 57 80 55 750540 46 26 29 202 344 458 750567 62 49 31 86 108 227 749984 47 42 N.D. 68 99 277 750009 31 39 22 65 92 217 749865 44 N.D. 12 201 N.D. 249 749860 36 22 9 285 265 263 750006 32 17 66 241 329 513 582468 102 79 33 120 33 62 141923 150 169 100 135 126 53

TABLE-US-00060 TABLE 59 Reduction of UBE3A-ATS RNA and increase of UBE3A RNA in Angelman Syndrome IPS cell derived neurons UBE3A-ATS RNA UBE3A RNA Com- (% UTC) (RTS4796) (% UTC) (RTS35984) pound 741 2222 6667 20000 741 2222 6667 20000 ID nM nM nM nM nM nM 749894 20 13 11 10 301 200 159 546 749969 47 44 45 51 125 88 85 205 750032 16 17 12 17 215 190 182 236 750140 26 33 13 22 252 429 160 222 1065690 18 7 11 9 313 295 203 254 1065868 32 17 9 9 249 303 211 143 1065579 20 19 25 7 155 134 133 231 1065858 22 24 25 17 190 239 229 249 1065859 55 57 36 29 117 141 199 193 1065812 25 29 9 10 133 120 185 163 749860 35 21 14 11 129 250 196 278 1065593 18 10 7 9 228 181 211 261 1065953 15 11 12 10 148 269 243 221 1065856 15 18 11 9 292 286 103 234 1065937 21 26 12 10 93 205 208 1065728 35 24 20 12 132 143 144 198 750139 38 28 13 14 176 229 248 315 617557 55 23 30 33 166 206 193 226 750519 15 15 5 3 173 101 220 324

TABLE-US-00061 TABLE 60 Reduction of UBE3A-ATS RNA and increase of UBE3A RNA in Angelman Syndrome IPS cell derived neurons UBE3A-ATS RNA UBE3A RNA Com- (% UTC) (RTS4796) (% UTC) (RTS35984) pound 741 2222 6667 20000 741 2222 6667 20000 ID nM nM nM nM nM nM 1066092 138 160 98 55 247 253 254 111 1065902 92 85 55 42 168 231 203 304 1065840 57 54 37 22 209 176 141 307 1066253 77 37 52 36 182 167 184 236 1065785 93 35 41 54 155 198 225 240 1065821 40 19 20 20 220 288 215 276 750006 78 33 33 56 132 119 168 185 750028 36 17 16 19 173 235 189 216 617557 37 26 13 23 194 166 214 241

TABLE-US-00062 TABLE 61 Reduction of UBE3A-ATS RNA and increase of UBE3A RNA in Angelman Syndrome IPS cell derived neurons UBE3A-ATS RNA UBE3A RNA (% UTC) (RTS4796) (% UTC) (RTS35984) 741 2222 6667 20000 741 2222 6667 20000 Compound ID nM nM nM nM nM nM nM nM nM nM nM 750519 41 26 11 8 138 158 148 158 1065671 106 81 108 61 213 189 387 235 617557 69 52 37 35 129 132 210 195 1065686 101 101 81 88 145 105 118 164 750359 121 70 79 48 108 119 134 131 1065817 75 78 40 N.D. 119 174 131 N.D. 750386 100 65 58 50 106 155 190 128 749894 58 40 19 20 176 187 159 244 1065591 68 51 52 34 138 132 227 155 750140 178 165 133 129 129 117 98 142 750032 95 56 40 31 109 144 126 172 1065599 73 45 42 33 135 131 188 210 1065690 66 48 35 29 182 175 211 185 1065868 132 125 107 90 93 125 113 99 1065645 83 47 50 25 169 144 287 158 1065858 78 46 22 18 175 192 162 227 1065856 76 66 49 35 138 195 167 165 1065667 137 127 137 94 115 116 163 134 1066092 133 126 87 75 100 113 104 106

Example 7: Design and Synthesis of Modified Oligonucleotides Complementary to a Human UBE3A-ATS Nucleic Acid

[0471] Modified oligonucleotides were synthesized as indicated in the tables below.

[0472] The compounds in Table 62 are 4-10-6 MOE gapmers. The gapmers are 20 nucleosides in length, wherein the central gap segment consists of ten 2'- β -D-deoxynucleosides, the 5' wing segment consists of four 2'-MOE nucleosides, and the 3' wing segment consists of six 2'-MOE nucleosides. The sugar motif of the gapmers is (from 5' to 3'): eeeeddddddddeeeeee; wherein 'd' represents a 2'- β -D-deoxyriboxyl sugar moiety, and 'e' represents a 2'-MOE sugar moiety. The gapmers have an internucleoside linkage motif of (from 5' to 3'): soooossssssooooss; wherein "s" represents a phosphorothioate internucleoside linkage and "o" represents a phosphodiester internucleoside linkage. All cytosine residues are 5-methylcytosines.

TABLE-US-00063 TABLE 62 4-10-6 MOE gapmers with a mixed PO/PS internucleoside linkages complementary to human

UBE3A-ATS SEQ SEQ ID ID NO: 1 NO: 1 SEQ Compound Start Stop ID ID Site Site SEQUENCE NO 1263473 460996 461015

TTTTTCCATTTTCTCTTAG 2745 1263474 460997 461016 GTTTTCCATTTTCTCTTA 2746 1263475 460999 461018
GTGTTTTTCCATTTTCTCT 598 1263476 461000 461019 TGTGTTTTTCCATTTTCTC 2747 1263477 461001 461020
GTGTGTTTTTCCATTTTCT 2748 1263478 461002 461021 TGTGTGTTTTTCCATTTTCT 2749 1263479 465235 465254
TCATCTCGGGTATATAAAT 2750 1263480 465236 465255 GTCATCTCGGGTATATAAAT 1132 1263481 465237 465256
GGTCATCTCGGGTATATAA 1207 1263482 465238 465257 TGGTCATCTCGGGTATATAA 318 1263483 465239 465258
TTGGTCATCTCGGGTATATA 2473 1263484 465240 465259 ATTGGTCATCTCGGGTATAT 1655 1263485 465241 465260
TATTGGTCATCTCGGGTATA 2548 1263486 468985 469004 TCACCATTTTGACCTTCTTA 2751 1263487 468986 469005
TTCACCATTTTGACCTTCTT 2752 1263488 468987 469006 CTTACCATTTTGACCTTCT 2753 1263489 468988 469007
GCTTCACCATTTTGACCTTC 377 1263490 468989 469008 TGCTTCACCATTTTGACCTT 2754 1263491 468990 469009
CTGCTTCACCATTTTGACCT 2755 1263492 468991 469010 ACTGCTTCACCATTTTGACC 2756 1263494 464526 464545
TCAGTTAGGTTAGTGCACAG 2757 1263495 464527 464546 CTCAGTTAGGTTAGTGCACA 2758 1263496 464528 464547
GCTCAGTTAGGTTAGTGCAC 1504 1263497 464529 464548 TGCTCAGTTAGGTTAGTGCAC 2759 1263498 464530 464549
CTGCTCAGTTAGGTTAGTGC 2729 1263499 464531 464548 TCTGCTCAGTTAGGTTAGTG 2760 1263500 479994 480013
GAGCTATCTGTACAAAATGG 2761 1263501 479995 480014 TGAGCTATCTGTACAAAATG 2743 1263502 479996 480015
GTGAGCTATCTGTACAAAAT 2744 1263503 479997 480016 CGTGAGCTATCTGTACAAAAT 1142 1263532 483970 483989
GCATACCCAGGGTAGGATTC 765 1263534 483971 483990 TGCATACCCAGGGTAGGATT 1445 1263536 483972 483991
GTGCATACCCAGGGTAGGAT 766 1263539 483973 483992 TGTGCATACCCAGGGTAGGA 2762 1263541 483974 483993
ATGTGCATACCCAGGGTAGG 2595 1273009 457735 457754 GCCAGGTGCTTATATCTAT 2852 1273010 474393 474412
GGTCAACCAATTTGCTATTC 1809 1273011 474394 474413 AGGTCAACCAATTTGCTATT 2853 1273012 474396 474415
TTAGGTCAACCAATTTGCTA 2854 1273013 457736 457755 AGCCAGGTGCTTATATCTA 2855 1273014 478535 478554
AACGCAATGTATCAGGCAAC 2856 1273015 478536 478555 AAACGCAATGTATCAGGCAA 2857 1273016 478730 478749
GATCACATTACCCATCCGTT 2858 1273017 478731 478750 TGATCACATTACCCATCCGT 2859 1273018 478732 478751
CTGATCACATTACCCATCCG 2860 1273019 474395 474414 TAGGTCAACCAATTTGCTAT 2861 1273020 478733 478752
GCTGATCACATTACCCATCC 1291 1273021 478734 478753 TGCTGATCACATTACCCATC 2862 1273022 478735 478754
TTGCTGATCACATTACCCAT 2863

[0473] The compounds in Table 63 are 5-10-5 MOE gapmers. The gapmers are 20 nucleosides in length, wherein the central gap segment consists of ten 2'- β -D-deoxyribonucleosides and the 5' and 3' wing segments each consist of five 2'-MOE nucleosides. The sugar motif for the gapmers is (from 5' to 3'): eeeedddddddeeeeee; wherein 'd' represents a 2'- β -D-deoxyribosyl sugar moiety, and 'e' represents a 2'-MOE sugar moiety. The gapmers have an internucleoside linkage motif of (from 5' to 3'): soooossssssooss, wherein "s" represents a phosphorothioate internucleoside linkage and "o" represents a phosphodiester internucleoside linkage. All cytosine residues are 5-methylcytosines.

TABLE-US-00064 TABLE 63 5-10-5 MOE gapmers with a mixed PO/PS internucleoside linkages complementary to human

UBE3A-ATS SEQ SEQ ID ID NO: 1 NO: 1 SEQ Compound Start Stop ID ID Site Site SEQUENCE NO 1263451 460996 461015

TTTTTCCATTTTCTCTTAG 2745 1263452 460997 461016 GTTTTCCATTTTCTCTTA 2746 1263453 461000 461019
TGTGTTTTTCCATTTTCTC 2747 1263454 461001 461020 GTGTGTTTTTCCATTTTCT 2748 1263455 461002 461021
TGTGTGTTTTTCCATTTTCT 2749 1263456 465235 465254 TCATCTCGGGTATATAAAT 2750 1263460 468985 469004
TCACCATTTTGACCTTCTTA 2751 1263461 468986 469005 TTCACCATTTTGACCTTCTT 2752 1263462 468987 469006
CTTCACCATTTTGACCTTCT 2753 1263463 468989 469008 TGCTTCACCATTTTGACCTT 2754 1263464 468990 469009
CTGCTTCACCATTTTGACCT 2755 1263465 468991 469010 ACTGCTTCACCATTTTGACC 2756 1263466 464526 464545
TCAGTTAGGTTAGTGCACAG 2757 1263467 464527 464546 CTCAGTTAGGTTAGTGCACA 2758 1263468 464529 464548
TGCTCAGTTAGGTTAGTGCA 2759 1263469 464531 464550 TCTGCTCAGTTAGGTTAGTG 2760 1263470 479994 480013
GAGCTATCTGTACAAAATGG 2761 1263471 479995 480014 TGAGCTATCTGTACAAAATG 2743 1263472 479996 480015
GTGAGCTATCTGTACAAAAT 2744 1263538 483973 483992 TGTGCATACCCAGGGTAGGA 2762 1272943 467048 467067
TTTCATCAGTTAGTCAAGTT 2786 1272944 465605 465624 CCTTCTATTTTACAGACCGAA 2787 1272945 465607 465626
TGCCCTTTCTATTTTACAGCCG 2788 1272946 465608 465627 GTGCCCTTTCTATTTTACAGACC 2789 1272947 465609 465628
AGTGCCCTTTCTATTTTACAGAC 2790 1272948 468919 468938 AGTATAGTGCCTCTCCCTCT 2791 1272949 468920 468939
AAGTATAGATGCCTCTCCTC 2792 1272950 468921 468940 TAAGTATAGATGCCTCTCCTC 2793 1272951 474796 474815
ATTGACACCTCCAACCTGTAA 2794 1272952 474798 474817 GTATTGACACCTCCAACCTGT 2795 1272953 474800 474819
TGGTATTGACACCTCCAAC 2796 1272954 474801 474820 TTGGTATTGACACCTCCAAC 2797 1272955 474802 474821
TTTGGTATTGACACCTCCAA 2798 1272956 476012 476031 GTTTTCGCCCCGTTACCTCAA 2799 1272957 476013 476032
AGTTTTCGCCCCGTTACCTCA 2800 1272958 476014 476033 CAGTTTTTCGCCCCGTTACCTC 2801 1272959 476017 476036
CTCCAGTTTTCGCCCCGTTAC 2802 1272960 467052 467071 ACACCTTCATCAGTTAGTCA 2803 1272961 476018 476037
TCTCCAGTTTTTCGCCCCGTTA 2804 1272962 478437 478456 GCTATAGGTGTCACATATTC 2805 1272963 478442 478461
TTGTAGCTATAGGTGTCACA 2806 1272964 478443 478462 TTTGATAGCTATAGGTGTCAC 2807 1272965 483106 483125
GCAATGGACTTAGTACACAA 2808 1272966 483107 483126 GGCAATGGACTTAGTACACA 2809 1272967 483110 483129
TTAGGCAATGGACTTAGTAC 2810 1272968 483111 483130 CTTAGGCAATGGACTTAGTA 2811 1272969 485766 485785
CAGATTCCTAAATACGCACA 2812 1272970 485767 485786 TCAGATTCCTAAATACGCAC 2813 1272971 485768 485787
GTCAGATTCCTAAATACGCA 2814 1272972 485771 485790 GTGGTCAGATTCCTAAATAC 2815 1272973 487601 487620
AGTGTCATATGTAGCAATTA 2816 1272974 487603 487622 TTAGTGTCATATGTAGCAAT 2817 1272975 501337 501356
TATGTAGCTCAGTCAATGT 2818 1272976 501339 501358 CTTATGTAGCTCAGTCAAT 2819 1272977 501342 501361
CTGCTTATGTAGCTCAGCTC 2820 1272978 468734 468753 AAAATCCATTTGTCCAGTCT 2821 1272979 505552 505571
TTTGCTTTTTCAGTCAGGTAC 2822 1272980 468735 468754 TAAATCCATTTGTCCAGTC 2823 1272981 468736 468755
CTAAAATCCATTTGTCCAGT 2824 1272982 506110 506129 GCATTGGCTTCATATTTCTC 2825 1272983 506112 506131
GAGCATTGGCTTCATATTTT 2826 1272984 508942 508961 CATTATTCTCTAGTGCCTAT 2827 1272985 508943 508962
TCATTATTCTCTAGTGCCTA 2828 1272986 508947 508966 GTCTTCATTATTCTCTAGTG 2829 1272987 508948 508967
AGTCTTCATTATTCTCTAGT 2830 1272988 458439 458458 AACTTCATCAATATTTCCCC 2831 1272989 458397 458416
TGTGGGCCTCTATTAAGATC 2832 1272990 458399 458418 CATGTGGGCCTCTATTAAGA 2833 1272991 458401 458420

TGCATGTGGGCGCTCTAATTAA 2834 1272992 458453 458472 TTTATCTTTTACCCCAACTTC 2835 1272993 458454 458473
CTTTTATACTTTACCCCAACTT 2836 1272994 458455 458474 GCTTTTATACTTTACCCCAACT 2837 1272995 458440 458459
CAACTTCATCAATATTTTCCC 2838 1272996 458457 458476 TGGCTTTTATACTTTACCCAA 2839 1272997 458458 458477
TTGGCTTTTATACTTTACCCA 2840 1272998 458459 458478 TTTGGCTTTTATACTTTACCC 2841 1272999 458569 458588
CTTTAGTATGTCGAGAATC 2842 1273000 458441 458460 CCAACTTCATCAATATTTCC 2843 1273001 461466 461485
ACTCATCATGATCTTGGTAA 2844 1273002 463053 463072 TCACGTAGTTTTTGTAGTTC 2845 1273003 463191 463210
TCTGGAATCTTGTAGAGGAT 2846 1273004 463192 463211 TTCTGGAATCTTGTAGAGGA 2847 1273005 467047 467066
TTCATCAGTTAGTCAGGTTA 2848 1273006 464395 464414 ATTCAGCACTTAGGTAGTTC 2849 1273007 464399 464418
TCCCATTTCAGCACTTAGGTA 2850 1273008 464400 464419 TTCCCATTTCAGCACTTAGGT 2851 1273061 468982 469001
CCATTTTGACCTTCTTAGGCC 2872 1273062 468983 469002 ACCATTTTGACCTTCTTAGC 2873 1273063 465232 465251
TCTCGGGTATATAAAATTAAT 2874 1273064 465233 465252 ATCTCGGGTATATAAAATTA 2875 1273065 468992 469011
AACTGCTTCACCAATTTTGAC 2876 1273066 468993 469012 TAACTGCTTCACCAATTTTGA 2877 1273067 468994 469013
TTAACTGCTTCACCAATTTTG 2878 1273068 464522 464541 TTAGGTTAGTGCACAGATA 2879 1273069 464523 464542
TTAGGTTAGTGCACAGATA 2880 1273070 464534 464553 GTCTGTCTCAGTTAGGTTA 2881 1273071 479991 480010
CTATCTGTACAAAATGGAAC 2882 1273072 479992 480011 GCTATCTGTACAAAATGGAA 2883 1273073 479993 480012
AGCTATCTGTACAAAATGGA 2884 1273084 457732 457751 AGGTGTCTTATATCTATGAT 2885 1273085 457742 457761
GAAACCAGCCAGGTGTCTTA 2886 1273086 474391 474410 TCAACCAATTTGCTATTCAT 2887 1273087 474392 474411
GTCAACCAATTTGCTATTCAT 2888 1273088 457733 457752 CAGGTGTCTTATATCTATGA 2889 1273089 474394 474413
AGGTCAACCAATTTGCTATT 2853 1273090 474395 474414 TAGGTCAACCAATTTGCTAT 2861 1273091 474396 474415
TTAGGTCAACCAATTTGCTA 2854 1273092 474397 474416 TTTAGGTCAACCAATTTGCT 2890 1273093 474398 474417
GTTTAGGTCAACCAATTTGC 2891 1273094 474399 474418 GGTTTAGGTCAACCAATTTG 2892 1273095 478527 478546
TTCATCAGGCAACAGAATCTC 2893 1273096 478528 478547 TGTATCAGGCAACAGAATCT 2894 1273097 478529 478548
ATGTATCAGGCAACAGAATC 2895 1273098 457734 457753 CCAGGTGTCTTATATCTATG 2896 1273099 478535 478554
AACGCAATGTATCAGGCAAC 2856 1273101 478537 478556 AAAACGCAATGTATCAGGCA 2897 1273102 478538 478557
TAAACGCAATGTATCAGGC 2898 1273103 457735 457754 GCCAGGTGTCTTATATCTAT 2852 1273104 478727 478746
CACATTACCCATCCGTTCTT 2899 1273105 478728 478747 TCACATTACCCATCCGTTCT 2900 1273106 478729 478748
ATCACATTACCCATCCGTTCT 2901 1273107 478730 478749 GATCACATTACCCATCCGTT 2858 1273108 478731 478750
TGATCACATTACCCATCCGT 2859 1273109 478732 478751 CTGATCACATTACCCATCCG 2860 1273110 478734 478753
TGCTGATCATCATCCCATC 2862 1273111 478735 478754 TTGCTGATCACATTACCCAT 2863 1273112 457736 457755
AGCCAGGTGTCTTATATCTA 2855 1273113 478737 478756 TCTTGCTGATCACATTACCC 2902 1273114 478738 478757
TTCTTGCTGATCACATTACC 2903 1273115 457737 457756 CAGCCAGGTGTCTTATATCT 2904

[0474] The compounds in Table 64 are 6-10-4 MOE gapmers. The gapmers are 20 nucleosides in length, wherein the central gap segment consists of ten 2'- β -D-deoxynucleosides, the 5' wing segment consists of six 2'-MOE nucleosides, and the 3' wing segment consists of four 2'-MOE nucleosides. The sugar motif of the gapmers is (from 5' to 3'): eeeeeeddddddddeeeee; wherein 'd' represents a 2'- β -D-deoxyriboxyl sugar moiety, and 'e' represents a 2'-MOE sugar moiety. The gapmers have an internucleoside linkage motif of (from 5' to 3'): sooooooooooooooss, wherein "s" represents a phosphorothioate internucleoside linkage and "o" represents a phosphodiester internucleoside linkage. All cytosine residues are 5-methylcytosines. TABLE-US-00065 TABLE 64 6-10-4 MOE gapmers with a mixed PO/PS internucleoside linkages complementary to human UBE3A-ATS SEQ SEQ ID ID NO: 1 NO: 1 SEQ Compound Start Stop ID ID Site Site SEQUENCE NO 1263457 479995 480014

TGAGCTATCTGTACAAAATG 2743 1263458 479996 480015 GTGAGCTATCTGTACAAAAT 2744 1263459 479997 480016
CGTGAGCTATCTGTACAAAA 1142 1263504 460996 461015 TTTTCCATTTTCTCTTAG 2745 1263505 460997 461016
GTTTTTCCATTTTCTCTTA 2746 1263506 460999 461018 GTGTTTTTCCATTTTCTCT 598 1263507 461000 461019
TGTGTTTTTCCATTTTCTC 2747 1263508 461001 461020 GTGTGTTTTTCCATTTTCT 2748 1263509 461002 461021
TGTGTGTTTTTCCATTTTTC 2749 1263510 465235 465254 TCATCTCGGGTATATAAATT 2750 1263511 465236 465255
GTCATCTCGGGTATATAAAT 1132 1263512 465237 465256 GGTCATCTCGGGTATATAAA 1207 1263513 465238 465257
TGGTCATCTCGGGTATATAA 318 1263514 465239 465258 TTGGTCATCTCGGGTATATA 2473 1263515 465240 465259
ATTGGTCATCTCGGGTATAT 1655 1263516 465241 465260 TATTGGTCATCTCGGGTATA 2548 1263517 468985 469004
TCACCATTTTGACCTTCTTA 2751 1263518 468986 469005 TTCACCATTTTGACCTTCTT 2752 1263519 468987 469006
CTTCACCATTTTGACCTTCT 2753 1263520 468988 469007 GCTTCACCATTTTGACCTTC 377 1263521 468989 469008
TGCTTCACCATTTTGACCTT 2754 1263522 468990 469009 CTGCTTCACCATTTTGACCT 2755 1263523 468991 469010
ACTGCTTCACCATTTTGACC 2756 1263524 464525 464544 CAGTTAGGTTAGTGCACAGA 2728 1263525 464526 464545
TCAGTTAGGTTAGTGCACAG 2757 1263526 464527 464546 CTCAGTTAGGTTAGTGCACA 2758 1263527 464528 464547
GCTCAGTTAGGTTAGTGCAC 1504 1263528 464529 464548 TGCTCAGTTAGGTTAGTGCA 2759 1263529 464530 464549
CTGCTCAGTTAGGTTAGTGC 2729 1263530 464531 464550 TCTGCTCAGTTAGGTTAGTG 2760 1263531 479994 480013
GAGCTATCTGTACAAAATGG 2761 1263533 483970 483989 GCATACCCAGGGTAGGATTC 765 1263535 483971 483990
TGCATACCCAGGGTAGGATT 1445 1263537 483972 483991 GTGCATACCCAGGGTAGGAT 766 1263540 483973 483992
TGTGCATACCCAGGGTAGGA 2762 1263542 483974 483993 ATGTGCATACCCAGGGTAGG 2595 1273023 457735 457754
GCCAGGTGTCTTATATCTAT 2852 1273024 457736 457755 AGCCAGGTGTCTTATATCTA 2855 1273025 474393 474412
GGTCAACCAATTTGCTATTC 1809 1273026 474394 474413 AGGTCAACCAATTTGCTATT 2853 1273027 474395 474414
TAGGTCAACCAATTTGCTAT 2861 1273028 474396 474415 TTAGGTCAACCAATTTGCTA 2854 1273029 478535 478554
AACGCAATGTATCAGGCAAC 2856 1273030 478536 478555 AAACGCAATGTATCAGGCAA 2857 1273031 478730 478749
GATCACATTACCCATCCGTT 2858 1273032 478731 478750 TGATCACATTACCCATCCGT 2859 1273033 478732 478751
CTGATCACATTACCCATCCG 2860 1273034 478733 478752 GCTGATCACATTACCCATCC 1291 1273035 478734 478753
TGCTGATCACATTACCCATC 2862 1273036 478735 478754 TTGCTGATCACATTACCCAT 2863

[0475] The compounds in Table 65 are 4-8-6 MOE gapmers. The gapmers are 18 nucleosides in length, wherein the central gap segment consists of eight 2'- β -D-deoxynucleosides, the 5' segment consists of four 2'-MOE nucleosides, and the 3' wing segment consists of six 2'-MOE nucleosides. The sugar motif of the gapmers is (from 5' to 3'): eeeeddddddeeeee; wherein 'd' represents a 2'- β -D-deoxyriboxyl sugar moiety, and 'e' represents a 2'-MOE sugar moiety. The gapmers have an internucleoside linkage motif of (from 5' to 3'): sooooooooooooo; wherein "s" represents a phosphorothioate internucleoside linkage and "o" represents a phosphodiester internucleoside linkage. All cytosine residues are 5-methylcytosines. TABLE-US-00066 TABLE 65 4-8-6 MOE gapmers with a mixed PO/PS internucleoside linkages complementary to human UBE3A-ATS SEQ SEQ ID ID NO: 1 NO: 1 SEQ Compound Start Stop ID ID Site Site SEQUENCE NO 1263403 465237 465254

TCATCTCGGGTATATAAA 2768 1263404 465238 465255 GTCATCTCGGGTATATAA 2769 1263405 465239 465256
GGTCATCTCGGGTATATA 2770 1263406 465240 465257 TTGGTCATCTCGGGTATAT 2771 1263407 465241 465258
TTGGTCATCTCGGGTATA 2772 1263408 468987 469004 TCACCATTTTGACCTTCT 2773 1263409 468988 469005
TTCACCATTTTGACCTTC 2774 1263410 468989 469006 CTTCACCATTTTGACCTT 2775 1263411 468991 469008
TGCTTCACCATTTTGACC 2776 1263412 464527 464544 CAGTTAGGTTAGTGCACA 2777 1263413 464528 464545

TCAGTTAGGTTAGTGCAC 2778 1263416 464529 464545 CTCAGTTAGGTTAGTGCAC 2779 1263415 464530 464547
GCTCAGTTAGGTTAGTGC 2780 1263416 464531 464548 TGCTCAGTTAGGTTAGTG 2781 1263417 479996 480013
GAGCTATCTGTACAAAAT 2782 1263418 479997 480014 TGAGCTATCTGTACAAA 2783 1263419 479998 480015
GTGAGCTATCTGTACAAA 2784 1263420 479999 480016 CGTGAGCTATCTGTACAA 2785 1263543 483971 483988
CATACCCAGGGTAGGATT 2763 1263546 483972 483989 GCATACCCAGGGTAGGAT 2764 1263549 483973 483990
TGCATACCCAGGGTAGGA 2765 1263552 483974 483991 GTGCATACCCAGGGTAGG 2766 1263557 483975 483992
TGTGCATACCCAGGGTAG 2767 1273037 457737 457754 GCCAGGTGTCTTATATCT 2864 1273038 478535 478552
CGCAATGTATCAGGCAAC 2865 1273039 478536 478553 ACGCAATGTATCAGGCAA 2866 1273040 478732 478749
GATCACATTACCCATCCG 2867 1273041 478733 478750 TGATCACATTACCCATCC 2868 1273042 478735 478752
GCTGATCACATTACCCAT 2869 1273043 457738 457755 AGCCAGGTGTCTTATATC 2870 1273060 474393 474410
TCAACCAATTTGCTATTC 2871

[0476] The compounds in Table 66 are 5-8-5 MOE gapmers. The gapmers are 18 nucleosides in length, wherein the central gap segment consists of eight 2'-β-D-deoxynucleosides and the 5' and 3' wing segments each consists of five 2'-MOE nucleosides. The sugar motif of the gapmers is (from 5' to 3'): eeeeeedddddddeeee; wherein 'd' represents a 2'-β-D-deoxyriboxyl sugar moiety, and 'e' represents a 2'-MOE sugar moiety. The gapmers have an internucleoside linkage motif of (from 5' to 3'): sooooooooooss; wherein "s" represents a phosphorothioate internucleoside linkage and "o" represents a phosphodiester internucleoside linkage. All cytosine residues are 5-methylcytosines.

TABLE-US-00067 TABLE 66 5-8-5 MOE gapmers with a mixed PO/PS internucleoside linkages complementary to human UBE3A-ATS SEQ SEQ ID ID NO: 1 NO: 1 SEQ Compound Start Stop ID ID Site Site SEQUENCE NO 1263421 465237 465254

TCATCTCGGGTATATAAA 2768 1263422 465238 465255 GTCATCTCGGGTATATAA 2769 1263423 465239 465256
GGTCATCTCGGGTATATA 2770 1263424 465240 465257 TGGTCATCTCGGGTATAT 2771 1263425 465241 465258
TTGGTCATCTCGGGTATA 2772 1263426 468987 469004 TCACCATTTTGACCTTCT 2773 1263427 464527 464544
TCAGTTAGGTTAGTGACA 2777 1263428 464528 464545 TCAGTTAGGTTAGTGAC 2778 1263429 464529 464546
CTCAGTTAGGTTAGTGCA 2779 1263430 464530 464547 GCTCAGTTAGGTTAGTG 2780 1263431 464531 464548
TGCTCAGTTAGGTTAGTG 2781 1263432 479996 480013 GAGCTATCTGTACAAAAT 2782 1263433 479997 480014
TGAGCTATCTGTACAAA 2783 1263434 479998 480015 GTGAGCTATCTGTACAAA 2784 1263435 479999 480016
CGTGAGCTATCTGTACAA 2785 1263544 483971 483988 CATACCCAGGGTAGGATT 2763 1263547 483972 483989
GCATACCCAGGGTAGGAT 2764 1263550 483973 483990 TGCATACCCAGGGTAGGA 2765 1263553 483974 483991
GTGCATACCCAGGGTAGG 2766 1263554 483975 483992 TGTGCATACCCAGGGTAG 2767 1273052 457737 457754
GTCAGGTGTCTTATATCT 2864 1273053 457738 457755 AGCCAGGTGTCTTATATC 2870 1273054 474393 474410
TCAACCAATTTGCTATTC 2871 1273055 478535 478552 CGCAATGTATCAGGCAAC 2865 1273056 478536 478553
ACGCAATGTATCAGGCAA 2866 1273057 478732 478749 GATCACATTACCCATCCG 2867 1273058 478733 478750
TGATCACATTACCCATCC 2868 1273059 478735 478752 GCTGATCACATTACCCAT 2869

[0477] The compounds in Table 67 are 6-8-4 MOE gapmers. The gapmers are 18 nucleosides in length, wherein the central gap segment consists of eight 2'-β-D-deoxynucleosides, the 5' wing segment consists of six 2'-MOE nucleosides, and the 3' wing segment consists of four 2'-MOE nucleosides. The sugar motif of the gapmers is (from 5' to 3'): eeeeeedddddddeeee; wherein 'd' represents a 2'-β-D-deoxyriboxyl sugar moiety, and 'e' represents a 2'-MOE sugar moiety. The gapmers have an internucleoside linkage motif of (from 5' to 3'): sooooooooooss; wherein "s" represents a phosphorothioate internucleoside linkage and "o" represents a phosphodiester internucleoside linkage. All cytosine residues are 5-methylcytosines.

TABLE-US-00068 TABLE 67 6-8-4 MOE gapmers with a mixed PO/PS internucleoside linkages complementary to human UBE3A-ATS SEQ SEQ ID ID NO: 1 NO: 1 SEQ Compound Start Stop ID ID Site Site SEQUENCE NO 1263436 465237 465254

TCATCTCGGGTATATAAA 2768 1263437 465238 465255 GTCATCTCGGGTATATAA 2769 1263438 465239 465256
GGTCATCTCGGGTATATA 2770 1263439 465240 465257 TGGTCATCTCGGGTATAT 2771 1263440 465241 465258
TTGGTCATCTCGGGTATA 2772 1263441 468987 469004 TCACCATTTTGACCTTCT 2773 1263442 464527 464544
CAGTTAGGTTAGTGACA 2777 1263443 464528 464545 TCAGTTAGGTTAGTGAC 2778 1263444 464529 464546
CTCAGTTAGGTTAGTGCA 2779 1263445 464530 464547 GCTCAGTTAGGTTAGTG 2780 1263446 464531 464548
TGCTCAGTTAGGTTAGTG 2781 1263447 479996 480013 GAGCTATCTGTACAAAAT 2782 1263448 479997 480014
TGAGCTATCTGTACAAA 2783 1263449 479998 480015 GTGAGCTATCTGTACAAA 2784 1263450 479999 480016
CGTGAGCTATCTGTACAA 2785 1263545 483971 483988 CATACCCAGGGTAGGATT 2763 1263548 483972 483989
GCATACCCAGGGTAGGAT 2764 1263551 483973 483990 TGCATACCCAGGGTAGGA 2765 1263555 483974 483991
GTGCATACCCAGGGTAGG 2766 1263556 483975 483992 TGTGCATACCCAGGGTAG 2767 1273044 457737 457754
GCCAGGTGTCTTATATCT 2864 1273045 457738 457755 AGCCAGGTGTCTTATATC 2870 1273046 474393 474410
TCAACCAATTTGCTATTC 2871 1273047 478535 478552 CGCAATGTATCAGGCAAC 2865 1273048 478536 478553
ACGCAATGTATCAGGCAA 2866 1273049 478732 478749 GATCACATTACCCATCCG 2867 1273050 478733 478750
TGATCACATTACCCATCC 2868 1273051 478735 478752 GCTGATCACATTACCCAT 2869

Example 8: Effect of Modified Oligonucleotides on Human UBE3A-ATS RNA In Vitro, Multiple Doses

[0478] Modified oligonucleotides selected from the examples above were tested at various doses in human IPS derived iCell GABANeurons (Cellular Dynamics), as described in Example 4. Reduction of UBE3A-ATS RNA is presented in the tables below as percent UBE3A-ATS RNA amount relative to untreated control (UTC) cells. Where possible, the half maximal inhibitory concentration (IC_{sub.50}) of each modified oligonucleotide was calculated using a linear regression on a log/linear plot of the data in excel. In some cases, an IC_{sub.50} could not be reliably calculated and the data point is marked as "NC.". Values marked with "N.D." indicate that a value was not determined in this experiment.

TABLE-US-00069 TABLE 68 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (40,000 cells/well) UBE3A-ATS expression (% UTC) Compound 250 740.7 2,222 6,666 20,000 IC_{sub.50} Number nM nM nM nM nM (μM) 750519 69 53 23 N.D. N.D. 0.7 617557 66 54 34 25 N.D. 0.9 749860 62 48 39 20 13 0.7 1065593 44 32 21 11 9 N.C. 1263517 55 47 35 17 18 0.4 1263519 52 39 25 17 16 0.2 1263533 51 34 29 28 23 0.1 1263540 39 29 29 29 24 N.C. 1272994 51 39 28 14 10 0.2 1272996 62 45 38 20 13 0.6 1272997 73 53 39 24 18 1.1 1272998 67 53 30 19 13 0.8 1273030 68 54 33 30 22 1.0 1273033 51 50 31 32 20 0.4 1273049 57 41 32 24 21 0.3

TABLE-US-00070 TABLE 69 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (40,000 cells/well) UBE3A-ATS expression (% UTC) Compound 250 740.7 2,222 6,666 20,000 IC_{sub.50} Number nM nM nM nM nM (μM) 750519 79 66 29 15 N.D. 1.1 1263426 64 48 34 29 25 0.7 1273042 92 72 58 43 40 5.5 1273051 70 47 34 29 22 0.9 1273055 71 58 47 40 27 1.9 1273057 73 71 50 37 39 3.4 1273058 80 69 58 48 34 4.8 1273087 55 34 26 14 20 0.2 1273107 63 59 36 26 1.0 1273113 75 68 58 47 28 3.6

TABLE-US-00071 TABLE 70 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (40,000 cells/well) UBE3A-ATS expression (% UTC) Compound 250 740.7 2,222 6,666 20,000 IC_{sub.50} Number nM nM nM nM nM (μM) 750519 56 38 22 10 25 0.2 749860 68 47 37 19 16 0.8 1065645 66 45 31 19 12 0.6 1263461 70 66 55 32 27 2.3 1263486 74 48 44 25 21 1.2

TABLE-US-00072 TABLE 71 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (40,000 cells/well) UBE3A-ATS expression (% UTC) Compound 250 740.7 2,222 6,666 20,000 IC_{sub.50} Number nM nM nM nM nM (μM) 750519 63 39 19 9 4 0.4 1263517 56 40 27 17 11 0.3 1263518 72 65 42 28 20 1.6 1263532 57 44 38 33 64 0.5 1263533 41 56 37 ND 26 NC

TABLE-US-00073 TABLE 72 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (40,000 cells/well) UBE3A-ATS expression (% UTC) Compound 250 740.7 2,222 6,666 20,000 IC.sub.50 Number nM nM nM nM nM (μM) 617557 59 56 37 23 23 0.8 1263537 70 86 82 77 59 N.C. 1272944 51 39 23 13 8 0.2 1273033 55 40 32 21 15 0.3 1273039 68 58 37 28 20 1.1

TABLE-US-00074 TABLE 73 Dose-dependent percent reduction of human UBE3A-ATS RNA in vitro (40,000 cells/well) UBE3A-ATS expression (% UTC) Compound 250 740.7 2,222 6,666 20,000 IC.sub.50 Number nM nM nM nM nM (μM) 617557 70 57 38 N.D. N.D. 1.0 1273050 81 75 76 62 41 16.5 1273055 59 48 40 26 18 0.6 1273062 54 37 28 18 13 0.2 1273090 76 64 71 56 55 N.C. 1273091 75 79 56 44 32 4.3

Example 9: Effect of Modified Oligonucleotides on Human UBE3A-ATS RNA and UBE3A RNA In Vitro, Multiple Doses

[0479] Modified oligonucleotides selected from the examples above were tested at various doses in 10-week differentiated human neuronal cells derived from Angelman Syndrome patient derived IPS cells as described in Example 6 above. Reduction of UBE3A-ATS RNA or induction of UBE3A RNA is presented in the tables below as percent UBE3A-ATS RNA amount or percent UBE3A RNA relative to untreated control (UTC) cells. Values marked with “N.D.” indicate that a value was not determined in this experiment.

[0480] Several modified oligonucleotides were found to reduce UBE3A-ATS RNA accompanied by concurrent increase in UBE3A RNA in Angelman's patient IPS cell derived neurons.

TABLE-US-00075 TABLE 74 Reduction of UBE3A-ATS RNA and increase of UBE3A RNA in Angelman Syndrome IPS cell derived neurons UBE3A-ATS RNA UBE3A RNA (% UTC) (RTS4796) (% UTC) (RTS35984) Compound 250 741 2222 6667 20000 250 741 2222 6667 20000 ID nM nM nM nM nM nM nM nM nM nM 750519 51 20 11 7 5 297 224 235 250 205 749860 47 21 18 15 8 148 119 245 307 324 1065645 41 15 13 11 13 213 185 298 315 353 1263461 49 42 34 25 16 141 185 248 235 221 1263486 64 47 33 29 16 136 174 215 323 264

TABLE-US-00076 TABLE 75 Reduction of UBE3A-ATS RNA and increase of UBE3A RNA in Angelman Syndrome IPS cell derived neurons UBE3A-ATS RNA UBE3A RNA (% UTC) (RTS4796) (% UTC) (RTS35984) Compound 250 741 2222 6667 20000 250 741 2222 6667 20000 ID nM nM nM nM nM nM nM nM nM nM 750519 46 22 13 7 7 259 274 520 482 272 1263517 37 29 16 12 12 230 261 462 521 283 1263518 66 39 29 21 18 187 183 269 496 277 1263532 39 26 21 17 24 192 162 393 390 357 1263533 34 25 24 18 29 296 367 389 371 548

TABLE-US-00077 TABLE 76 Reduction of UBE3A-ATS RNA and increase of UBE3A RNA in Angelman Syndrome IPS cell derived neurons UBE3A-ATS RNA UBE3A RNA (% UTC) (RTS4796) (% UTC) (RTS35984) Compound 250 741 2222 6667 20000 250 741 2222 6667 20000 ID nM nM nM nM nM nM nM nM nM nM 617557 68 44 28 24 20 120 210 319 453 396 1263537 29 26 21 26 16 139 203 414 440 480 1272944 35 25 13 11 7 148 235 469 702 677 1273033 38 21 21 16 18 119 244 487 422 514 1273039 64 41 29 20 17 183 237 395 376 418

TABLE-US-00078 TABLE 77 Reduction of UBE3A-ATS RNA and increase of UBE3A RNA in Angelman Syndrome IPS cell derived neurons UBE3A-ATS RNA UBE3A RNA (% UTC) (RTS4796) (% UTC) (RTS35984) Compound 250 741 2222 6667 20000 250 741 2222 6667 20000 ID nM nM nM nM nM nM nM nM nM nM 617557 63 50 33 28 27 126 230 347 335 368 1273050 64 48 59 41 28 165 152 240 457 483 1273055 44 31 38 22 17 541 664 1014 1094 887 1273062 34 54 51 47 27 253 296 421 705 997 1273090 70 54 30 39 23 155 165 259 487 692 1273091 91 86 54 37 31 96 121 135 156 284

Example 10: Tolerability of Modified Oligonucleotides Complementary to Human UBE3A-ATS in Wild-Type Mice. 3 Hour Study

[0481] Modified oligonucleotides described above were tested in wild-type female C57/B16 mice to assess the tolerability of the oligonucleotides. Wild-type female C57/B16 mice each received a single ICV dose of 700 μg of modified oligonucleotide listed in the table below. Each treatment group consisted of 4 mice. A group of 4 mice received PBS as a negative control for each experiment (identified in separate tables below). At 3 hours post-injection, mice were evaluated according to seven different criteria. The criteria are (1) the mouse was bright, alert, and responsive; (2) the mouse was standing or hunched without stimuli; (3) the mouse showed any movement without stimuli; (4) the mouse demonstrated forward movement after it was lifted; (5) the mouse demonstrated any movement after it was lifted; (6) the mouse responded to tail pinching; (7) regular breathing. For each of the 7 criteria, a mouse was given a subscore of 0 if it met the criteria and 1 if it did not (the functional observational battery score or FOB). After all 7 criteria were evaluated, the scores were summed for each mouse and averaged within each treatment group. The results are presented in the tables below.

TABLE-US-00079 TABLE 78 Tolerability scores in mice at 700 μg dose Compound Number 3 hr FOB PBS 0.0 749860 0.0 749865 4.0 749984 4.8 750006 3.5 750009 3.8 750540 3.5 750544 2.8 750567 5.0

TABLE-US-00080 TABLE 79 Tolerability scores in mice at 700 μg dose Compound Number 3 hr FOB PBS 0 1065272 5.0 1065295 2.0 1065296 5.5 1065324 2.8 1065330 6.8 1065369 6.8 1065438 5.8 1065465 5.0 1065513 5.8 1065576 7.0 1065578 3.2 1065579 3.8 1065582 4.2 1065586 3.0 1065590 6.0 1065591 4.0 1065593 1.0 1065595 4.8 1065597 4.5 1065599 6.0 1065600 4.2 1065605 3.8 1065607 5.2 1065608 3.0

TABLE-US-00081 TABLE 80 Tolerability scores in mice at 700 μg dose Compound Number 3 hr FOB PBS 0.0 1065609 0.0 1065613 0.0 1065616 5.5 1065619 5.0 1065621 2.0 1065623 3.5 1065624 3.5 1065631 1.0 1065635 1.0 1065641 1.5 1065642 4.0 1065644 6.8 1065645 3.0 1065646 2.0 1065651 4.0 1065654 7.0 1065667 4.0 1065669 1.0 1065671 4.0 1065672 6.8 1065674 4.0 1065676 2.0 1065678 5.0 1065680 1.0

TABLE-US-00082 TABLE 81 Tolerability scores in mice at 700 μg dose Compound Number 3 hr FOB PBS 0 1065685 5.0 1065686 4.8 1065690 3.8 1065696 6.2 1065708 5.5 1065709 4.2 1065710 2.8 1065712 6.8 1065713 5.0 1065719 4.0 1065728 1.0 1065735 3.5 1065750 6.2 1065754 3.2 1065765 4.2 1065766 5.2 1065768 5.5 1065785 3.8 1065795 1.0 1065799 1.0 1065810 2.2 1065812 3.8 1065813 5.0 1065817 5.0

TABLE-US-00083 TABLE 82 Tolerability scores in mice at 700 μg dose Compound Number 3 hr FOB PBS 0.0 1065821 6.5 1065823 4.0 1065826 3.0 1065829 4.8 1065840 4.5 1065856 6.2 1065857 6.0 1065858 4.0 1065859 0.0 1065863 4.8 1065868 3.2 1065889 1.0 1065894 3.8 1065899 4.5 1065901 4.0 1065902 4.0 1065903 5.5 1065914 5.0 1065920 2.0 1065932 4.5 1065937 1.0 1065947 1.2 1065953 6.5 1065954 5.5

TABLE-US-00084 TABLE 83 Tolerability scores in mice at 700 μg dose Compound Number 3 hr FOB PBS 0.0 1065955 5.0 1065977 6.2 1066009 2.0 1066037 4.2 1066046 1.0 1066072 4.0 1066076 5.0 1066089 1.0 1066092 1.0 1066097 1.2 1066119 11.0 1066217 6.0 1066221 3.8 1066249 4.5 1066253 6.0 1066311 2.0 1066350 4.0 1066375 1.0 1066377 6.2 1066378 5.2 1066396 6.0 1066420 4.0 1066423 2.8 1066429 5.8

TABLE-US-00085 TABLE 84 Tolerability scores in mice at 700 μg dose Compound Number 3 hr FOB PBS 0.0 749931 0.0 749933 4.0 749937 0.5 749944 3.8 749956 1.8 749964 2.0 749969 5.2 749991 2.2 750028 4.2 750030 3.2 750032 2.2 750040 4.2 750051 2.5 750092 0.8 750100 2.5 750131 0.5 750139 3.0 750140 1.2 750196 4.0 750210 3.5 750214 3.25 750228 0.0 750270 1.0 750292 0.0 750312 4.8 750325 1.0 750326 0.0 750329 5.5 750344 6.2 750350 2.8 750359 0.0 750360 3.2 750365 5.2 750366 0.0 750386 0.0 750413 0.0 750416 7.0 750418 1.8 750430 6.2 750431 5.0 750439 5.5 750452 2.5 750517 3.0 750519 1.0 750542 3.5 750549 0.0

TABLE-US-00086 TABLE 85 Tolerability scores in mice at 700 μg dose Compound Number 3 hr FOB PBS 0.0 617456 4.8 617459 7.0 617470 6.5 617473 6.8 617536 6.5 617547 6.8 617557 5.2 617593 6.5 749794 5.0 749796 4.5 749816 5.8 749861 3.0 749863 5.5 749869 3.8 749882 6.8 749885 0.5 749893 1.2 749894 0.0 749907 5.0 749921 6.0

TABLE-US-00087 TABLE 86 Tolerability scores in mice at 700 μg dose Compound Number 3 hr FOB PBS 0.0 1065473 3.2 1065561 6.0 1065592 4.0 1065625 7.0 1065791 1.0 1065924 1.0 1066003 4.0

TABLE-US-00088 TABLE 87 Tolerability scores in mice at 700 μg dose Compound ID 3 hr FOB PBS 0.0 1263471 6.2 1263472 3.2 1263473 1.0 1263474 0.0 1263475 0.0 1263476 1.0 1263477 0.0 1263478 1.0 1263479 4.0 1263480 0.0 1263481 5.5 1263482 4.0 1263483 4.0 1263484 4.0 1263485 4.0 1263486 2.0 1263487 1.0 1263488 1.0 1263489 0.0 1263490 0.0 1263491 0.0 1263492 1.0 1263494 4.0 1263495 4.0 1263496 4.0 1263497 4.0 1263498 4.0 1263499 4.0 1263500 4.7 1263501 5.5 1263502 4.0 1263503 4.0 1263504 1.0 1263505 0.0 1263506 0.0 1263507 0.0 1263508 0.0 1263509 0.0 1263510 2.0

TABLE-US-00089 TABLE 88 Tolerability scores in mice at 700 μg dose Compound ID 3 hr FOB PBS 0.0 1165878 4.0 1263511 4.0 1263512 6.0 1263513 6.0 1263514 6.5 1263515 4.0 1263516 4.0 1263517 1.0 1263518 0.0 1263519 1.0 1263520 1.0 1263521 0.0 1263522 0.0 1263523 1.0 1263524 5.2 1263525 4.8 1263526 4.0 1263527 4.0 1263528 4.0 1263529 3.0 1263530 4.0 1263531 5.2 1263532 1.0 1263533 1.0 1263534 1.0

1263535 2.0 1263536 2.0 1263537 1.0 1263538 3.0 1263539 1.0 1263540 1.0
TABLE-US-00090 TABLE 89 Tolerability scores in mice at 700 µg dose Compound ID 3 hr FOB PBS 0.0 1263541 1.8 1263542 3.5 1263543 1.0
1263544 1.0 1263545 2.0 1263546 0.0 1263547 1.0 1263548 1.0 1263549 0.0 1263550 1.0 1263551 2.0 1263552 1.0 1263553 5.5 1263554 1.0
1263555 4.0 1263556 1.0 1263557 1.0
TABLE-US-00091 TABLE 90 Tolerability scores in mice at 700 µg dose Compound ID 3 hr FOB PBS 0.0 1263403 5.3 1263404 4.3 1263405 6.3
1263406 4.7 1263407 2.3 1263408 0.3 1263409 0.0 1263410 0.0 1263411 0.0 1263412 2.0 1263413 2.3 1263414 5.3 1263415 4.0 1263416 7.0
1263417 3.3 1263418 3.0 1263419 1.7 1263420 1.0 1263421 3.0 1263422 4.3 1263423 6.0 1263424 6.3 1263425 5.0
TABLE-US-00092 TABLE 91 Tolerability scores in mice at 700 µg dose Compound ID 3 hr FOB PBS 0.0 749952 6.0 749953 3.3 1165583 5.7
1165584 3.3 1263426 2.0 1263427 4.0 1263428 4.3 1263429 1.3 1263430 2.3 1263431 6.3 1263432 6.0 1263433 4.7 1263434 1.0 1263435 1.3
1263436 2.3 1263437 4.7 1263438 6.0 1263439 4.7 1263440 6.7 1263441 0.3 1263442 3.7 1263443 3.0 1263444 6.3 1263445 5.0 1263446 5.7
1263447 4.7 1263448 5.0 1263449 4.7 1263450 2.0 1263451 1.3 1263452 0 1263453 0
TABLE-US-00093 TABLE 92 Tolerability scores in mice at 700 µg dose Compound ID 3 hr FOB PBS 0 1065786 2.0 1165526 0.7 1165577 3.7
1263454 0.0 1263455 0.0 1263456 3.3 1263457 5.3 1263458 1.3 1263459 1.0 1263460 1.0 1263461 1.0 1263462 1.0 1263463 1.0 1263464 1.3
1263465 1.7 1263466 5.0 1263467 6.0 1263468 4.3 1263469 7.0 1263470 6.7 1272943 2.0 1272944 2.3 1272945 2.7 1272946 0.0 1272947 0.0
1272948 3.3 1272949 4.0 1272950 3.7 1272951 0.7 1272952 2.0 1272953 0.3 1272954 1.0
TABLE-US-00094 TABLE 93 Tolerability scores in mice at 700 µg dose Compound ID 3 hr FOB PBS 0.0 1272955 3.7 1272956 5.3 1272957 4.0
1272958 5.0 1272959 3.3 1272960 1.0 1272961 4.7 1272962 2.0 1272963 4.0 1272964 2.3 1272965 1.0 1272966 2.3 1272967 1.0 1272968 1.0
1272969 0.0 1272970 0.0 1272971 0.0 1272972 0.0 1272973 4.0 1272974 1.0 1272975 1.0 1272976 2.0 1272977 1.0 1272978 2.3 1272979 1.0
TABLE-US-00095 TABLE 94 Tolerability scores in mice at 700 µg dose Compound ID 3 hr FOB PBS 0.0 1272980 3.3 1272981 3.3 1272982 1.0
1272983 1.0 1272984 4.0 1272985 1.0 1272986 4.7 1272987 1.0 1272988 1.0 1272989 1.0 1272990 4.0 1272991 4.0 1272992 0.0 1272993 0.0
1272994 1.0 1272995 1.0 1272996 0.7 1272997 1.0 1272998 0.0 1272999 5.0 1273000 0.0 1273001 3.0 1273002 3.7 1273003 2.7 1273004 3.0
1273005 4.0 1273006 3.3 1273007 3.0 1273008 3.3 1273009 1.0 1273010 4.7 1273011 1.0
TABLE-US-00096 TABLE 95 Tolerability scores in mice at 700 µg dose Compound ID 3 hr FOB PBS 0.0 699781 0.0 1065707 2.0 1065711 4.0
1065921 5.7 1165585 3.7 1165621 1.0 1273012 0.0 1273013 4.0 1273014 2.0 1273015 2.0 1273016 0.0 1273017 1.0 1273018 0.0 1273019 0.0
1273020 0.0 1273021 0.0 1273022 0.0 1273023 0.0 1273024 0.0 1273025 5.0 1273026 5.3 1273027 5.0 1273028 0.0 1273029 4.7 1273030 3.0
1273031 0.0 1273032 0.0 1273033 0.0
TABLE-US-00097 TABLE 96 Tolerability scores in mice at 700 µg dose Compound ID 3 hr FOB PBS 0.0 1273034 0.0 1273035 1.0 1273036 1.0
1273037 1.0 1273038 1.0 1273039 1.0 1273040 0.0 1273041 0.0 1273042 0.0 1273043 6.0 1273044 0.0 1273045 0.0 1273046 3.0 1273047 3.0
1273048 0.0 1273049 0.0 1273050 0.0 1273051 0.0 1273052 0.0 1273053 6.0 1273054 3.0 1273055 2.0 1273056 2.3 1273057 2.0 1273058 1.0
1273059 1.0 1273060 0.7 1273061 1.0 1273062 0.0 1273063 0.0 1273064 0.0 1273065 0.0 1273066 0.0 1273067 1.0 1273068 6.0 1273069 6.0
1273070 1.0 1273071 0.0 1273072 0.0 1273073 4.0 1273084 0.0 1273085 3.0 1273086 2.0 1273087 1.0 1273088 0.0 1273089 5.3 1273090 0.0
1273091 0.0 1273092 0.0 1273093 1.0 1273094 6.0 1273095 4.3 1273096 4.7 1273097 5.0 1273098 1.0 1273099 1.0 1273101 0.7 1273102 1.0
1273103 0.0 1273104 0.0 1273105 0.0 1273106 0.0 1273107 0.0 1273108 0.0 1273109 1.0 1273110 0.0 1273111 0.0 1273112 2.3 1273113 0.0
1273114 0.0 1273115 2.3
TABLE-US-00098 TABLE 97 Tolerability scores in mice at 700 µg dose Compound ID 3 hr FOB PBS 0.0 1165523 2.0 1165533 2.3 1165538 3.0
1165550 1.3 1165563 1.3 1165586 2.3 1165596 2.3 1165608 4.0 1165616 3.0 1165694 3.3 1165737 3.7 1165827 3.3 1165855 3.0 1165897 0.3
1179808 2.3 1179839 6.7 1179841 2.0 1179843 2.0 1273100 7.0
TABLE-US-00099 TABLE 98 Tolerability scores in mice at 700 µg dose Compound Number 3 hr FOB PBS 0.0 1165521 3.3 1165524 4.7 1165536
4.0 1165545 6.0 1165552 5.7 1165553 6.0 1165554 7.0 1165555 6.0 1165562 5.7 1165588 4.7 1165590 5.7 1165611 6.0
Example 11: Tolerability of Modified Oligonucleotides Complementary to Human UBE3A-ATS in Rats, 3 Hour Study
[0482] Modified oligonucleotides described above were tested in rats to assess the tolerability of the oligonucleotides. Sprague Dawley rats each
received a single intrathecal (IT) dose of 3 mg of oligonucleotide listed in the table below. Each treatment group consisted of 4 rats. A group of four
rats received PBS as a negative control. At 3 hours post-injection, movement in 7 different parts of the body were evaluated for each rat. The 7 body
parts are (1) the rat's tail; (2) the rat's posterior posture; (3) the rat's hind limbs; (4) the rat's hind paws; (5) the rat's forepaws; (6) the rat's anterior
posture; (7) the rat's head. For each of the 7 different body parts, each rat was given a sub-score of 0 if the body part was moving or 1 if the body part
was paralyzed (the functional observational battery score or FOB). After each of the 7 body parts were evaluated, the sub-scores were summed for
each rat and then averaged for each group. For example, if a rat's tail, head, and all other evaluated body parts were moving 3 hours after the 3 mg IT
dose, it would get a summed score of 0. If another rat was not moving its tail 3 hours after the 3 mg IT dose but all other evaluated body parts were
moving, it would receive a score of 1. Results are presented as the average score for each treatment group.
TABLE-US-00100 TABLE 99 Tolerability scores in rats at 3 mg dose Compound No. FOB 3 hr PBS 0.0 749860 0.0 749861 4.0 749869 4.8 749885
0.8 749893 0.5 749931 0.5 750006 4.5
TABLE-US-00101 TABLE 100 Tolerability scores in rats at 3 mg dose Compound No. FOB 3 hr PBS 0.0 750030 4.0 750051 4.0 750092 2.5 750100
4.0 750139 5.3 750140 3.5 750270 3.0 750292 0.0 750325 1.8 750386 0.8
TABLE-US-00102 TABLE 101 Tolerability scores in rats at 3 mg dose Compound No. FOB 3 hr PBS 0.0 1065578 4.0 1065586 3.0 1065609 2.0
1065613 3.8 1065635 1.8 1065641 3.3 1065645 3.0 1065646 2.5
TABLE-US-00103 TABLE 102 Tolerability scores in rats at 3 mg dose Compound FOB No. 3 hr PBS 0.0 750413 0.3 1065669 2.8 1065680 4.0
1065710 5.5 1065754 5.3 1065785 6.0 1065795 3.3 1065799 2.0 1065812 4.0 1065826 2.0
TABLE-US-00104 TABLE 103 Tolerability scores in rats at 3 mg dose Compound FOB No. 3 hr PBS 0.0 1065920 3.5 1065937 3.8 1066092 3.3
1066221 4.0 1066350 2.3
TABLE-US-00105 TABLE 104 Tolerability scores in rats at 3 mg dose Compound FOB No. 3 hr PBS 0.3 1263408 2.0 1263409 2.5 1263410 2.5
1263411 1.3 1263412 3.0 1263419 1.8 1263429 2.5 1263434 2.0
TABLE-US-00106 TABLE 105 Tolerability scores in rats at 3 mg dose Compound FOB No. 3 hr PBS 0.3 1263435 3.0 1263441 2.3* 1263450 2.0
1263451 4.0 1263452 2.0 1263453 1.8 1263454 1.5 1263455 3.3 1263458 0.3 *One of the four rats in this group died during surgery, and so is
excluded from this value.
TABLE-US-00107 TABLE 106 Tolerability scores in rats at 3 mg dose Compound FOB No. 3 hr PBS 0.5 1263476 0.3 1263478 1.0 1263488 3.0
1263489 2.3 1263507 2.0 1263517 1.3 1263518 1.8 1263519 1.0 1263521 2.0 1263533 1.3
TABLE-US-00108 TABLE 107 Tolerability scores in rats at 3 mg dose Compound FOB No. 3 hr PBS 0.0 1165563 3.0 1179841 3.0 1179843 3.3
1263407 3.8 1263413 4.0 1263430 4.3 1263436 2.8 1263540 2.5 1272944 2.0 1272994 1.0 1272996 1.0
TABLE-US-00109 TABLE 108 Tolerability scores in rats at 3 mg dose Compound FOB No. 3 hr PBS 0.0 1272998 0.3 1273034 0.0 1273035 0.8
1273036 0.0 1273049 0.3 1273051 0.0 1273055 2.0 1273056 3.7 1273057 1.0 1273058 0.8 1273062 2.5 1273065 2.3 1273087 2.0 1273099 2.5
1273102 4.5 1273106 2.0 1273107 2.0 1273108 2.0 1273113 1.0 1273115 3.3 1165523 5.0 1165550 3.0
TABLE-US-00110 TABLE 109 Tolerability scores in rats at 3 mg dose Compound FOB No. 3 hr PBS 0.0 1263426 2.3 1263462 2.8 1272974 3.8
1272997 0.8 1273020 0.5 1273033 1.8 1273038 3.0 1273039 3.3 1273042 2.0 1273048 3.0 1273104 3.0 1273105 3.0 1273109 3.8 1273110 1.0
1273111 2.7 1065593 1.5 1263547 4.0 1165586 0.0 1263537 1.5
TABLE-US-00111 TABLE 110 Tolerability scores in rats at 3 mg dose Compound FOB No. 3 hr PBS 0.0 749953 4.5 1065711 3.5 1263417 6.3

1263418 3.3 1263421 5.8 1263443 5.0 1263456 5.0 1263529 4.5 1263538 4.7 1273030 0.0 1273047 4.0 1273085 4.0

TABLE-US-00112 TABLE 111 Tolerability scores in rats at 3 mg dose Compound FOB No. 3 hr PBS 0.0 749991 1.3 1263460 3.0 1263461 2.0 1263486 3.0 1263532 1.5 1263551 4.0 1273050 1.0 1273090 1.0 1273091 2.0 1273101 3.5 1263522 1.7

Example 12: Tolerability of Modified Oligonucleotides Complementary to Human UBE3A-ATS in Wild-Type Mice, Two Week Study

[0483] Modified oligonucleotides described above were tested in wild-type female C57/B16 mice to assess the tolerability of the oligonucleotides. Wild-type female C57/B16 mice each received a single ICV dose of 700 µg of modified oligonucleotide listed in the table below. Each treatment group consisted of 4 mice. A group of 4 mice received PBS as a negative control for each experiment (identified in separate tables below). At 2 weeks post-injection, mice were evaluated according to seven different criteria. The criteria are (1) the mouse was bright, alert, and responsive; (2) the mouse was standing or hunched without stimuli; (3) the mouse showed any movement without stimuli; (4) the mouse demonstrated forward movement after it was lifted; (5) the mouse demonstrated any movement after it was lifted; (6) the mouse responded to tail pinching; (7) regular breathing. For each of the 7 criteria, a mouse was given a subscore of 0 if it met the criteria and 1 if it did not (the functional observational battery score or FOB). After all 7 criteria were evaluated, the scores were summed for each mouse and averaged within each treatment group. The results are presented in the table below.

TABLE-US-00113 TABLE 112 Tolerability scores in mice at 700 µg dose Compound 2-week Number FOB PBS 0.0 1065645 0.0 1263517 0.0 1263518 0.0 1263533 0.0 1273039 0.0 1273062 0.0

Example 13: Activity of Modified Oligonucleotides Complementary to Human UBE3A-ATS in Transgenic Mice

[0484] Modified oligonucleotides described above were tested in a UBE3A-ATS BAC transgenic mouse model.

[0485] The transgenic mouse model was developed at the University of Michigan Bacterial Artificial Chromosome Recombineering Core and contains region 25163935-25348867 from human chromosome 15 (GRCh38/hg38 assembly) (from BAC clone RP11-664B13). UBE3A-ATS expression is driven by the ENO2 promoter and terminated with the BGH poly(A) signal. The gene fragment was introduced into fertilized eggs from C57BL/6 mice by pronuclear injection to produce line RP 11-748 that is used in the experiments described below.

Treatment

[0486] The UBE3A-ATS transgenic mice were divided into groups of 2-4 mice each. Each mouse received a single ICV bolus of 350 µg of modified oligonucleotide. A group of 4 mice received PBS as a negative control.

RNA Analysis

[0487] After two weeks, mice were sacrificed and RNA was extracted from cortical brain tissue, hippocampal brain tissue, and spinal cord for real-time PCR analysis of measurement of RNA expression of UBE3A-ATS using primer probe set RTS4796 (described herein above) and/or primer probe set RTS40595 (forward sequence TCCTTCCCTACCTTAGTCTTGA, designated herein as SEQ ID NO: 14; reverse sequence CCCTCTTGAACCAGAAACA, designated herein as SEQ ID NO: 15; probe sequence AGATGGCAGCCCACATTTCTACTGT, designated herein as SEQ ID NO: 16). Primer probe set RTS4796 is less reliable when the oligonucleotide binds at a site far from the primer probe binding site. Results are presented as percent change of RNA, relative to PBS control, normalized to mouse GAPDH. Mouse GAPDH was amplified using primer probe set RTS108 (forward sequence GGCAAATTCAACGGCACAGT, designated herein as SEQ ID NO: 11; reverse sequence GGGTCTCGTCTCTGGAAGAT, designated herein as SEQ ID NO: 12; probe sequence AAGGCCGAGAATGGGAAGCTTGTCATC, designated herein as SEQ ID NO: 13). In some cases, RTPCR value is not defined for a certain sample, and is labeled N.D. (Not Defined).

[0488] As shown in the table below, treatment with modified oligonucleotides resulted in reduction of UBE3A-ATS RNA in comparison to the PBS control.

TABLE-US-00114 TABLE 113 Reduction of human UBE3A-ATS RNA in transgenic mice UBE3A-ATS RNA UBE3A-ATS RNA (% control)

RTS4796 (% control)	RTS40595	Compound	SPINAL	SPINAL ID	CORD	HIPPOCAMPUS	CORTEX	CORD	HIPPOCAMPUS	CORTEX	PBS																		
100	100	100	100	100	749860	14	N.D.	26	12	24	23	749861	27	N.D.	51	26	48	49	749863	39	N.D.	57	36	45	54	749865	61		
N.D.	92	66	97	85	749869	31	37	51	29	38	47	749882	31	28	35	30	30	32	749885	59	65	85	61	68	83	749893	28	35	
44	27	35	40	749894	5	7	9	3	6	7	749907	53	48	70	51	47	63	749931	35	24	33	15	22	28	749969	17	14	14	11
13	10	749991	27	26	38	22	24	31	750006	24	21	24	19	19	18	750028	26	22	34	17	19	26	750030	40	30	44	32	26	
34	750032	17	9	12	10	6	6	750040	55	42	65	50	44	59	750051	35	27	35	27	26	29	750092	55	43	63	47	40	53	
750139	24	22	30	17	20	23	750140	22	16	22	15	13	15	750214	48	46	60	45	41	55	750292	27	35	40	21	31	34		
750312	59	48	47	53	43	22	750359	30	23	25	25	19	20	750360	44	40	54	39	37	47	750365	58	69	88	56	65	82		
750386	34	24	27	26	19	19	750413	45	37	38	36	34	30	750430	48	42	41	41	42	33	750439	53	46	52	34	44	46		
1065296	34	34	39	31	34	34	1065578	35	42	49	34	42	42	1065579	19	18	19	17	18	14	1065586	45	55	74	43	55	68		
1065590	43	53	67	42	51	63	1065591	21	24	29	19	21	25	1065595	25	34	49	13	30	41	1065599	19	22	27	18	22	24		
1065600	31	35	42	32	35	39	1065605	44	64	67	44	62	63	1065607	28	36	39	29	33	34	1065608	18	42	53	18	38	49		
1065623	16	28	34	15	29	30	1065635	32	39	41	30	38	36	1065641	32	55	50	30	49	44	1065642	43	58	65	42	52	59		
1065645	18	30	24	15	24	18	1065646	39	60	50	39	54	45	1065651	29	47	38	28	43	34	1065667	23	30	26	19	24	22		
1065671	21	29	30	18	25	26	1065685	71	76	82	40	68	71	1065690	12	16	20	9	12	15	1065708	28	38	33	24	31	27		
1065710	29	34	32	27	29	26	1065713	31	34	40	29	29	34	1065765	25	27	33	21	22	28	1065813	38	49	67	34	41	62		
1065823	37	33	36	30	24	28	1065858	23	25	21	15	14	13	1065859	29	20	22	19	11	13	1065863	52	45	39	39	34	29		
1065868	22	19	19	14	13	12	1065920	47	60	56	45	51	44	1065932	28	33	35	23	25	26	1065947	34	35	40	28	25	30		
1066221	44	74	76	43	69	69	1066420	55	74	71	52	71	61	1066423	43	42	39	37	35	32	1066429	75	74	83	70	65	72		

TABLE-US-00115 TABLE 114 Reduction of human UBE3A-ATS RNA in transgenic mice UBE3A-ATS RNA UBE3A-ATS RNA (% control)

RTS4796 (% control)	RTS40595	Compound	SPINAL	SPINAL ID	CORD	HIPPOCAMPUS	CORTEX	CORD	HIPPOCAMPUS	CORTEX	PBS																		
100	100	100	100	100	1065272	40	29	39	37	31	39	1065582	45	28	52	43	31	51	1065616	38	25	45	36	25	42	1065619	33	23	
54	31	27	55	1065621	42	33	57	43	41	54	1065624	20	40	62	20	42	61	1065631	24	32	71	24	34	65	1065669	38	33	89	34
30	72	1065674	37	36	111	35	34	101	1065678	33	36	105	32	35	96	1065680	43	61	134	40	59	121	1065686	21	25	63	19	22	
55	1065696	44	36	117	40	32	108	1065719	63	36	122	56	32	109	1065728	20	11	53	14	9	42	1065754	47	25	112	41	24	105	
1065766	29	41	110	24	41	98	1065768	40	41	99	32	38	84	1065799	44	43	104	38	40	99	1065812	19	15	35	12	12	26		
1065817	29	23	60	22	20	52	1065821	23	19	42	16	17	34	1065829	36	19	80	27	18	58	1065840	27	15	61	21	14	45		
1065856	22	19	44	14	13	30	1065857	35	28	58	26	21	38	1065899	37	30	77	30	24	58	1065902	21	17	55	17	14	41		
1065937	22	12	91	17	11	65	1065953	19	13	58	13	10	41	1065955	34	19	69	26	16	53	1066046	58	35	122	50	32	106		
1066076	46	27	76	41	25	N.D.	1066092	27	15	56	21	13	47	1066217	37	26	68	32	26	62	1066253	27	15	50	22	14	39		
1066377	50	30	79	40	28	73																							

TABLE-US-00116 TABLE 115 Reduction of human UBE3A-ATS RNA in transgenic mice UBE3A-ATS RNA (% control) RTS4796 Compound

SPINAL	ID	CORD	HIPPOCAMPUS	CORTEX	PBS																																																						
100	100	100	1263473	43	71	66	1263474	8	15	10	1263475	6	10	8	1263476	6	12	14																																									
1263477	4	10	11	1263478	13	18	17	1263486	27	27	32	1263487	31	43	43	1263488	23	24	14	1263489	16	17	16	1263490	24	28	23	1263491	22	25	21																												
1263492	29	50	58	1263504	9	11	12	1263505	6	11	7	1263506	5	10	6	1263507	5	14	6	1263508	7	15	9	1263509	6	17	7	1263510	37	70	72																												
1263517	13	22	17	1263518	19	36	30	1263519	20	25	15	1263520	15	19	11	1263521	20	44	20	1263522	22	29	35	1263523	21	30	32	1263532	29	49	40																												
1263533	18	24	17	1263534	27	52	37	1263536	30	45	36	1263537	29	43	34	1263539	51	59	88	1263540	22*	21*	29*	1263541	52	66	80	1263543	49	362	84	1263544	53	100	104	1263545	36	54	57	1263546	45	45	88	1263547	26	28	30	1263548	36	45	47	1263549	66	87	84	1263550	47	75	76

1263551 30 45 39 1263552 66 65 93 1263554 72 75 88 1263556 60 59 83 1263557 86 91 109 1263400 29 70 46 1263409 29 44 33 1263410 29 64 42 1263411 42 72 58 1263419 47 126 65 1263420 73 130 72 1263429 27 33 36 1263434 44 50 42 1263435 47 42 61 1263441 22 17 29 1263451 21 16 28 1263452 14 8 15 1263453 13 9 9 *Only 1 animal in group

TABLE-US-00117 TABLE 116 Reduction of human UBE3A-ATS RNA in transgenic mice UBE3A-ATS RNA (% control) Compound SPINAL ID CORD HIPPOCAMPUS CORTEX PBS 100 100 100 1263454 6 8 13 1263455 5 19 35 1263458 39 62 88 1263459 28 45 52 1263460 11 28 34 1263461 24 30 35 1263462 25 24 25 1263463 36 40 66 1263464 25 22 38 1263465 32 35 39 1065786 121 75 32 1165526 116 78 46 1272943 41 73 32 1272946 37 69 33 1272947 55 94 54 1272951 70 89 75 1272952 125 85 109 1272953 86 78 51 1272954 110 89 70 1272955 137 136 70 1272960 59 77 40 1272962 45 52 26 1272964 74 58 62 1272965 43 39 25 1272966 52 68 67 1272967 65 69 62 1272968 76 62 66 1272969 83 60 147 1272970 143 57 60 1272971 81 58 48 1272972 74 69 38 1272974 40 29 18 1272975 124 86 96 1272976 93 62 50 1272977 58 59 47 1272978 39 46 40 1272979 61 69 46 1272982 60 58 42 1272983 84 83 70 1272985 72 56 38 1272987 181 37 28 1272988 66 34 31 1272989 87 60 68 1272992 77 45 42 1272993 57 36 60 1272994 16 12 10 1272995 182 35 93 1272996 26 9 18 1272997 24 12 27 1272998 26 14 17 1273000 104 45 56 1273003 93 28 60 1273009 39 8 19 1273011 145 78 80 1273012 110 36 71 1273014 46 19 37 1273015 79 35 68 1273016 58 34 37 1273017 46 39 56 1273018 41 24 60 1273019 98 39 80 1273020 44 10 19 1273021 59 27 47 1273022 73 23 45 1273023 54 23 31 1273024 29 21 25 1273028 46 50 28 1273031 45 17 28 1273032 40 21 35 1273033 36 12 26 699781 66 25 37 1065707 78 44 70 1165621 107 41 71

TABLE-US-00118 TABLE 117 Reduction of human UBE3A-ATS RNA in transgenic mice UBE3A-ATS RNA (% control) Compound SPINAL ID CORD HIPPOCAMPUS PBS 100 100 1273034 25 23 1273035 16 20 1273036 24 26 1273061 33 22 1273062 28 10 1273063 32 29 1273064 38 25 1273065 27 16 1273066 34 36 1273067 44 42 1273070 25 23 1273071 57 71 1273072 38 48 1273037 12 25 1273038 16 29 1273039 15 29 1273040 39 33 1273041 48 29 1273042 51 17 1273044 45 27 1273045 48 24 1273048 30 18 1273049 19 15 1273050 30 31 1273051 17 21 1273052 14 15 1273055 14 20 1273056 16 22 1273057 16 18 1273058 20 22 1273059 16 17 1273060 36 35 1273084 41 35 1273087 12 14 1273088 33 28 1273090 24 41 1273091 26 31 1273092 37 38 1273093 28 23 1273098 17 23 1273099 14 10 1273101 29 32 1273102 20 18 1273103 12 5 1273104 57 16 1273105 34 9 1273106 30 11 1273107 24 8 1273108 31 13 1273109 30 13 1273110 37 8 1273111 46 12 1273113 25 10 1273114 46 26 1273115 15 7 1165523 38 17 1165533 31 19 1165538 64 25 1165550 47 11 1165563 30 12 1165586 35 19 1165596 69 49 1165616 109 25 1165855 49 16 1165897 39 28 1179808 83 36 1179841 22 9 1179843 22 11

Example 14: Activity of Modified Oligonucleotides Complementary to Human UBE3A-ATS in Transgenic Mice, Multiple Doses

[0489] Modified oligonucleotides described above were tested in the UBE3A-ATS BAC transgenic mouse model RP11-748 line.

Treatment

[0490] The UBE3A-ATS transgenic mice were divided into groups of 3 mice each. Each mouse received a single ICV bolus of 10, 30, 100, 300, or 700 µg of modified oligonucleotide, and sacrificed two weeks later. A group of 8 mice received PBS as a negative control.

RNA Analysis

[0491] After two weeks, mice were sacrificed, and RNA was extracted from cortical brain tissue, hippocampal brain tissue, and spinal cords for real-time PCR analysis of measurement of RNA expression of UBE3A-ATS using primer probe set RTS40595 (forward sequence TCCTTCCCTACCTTAGTCTTGA, designated herein as SEQ ID NO: 14; reverse sequence CCCTCTTGAACCAGGAAACA, designated herein as SEQ ID NO: 15; probe sequence AGATGGCAGCCCACATTTCTACTGT, designated herein as SEQ ID NO: 16). Results are presented as percent change of RNA, relative to PBS control, normalized to mouse GAPDH (measured by primer-probe set RTS108).

[0492] As shown in the table below, treatment with modified oligonucleotides resulted in dose-dependent reduction of UBE3A-ATS RNA in comparison to the PBS control.

TABLE-US-00119 TABLE 118 Dose-dependent percent reduction of human UBE3A-ATS RNA in transgenic mice Cortex Hippocampus Spinal Cord UBE3A- UBE3A- UBE3A- Compound Dose ATS RNA ED50 ATS RNA ED50 ATS RNA ED50 ID (µg) (% control) (µg) (% control) (µg) (% control) (µg) PBS 0 100 — 100 — 100 — 749860 10 98.16531 138 81.99485 109 59.65406 17 30 84.95562 91.16468 45.18495 100 67.32992 52.07824 25.79271 300 26.23195 33.3197 10.36557 700 11.97921 16.72183 9.517061 1065645 10 90.14288 74 88.23692 71 37.14751 6 30 72.43123 66.81222 16.11173 100 49.12943 51.83015 10.87515 300 16.34235 25.55912 7.843655 1263461 10 88.27412 94 86.23292 126 60.06681 14 30 81.19946 84.36883 33.7366 100 55.94376 68.69458 21.85816 300 16.25568 23.08811 28.43356 700 11.55421 20.98394 18.26012 1263486 10 84.88829 91 88.16814 110 80.05438 49 30 73.65319 88.23227 67.24883 100 53.51549 46.18175 35.76668 300 29.17494 39.41106 25.80888 700 13.16333 21.58236 14.67831 1263517 10 61.99356 27 78.62965 38 28.43223 5 30 57.35782 64.77742 4.093424 100 28.22543 31.27599 3.855931 300 13.48909 11.49965 2.253222 700 6.218008 9.114471 0.845911 1263518 10 83.49507 47 74.9207 36 7.813722 N.C. 30 63.00462 58.90944 6.925703 100 36.83505 35.79351 9.984906 300 10.60636 16.33499 22.15675 1263532 10 87.76102 437 104.2908 508 85.92361 126 30 93.21178 107.4279 86.97925 100 82.08274 79.80543 56.10128 300 63.11973 71.02516 37.9068 1263533 10 85.64712 62 91.08657 93 79.33594 35 30 74.82061 92.99996 56.07265 100 37.83251 50.48182 32.95345 300 17.85169 15.11169 16.13775 700 13.22447 16.26845 14.77717 1263537 10 99.27557 167 101.6452 149 58.69295 32 30 61.11154 67.12889 57.56289 100 70.65222 60.23925 35.5374 300 35.35647 38.67746 31.30235 700 37.55784 42.24916 26.47357 1272944 10 85.26636 171 98.97738 241 86.06921 66 30 83.3624 91.08809 60.22092 100 64.40395 69.68625 45.39798 300 47.00232 54.90247 34.02307 700 18.19851 26.34956 26.27892 1273033 10 92.81903 159 104.9703 202 71.92708 27 30 86.45458 92.23621 47.44761 100 71.94214 74.17289 34.27954 300 28.94805 34.12991 17.19729 700 19.14991 35.15452 16.60517 1273039 10 84.62214 63 93.89685 77 52.24285 33 30 70.19074 70.45983 45.93645 100 40.06878 45.44749 30.60229 300 22.18407 28.77613 27.68067 700 16.46246 23.35814 20.2156 1273050 10 60.65157 34 76.1426 178 108.4293 45 30 46.72804 76.65907 59.53868 100 42.13233 63.21802 30.24442 300 35.42049 42.84989 16.76795 700 27.88596 44.42969 16.65603 1273055 10 54.09476 104 101.4753 139 95.64317 46 30 75.60993 80.40454 46.396 100 62.55598 61.10855 43.64926 300 37.44794 36.83808 26.42752 700 18.68964 24.28962 17.02671 1273062 10 83.90397 65 91.25815 65 96.79799 85 30 71.94265 68.4521 78.51707 100 41.74674 43.10219 45.07505 300 23.62598 26.78065 25.9929 700 8.048482 17.65158 20.9903 1273090 10 51.37967 25 83.05922 143 94.48769 120 30 53.46583 70.46917 79.90936 100 41.09529 69.46586 53.85753 300 25.47022 49.33954 51.06804 1273091 10 48.66948 28 86.34518 195 126.1334 240 30 49.31716 78.33165 118.9585 100 48.19272 67.14459 75.20145 300 31.19941 48.97407 42.91308 700 28.63277 34.46767 43.51426

Example 15: Tolerability of Modified Oligonucleotides Complementary to Human UBE3a-ATS in Wild-Type Mice. 3 Hour and 2 Week Study

[0493] Modified oligonucleotides closely matched to LNA/DNA oligonucleotides described in WO2017/081223 were tested in female wild-type C57/B316 mice to assess the tolerability of the oligonucleotides. See, Certain Comparator Compounds, hereinabove. Wild-type female C57/1B16 mice each received a single ICV dose of 700 g of modified oligonucleotide listed in the table below. Each treatment group consisted of 4 mice. A group of 4 mice received PBS as a negative control for the experiment. At 3 hours post-injection and 2 weeks post-injection, mice were evaluated according to seven different criteria. The criteria are (1) the mouse was bright, alert, and responsive; (2) the mouse was standing or hunched without stimuli; (3) the mouse showed any movement without stimuli; (4) the mouse demonstrated forward movement after it was lifted; (5) the mouse demonstrated any movement after it was lifted; (6) the mouse responded to tail pinching; (7) regular breathing. For each of the 7 criteria, a mouse was given a subscore of 0 if it met the criteria and 1 if it did not (the functional observational battery score or FOB). After all 7 criteria were evaluated, the scores were summed for each mouse and averaged within each treatment group. The results are presented in the table below.

TABLE-US-00120 TABLE 119 Tolerability scores in mice at 700 µg dose SEQ ID SEQ NO: 1 ID NO: Chemistry SEQ Compound 3 hr 2 week Start 1 Stop Notation ID ID FOB FOB Site Site (5' to 3') NO PBS 0.0 0.0 1219022 7.0 n.d. 468275 468294 A.sub.lsA.sub.lsA.sub.dsT.sub.dST.sub.ls 2905 A.sub.dsT.sub.lsT.sub.dsT.sub.dsA.sub.dS T.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ds .sup.mC.sub.dsA.sub.lsT.sub.ls.sup.mC.sub.lsA.sub.l 1219023 1.0 3.5 462549 462566 T.sub.lsT.sub.lsT.sub.lsA.sub.dsT.sub.ds 2906 .sup.mC.sub.lsA.sub.dsA.sub.dsT.sub.dsA.sub.ds T.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ds T.sub.ls.sup.mC.sub.lsA.sub.l 1219024 5.3

6.5 471876 471894 G.sub.ls.sup.mC.sub.dsA.sub.ls.sup.mC.sub.dsA.sub.ls 2907 T.sub.dST.sub.ds.sup.mC.sub.dsT.sub.dsT.sub.ds
T.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.dsT.sub.ds A.sub.ds.sup.mC.sub.ds.sup.mC.sub.lsT.sub.l 1219025 4.0 6.0 461987 462003
T.sub.lsT.sub.lsA.sub.dsT.sub.lsA.sub.ls 2908 G.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsA.sub.ds T.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.ds
T.sub.ds.sup.mC.sub.lsT.sub.l 1219026 1.0 6.0 487371 487387 .sup.mC.sub.lsT.sub.ds.sup.mC.sub.lsA.sub.ls 2909
A.sub.lsA.sub.dsG.sub.dsA.sub.dST.sub.dS .sup.mC.sub.dsA.sub.dsT.sub.dsT.sub.ds .sup.mC.sub.lsT.sub.ls.sup.mC.sub.lsA.sub.l 1219027 1.0 0.0
487430 487449 T.sub.lsT.sub.dsA.sub.ls.sup.mC.sub.dsA.sub.ls 2910 .sup.mC.sub.dsT.sub.lsT.sub.dsA.sub.dsA.sub.ds
T.sub.dsT.sub.dsA.sub.dsT.sub.dsA.sub.ds .sup.mC.sub.dsT1ST.sub.ds.sup.mC.sub.ls.sup.mC.sub.l 1219028 6.0 6.0 496570 496587
G.sub.lsT.sub.dsT.sub.dsT.sub.ds.sup.mC.sub.ds 2911 .sup.mC.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.dsT.sub.ds
A.sub.ds.sup.mC.sub.dsT.sub.lsA.sub.lsT.sub.ds T.sub.lsA.sub.lsA.sub.l 1219029 6.3 6.0 503154 503170
.sup.mC.sub.lsT.sub.lsG.sub.dST.sub.dsA.sub.ls 2912 T.sub.dsA.sub.ds.sup.mC.sub.dsA.sub.ds.sup.mC.sub.ds
.sup.mC.sub.dsA.sub.dsT.sub.ds.sup.mC.sub.ds.sup.mC.sub.d .sub.s.sup.mC.sub.lsA.sub.l 1219030 5.0 5.0 487478 487494
A.sub.lsG.sub.dsT.sub.dsT.sub.ls.sup.mC.sub.ls 2913 T.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.dsA.sub.ds
T.sub.dsA.sub.ds.sup.mC.sub.dsT.sub.lsT.sub.ls T.sub.ls.sup.mC.sub.l 1219031 5.3 6.0 464879 464898
T.sub.lsA.sub.lsT.sub.dsA.sub.ds.sup.mC.sub.ls 2914 .sup.mC.sub.dsT.sub.lsT.sub.dsT.sub.ds.sup.mC.sub.ds
T.sub.dsT.sub.dsT.sub.dsA.sub.dsA.sub.ds .sup.mC.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.lsT.sub.l A subscript “l” indicates a 4’-2’ LNA

modified sugar moiety, a subscript “d” indicates a 2’-β-D-deoxyribosyl sugar moiety, a subscript “s” indicates a phosphorothioate internucleoside linkage, and a superscript “m” before a C indicates 5-methyl cytosine. n.d. means no data; not tested.

Example 16: Tolerability of Modified Oligonucleotides Complementary to Human UBE3a-ATS in Rats, 8 Week Study

[0494] Modified oligonucleotides described above were tested in rats to assess the tolerability of the oligonucleotides. Groups of male, 6-8 week old Sprague Dawley rats each received a single intrathecal (IT) dose of 3 mg of oligonucleotide as listed in the table below. Each treatment group consisted of 4 rats, approximately matched for starting weights. A group of four rats received PBS as a negative control. The rats were weighed before dosing, and were weighed again at 8 weeks post-dose. Rats are expected to gain weight over the course of the study due to growth; too little weight gain is a sign of compound toxicity. Absolute values of body weight change were normalized to the body weight change observed in PBS-treated groups to allow comparison between two studies with different average starting weights for the rats, with the body weight change of the PBS group set to 100%. Statistical significance (p-value) of the change in body weight compared to the PBS-treated group was calculated by a two-tailed Welch’s T-test in Excel (two sample, unequal variance). A p-value of <0.05 indicates that there is less than a 5% chance that the observed differences are due to random sampling errors.

TABLE-US-00121 TABLE 120 Body Weight Change in Rats at 8 weeks post-dose Compound B.W. change ID as % PBS p-value PBS 100 n/a
1219027 87 0.03 1065645 101 0.83 1263517 100 0.91 1263518 103 0.50 1263533 100 0.99 1273039 105 0.26 1273062 100 0.98

Claims

1.-89. (canceled)

90. A modified oligonucleotide according to the following chemical structure: ##STR00031##

91. The modified oligonucleotide of claim 90 which is the sodium salt or the potassium salt.

92. A modified oligonucleotide corresponding to the following chemical structure: ##STR00032##

93. A compound comprising a modified oligonucleotide according to the following chemical notation:

T.sub.es.sup.mC.sub.eoA.sub.eo.sup.mC.sub.eoA.sub.eoT.sub.dsT.sub.dsT.sub.dsG.sub.dsA.sub.ds.sup.mC.sub.ds.sup.mC.sub.dsT.sub.ds
(SEQ ID NO: 2918), wherein: A=an adenine nucleobase, mC=a 5-methyl cytosine nucleobase, G=a guanine nucleobase, T=a thymine nucleobase, e=a 2’-O(CH.sub.2).sub.2OCH.sub.3 β-D-ribose sugar moiety, d=a 2’-β-D-deoxyribose sugar moiety, s=a phosphorothioate internucleoside linkage, and o=a phosphodiester internucleoside linkage.

94. The compound of claim 93, comprising the modified oligonucleotide covalently linked to a conjugate group.

95. A pharmaceutical composition comprising the modified oligonucleotide of claim 90 and a pharmaceutically acceptable carrier or diluent.

96. The pharmaceutical composition of claim 95, wherein the pharmaceutically acceptable diluent is artificial cerebrospinal fluid.

97. The pharmaceutical composition of claim 96, wherein the pharmaceutical composition consists essentially of the modified oligonucleotide and artificial cerebrospinal fluid.

98. A pharmaceutical composition comprising the modified oligonucleotide of claim 92 and a pharmaceutically acceptable carrier or diluent.

99. The pharmaceutical composition of claim 98, wherein the pharmaceutically acceptable diluent is artificial cerebrospinal fluid.

100. The pharmaceutical composition of claim 99, wherein the pharmaceutical composition consists essentially of the modified oligonucleotide and artificial cerebrospinal fluid.

101. A pharmaceutical composition comprising the compound of claim 93 and a pharmaceutically acceptable diluent or carrier.

102. The pharmaceutical composition of claim 101, wherein the pharmaceutically acceptable diluent is artificial cerebrospinal fluid.

103. The pharmaceutical composition of claim 102, wherein the pharmaceutical composition consists essentially of the compound and artificial cerebrospinal fluid.

104. A pharmaceutical composition comprising the compound of claim 94 and a pharmaceutically acceptable diluent or carrier.

105. The pharmaceutical composition of claim 104, wherein the pharmaceutically acceptable diluent is artificial cerebrospinal fluid.

106. The pharmaceutical composition of claim 105, wherein the pharmaceutical composition consists essentially of the compound and artificial cerebrospinal fluid.

107. A method comprising administering to a subject the pharmaceutical composition of claim 95.

108. A method of treating Angelman Syndrome, comprising administering to an individual having or at risk for developing Angelman Syndrome a therapeutically effective amount of the pharmaceutical composition of claim 95.
